Switching/protection devices

3VL molded-case circuit breakers

System Manual · 03/2009

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Switching/protection devices
3VL molded-case circuit breakers

System Manual

03/2009
110 0110 - 02 DS 01

Legal information

Warning notice system

This manual contains notices you have to observe in order to ensure your personal safety, as well as to prevent damage to property. The notices referring to your personal safety are highlighted in the manual by a safety alert symbol, notices referring only to property damage have no safety alert symbol. These notices shown below are graded according to the degree of danger.

⚠ DANGER
indicates that death or severe personal injury **will** result if proper precautions are not taken.

⚠ WARNING
indicates that death or severe personal injury **may** result if proper precautions are not taken.

⚠ CAUTION
with a safety alert symbol, indicates that minor personal injury can result if proper precautions are not taken.

CAUTION
without a safety alert symbol, indicates that property damage can result if proper precautions are not taken.

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indicates that an unintended result or situation can occur if the corresponding information is not taken into account.

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The device/system may only be set up and used in conjunction with this documentation. Commissioning and operation of a device/system may only be performed by **qualified personnel**. Within the context of the safety notes in this documentation qualified persons are defined as persons who are authorized to commission, ground and label devices, systems and circuits in accordance with established safety practices and standards.

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Disclaimer of Liability

We have reviewed the contents of this publication to ensure consistency with the hardware and software described. Since variance cannot be precluded entirely, we cannot guarantee full consistency. However, the information in this publication is reviewed regularly and any necessary corrections are included in subsequent editions.

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Ordernumber: 3ZX1012-0VL10-0AC1
℗ 03/2009

Table of contents

1 About this document

1.1 Introduction

Purpose of this manual

This manual is intended for reference purposes. The information in this manual enables you to configure and operate the SENTRON VL system.

Audience

This manual is aimed at people with the required qualifications to commission and operate the SENTRON VL system.

2 Product-specific information

2.1 Important notes

Validity

This manual applies to SENTRON circuit breakers with the following designations:

- VL160X
- VL160
- VL250
- VL400
- VL630
- VL800
- VL1250
- VL1600

Standards and certifications

The SENTRON VL circuit breakers comply with the standards:

- IEC 60947-2 / DIN EN 60947-2 (VDE 0660-101)
- IEC 60947-1 / DIN EN 60947-1 (VDE 0660-100)
- Isolating features in accordance with IEC 60947-3 / EN 60947-3 (VDE 0660-107)
- Network disconnection features for stopping and shutting down in an emergency (main switch and EMERGENCY-OFF switch) in accordance with IEC 60204-1 / DIN EN 60204-1 (VDE 0113-1)

Disclaimer of liability

The products described here were developed to perform safety-oriented functions as part of an overall installation or machine. A complete safety-oriented system generally features sensors, evaluation units, signaling units, and reliable shutdown concepts. It is the responsibility of the manufacturer to ensure that a system or machine is functioning properly as a whole. Siemens AG, its regional offices, and associated companies (hereinafter referred to as "Siemens") cannot guarantee all the properties of a whole plant or machine that has not been designed by Siemens.

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Technical Assistance: Telephone: +49 (0) 911-895-5900 (8°° - 17°° CET)

Fax: +49 (0) 911-895-5907

or on the Internet at:

e-mail: technical-assistance@siemens.com

Internet: www.siemens.de/lowvoltage/technical-assistance

Technical Support:

Telephone: +49 (0) 180 50 50 222

Correction sheet

A correction sheet is included at the end of the manual. Please use it to record your suggestions for improvements, additions and corrections, and return the sheet to us. This will help us to improve the next edition of the manual.

Product description 3

3.1 SENTRON VL overview

SENTRON VL circuit breakers are resistant to extreme climates. They are designed for use in closed rooms where no onerous operating conditions prevail (e.g. dust, caustic vapors, hazardous gases).

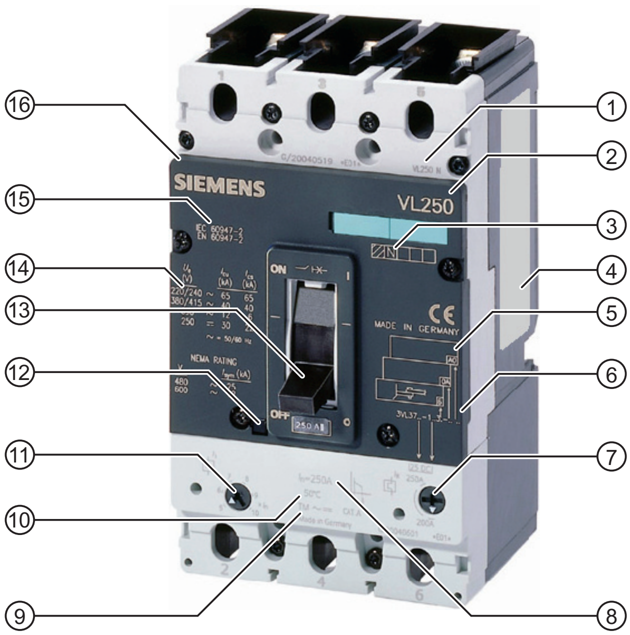
SENTRON VL types

The type designations of all available circuit breakers are oriented around the rated current

Type designation	Maximum rated current (A)
VL160X	160
VL160	160
VL250	250
VL400	400
VL630	630
VL800	800
VL1250	1250
VL1600	1600

Type plate and ID number

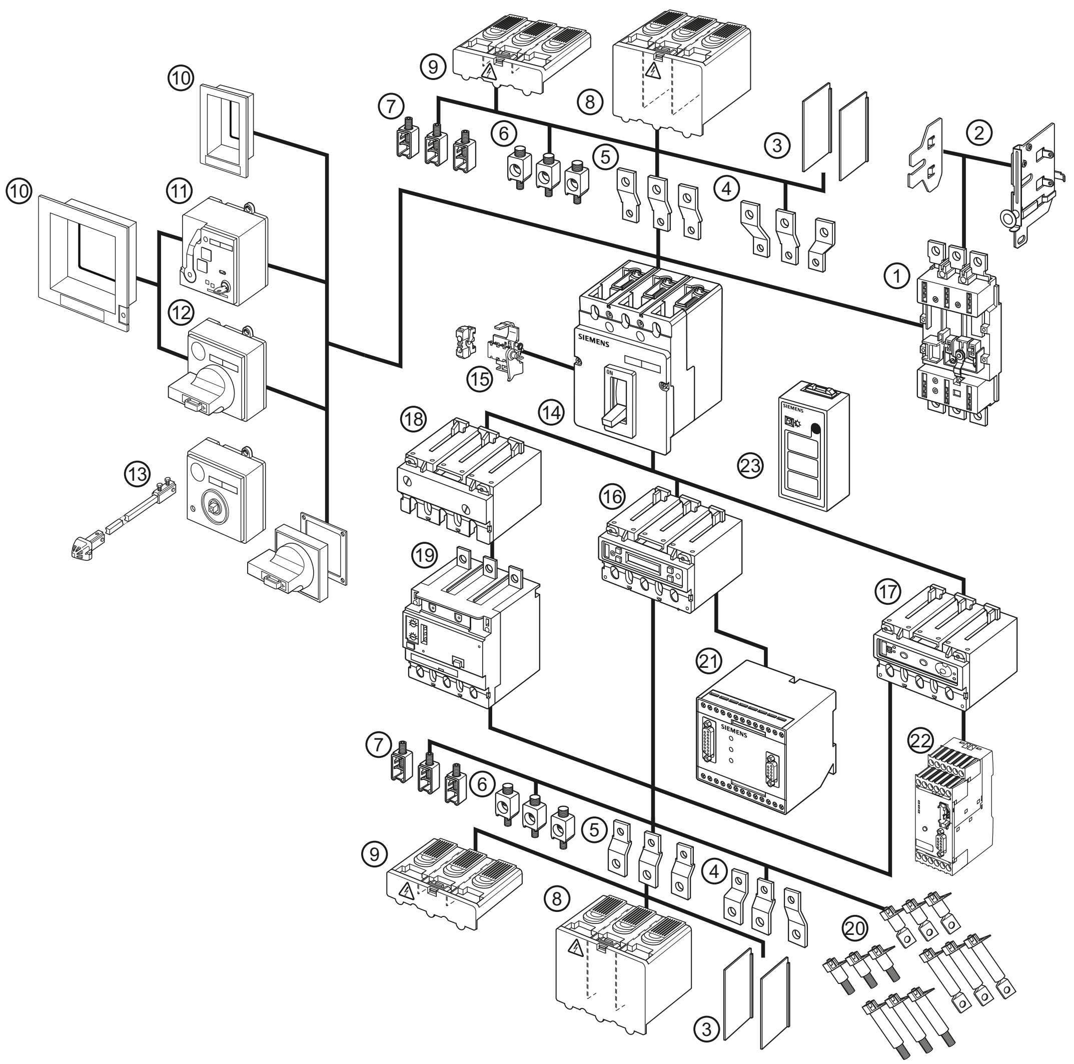
The figure shows all the operator elements, setting options and names corresponding to the precise specified use of the circuit breaker.



(1)	Specified frame size
(2)	Circuit breaker type
(3)	Display of switching capacity
(4)	Rating plate
(5)	Accessories ID fields
(6)	Catalog number (machine-readable product code)
(7)	Overcurrent adjustment
(8)	I_n rated current of the circuit breaker
(9)	Overcurrent tripping type TM (thermomagnetic)
(10)	Reference temperature
(11)	Short-circuit tripping/adjustment
(12)	Test key
(13)	Toggle handle with 3 positions
(14)	Switching capacity
(15)	Standards
(16)	Accessories cover (removable)

Figure 3-1 SENTRON VL circuit breakers - labeling and operator elements

SENTRON VL accessories



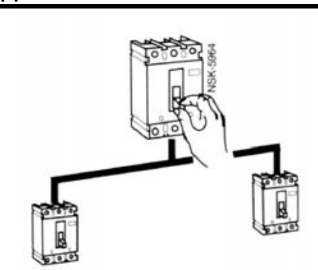
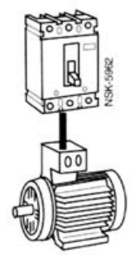
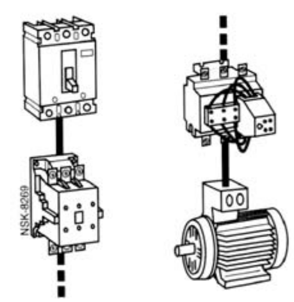
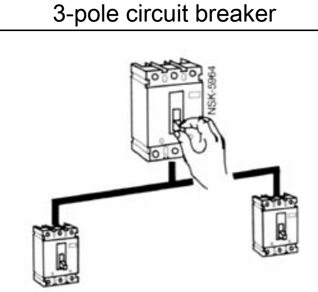
(1) Withdrawable/plug-in socket
(2) Withdrawable side panels
(3) Phase barriers
(4) Flared busbar extensions
(5) Straight connecting bars
(6) Multiple feed-in terminal for Al/Cu
(7) Box terminal for Cu
(8) Extended terminal cover
(9) Standard terminal cover
(10) Masking/cover frame for door cutout
(11) Stored-energy motorized operating mechanism
(12) Front rotary operating mechanism
(13) Door-coupling rotary operating mechanism
(14) SENTRON 3VL circuit breaker
(15) Internal accessories
(16) Electronic overcurrent trip unit LCD ETU
(17) Electronic overcurrent trip unit with communication function
(18) Thermal/magnetic overcurrent trip unit
(19) RCD module
(20) Rear terminals – flat and round
(21) COM10 communication module for PROFIBUS-DP
(22) COM20 communication module for PROFIBUS-DP
(23) Battery power supply with test function

Figure 3-2 SENTRON VL accessories

3.2 Application overview

The following overview shows the most frequently occurring applications

Application overview

Application	Type	Description
 3- and 4-pole circuit breakers	VL160X VL160 VL250 VL400 VL630 VL800 VL125 VL1600	**Line protection** The trip units for line protection are designed to protect cables and non-motorized loads against overload and short-circuit.
 3-pole circuit breaker	VL160 VL250 VL400 VL630	**Motor/generator protection** The overload and short-circuit releases are designed for optimum protection and direct-online starting of AC squirrel-cage motors. The circuit breakers for motor protection have phase-failure sensitivity and a thermal memory that protects the motor against overheating. The adjustable time lag class enables users to adjust the overload release to the startup conditions of the motor to be protected.
 3-pole circuit breaker	VL160 VL250 VL400 VL630	**Starter combination** Starter combinations consist of: Circuit breaker + contactor + overload relay. The circuit breaker handles short-circuit protection and the isolating function. The contactor has the task of switching the load feeder normally. The overload relay handles overload protection that can be specially matched to the motor. The circuit breaker for starter combination is therefore equipped with an adjustable and non-delayed short-circuit release.
 3- and 4-pole circuit breakers	VL160X VL160 VL250 VL400 VL630 VL800 VL1250 VL1600	**Non-automatic air circuit breakers** These circuit breakers are used as incoming circuit breakers, main switches or isolating switches without overload protection. They have fixed short-circuit releases so that back-up fuses are not necessary.

3.3 Configuration

3.3.1 Functional principle

Design - mechanical principle

All SENTRON VL circuit breakers have a trip-free mechanism that ensures the trip process is not prevented even if the operating mechanism is blocked or manually held in the "ON" position.

The contacts are opened and closed by a toggle handle positioned in the center. This is attached to the front side on all circuit breakers.

All SENTRON VL circuit breakers are "joint trip units". This means all contacts open or close simultaneously when the circuit breaker toggle handle is moved from "OFF" to "ON" or from "ON" to "OFF", or when the tripping mechanism is activated by an overcurrent or with the help of an auxiliary trip (shunt release or undervoltage release).

Current limiting

The SENTRON VL circuit breakers are designed on the principle of magnetic repulsion of the contacts. The contacts open before the expected peak-value of the short-circuit current is reached. Magnetic repulsion of the contacts very significantly reduces the thermal load I^2t as well as the mechanical load resulting from the impulse short-circuit current I_P of the system components that occur during a short-circuit.

You can find more information in Chapter Use in the motor protection area (Page 74).

3.3.2 Subdivision according to power ranges

VL160X circuit breakers

The most important components of the VL160X circuit breakers are the three current paths with the incoming and outgoing terminals. The fixed and movable contacts are arranged in such a way as to guarantee magnetic repulsion of the contacts. In conjunction with the arc splitter chambers, a dynamic impedance is created that causes a current limitation through the reduction in the harmful effects of I^2t and the I_p energy resulting from short-circuits.

The overcurrent trip unit is a thermomagnetic device installed at the factory. It is equipped with fixed or adjustable overload releases and a fixed short-circuit release in each pole.

To the right and left of the centrally positioned toggle handle of every SENTRON VL circuit breaker is a double-insulated accessories compartment for installing auxiliary switches or alarm switches as well as voltage and undervoltage releases.

VL160 to VL630 circuit breakers

The arrangement of current paths, contact configuration and switch mechanism of the VL160 to VL630 circuit breakers corresponds to that of the VL160X circuit breaker. The designs diverge with regard to the overcurrent trip unit.

- The overcurrent trip units are available in a thermomagnetic version and in an electronic version.
- The overcurrent trip units can be installed or replaced on-site without special tools.
- Thermomagnetic overcurrent trip units are available with adjustable overload releases and short-circuit releases.

VL800 bis VL1600 circuit breakers

The arrangement of the current paths and switch mechanisms is identical to that of the VL160X to VL630 circuit breakers.

However, the VL800 to VL1600 circuit breakers are only available in the version with electronic overcurrent trip unit. As with all electronic overcurrent trip units for the SENTRON VL circuit breakers from Siemens, the current transformers (one per phase) are accommodated within the overcurrent trip unit housing. They transmit a signal proportional to the load current to the electronic trip unit.

All SENTRON VL circuit breakers with electronic trip units measure the actual effective current. This method is the most accurate way of measuring currents in electrical distribution systems with extremely high harmonics.

3.3.3 Thermomagnetic overcurrent trip units

Thermomagnetic overcurrent trip units

A thermomagnetic overcurrent trip unit consists of two components - a thermal release for protecting against overload, and a magnetic release for protecting against short-circuit. Both trip unit components are switched in series.

The thermal trip unit

consists of a temperature-dependent bimetal that heats up as a result of the flow of current. This means the release is current-dependent. The heating of the bimetal strip depends on the ambient temperature of the circuit breaker. All current values specified for 3VL for thermomagnetic trip units refer to an ambient temperature of 40°C. Where ambient temperatures deviate from this, the values in the tables in Chapter Derating factors (Page 159) are to be used.

The magnetic trip unit

comprises a yoke mounting through which a current path runs, and a flap armature that is kept at a distance from the yoke mounting by a tension spring. If a short-circuit current now flows along the current path, the magnetic field thus generated causes the flap armature to be moved towards the yoke mounting against the opposite force of the tension spring. The release time is almost current-independent and instantaneous. The flap armature releases the switching lock and thus opens the switching contacts before the short-circuit current can reach its maximum; a current limiting effect is thus achieved. Immediately after release, the flap armature is moved back to its starting position by the opposite force of the tension spring.

3.3.4 Electronic overcurrent trip unit (ETU)

Electronic trip units (ETU)

In contrast to thermomagnetic trip units (TMTUs) where the overcurrent trip is unit caused by a bimetal strip or magnetic release, electronic overcurrent trip units (ETUs) use electronics with current transformers. The ETU captures the actual currents and compares them with the default specifications.

All SENTRON 3VL circuit breakers with electronic overcurrent trips measure the actual effective current (true RMS). This is the most accurate method of measuring.

ETUs are available from the VL160 circuit breaker up to and including the VL1600. The SENTRON VL800, VL1250 und VL1600 circuit breakers are only available in the version with electronic overcurrent trip unit.

The trip units can be replaced by the customer without special tools. Replacement is described precisely in the operating instructions included with the ETUs. After installing the electronic overcurrent trip unit in the relevant circuit breaker, the battery supply must be tested with test function 3VL9000-8AP00.

Configuration

The electronic overcurrent tripping system consists of:

- 3 to 4 (3-pole or 4-pole) current transformers that also provide their own power supply.

  This means an external auxiliary voltage is not required.
- Evaluation electronics with microprocessor
- Tripping solenoid

In all versions with electronic trip units for the SENTRON 3VL circuit breakers, the current transformers are located in the same housing as the trip unit. At the output of the electronic overcurrent tripping module, there is a tripping solenoid that releases the circuit breaker in the event of an overload or short-circuit.

Power supply

The protection functions of the electronic overcurrent trip unit are guaranteed without additional auxiliary voltage. The overcurrent trip units are supplied with energy via internal current transformers. The protection function is parameterized via rotary encoding switches on the ETU or via an LCD display. In the case of an LCD display, the electronic overcurrent trip unit must be activated. This requires a 3-phase (3-pole) load current of at least 20% or, in the case of a single-phase (single-pole) load, 30% of the relevant rated current of the circuit breaker. If this load current is not available, the necessary auxiliary energy can be supplied via a battery power supply (order no. 3VL9000-8AP00). With communication-capable circuit breakers, the trip unit is supplied with energy via the COM10/COM20 module.

4-pole circuit breakers

The four-pole circuit breakers for system protection can be supplied in all 4 poles with or without current transformers. The trip units in the 4th pole (N) can be set to 50% or 100% of the current in the 3 main current paths dependent on the frame size, so that safe protection of the neutral conductor can be guaranteed even with a reduced cross-section. In the case of LCD-ETUs, the neutral conductor protection can be adjusted in steps from 50% to 100% or switched off.

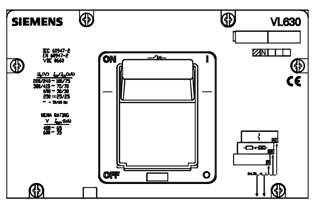
3.4 Mechanical operating mechanisms

3.4.1 Toggle handle operating mechanism

In the basic version, the SENTRON VL circuit breakers have a toggle handle as an operating mechanism. This also functions as an indicator of the switching position. The "Tripped" position is also displayed in addition to the "ON" and "OFF" positions.

The toggle handle goes to the "tripped" position when the internal trip mechanism is activated by an overcurrent situation, e.g. overload or short-circuit.

Activation by an undervoltage release or shunt release will also cause the toggle handle to move to the "Tripped" position.

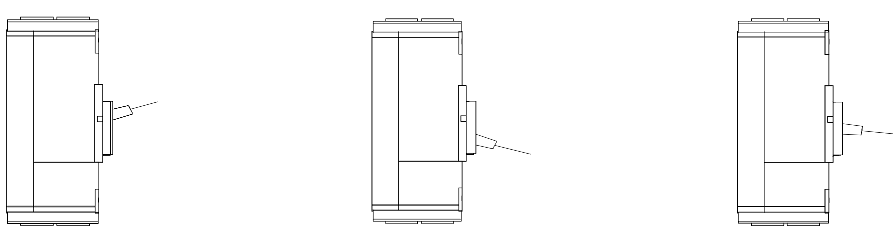


Toggle handle in the "ON" position

The toggle handle must be returned to the "OFF/RESET" position before the circuit breaker can be turned back on again. This enables the internal release mechanism to be reset. SENTRON VL circuit breakers with toggle handle operation comply with the "Network disconnecting device" condition (5.3.2 Section c) and 5.3.3) according to DIN EN 60204-1 (VDE 0113-1).

Toggle handle positions

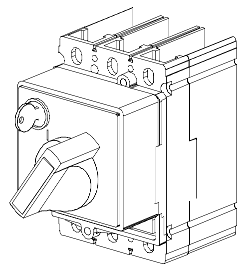
ON | OFF RESET | Tripped



Toggle handle positions

3.4.2 Rotary mechanism on front (optional)

The rotary mechanism on the front converts the vertical movement of the toggle handle into rotary motion. The circuit breaker is switched on/off or tripped with the help of the rotary mechanism on the front. The rotary motion on the switching knob is converted to vertical motion on the toggle handle.



Rotary mechanism

The rotary mechanism on the front is mounted direct on the circuit breaker. SENTRON VL circuit breakers with rotary mechanism comply with the "Network disconnecting device" condition of DIN EN 60204-1 (VDE 0113-1).

Degree of protection

The rotary mechanism on the front offers degree of protection IP30

Interlocking

Lockable in the "OFF" position with up to 3 padlocks.

A safety lock can also be used.

Application

Standard application:

- Black knob
- Gray indicator plate

Network disconnector facility with features for stopping and shutting down in an emergency:

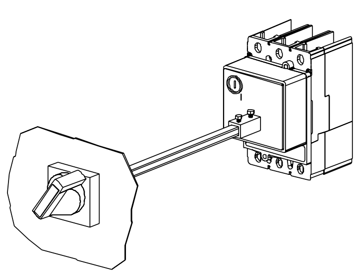
- Red knob
- Yellow indicator plate

Accessories

Optionally, up to 4 changeover contacts can be used. Two contacts can be used as leading NO contacts and two contacts as leading NC contacts. These are equipped with 1.5 m long connection cables.

3.4.3 Door-coupling rotary operating mechanism (optional)

The door-coupling rotary operating mechanism is available for installation in control cabinets and distribution boards.



Door-coupling rotary operating mechanism

SENTRON VL circuit breakers with door-coupling rotary mechanisms comply with the "Network disconnecting device" condition of DIN EN 60204-1 (VDE 0113-1).

The door-coupling rotary operating mechanism is designed as follows:

- Rotary mechanism on the front with shaft stub (without knob)
- Shaft coupling
- 300 mm extension shaft (600 mm optional, clip required)
- Actuator

Degree of protection

This mechanism offers degree of protection IP65

Interlocking

Lockable in the "OFF" position with up to 3 padlocks. A safety lock can also be used.

Application

Standard application:

- Black knob
- Gray indicator plate

Network disconnector facility with features for stopping and shutting down in an emergency:

- Red knob
- Yellow indicator plate

Accessories

Optionally, up to 4 changeover contacts can be used:

Two contacts can be used as leading NO contacts and two contacts as leading NC contacts. These are equipped with 1.5 m long connection cables

3.5 Motorized operating mechanisms (optional)

Motorized operating mechanisms enable the circuit breaker to be switched on/off locally or on-site or by remote control. For electrical and mechanical locking of the operating mechanism, they are equipped with a locking device for padlocks (standard) and an (optional) safety lock. Motorized operating mechanisms can also be actuated manually. Two types of mechanisms are offered.

Note

SENTRON circuit breakers with motorized operating mechanisms **cannot** be used as network disconnection devices in accordance with DIN EN 60204-1 (VDE 0113-1).

Designation of the connecting cables

Internal terminal strip		Identifier
Internally wired	Externally wired	
PE	X20.5	Green/yellow
L2-	X20.1	N
S2A	X20.2	S2A
S2B	X20.3	S2B
L1+	X20.4	L1

3.5.1 Stored-energy motorized operating mechanism

Motorized operating mechanism for VL160X-VL800

- The stored-energy motorized operating mechanism is suitable for synchronization tasks.
- The motor charges a stored-energy spring mechanism and moves the SENTRON VL toggle handle to the "OFF/RESET" position.
- The stored-energy spring mechanism discharges when actuated, quickly switching the SENTRON VL toggle handle to the "ON" position.
- A changeover switch allows local (Manual) or remote (Auto) operation to be selected.
- The manual actuator handle is located on the front of the operating mechanism cover.

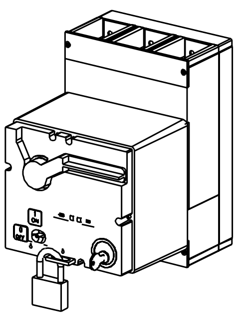


Figure 3-3 Stored-energy motorized operating mechanism

3.5.2 Motorized operating mechanism without stored-energy mechanism

Motorized operating mechanism for VL1250-1600

- The motor drives a mechanism that switches the SENTRON VL toggle handle to the "ON" and "OFF/RESET" positions.
- The manual actuator handle is located on the front of the operating mechanism cover.

4 System overview

4.1 Possible applications

Thanks to its universal connection and switching configuration, the SENTRON VL circuit breaker offers a diverse range of possible applications:

Table 4- 1 Possible applications

Area of application	Function
Plant	Current limiting
	Controller monitoring
	Ground-fault protection
	Undervoltage protection
Motor/generator	Overload protection
	Phase-failure protection
	Thermodynamic winding protection
Converter	Harmonic protection
	Frequency-independent load protection
Switch disconnectors	Remote trip units
	Auxiliary contact/alarm trip unit

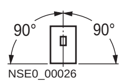
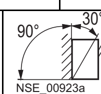
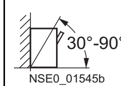
4.2 Key data

4.2.1 General data - 3VL molded-case circuit breakers

Type	VL160X 3VL1	VL160 3VL2	VL250 3VL3	VL400 VL4	VL630 3VL5	VL800 VL6	VL1250 3VL7	VL1600 3VL8
Max. rated current I_n [A]	160	160	250	400	630	800	1250	1600
N pole [A]	160	160	250	400	630	800	1250	1600
Rated insulation voltage V_i in accordance with IEC 60947-2								
Main current paths [V AC]	800	800	800	800	800	800	800	800
Auxiliary circuits [V AC]	690	690	690	690	690	690	690	690
Rated impulse withstand voltage V_{imp}								
Main current paths [kV]	8	8	8	8	8	8	8	8
Auxiliary circuits [kV]	4	4	4	4	4	4	4	4
Rated operating voltage U_e								
IEC 50/60 Hz [V AC]	690	690	690	690	690	690	690	690
IEC 50/60 Hz (V DC[2])	500	600	600	600	600	-[1]	-[1]	-[1]
NEMA 60 Hz (V AC)	600	600	600	600	600	600	600	600
Utilization category (IEC 60947-2)	A	A	A	A	A B[3]	A B[3]	A B[3]	A B[3]
Permissible ambient temperature [4]								
Operation [°C]	-25 to +70	-25 to +70	-25 to +70	-25 to +70	-25 to +70	-25 to +70	-25 to +70	-25 to +70
Storage [°C]	-40 to +80	-40 to +80	-40 to +80	-40 to +80	-40 to +80	-40 to +80	-40 to +80	-40 to +80
Permissible load at different ambient temperatures in the immediate vicinity of the circuit breaker, related to the rated current of the circuit breaker								
• Circuit breaker for system protection								
TM/ETU up to 50 °C [%]	100 /-	100 / 100	100 / 100	100 / 100	100 / 100	- / 100	- / 100	- / 100
TM/ETU up to 60 °C [%]	93 / -	93 / 95	93 / 95	93 / 95	93 / 95	- / 95	- / 95	- / 95
TM/ETU up to 70 °C [%]	86 / -	86 / 80	86 / 80	86 / 80	86 / 80	- / 80	- / 80	- / 80
• Circuit breakers for motor protection								
Up to 50 °C [%]	-	100	100	100	100	-	-	-
At 60 °C [%]	-	95	95	95	95	-	-	-
At 70 °C [%]	-	80	80	80	80	-	-	-
• Circuit breakers for starter combinations and non-automatic air circuit breakers								
up to 50 °C [%]	100	100	100	100	100	100	100	100
At 60 °C [%]	93	93	93	93	93	93	93	93
At 70 °C [%]	86	86	86	86	86	86	86	86

Type	VL160X 3VL1	VL160 3VL2	VL250 3VL3	VL400 VL4	VL630 3VL5	VL800 VL6	VL1250 3VL7	VL1600 3VL8
Weights of 3-pole circuit breakers [kg]								
• Basic switch without overcurrent trip unit	-	1.5	1.6	4.2	7.8	14.2	21	27.3
• Thermomagnetic overcurrent trip unit	-	0.7	0.7	1.5	1.2	-	-	-
• Electronic overcurrent trip unit	-	0.9	0.9	1.7	1.5	1.8	4.0	4.0
• Basic switch with thermomagnetic overcurrent trip unit	2.0	2.2	2.3	5.7	9.0	-	-	-
• Basic switch with electronic overcurrent trip unit	-	2.4	2.5	5.9	9.3	16.0	25.0	31.3
Weights of 4-pole circuit breakers [kg]								
• Basic switch without overcurrent trip unit	-	2.0	2.2	5.5	9.7	18.2	27.5	34.8
• Thermomagnetic overcurrent trip unit	-	1.0	1.0	1.9	1.5	-	-	-
• Electronic overcurrent trip unit	-	1.1	1.1	2.1	2.0	2.3	6.0	6.0
• Basic switch with thermomagnetic overcurrent trip unit	2.5	3.0	3.2	7.4	11.2	-	-	-
• Basic switch with electronic overcurrent trip unit	-	3.1	3.3	7.6	11.7	20.5	33.5	40.8
Rated short-circuit breaking capacity in accordance with IEC 60947-2								
See Chapter Technical overview (Page 145)								
• Service life make-break operations	20000	20000	20000	20000	10000	10000	3000	3000
• Service life electrical make-break operations	10000	10000	10000	10000	5000	3000	1500	1500
• Max. switching frequency [1/h]	120	120	120	120	60	60	30	30
• Connection types	See Chapter Connecting (Page 97)							
Connection cross-sections								
Box terminal[8)]								
• Solid or stranded cable; copper only [mm²]	2.5 to 95	2.5 to 95	25 to 185	50 to 300	-	-	-	-
• Finely stranded with end sleeve [mm²]	2.5 to 50	2.5 to 50	25 to 120	50 to 240	-	-	-	-
• Flexible power rail [mm]	12 x 10	12 x 10	17 x 10	25 x 10	-	-	-	-
• Terminal plate for flexible power rail[7)] [mm]	-	-	-	-	Qty. 2 10 × 32	-	-	-

Type	VL160X 3VL1	VL160 3VL2	VL250 3VL3	VL400 VL4	VL630 3VL5	VL800 VL6	VL1250 3VL7	VL1600 3VL8
Round conductor terminal for cable								
• Solid or stranded cable; copper or Al [mm²]	16 to 70	16 to 70	25 to 185	50 to 300	-	-	-	-
• Finely stranded with end sleeve [mm²]	16 to 50	16 to 50	25 to 120	50 to 240	-	-	-	-
Multiple feed-in terminal[8)]								
• Solid or stranded cable; copper or Al [mm²]	-	-	-	Qty. 2 50 to 120	Qty. 2 50 to 240	Qty. 3 50 to 240	Qty. 4 50 to 240	-
• Finely stranded with end sleeve	-	-	-	2 pieces 50 to 95	2 pieces 50 to 185	3 pieces 50 to 185	4 pieces 50 to 185	-
• Direct connection of busbars; Cu or Al [mm]	17 x 7	22 x 7	24 x 7	32 x 10	40 x 10	2 x 40 x 10	2 x 50 x 10	3 x 60 x 10
• Screw for screw-type connection	M6	M6	M8	M8	M6	M8	M8	-
Connection cross-sections for control circuits with terminal connection								
Screw-type terminals								
• Solid [mm²]	0.75 to 1.5	0.75 to 1.5	0.75 to 1.5	0.75 to 1.5	0.75 to 1.5	0.75 to 1.5	0.75 to 1.5	0.75 to 1.5
• Finely stranded with end sleeve [mm²]	0.75 to 1.0	0.75 to 1.0	0.75 to 1.0	0.75 to 1.0	0.75 to 1.0	0.75 to 1.0	0.75 to 1.0	0.75 to 1.0
See installation instructions for details.								

Type	VL160X 3VL1	VL160 3VL2	VL250 3VL3	VL400 VL4	VL630 3VL5	VL800 VL6	VL1250 3VL7	VL1600 3VL8
Power losses per circuit breaker at max. rated current								
• System protection TM 0.8 to 1.0 [W]	12 to 70	15 to 48	32 to 80	60 to 175	85 to 230	-	-	-
• System protection ETU or LCD-ETU [W]		40	60	90	160	250	210	260
• for starter combinations or non-automatic air circuit breakers [W]	40	40	60	90	160	250	210	260
• for motor protection [W]		40	60	90	160	-	-	-
• Permissible position of use 5)								

1) Breaker cannot be used for direct current.

2) The values apply for at least 3 current paths in series and extremely high switching capacity L. For switching direct current, the maximum permissible direct voltage per current path must be observed, see Chapter Use in DC systems (Page 69) (switching suggestions for direct current systems)

3) On request.

4) Exception: 3VL molded-case circuit breaker with TM TU: 0 °C to 75 °C

5) For VL800 to VL1600 circuit breakers with guide frame in lateral installation position. Adapter set on request.

6) Permissible current load factor 0.9; with internal accessories only.

7) Not for 690 V AC/600 V DC.

8) Cross-sections in accordance with IEC 60999

4.2.2 General data - auxiliary and alarm switches

Type	VL160X 3VL1	VL160 3VL2	VL250 3VL3	VL400 VL4	VL630 3VL5	VL800 VL6	VL1250 3VL7	VL1600 3VL8
Conventional free air thermal current I_{th} [A]	10	10	10	10	10	10	10	10
Rated making capacity [A]	10	10	10	10	10	10	10	10
AC								
Rated operating voltage [V]	24	48	110	230	400	600	-	-
Rated operating current [A] AC-12	10	10	10	10	10	10	-	-
Rated operating current [A] AC-15	6	6	6	6	3	1	-	-
DC								
Rated operating voltage [V]	24	48	110	230	-	-	-	-
Rated operating current [A] DC-12	10	5	2.5	1	-	-	-	-
Rated operating current [A] DC-13	3	1.5	0.7	0.3	-	-	-	-
Backup fuse/miniature circuit breaker [A]	10TDz/ 10	10TDz/ 10	10TDz/ 10	10TDz/ 10	10TDz/ 10	10TDz/ 10	10TDz/ 10	10TDz/ 10
Leading auxiliary switch in rotary mechanism								
Thermal rated current I_{th} [A]	2	2	2	2	2	2	2	2
Rated making capacity, resistive, cos φ = 0.7 [A]	2 (ind. 0.5)	2 (ind. 0.5)	2 (ind. 0.5)	2 (ind. 0.5)	2 (ind. 0.5)	2 (ind. 0.5)	2 (ind. 0.5)	2 (ind. 0.5)
Rated operating voltage [V AC]	230	230	230	230	230	230	230	230
Rated operating current [A]	2	2	2	2	2	2	2	2
Rated breaking capacity, resistive, cos φ = 0.7 [A]	2 (ind. 0.5)	2 (ind. 0.5)	2 (ind. 0.5)	2 (ind. 0.5)	2 (ind. 0.5)	2 (ind. 0.5)	2 (ind. 0.5)	2 (ind. 0.5)
Quick-response short-circuit fuse	2	2	2	2	2	2	2	2
Position signaling switch								
Thermal rated current I_{th} [A]	16	16	-	-	-	-	-	-
Rated making capacity [A]	16	16	-	-	-	-	-	-
Rated operating voltage [V AC]	250	400	-	-	-	-	-	-
Rated operating current [A]	16	10	-	-	-	-	-	-
Rated breaking capacity, inductive, cos φ = 0.7 [A]	4	4	-	-	-	-	-	-
Rated breaking capacity, resistive [A]	16	10	-	-	-	-	-	-
Quick-response short-circuit fuse [A]	16	10	-	-	-	-	-	-

Type	VL160X 3VL1	VL160 3VL2	VL250 3VL3	VL400 VL4	VL630 3VL5	VL800 VL6	VL1250 3VL7	VL1600 3VL8
Tripped signaling switch in the RCD module [1]								
Thermal rated current I_{th} [A]	-	2	2	2	-	-	-	-
Rated making capacity [A]	-	2	2	2	-	-	-	-
Rated operating voltage [V AC]	-	250	250	250	-	-	-	-
Rated operating current [A]	-	2	2	2	-	-	-	-
Rated breaking capacity, inductive, cos φ = 0.7 [A]	-	0.5	0.5	0.5	-	-	-	-
Rated breaking capacity, resistive [A]	-	2	2	2	-	-	-	-
Quick-response short-circuit fuse [A]	-	2	2	2	-	-	-	-

[1] DC rated operating voltage max. 125 V, minimum load 50 mA at 5 V DC.

4.2.3 General data - Trip units

	Group No. 1 VL160X to VL400	Group No. 2 VL630 to VL1600
Undervoltage release		
Response voltage:		
• Drop (switch tripped) [V]	0.35 to 0.70 x U_s	0.35 to 0.70 x U_s
• Pick-up (switch can be switched on) [V]	0.85 to 1.1 x U_s	0.85 to 1.1 x U_s
Power consumption (continuous operation) at:		
• 50 / 60 Hz 24 V AC [VA]	1.4	1.2
• 50 / 60 Hz 110 to 127 V AC [VA]	1.5	1.1
• 50 / 60 Hz 220 to 250 V AC [VA]	1.5	2.1
• 50 / 60 Hz 208 V AC [VA]	1.8	2.2
• 50 / 60 Hz 277 V AC [VA]	2.1	1.6
• 50 / 60 Hz 380 to 415 V AC [VA]	1.6	2.0
• 50 / 60 Hz 440 to 480 V AC [VA]	1.8	2.3
• 50 / 60 Hz 500 to 525 V AC [VA]	2.5	2.9
• 50 / 60 Hz 600 V AC [VA]	2.4	--
• 12 V DC (W)	0.75	1.2
• 24 V DC (W)	0.8	1.4
• 48 V DC (W)	0.8	1.5
• 60 V DC (W)	0.8	1.6
• 110 to 127 V DC (W)	0.8	1.2
• 220 to 250 V DC (W)	0.8	1.5
Max. opening (release) time [ms]:	50	80

	Group No. 1 VL160X to VL400	Group No. 2 VL630 to VL1600
Shunt release		
Response voltage:		
• Pick-up (switch tripped)	0.7 ... 1.1 x U_s	0.7 ... 1.1 x U_s
Power consumption (briefly) at:		
• 50 / 60 Hz 24 V AC [VA]	310	330
• 50 / 60 Hz 48 to 60 V AC [VA]	158...200	380...480
• 50 / 60 Hz 110 to 127 V AC [VA]	136...158	302...353
• 50 / 60 Hz 208 to 277 V AC [VA]	274...350	330...439
• 50 / 60 Hz 380 to 600 V AC [VA]	158...237	243...384
• 12 V DC [W]	110	50
• 24 V DC [W]	110	360
• 48 to 60 V DC [W]	110...172	512...820
• 110 to 127 V DC [W]	220...254	302...353
• 220 to 250 V DC [W]	97...110	348...397
Max. opening (release) time [ms]:	50	50
Max. in-service period [s]	Automatic interruption, less than 10 ms automatic interruption, less than 10 ms	
Delay unit for undervoltage releases		
Rated control supply voltage V_s [V AC / DC]	220 ... 250	220 ... 250
Control voltage for undervoltage release [V DC]	220 ... 250	220 ... 250
Connection cross-sections		
• Finely stranded with core end sleeve [mm^2]	2 x (0.5 to 1.5)	2 x (0.5 to 1.5)
• Solid conductor [mm^2]	2 x (0.5 to 1.5)	2 x (0.5 to 1.5)
Delay time/RC circuit		
• Undervoltage release [s]	3 / -	1.5 / -
	6 / jumper Y2-Y1	3 / jumper Y2-Y1
• Undervoltage release and auxiliary relay (3RH11) [s]	0.6 / -	0.3 / -
	1.2 / jumper Y2-Y1	0.6 / jumper Y2-Y1

4.2.4 General data - motorized operating mechanisms

Type	VL160X 3VL1	VL160 3VL2	VL250 3VL3	VL400 VL4	VL630 3VL5	VL800 VL6	VL1250 3VL7	VL1600 3VL8
Motorized operating mechanism	x	x	x	-	-	-	x	x
Stored-energy motorized operating mechanism (synchronization-enabled)	x	x	x	x	x	x	-	-
Motorized operating mechanism								
Power consumption [VA / W]	< 100	< 100	< 100	-	-	-	< 250	
Rated control supply voltage V_s [50 / 60 Hz V AC]	42 / 110-127 / 220-240			-	-	-	42-48 / 60 / 110-127 / 220-250	
Rated control supply voltage V_s [V DC]	24 / 48 / 60 / 110-127 / 220			-	-	-	24 / 42-48 / 60 110-127 / 220-250	
DIAZED fuse (performance class gG, characteristic time lag) [A]	4	2		-	-	-	4	2
Miniature circuit breaker (C characteristic in accordance with DIN VDE 0641)	4	2		-	-	-	4	2
Operating range [V]	0.85 to 1.1 x V_s	0.85 to 1.1 x V_s	0.85 to 1.1 x V_s	-	-	-	0.85 to 1.1 x V_s	0.85 to 1.1 x V_s
Minimum command duration at V_s [ms]	50	50	50	-	-	-	50	50
Max. command duration, connection-dependent[1)]	Jog/pushbutton or continuous command			-	-	-	Jog/pushbutton or continuous command	
Total closing time [s]	< 1	< 1	< 1	-	-	-	< 5	< 5
Break time [s]	< 3	< 3	< 3	-	-	-	< 5	< 5
Pause between the commands OFF and ON [s]	> 3	> 3	> 3	-	-	-	> 5	> 5
Pause between the commands ON and OFF [s]	> 3	> 3	> 3	-	-	-	> 5	> 5
Max. permissible switching frequency [1/h]	120	120	120	-	-	-	30	30

Type	VL160X 3VL1	VL160 3VL2	VL250 3VL3	VL400 VL4	VL630 3VL5	VL800 VL6	VL1250 3VL7	VL1600 3VL8
Stored-energy motorized operating mechanism (synchronization-enabled)								
Power consumption [VA / W]	< 100	< 100	< 100	< 200	< 250	< 250	-	-
Rated control supply voltage V_s [50 / 60 Hz V AC]	42-48 / 60			110-127 / 220-250			-	-
Rated control supply voltage V_s [V DC]	24 / 42-48 / 60			110-127 / 220-250			-	-
DIAZED fuse (performance class gG, characteristic slow) [A]	4			2			-	-
Miniature circuit breaker (C characteristic in accordance with DIN VDE 0641) [A]	4			2			-	-
Operating range [V]	0.85 to 1.1 x V_s	0.85 to 1.1 x V_s	0.85 to 1.1 x V_s	0.85 to 1.1 x V_s	0.85 to 1.1 x V_s	0.85 to 1.1 x V_s	-	-
Minimum command duration with V_s [ms]	50	50	50	50	50	50	-	-
Max. command duration, connection-dependent[1)]	Jog/pushbutton or continuous command						-	-
Total closing time [ms]	< 100	< 100	< 100	< 100	< 100	< 100	-	-
Break time [s]	< 5	< 5	< 5	< 5	< 5	< 5	-	-
Pause between the commands OFF and ON [s]	> 5	> 5	> 5	> 5	> 5	> 5	-	-
Pause between the commands ON and OFF [s]	> 1	> 1	> 1	> 1	> 1	> 1	-	-
Max. permissible switching frequency [1/h]	120	120	120	120	60	60	-	-

x: Available

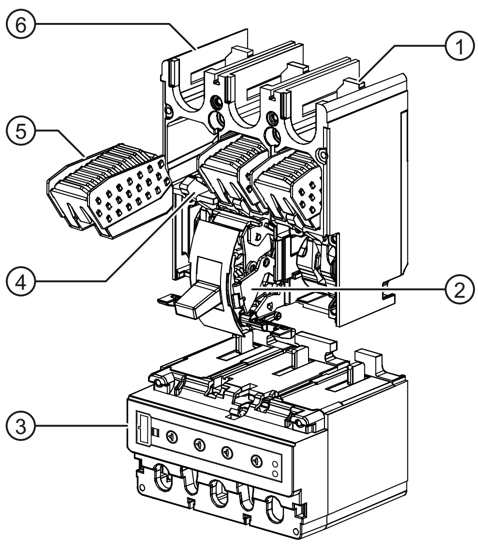
1) Changeover contact also permissible, but note the pause times between ON and OFF.

Functions 5

5.1 Current protection

5.1.1 Overcurrent trip unit

The SENTRON VL circuit breakers are designed on the principle of magnetic repulsion of the contacts. The contacts open before the expected peak-value of the short-circuit current is reached. Magnetic repulsion of the contacts very significantly reduces the thermal load I^2t as well as the mechanical load resulting from the impulse short-circuit current IP of the system components that occur during a short-circuit.



(1) Main connections

(2) Breaker mechanism

(3) Overcurrent trip unit

(4) Movable contact arm

(5) Arc chute

(6) Enclosure

Figure 5-1 Interior view MCCB

The circuit breaker as an overload current tripping system

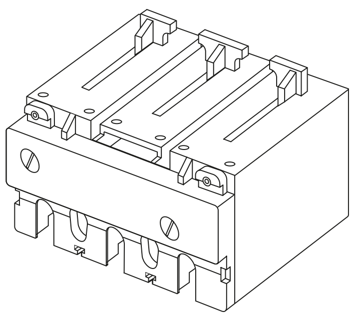
1. Overcurrent trip unit of the SENTRON VL160X to VL630 circuit breaker, thermomagnetic, TM.

The overcurrent trip unit and short-circuit release work with bimetals and magnetic coils. They are available with fixed settings or adjustable.

The 4-pole circuit breakers for system protection can be supplied with overcurrent trip units in all 4 poles or without an overcurrent trip unit in the 4th pole (N). From 100 A, the trip units in the 4th pole (N) can be set to 60% of the current in the 3 main current paths, so that safe protection of the neutral conductor can be guaranteed with a reduced cross-section.

The circuit breakers for starter combination applications are usually combined with motor contactor and suitable overload relays.

The non-automatic circuit breakers have integrated self-protection against short-circuit so that back-up fuses can be omitted. These circuit breakers have no overload protection. Four-pole circuit breakers have no short-circuit release in the 4th pole (N).



2. Overcurrent trip unit of the SENTRON VL160 to VL1600 circuit breakers, electronic, ETU / LCD-ETU

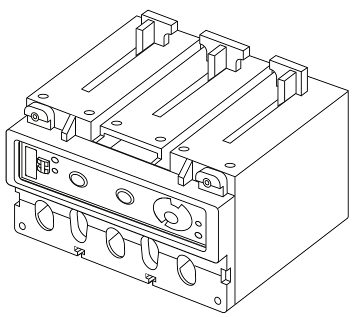
The electronic overcurrent tripping system consists of:

- current transformers
- Evaluation electronics with microprocessor
- Tripping solenoid

No auxiliary power supply is necessary for the tripping system.

Activation of the microprocessor trip unit requires a minimum load current of approximately 20% of the relevant rated current I_n of the circuit breaker.

At the output of the electronic overcurrent trip unit, there is a tripping solenoid that releases the circuit breaker in the event of an overload or short-circuit.



5.1.2 Function overview

Table 5- 1 Function overview

Order No. supplement	Trip unit	System protection	Motor Protection	Starter protection	Generator protection	Function Release type
DK	M	-	-	✓	-	I
DC	TM2)	✓	-	-	-	LI
EJ	TM2)	✓	-	-	-	LI
EC	TM2)	✓	-	-	-	LIN
EM	TM2)	✓	-	-	-	LIN
SP	ETU10M3)	-	✓	-	✓	LI
MP	ETU10M3)	-	✓	-	✓	LI
SB	ETU10	✓	-	-	-	LI
MB	ETU10	✓	-	-	-	LI
TA	ETU10	✓	-	-	-	LIN
NA	ETU10	✓	-	-	-	LIN
TB	ETU10	✓	-	-	-	LI
NB	ETU10	✓	-	-	-	LI
SL	ETU12	✓	-	-	-	LIG
ML	ETU12	✓	-	-	-	LIG
SF	ETU12	✓	-	-	-	LING
MF	ETU12	✓	-	-	-	LING
TN	ETU12	✓	-	-	-	LING
NN	ETU12	✓	-	-	-	LING
SE	ETU20	✓	-	-	✓	LSI
ME	ETU20	✓	-	-	✓	LSI
TE	ETU20	✓	-	-	✓	LSI
NE	ETU20	✓	-	-	✓	LSI
TF	ETU20	✓	-	-	✓	LSIN
NF	ETU20	✓	-	-	✓	LSIN

Order No. supplement	Trip unit	System protection	Motor Protection	Starter protection	Generator protection	Function Release type
SG	ETU22	✓	-	-	✓	LSIG
MG	ETU22	✓	-	-	✓	LSIG
SH	ETU22	✓	-	-	✓	LSING
MH	ETU22	✓	-	-	✓	LSING
TH	ETU22	✓	-	-	✓	LSING
NH	ETU22	✓	-	-	✓	LSING
SS	ETU30M[3)]	-	✓	-	✓	LI
MS	ETU30M[3)]	-	✓	-	✓	LI
CP	LCD-ETU40M[3)]	-	✓	-	✓	LI
CH	LCD-ETU40	✓	-	-	-	LSI
CJ	LCD-ETU40	✓	-	-	-	LSIN
CL	LCD-ETU42	✓	-	-	-	LSIG
CM	LCD-ETU42	✓	-	-	-	LSIG
CN	LCD-ETU42	✓	-	-	-	LSING

1) Dependent on frame size

2) TM to I_n = 630 A

3) Motor protection to I_n = 500 A

L: Long time delay

S: Short time delay

I: Instantaneous

N: Neutral protection

G: Ground fault

5.1.3 Setting options

In view of the large number of setting options of the individual overcurrent trip units, an overview in table form is useful for calculating the optimal operating point.

Table 5- 2 Overcurrent tripping method - setting options

Order No. supplement	Trip unit	Setting options					
		L Overload protection	S1) Short-circuit protection (short time delay)		I1) Short-circuit protection (instantaneous)	G Ground fault protection	
		$I_r = x\ I_n$	$I_{sd} = x\ I_r$	t_{sd} [s]	$I_i = x\ I_n$	$I_g = I_n$	t_g [s]
DK	M	-	-	-	7 ... 15	-	-
DC	TM2)	0,8 ... 1	-	-	5 ... 10	-	-
EJ	TM2)	0,8 ... 1	-	-	5 ... 10	-	-
EC	TM2)	0,8 ... 1	-	-	5 ... 10	-	-
EM	TM2)	0,8 ... 1	-	-	5 ... 10	-	-
SP	ETU10M3)	0,4 ... 1	-	-	1,25 ... 11	-	-
MP	ETU10M3)	0,4 ... 1	-	-	1,25 ... 11	-	-
SB	ETU10	0,4 ... 1	-	-	1,25 ... 11	-	-
MB	ETU10	0,4 ... 1	-	-	1,25 ... 11	-	-
TA	ETU10	0,4 ... 1	-	-	1,25 ... 11	-	-
NA	ETU10	0,4 ... 1	-	-	1,25 ... 11	-	-
TB	ETU10	0,4 ... 1	-	-	1,25 ... 11	-	-
NB	ETU10	0,4 ... 1	-	-	1,25 ... 11	-	-
SL	ETU12	0,4 ... 1	-	-	1,25 ... 11	0.6 ... 1 OFF	0.1/0.3/Off
ML	ETU12	0,4 ... 1	-	-	1,25 ... 11	0.6 ... 1 OFF	0.1/0.3/Off
SF	ETU12	0,4 ... 1	-	-	1,25 ... 11	0.6 ... 1 OFF	0.1/0.3/Off
MF	ETU12	0,4 ... 1	-	-	1,25 ... 11	0.6 ... 1 OFF	0.1/0.3/Off
TN	ETU12	0,4 ... 1	-	-	1,25 ... 11	0.6 ... 1 OFF	0.1/0.3/Off
NN	ETU12	0,4 ... 1	-	-	1,25 ... 11	0.6 ... 1 OFF	0.1/0.3/Off
SE	ETU20	0,4 ... 1	1,5 ... 10	0 ... 0,5	11	-	-
ME	ETU20	0,4 ... 1	1,5 ... 10	0 ... 0,5	11	-	-
TE	ETU20	0,4 ... 1	1,5 ... 10	0 ... 0,5	11	-	-
NE	ETU20	0,4 ... 1	1,5 ... 10	0 ... 0,5	11	-	-
TF	ETU20	0,4 ... 1	1,5 ... 10	0 ... 0,5	11	-	-
NF	ETU20	0,4 ... 1	1,5 ... 10	0 ... 0,5	11	-	-

Order No. supplement	Trip unit	Setting options					
		L Overload protection	S[1)] Short-circuit protection (short time delay)		I[1)] Short-circuit protection (instantaneous)	G Ground fault protection	
		I_r = x I_n	I_{sd} = x I_r	t_{sd} [s]	I_i = x I_n	I_g = I_n	t_g [s]
SG	ETU22	0,4 ... 1	1,5 ... 10	0 ... 0,5	11	0.6 ... 1 OFF	0.1/0.3/Off
MG	ETU22	0,4 ... 1	1,5 ... 10	0 ... 0,5	11	0.6 ... 1 OFF	0.1/0.3/Off
SH	ETU22	0,4 ... 1	1,5 ... 10	0 ... 0,5	11	0.6 ... 1 OFF	0.1/0.3/Off
MH	ETU22	0,4 ... 1	1,5 ... 10	0 ... 0,5	11	0.6 ... 1 OFF	0.1/0.3/Off
TH	ETU22	0,4 ... 1	1,5 ... 10	0 ... 0,5	11	0.6 ... 1 OFF	0.1/0.3/Off
NH	ETU22	0,4 ... 1	1,5 ... 10	0 ... 0,5	11	0.6 ... 1 OFF	0.1/0.3/Off
SS	ETU30M[3)]	0,4 ... 1	-	-	6/8/11	-	-
MS	ETU30M[3)]	0,4 ... 1	-	-	6/8/11	-	-
CP	LCD-ETU40M[3)]	0,4 ... 1	-	-	1,25 ... 11	-	-
CH	LCD-ETU40	0,4 ... 1	1,5 ... 10	0 ... 0,5	1,25 ... 11	-	-
CJ	LCD-ETU40	0,4 ... 1	1,5 ... 10	0 ... 0,5	1,25 ... 11	-	-
CL	LCD-ETU42	0,4 ... 1	1,5 ... 10	0 ... 0,5	1,25 ... 11	0,4 ... 1	0,1 ... 0,5
CM	LCD-ETU42	0,4 ... 1	1,5 ... 10	0 ... 0,5	1,25 ... 11	0,4 ... 1	0,1 ... 0,5
CN	LCD-ETU42	0,4 ... 1	1,5 ... 10	0 ... 0,5	1,25 ... 11	0,4 ... 1	0,1 ... 0,5

[1)] Dependent on frame size, refer to chapter "Dimensioning short-circuit protection according to frame size (Page 47)"

[2)] TM to I_n = 630 A

[3)] Motor protection to I_n = 500 A

5.1.4 Dimensioning short-circuit protection according to frame size

Short-circuit protection (instantaneous) I_i = x I_n)											
63 A	100 A	160 A	200 A	250 A	315 A	400 A	630 A	800 A	1000 A	1250 A	1600 A
11	11	11	11	11	11	11	10	8	11	11	9

Short-circuit protection (instantaneous) I_{SD} = x I_r)											
63 A	100 A	160 A	200 A	250 A	315 A	400 A	630 A	800 A	1000 A	1250 A	1600 A
1,5 ... 10	1,5 ... 10	1,5 ... 10	1,5 ... 10	1,5 ... 10	1,5 ... 10	1,5 ... 10	1,5 ... 9	1,5 ... 7	1,5 ... 10	1,5 ... 10	1,5 ... 8

5.1.5 General technical specifications

Table 5- 3 General data I

Order No. supplement	Trip unit	Thermal memory	Phase failure	Communications capability	Ground-fault protection	Number of poles	N pole protected[1)]
DK	M	-	-	-	-	3	-
DC	TM[2)]	✓	-	-	-	3	-
EJ	TM[2)]	✓	-	-	-	4	-
EC	TM[2)]	✓	-	-	-	4	60 %
EM	TM[2)]	✓	-	-	-	4	100 %
SP	ETU10M[3)]	✓	40% I_R	-	-	3	-
MP	ETU10M[3)]	✓	40% I_R	✓[4)]	-	3	-
SB	ETU10	✓	-	-	-	3	-
MB	ETU10	✓	-	✓[4)]	-	3	-
TA	ETU10	✓	-	-	-	4	50 / 100 %
NA	ETU10	✓	-	✓[4)]	-	4	50 / 100 %
TB	ETU10	✓	-	-	-	4	50 / 100 %
NB	ETU10	✓	-	✓[4)]	-	4	50 / 100 %
SL	ETU12	✓	-	-	①	3	-
ML	ETU12	✓	-	✓[4)]	①	3	-
SF	ETU12	✓	-	-	①	3	-
MF	ETU12	✓	-	✓[4)]	①	3	-
TN	ETU12	✓	-	-	②	4	50 / 100 %
NN	ETU12	✓	-	✓[4)]	②	4	50 / 100 %
SE	ETU20	✓	-	-	-	3	-
ME	ETU20	✓	-	✓[4)]	-	3	-
TE	ETU20	✓	-	-	-	4	-
NE	ETU20	✓	-	✓[4)]	-	4	-
TF	ETU20	✓	-	-	-	4	50 / 100 %
NF	ETU20	✓	-	✓[4)]	-	4	50 / 100 %

Order No. supplement	Trip unit	Thermal memory	Phase failure	Communications capability	Ground-fault protection	Number of poles	N pole protected[1)]
SG	ETU22	✓	-	-	①	3	-
MG	ETU22	✓	-	✓[4)]	①	3	-
SH	ETU22	✓	-	-	①	3	-
MH	ETU22	✓	-	✓[4)]	①	3	-
TH	ETU22	✓	-	-	②	4	50 / 100 %
NH	ETU22	✓	-	✓[4)]	②	4	50 / 100 %
SS	ETU30M[3)]	✓	40% I_R	-	-	3	-
MS	ETU30M[3)]	✓	40% I_R	✓[4)]	-	3	-
CP	LCD-ETU40M[3)]	✓	5 to 50% I_R	✓[5)]	-	3	-
CH	LCD-ETU40	✓	-	✓[5)]	-	3	-
CJ	LCD-ETU40	✓	-	✓[5)]	-	4	50 /100 %, OFF
CL	LCD-ETU42	✓	-	✓[5)]	①	3	-
CM	LCD-ETU42	✓	-	✓[5)]	① / ③	3	-
CN	LCD-ETU42	✓	-	✓[5)]	②	4	50 / 100%, OFF

1) Dependent on frame size

2) TM to I_n = 630 A

3) Motor protection to I_n = 500 A

4) With COM20/COM21

5) With COM10/COM11

① Vectorial summation current formation (3-conductor system), see 1st figure in Chapter Ground-fault protection (Page 58)

② Vectorial summation current formation (4-conductor system), see 2nd and 3rd figures in Chapter Ground-fault protection (Page 58)

③ Direct capture of the ground fault current in the neutral point of the transformer, see 4th figure in Chapter Ground-fault protection (Page 58)

Table 5- 4 General data II

Order No. supplement	Trip unit	I^2t (ON/OFF)	Trip class (t_C)	Time-lag class (t_R)	Thermomagnetic trip unit	Magnetic trip unit	Electronic trip unit	LCD display
DK	M	-	-	-	-	✓	-	-
DC	TM2)	-	-	-	✓	-	-	-
EJ	TM2)	-	-	-	✓	-	-	-
EC	TM2)	-	-	-	✓	-	-	-
EM	TM2)	-	-	-	✓	-	-	-
SP	ETU10M3)	-	10	-	-	-	✓	-
MP	ETU10M3)	-	10	-	-	-	✓	-
SB	ETU10	-	-	2,5 ... 30	-	-	✓	-
MB	ETU10	-	-	2,5 ... 30	-	-	✓	-
TA	ETU10	-	-	2,5 ... 30	-	-	✓	-
NA	ETU10	-	-	2,5 ... 30	-	-	✓	-
TB	ETU10	-	-	2,5 ... 30	-	-	✓	-
NB	ETU10	-	-	2,5 ... 30	-	-	✓	-
SL	ETU12	✓	-	2,5 ... 30	-	-	✓	-
ML	ETU12	✓	-	2,5 ... 30	-	-	✓	-
SF	ETU12	✓	-	2,5 ... 30	-	-	✓	-
MF	ETU12	✓	-	2,5 ... 30	-	-	✓	-
TN	ETU12	✓	-	2,5 ... 30	-	-	✓	-
NN	ETU12	✓	-	2,5 ... 30	-	-	✓	-
SE	ETU20	✓	-	-	-	-	✓	-
ME	ETU20	✓	-	-	-	-	✓	-
TE	ETU20	✓	-	-	-	-	✓	-
NE	ETU20	✓	-	-	-	-	✓	-
TF	ETU20	✓	-	-	-	-	✓	-
NF	ETU20	✓	-	-	-	-	✓	-
SG	ETU22	✓	-	-	-	-	✓	-
MG	ETU22	✓	-	-	-	-	✓	-
SH	ETU22	✓	-	-	-	-	✓	-
MH	ETU22	✓	-	-	-	-	✓	-
TH	ETU22	✓	-	-	-	-	✓	-
NH	ETU22	✓	-	-	-	-	✓	-
SS	ETU30M3)	-	10, 20, 30	-	-	-	-	-
MS	ETU30M3)	-	10, 20, 30	-	-	-	-	-
CP	LCD-ETU40M3)	-	5, 10, 15, 20, 30	-	-	-	-	✓
CH	LCD-ETU40	✓	-	2,5 ... 30	-	-	✓	✓
CJ	LCD-ETU40	✓	-	2,5 ... 30	-	-	✓	✓
CL	LCD-ETU42	✓	-	2,5 ... 30	-	-	✓	✓
CM	LCD-ETU42	✓	-	2,5 ... 30	-	-	✓	✓
CN	LCD-ETU42	✓	-	2,5 ... 30	-	-	✓	✓

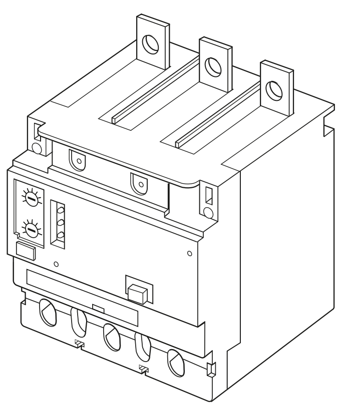
5.1.6 Differential current protection with RCD module

The circuit breakers with differential current protection are used in a variety of ways to implement a double function:

- Protection of systems against overload and short-circuit currents.
- Protection of cables and electrical equipment against damage from ground faults.

The SENTRON VL RCD modules are supplied as accessories for the VL160X, VL160, VL250 and VL400 circuit breakers with thermomagnetic overcurrent trip units. This combination is referred to a circuit breaker with differential current protection of type A. Type A means tripping is guaranteed both in the case of faults in sinusoidal alternating currents and in the case of faults in pulsating direct currents. These units have an adjustable trip time delay Δt. The values for the rated fault current $I_{\Delta n}$ can also be adjusted.

In a fault-free system, the sum of the currents in the summation current converter of the RCD module is zero. A ground fault current occurring in the protected circuit as the result of an insulation fault produces a differential current that induces a voltage in the secondary winding of the current transformer. The evaluation electronics monitor the induced voltage and transmit a trip command to the RCD trip unit if the trip criterion is met. The combination of circuit breaker with differential current protection is designed in such a way as to cause the circuit breaker contacts to open if the differential current reaches a given value.



Standard features

- Mechanical trip display:

  The Reset button pops out when the RCD module trips the circuit breaker.
- Reset button:

  This must be manually reset after the circuit breaker has been tripped by the RCD module. The circuit breaker can only be reset and switched on again after the RCD module has been reset.
- Cover:

  Modifiable settings for Δt and $I_{\Delta n}$.

  A sealable transparent cover is available for preventing modification.
- LED displays:
  – 3 LEDs (green/yellow/red) indicate the level of the leakage/fault current. The blinking indicator signals that the SENTRON VL RCD module is ready for operation.
  – Green: I_{Δ} = 25% of the set value, the cable is live
  – Green + yellow: 25% < I_{Δ}< 50 of the set $I_{\Delta n}$value
  – Green + yellow + red: I_{Δ} = 50% of the set $I_{\Delta n}$-value
- Test button:

  The functionality of the RCD module is checked with the test button. If the test button is pressed, differential current is simulated on a test winding attached to the summation current converter. If it is functioning correctly, the RCD module must trip the circuit breaker.

  The test button must remain pressed for at least the period of the set delay time Δt.
- A network disconnection device:
  – makes it possible to disconnect the evaluation electronics of the RCD module from the circuit without removing the primary cable or the busbars (e.g. before insulation tests).
  – Limitation of the maximum r.m.s. withstand voltage to an r.m.s. value of 3500 V AC for this feature.
- Protection function up to 50 V AC between phase and neutral conductor
- The RCD module has a surge withstand strength of I_{peak} = 2000 A. The standard surge wave is defined as 8 / 20-μs waveform.
- The RCD module does not trip in the case of making currents.

  $\Delta t \geq 0$ I_{rms} = 3000 A

  $\Delta t \geq$ 60ms $I_{peak} = 20 \times I_n \times \sqrt{2}$
- The circuit breaker combination with differential current protection can be supplied from both sides.
- Suitable for circuit breaker standard accessories – covers, phase barriers, wire connectors.

Special features of the VL160X

- The circuit breaker is tripped via an electromagnetic trip relay installed in the compartment for circuit breaker accessories to the left of the toggle handle. The trip unit is connected to the SENTRON VL RCD module and receives a trip command when the preset fault currents are reached.
- Internal accessories can also be installed in the recess for SENTRON VL accessories to the right of the toggle handle.
- The Reset button functions in exactly the same way as on the RCD modules VL160 to 400 and is accessible via the circuit breaker accessories cover supplied with this module.
- A special kit is available for mounting the RCD module and the VL160X next to each other. The mounting adapter enables installation on a DIN 50023 mounting rail. The collar of the combination is 45 mm wide along its entire length.

Note

Stored-energy motorized operating mechanisms and rotary operating mechanisms cannot be installed with this product.

Special features of VL160, VL250, VL400

- The circuit breaker is released with immediate effect via a tappet from the RCD module to the line protection switch. The electromagnetic trip unit is integrated into the RCD module.
- The Reset button pops out beyond the surface of the RCD module cover to indicate that the RCD module has tripped the system protection switch. This unit prevents the system protection switch contacts from closing before the Reset button of the RCD module has been manually reset.
- This design is compatible with the line protection switch accessories including the accessories for external operating mechanisms as well as for fixed-mounted assembly, plug-in assembly and withdrawable assembly.
- An auxiliary switch (changeover contact) is available. The contacts change status when the RCD module trips the system protection switch. The contact is suitable for
  - 2 A 250 V AC applications (0.5 A inductive)
  - 0.5 A 125 V DC.

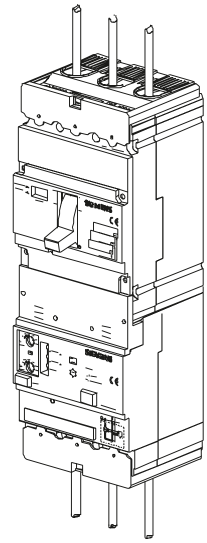
The lowest switching capacity is 50 mA at 5 V AC/DC.

- Remote tripping is supported. The customer connects a switch (NO contact) to terminals X13.1 and X13.3 via a twisted-pair cable. The switching contact must have a minimum switching capacity of 5 V/1 mA (e.g. SIEMENS 3SB3). If the NO contact is actuated, the RCD module trips. The connection terminals X13.1 and X13.3 are galvanically isolated from the system by means of a transformer (functional extra low voltage, FELV). The maximum trip time of the circuit breaker with differential current protection is 50 ms regardless of the set trip time delay Δt. In special cases, such as routing of the cable outside, care must be taken by means of suitable routing or protection circuits that the amplitude of overvoltages (e.g. storm overvoltages) between the conductor and ground is limited to 2.5 kV.

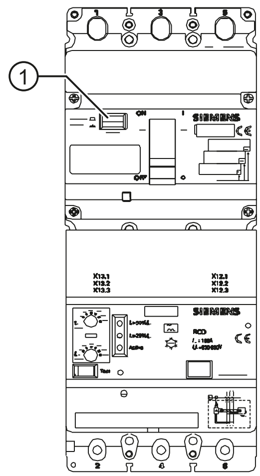
Special requirements

- Every RCD module requires a separate cable for remote tripping. It is not possible to use one cable when switching two or more RCD modules in parallel. It is possible to use two or more switches in parallel for remote tripping of an RCD module.
- The customer provides an unshielded or shielded twisted-pair cable with a maximum capacity of 36 nF as well as a maximum resistance of 50 Ohms (total length = out and back).
- Example: The maximum cable length with a cable capacity of 120 nF/km is 330 m. With a shielded cable, the shield must not be applied to the PE conductor of the system.
- A separate conductor must connect terminal X13.2 with the ground busbar (E or PE). This connection is recommended for the prevention of electrostatic charge on the remote tripping cable. This applies in particular when long cables (> 10 m) are used. Otherwise, the remote tripping cable is isolated.

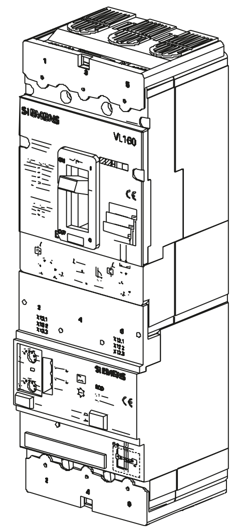
Design of the RCD module



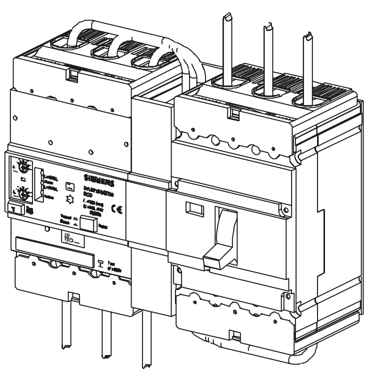
VL160X with RCD module I (angled view with cable connection)



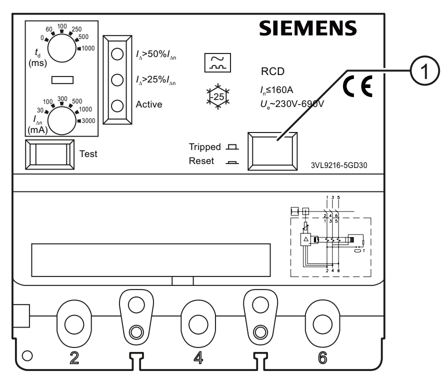
VL160X with RCD module II (front view, reset)



VL160 with RCD module (angled view without cable connection)



VL160X with RCD module, mounted on left



RCD module for VL160

(1) Reset

5.1.7 Single-pole operation with RCD module

Connection of the RCD module for single-pole operation

In principle, all 3-pole or 4-pole circuit breakers with RCD module can be operated with 2 poles (L to N), since the power supply of the RCD module is supplied from all three external conductors, and on 4-pole devices additionally from the N conductor.

Apart from the test current circuit, the RCD module is unrestricted in functionality if at least 2 conductors are connected.

When connecting the RCD module, you only have to ensure that the test current circuit connected to current path 1-2 and 3-4 (marking) is functioning or is supplied with power.

The following connections are possible in 2-pole operation:

On 3-pole circuit breakers

- Connection of the network to current path 1-2 and 3-4 (any incoming supply side)

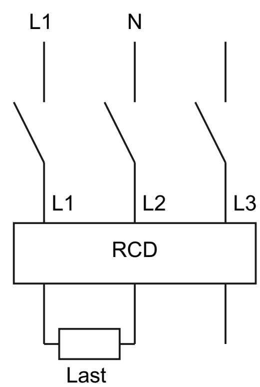


Figure 5-2 3-pole RCD

On 4-pole circuit breakers

- Connection of the network to current path 1-2 and 3-4 (any incoming supply side) or
- connection of the network to current path 1-2 and N; however, a jumper is required here from N to current path 3-4 (on the input or output side)

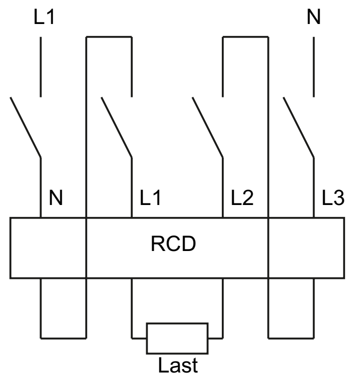


Figure 5-3 4-pole RCD

Note

Single-pole load

Series connection of the current paths is not necessary in the case of single-pole load

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5.1.8 Ground-fault protection

Ground fault trip "G" (ground fault overcurrent protection) captures fault currents escaping to ground that can cause fires in the plant. Several circuit breakers connected in series can be assigned time-graded discrimination by means of the adjustable delay time.

Measurement method 1: Vectorial summation current formation

Ground fault detection in balanced systems

The three phase currents are evaluated using vectorial summation current formation.

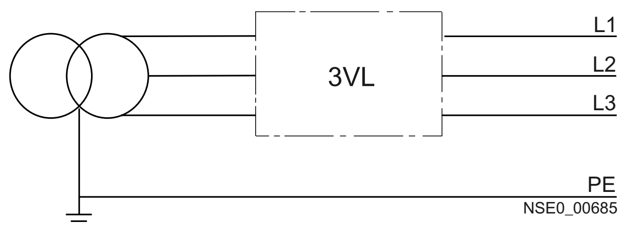


Figure 5-4 Circuit breaker in balanced system

Ground fault detection in unbalanced systems

The neutral conductor current is measured directly. Only the ground-fault current is evaluated for the 3-pole circuit breakers. In the case of the 4-pole circuit breakers, the neutral conductor overload protection is also evaluated.

The overcurrent trip unit calculates the ground-fault current using the vectorial summation of the three phase currents and the neutral conductor current.

The 4th current converter of the neutral conductor is installed internally in the case of 4-pole circuit breakers.

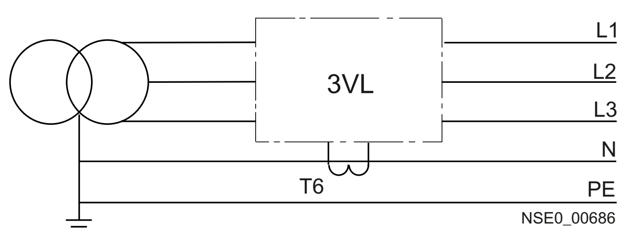


Figure 5-5 3-pole circuit breaker, current converter in neutral conductor current

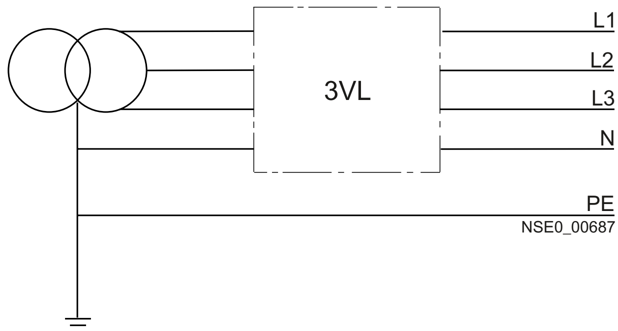


Figure 5-6 4-pole circuit breaker, current converter installed internally

Measurement method 2: Direct detection of the ground-fault current via a current transformer at the grounded star point of the transformer

The current converter is installed direct at the grounded star point of the transformer.

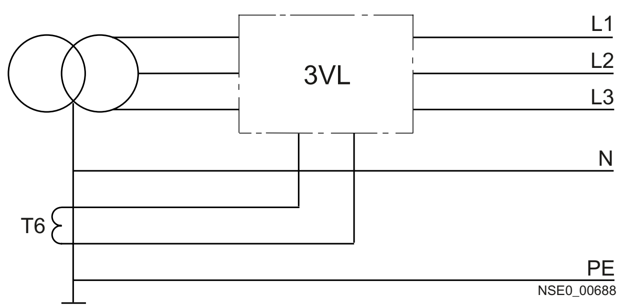
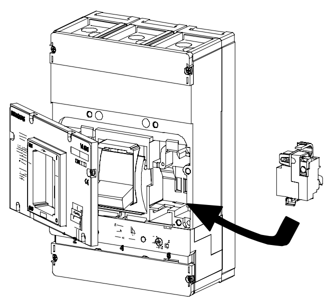


Figure 5-7 3-pole circuit breaker, current converter at the grounded star point of the transformer

5.2 Voltage protection

5.2.1 Undervoltage release

As an undervoltage release, the circuit breaker can protect certain electrical components when the voltage falls below a given level.



Undervoltage release

The undervoltage release trips the circuit breaker when the voltage fails or falls to an operating level between 70 and 35% x U_S. Re-closure of the circuit breaker contacts is only possible once the voltage has reached a value of at least 85% x U_S. Undervoltage releases can be installed for electronic locking.

Undervoltage releases are installed in the right accessory compartment of the SENTRON VL circuit breakers.

Possible configuration of the insulated accessory compartments

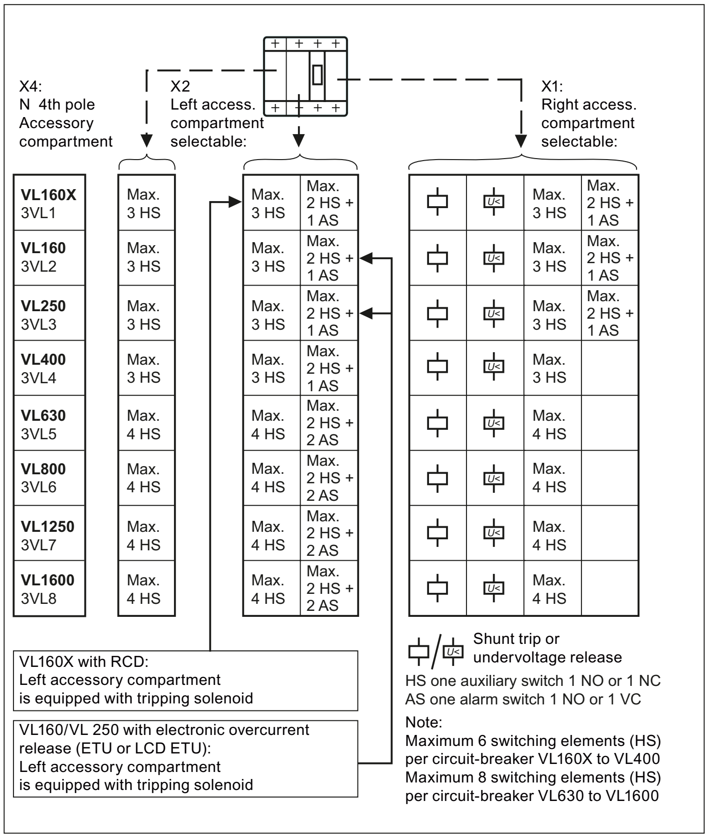
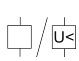


Figure 5-8 Possible configuration of the insulated accessory compartments



Shunt release or undervoltage release,
HS: Auxiliary switch, AS: Alarm switch (1 NO or 1 NC contact respectively)

Note

If a communication-enabled ETU is used, the left-hand accessory compartment X2 contains an auxiliary switch and an alarm switch!

Note

Max. no. of contact blocks

Maximum 6 contact blocks (HS) per circuit breaker VL160X to VL400

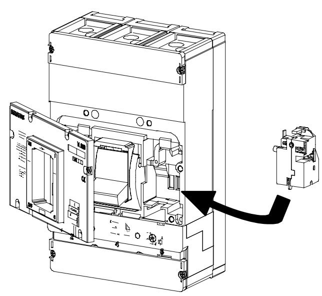
Maximum 8 contact blocks (HS) per circuit breaker VL 630 to VL 1600

Designation of the connecting cables

All types	Pos. 1	Pos. 2
Cable designations	D1	D2

5.2.2 Shunt release

As a shunt release, the circuit breaker is used for remote protection



Shunt release

It is designed for short-time operation and is therefore equipped with an interrupt contact for self-protection. Shunt releases are installed in the right-hand accessory compartment of the SENTRON VL circuit breakers.

Designation of the connecting cables

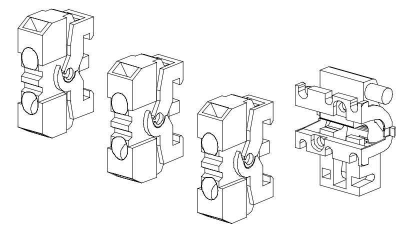
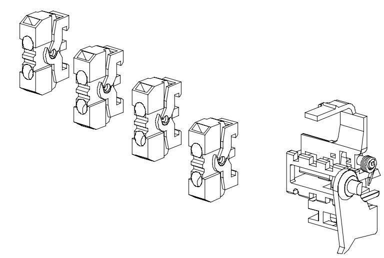
All types	Pos. 1	Pos. 2
Identifier	C1	C2

5.2.3 Auxiliary switches and alarm switches

Auxiliary and alarm switches are used to indicate the switching status of the circuit breaker.

Auxiliary switches show the position of the main contacts ("ON" or "OFF").

Alarm switches transmit a signal when the circuit breaker trips due to a short-circuit or overcurrent, or when the shunt release, undervoltage release, test button, or RCD module trips.

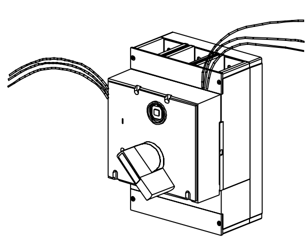
Group 1				Group 2			
VL160X	VL160	VL250	VL400	VL630	VL800	VL1250	VL1600
							

Possible configuration of the insulated accessory compartments

Auxiliary and alarm switches, shunt and undervoltage releases are used and wired in the available cutouts behind the front cover of every circuit breaker.

Leading auxiliary switches for switching on and off

The leading auxiliary switches (changeover switches) are available as accessories for front rotary operating mechanisms and door-coupling rotary operating mechanisms.



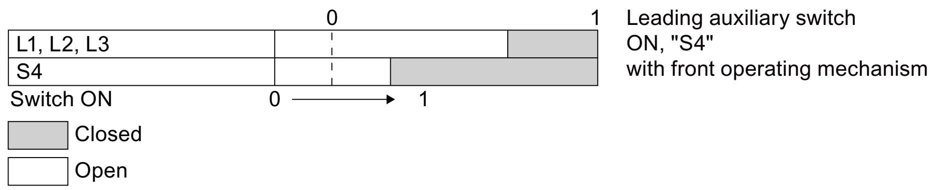
The following applications are possible:

Leading auxililary switch for switching from "ON" to "OFF"

Leading auxililary switch for switching from "OFF" to "ON"

Each version, leading auxiliary switch for switching on and off, can be equipped with one or two changeover switches. The connecting cables of the auxiliary switches are 1.5 m long.

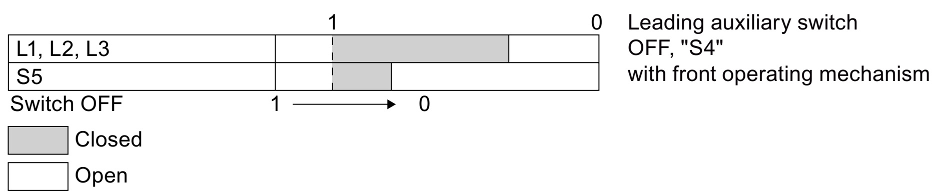
Leading auxililary switch for switching from "OFF" to "ON" (leading NO contact)



Application example:

If the circuit breaker is equipped with an undervoltage release, and the leading auxiliary switch is installed in the rotary operating mechanism, the leading NO contacts make it possible to supply the undervoltage release with power before the main contacts can be closed.

Leading auxiliary switch for switching off (leading NC contact)



Application example:

In applications with thyristors, it is necessary to reset the power electronics of the converter before the main circuit is switched off.

Circuit breakers with leading auxiliary switches create a leading signal that enables selective deceleration of the thyristor.

Designation of the auxiliary switches and the alarm switches in the circuit breaker, and designation of the connecting cables

MLFB	Pos. 1	Pos. 2	Pos. 3
3VL9400-2AB00/01	NC	NO	-
3VL9400-2AD00/01	NC	NO	NO (AS)
Cable designations	HS1/2	HS3/4	AS

MLFB	Pos. 1	Pos. 2	Pos. 3	Pos. 3
3VL9800-2AC00/01	NC	NO	NC	NO
3VL9800-2AE00/01	NC	NO	-	NO (AS)
Cable designations	HS1/2	HS3/4	HS5/6	HS7/8 or AS

Application planning

6

6.1 Use with frequency converters

Frequency converter and SENTRON VL circuit breaker combination

SENTRON VL circuit breakers can be used as primary connection protection devices in systems in which frequency converters, variable-speed drives, and electronic motor control devices are used. The thermomagnetic and electronic trip units of the SENTRON VL circuit breakers can be used in these applications. The SENTRON VL trip units are not influenced by harmonic effects due to the r.m.s. measurement.

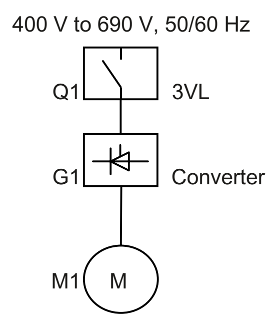


Figure 6-1 Frequency converters

Table 6- 1 Circuit breaker ↔ converter

Upstream: Circuit breaker		Downstream: Converter
Application	Trip unit	
Motor Protection	Electronic	Yes
System protection	Electronic	Yes
	Thermo-magnetic	Yes

Note

Alternative circuit breakers

SIRIUS 3RV circuit breakers can be used as an alternative for applications up to approximately 45 kW .

SIRIUS soft starters and SENTRON VL circuit breakers

For more detailed information, please refer to the soft starter catalogs and the selection guides.

Visit our site on the Internet at: http://www.siemens.de/sanftstarter.

Frequency converters/variable-speed drives and SENTRON VL circuit breakers

Please refer to the respective catalogs for information about the new SINAMICS series (Catalogs D11, D11.1, D21.2 and D21.3), the MICROMASTER 4 (Catalog DA51.2) and the SIMOVERT MASTERDRIVES (DA65.10 and DA65.11).

6.2 Use of capacitor banks

In general, reactive power compensation is used in order to reduce system losses and voltage drops in the power distribution system. As a result, the power fed into the system is used as active power and costs will be saved through a reduction in the capacitive and inductive power factors.

A combination of fixed and central compensations are used depending on the design of the low-voltage system and the loads involved.

Circuit breaker for protecting and switching capacitor banks

According to the relevant standards DIN VDE 0560 Part 41 / EN 60831-1 / IEC 70, capacitors must function under normal operating conditions with the current having an r.m.s. value up to 1.3 times the rated current of the capacitor. In addition, a further tolerance of up to 15% of the real value of the power must be taken into consideration.

The maximum current with which the selected circuit breaker can be constantly loaded, and which it must also be able to switch, is calculated as follows:

$I_{N\ max} = I_N \times 1.5$ (r.m.s. value, r.m.s. current)

Important values for selecting circuit breakers

More detailed information in the technical data: Capacitor banks (Page 173)

Abbr.	Designation
Q_n	Capacitor bank rated power in kVA
V_N	Rated voltage of the capacitor
I_N	Rated current of the capacitor bank
$I_{N\ max}$	Maximum expected rated current
I_i	Value for setting the instantaneous short-circuit release
I_R	Value for setting the current-dependent delayed overload release

The following applies:

$I_N = Q_N / \sqrt{3} \times V_N$

$I_R = I_{Nmax} = I_N \times 1.5$

$I_i > 9 \times I_R$ (minimum)

6.3 Primary-side transformer protection

The circuit breaker as primary-side transformer protection

When switching on low-voltage AC transformers, the extremely high inrush current peaks place special demands on the trip unit or on the making capacity of the circuit breakers if these are also used to switch the transformer.

For most applications, an inrush current of 20 to 30 times the rated operating current is expected in practice and must be taken into account when selecting the circuit breakers.

The maximum short-circuit current I_k of the 3VL circuit breakers is 11 x I_e (rated operating current). A circuit breaker in the lower setting range must therefore be operated for primary-side transformer protection.

Example: A transformer with 500 A rated current; 20 times the inrush current

Selected: ETU with I_n = 1000 A; setting range 0.4 - 1 x I_n = 400 A to 1000 A

50% of I_n = 500 A; I_i = 11 x I_n = 1000 A x 11 = 11000 A = 22 x current setting

---

Note

Switching the circuit breaker off

It is imperative to note that the minimum short-circuit current I_{kmin} in accordance with VDE 0100 is switched off in every case using a protection facility (e.g. circuit breaker).

With 3VL, the circuit breaker can be shut down using the time-delayed short-circuit release (S), e.g. a 3VL with an ETU20, where it is possible to set the delay time to up to 500 ms depending on the duration of the inrush current.

The short delay "bridges" the inrush current peak and the short-circuit protection can then respond at low current values after a delay.

CAUTION

Circuit breakers with phase failure protection

Circuit breakers with phase failure protection must not be used. Their trip units have protection against unbalanced network load. This cannot be switched off and could lead to unintentional trips.

6.4 Use in DC systems

The SENTRON VL circuit breakers for system protection with thermal overload and magnetic short-circuit releases are suitable for use in DC networks.

The SENTRON VL circuit breakers with electronic overcurrent trip units are **not** suitable for switching direct current.

Selection criteria for circuit breakers

The following are the most important criteria for selecting the optimal circuit breaker for protecting a DC system:

- The rated current determines the power rating and the frame size of the circuit breaker
- The rated voltage determines the number of series-connected poles required for switching off
- The maximum short-circuit current at the connection point determines the breaking capacity
- The type of network determines the circuit design

Ampacity of current path

The rated current values are the same for both DC and AC applications.

DC switching capacity

In AC circuits, arc quenching is facilitated by the fact that the current flows through the zero point. These preconditions aren't true for DC.

For this reason, a high arc voltage must be developed to interrupt the direct current.

Therefore, the switching capacity depends on the arc quenching method and the network voltage. Several switching contacts can be connected in series in order to achieve a higher arc voltage.

Furthermore, the kind of effects that are to be expected in the event of a ground fault or double ground fault must also be taken into consideration.

Setting range of the trip values

- Thermal overload release:

Same setpoints as in 50 / 60 Hz networks.

- Instantaneous short circuit trip unit:

The response threshold increases by 30 to 40%.

Example:

At the I_i = 4000 A setting, the overcurrent trip unit responds at approx. 5200 A ± 20%.

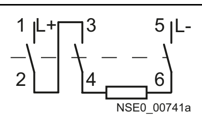
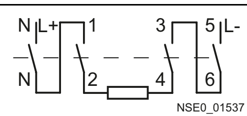
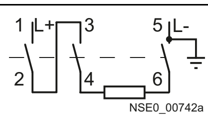
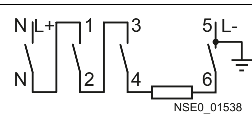
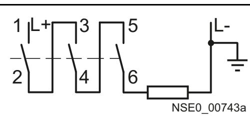
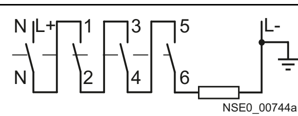
Depending on the voltage, a series connection of 2, 3 or 4 current paths is required.

The following circuits are recommended since the current must flow through all current paths in order to conform to the thermal tripping characteristic curves.

The thresold values of the instantaneous short-circuit release ("I" release) increase by 30 to 40%.

Suggested circuits for DC networks

Table 6- 2 Max. DC voltage Max. DC voltage for 3-pole circuit breakers[1] for 4-pole circuit breakers[2]

 250 V DC	 500 V DC	2-pole switching (non-grounded system) If ground faults have been excluded or if every ground fault is corrected immediately (ground fault monitoring), the maximum permissible direct voltage can be 600 V in both circuits.
 500 V DC	 600 V DC	2-pole switching (grounded system) The grounded pole should always be assigned to the single conducting path so that if a short-circuit occurs, 2 conducting paths are always connected in series in the case of 3-pole breakers, or 3 conducting paths in the case of 4-pole breakers.
 600 V DC	 600 V DC	1-pole switching (grounded system) The grounded pole is assigned to the ungrounded conducting path.

1) The max. direct voltage applies for the circuit breakers VL160 to VL630 with extremely high switching capacity (L). Otherwise a conversion is necessary in accordance with the permissible maximum voltage for 3 conducting paths, e.g. VL160X with switching capacity "N" with factor 250V / 600 V = 0.42.

2) In the case of 4 conducting paths in series, the 4th pole must be equipped either with a 100% trip unit or no trip unit at all. Moreover, the additional warming of the 4th conducting path means the maximum operating current must be reduced by 25%, and the trip times of the thermal overload release can change.

6.5 Use in IT networks

Use of the 3-VL circuit breakers in IT networks

The 3VL circuit breakers up to frame size VL1250 have been tested in accordance with IEC / EN 60947-2, Annex H (testing sequence for circuit breakers for IT systems) up to a maximum (V_i max.) voltage of 690 V AC. The switching capacities can be found in the certificates of conformance testing (Page 346). The 3VL8 cannot be used in the IT network.

The SIEMENS SENTRON VL circuit breakers for system protection, optionally with thermal overload and electromagnetic short-circuit releases, or electronic overcurrent trip units, are suitable for use in IT networks. The circuit breakers also meet the requirements of the IEC 60947-2 standard Annex H (DIN EN 60947-2, Annex H). The respective options are required here, and the necessary safety clearances (ventilation clearances) must be observed.

Selection criteria for circuit breakers

The devices are always dimensioned and selected independently of the relevant network type. The circuit breaker is always selected in accordance with the maximum occurring short-circuit current in the IT network. The device is selected in accordance with the relevant I_{cu} values of the 3VL circuit breaker.

Table 6- 3 I_{cu} values depending on V_e

V_e	3VL1	3VL2	3VL3	3VL4	3VL5	3VL6	3VL7	3VL8
I_{cu} at 240 V	65 kA	65 kA	65 kA	65 kA	65 kA	65 kA	65 kA	65 kA
I_{cu} at 415 V	55 kA	55 kA	55 kA	55 kA	45 kA	50 kA	50 kA	50 kA
I_{cu} at 690 V	8 kA	12 kA	12 kA	15 kA	20 kA	20 kA	20 kA	20 kA

The values in the above table apply provided one of the subsequent requirements is met:

- The IT network is operated with a grounded neutral cable,
- the plant operator takes the necessary precautions to prevent the occurrence of a double ground fault on the incoming or outgoing side of the circuit breaker.

Fault situation

The most critical fault for circuit breakers in ungrounded IT networks is a double ground fault on the incoming or load side of the circuit breaker. If this fault occurs, the entire phase-to-phase voltage is applied via one pole of the circuit breaker.

Illustration of a double ground fault

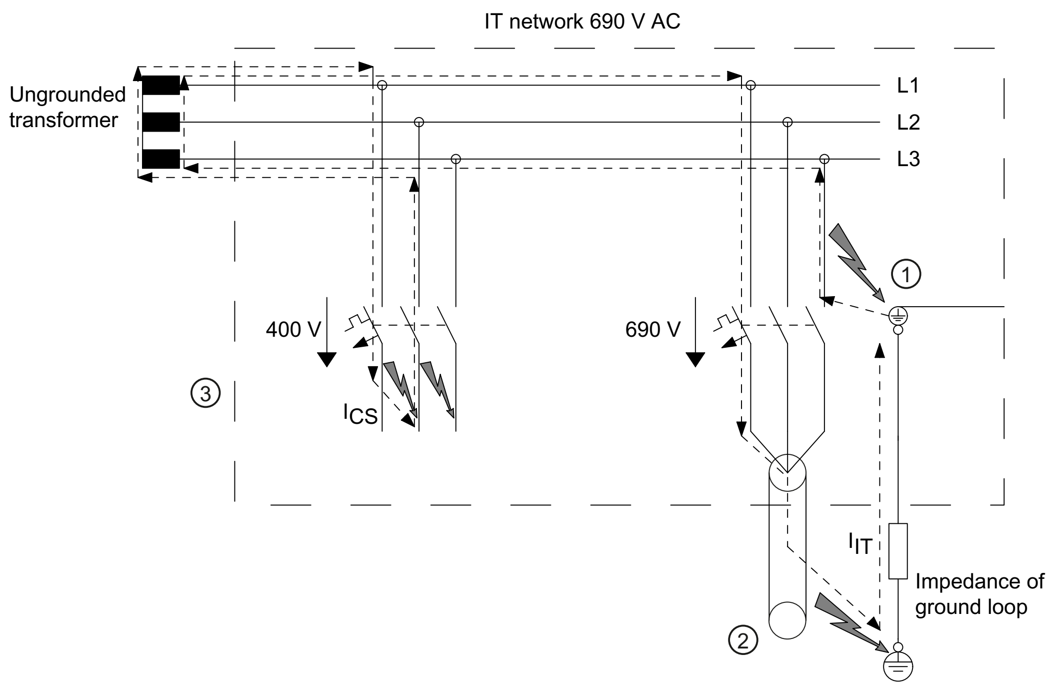


Figure 6-2 Ground and housing fault

Explanation of the illustration

Faults ① and ② simultaneously:

- Double ground fault on the load and incoming side
- Single-pole short-circuit, the full phase-to-phase voltage of 690 V is applied to main contact L1
- Selection of the circuit breaker according to their suitability as defined in IEC 60947-2, Annex H
- No calculation of the IT value possible or necessary

Fault ③

- 2 or 3-pole short-circuit
- Multi-pole short-circuit, a voltage of 690 V / 1.73 = 400 V is applied at the main contacts
- The circuit breaker is dimensioned as in other networks

If the above-listed requirements cannot be met by the customer, the values of the table below apply:

Table 6- 4 I_{cu} values depending on V_e in the event of a fault

V_e	3VL1	3VL2	3VL3	3VL4	3VL5	3VL6	3VL7	3VL8
I_{IT} at 240 V	3 kA	3 kA	3,5 kA	4,8 kA	8,7 kA	9,6 kA	15 kA	--
I_{IT} at 415 V	3 kA	3 kA	3,5 kA	4,8 kA	8,7 kA	9,6 kA	15 kA	--
I_{IT} at 690 V	3 kA	3 kA	3,5 kA	4,8 kA	8,7 kA	9,6 kA	15 kA	--

6.6 Use in the motor protection area

The overload and short-circuit releases are designed for optimal protection and direct-starting three-phase AC squirrel-cage motors. The motor protection circuit breakers are sensitive for phase failures and have an adjustable trip class.

The overcurrent trip units operate with a microprocessor.

Operating principle of the overcurrent trip units

The tripping characteristic curves of the current-dependent delayed overload releases are specially designed for overload protection of 3-phase AC motors.

In the case of the current-dependent delayed overload release "L", the value I_R can be set to be 0.4 to 1.0 times the rated current I_n of the circuit breaker. This occurs in 0.01 increments (e.g. 0.40 / 0.41 / 0.42 to 0.99 / 1.0 x I_n), so that the circuit breaker exactly matches the nominal current of the motor to provide optimal protection.

The current converters in the SENTRON VL circuit breaker don't only measure the load current, they also supply power to the electronic overcurrent trip unit.

This independence from an external energy supply guarantees a high standard of safety.

Area of application

Machine tools, manufacturing systems, presses, fans, air-conditioning units and packaging machines all require motors that must be protected. This is the main area of application of the SENTRON VL circuit breakers for motor protection.

Trip class

The SENTRON VL circuit breakers offer the option of selecting from various trip units with fixed or adjustable trip classes that are suitable for differing motor applications.

ETU 10 M

This version is equipped with a thermal memory, phase failure sensitivity and the fixed trip class 10.

ETU 30 M

This version is equipped with an adjustable trip class of 10A to bis 30 in addition to the thermal memory and phase failure sensitivity.

ETU 40 M

This version enables the parameters and the trip class to be configured step by step using a menu on the LCD display that is built into the trip unit.

Trip classes

Trip class 10A is used for motors that have very simple start-up characteristics (those with a short start-up time and a small moment of inertia). The class 30 releases are used to protect motors that have to withstand difficult start-up characteristics (long start-up time and large moment of inertia). The motor must be suitable for difficult start-ups.

The trip class must be selected so that it corresponds to the overload factor of the motor under operating conditions. See figure: Current-time curve before and after overload, with thermal memory.

Definition of the trip class

The trip class specifies the release time for balanced 3-pole loads, starting from the cold state, with 7.2 times the set current I_r according to IEC 60947-4-1. Combinations with class 10 are generally used.

Applications that require a longer start-up time, such as fans with large blades, require a higher trip class.

Thermal memory

All SENTRON VL circuit breakers possess a "thermal memory" which takes the pre-loading of the AC motor into consideration. The tripping times of the current-dependent delayed overload releases are only valid for the uncharged (cold) state.

The pre-loading of the 3-phase AC motor must be taken into consideration in order to prevent damage to the motor, e.g. after being frequently switched on without sufficient cooling time.

Siemens offers the SENTRON VL circuit breakers with fixed thermal memory to provide maximum protection for motors.

Functional principle of the thermal memory

During operation, a thermal image of the motor is simulated in the ETU. This reduces the response time of the circuit breaker with thermal memory to such an extent that further overloads cannot damage the motor windings. The motor is switched off within a time limit that is specified by the pre-loading.

The current required to switch the motor on again could also be considered to be an overload.

After an overcurrent tripping, the tripping times are reduced in accordance with the tripping characteristic curves.

A cooling time defined by the size of the motor is required before the motor can be switched on again. The circuit breaker prevents the motor from being turned on again during this time interval. This prevents the motor from being excessively thermally loaded by a current immediately after an overload release occurs.

Phase failure sensitivity

The "phase failure sensitivity" function is also integrated into the SENTRON VL circuit breakers for motor protection. This ensures that the motor is reliably protected against overheating if a phase interruption or a large fluctuation occurs.

The specified operational current IR is automatically reduced to 80% of the set value if the r.m.s. values of the operational currents in the three phases differ by more than 50%.

Deviations of more than 50% mean the value of the current in the least loaded phase drops to a level below 50% of the maximum loaded phase.

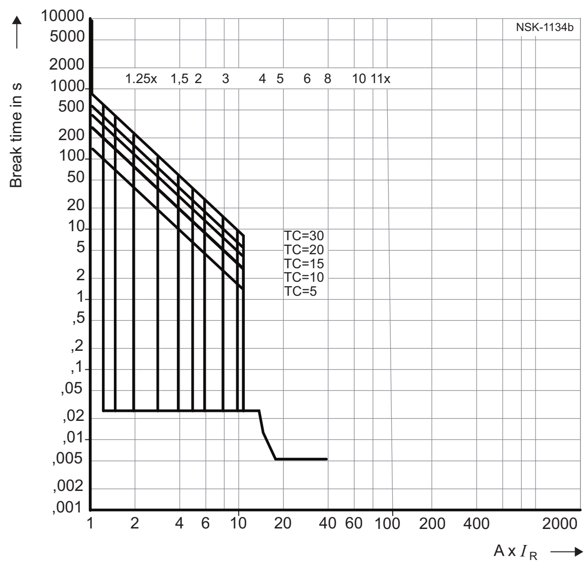


Figure 6-3 ETU with trip classes 5, 10, 15, 20, 30

Tripping characteristic curve for circuit breakers with electronic overcurrent trip unit.

I_{cu} 100 kA maximum at 415 V

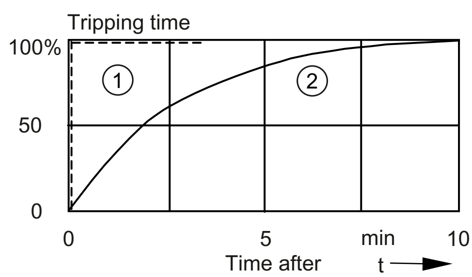


Figure 6-4 Response time of the trip unit after overload release

① Without "thermal memory"
② with "thermal memory"

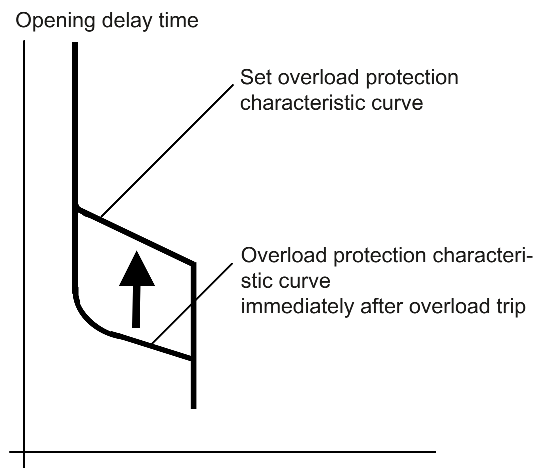


Figure 6-5 Current-time curve before and after overload, with thermal memory.

6.7 Use in harsh environments:

If the SENTRON VL are to be used outside closed rooms or in harsh operating conditions, the following information must be taken into account at the planning stage:

Derating factors under special ambient conditions

Harsh operating conditions include:

- Altitude in excess of 2000 meters
- Temperature above/below 50 °C
- Frequencies outside the 50 / 60 Hz range
- Humidity

Use at altitudes above 2,000 meters

The lower air density at altitudes above 2,000 meters affects the key electrical data of molded-case circuit breakers. The table in the Technical data (Page 159) shows the derating factors that have to be taken into account when using the circuit breakers at altitudes above 2,000 m.

Use at different ambient temperatures

A reduction (derating) of the rated operational current of the SENTRON VL circuit breaker is necessary if the ambient temperature exceeds 50 °C. The reference temperature is 40 °C for circuit breakers with RCD modules or plug-in/withdrawable versions.

The permissible loads for various ambient temperatures in relation to the rated operational current of the circuit breaker are shown in the technical data.

Furthermore, the following points must be taken into consideration, because each one of these factors can influence the rated operational current and permissible load.

- Type of circuit breaker (fixed-mounted, plug-in, or withdrawable version)
- Type of main connection (vertical/horizontal busbar, cable)
- Ambient temperature around the circuit breaker
- Altitude derating factors
- Temperature derating factors based on different trip units and connections
- Degree of protection

Thermomagnetic overcurrent trip unit

Thermomagnetic overcurrent trip units are calibrated to 50 °C. As a result, the tripping times of the thermal overcurrent trip unit increase for a constant current at low temperatures.

To correct the tripping times, the thermal overcurrent trip unit settings must be changed by the factor from the table "Derating factors for thermomagnetic overcurrent trip units" in the Technical data (lower settings).

Use in networks with different frequencies

If low-voltage switching devices designed for 50 / 60 Hz are to be used at other network frequencies, the following points must be taken into consideration:

- Thermal effects on the system components
- Switching capacity
- Service life of the contact system
- Tripping characteristics of the overcurrent trip unit
- Behavior of the accessories

Thermal rating of the system components and conductors depending on the network frequency

Circuit breakers designed for alternating current of 50 / 60 Hz can be used at lower frequencies for at least the same rated currents. However, in contrast to this, the permissible operating current must be reduced at frequencies higher than 100 Hz to ensure the specified temperature increase limits are not exceeded.

Circuit breakers for 400 Hz applications are available on request.

Use in 16 2/3 Hz networks

Circuit breakers must be selected according to their DC switching capacities for frequencies up to 16 2/3 Hz. These values can be found in the relevant table in the Technical data (Page 157). The rated operational current of the circuit breaker is the same at 16 2/3 Hz and 380/400 V as it is at 50/60 Hz – 3-pole, with two poles used in series. At 16 2/3 Hz and 500 V, all three poles must be used in series.

When used in 50/60 Hz networks, selection can be made in the relevant tables in the Technical data (Page 157) depending on the ambient temperature, switching capacity, etc.

Influence of temperature and humidity on overcurrent trip units

The relevant reduction in the rated operating current (derating) of the SENTRON VL circuit breakers is also necessary if the operating temperature of 50 °C or 70 °C is exceeded at a non-condensing humidity level of 95%.

Thermomagnetic TM trip units

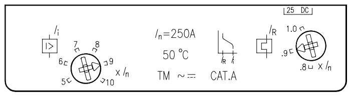


Figure 6-6 Thermomagnetic TM
–25 °C to +50 °C, ☂ 95%

The SENTRON VL thermomagnetic trip units are designed for use in ambient temperatures up to 70 °C and a non-condensing humidity level up to 95%. The appropriate correction factors must be applied for ambient temperatures above 50 °C.

Electronic trip unit ETU

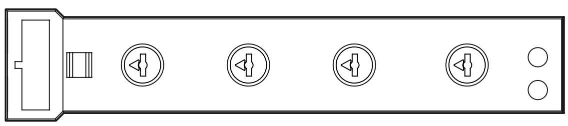


Figure 6-7 Standard ETU
-25 °C to +70 °C, ☂ 95%

The SENTRON VL electronic trip units are designed for use in ambient temperatures up to 70 °C and a non-condensing humidity level up to 95%. The appropriate correction factors must be applied for ambient temperatures above 50 °C.

Electronic trip unit LCD ETU

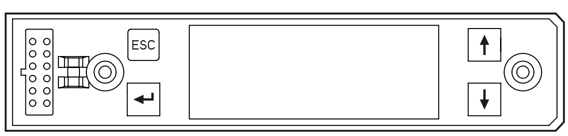


Figure 6-8 LCD-ETU
-25 °C to +70 °C, ☂ 95%

The high-quality SENTRON VL electronic trip units are designed for use in ambient temperatures up to 70 °C and a non-condensing humidity level up to 95%. The appropriate correction factors must be applied for ambient temperatures above 50 °C.

6.8 Use in series connection

In the case of circuit breakers connected in series, the overload and short-circuit protection is described as "selective" when, from the point of view of the energy flow, only the circuit breaker immediately upstream of the fault switches off.

Current selectivity

The selectivity can be calculated in the **overload range** by comparing the current characteristic curves and the time characteristic curves. In the short-circuit range, this comparison leads to values that are too low. The reason for this is that the trip unit behaves differently in the case of short-circuit currents compared to its long-term behavior, e.g. in the case of overloads.

If the **short-circuit currents differ sufficiently** at the points where two circuit breakers are mounted, the instantaneous short-circuit releases can normally be set such that if a short-circuit occurs downstream from the circuit breaker which itself is downstream from the short-circuit release, only the downstream one trips.

If the **short-circuit currents are approximately the same** at the points where the circuit breakers are mounted, the time grading of the tripping currents of the short-circuit releases only enables selectivitiy up to a specific short-circuit current .

This current is referred to as the selectivity limit.

If the values determined by the short-circuit current calculation (e.g. according to DIN VDE 0102) at the mounting point of the downstream circuit breaker lie below the selectivitiy limit listed in the respective table for the selected combination, selectivity is guaranteed for all possible short-circuit reductions at the mounting point.

If the calculated short-circuit current at the mounting point is higher than the selectivity limit, selective tripping by the downstream circuit breaker is only ensured up to the value listed in the table. The configuring engineer must judge whether the value can be considered to be sufficient because the probability of, for example, the maximum short-circuit occurring is low. Otherwise, a circuit breaker combination should be chosen whose selectivity limit lies above the maximum short-circuit current.

Time selectivity

Time selectivity is an alternative possibility for securing selectivity if the short-circuit currents are approximately the same at the mounting points. To achieve this, the upstream circuit breaker requires delayed short-circuit releases, so that if a fault occurs, only the downstream circuit breaker will disconnect the affected part of the system from the network.

Both the tripping delays and the tripping currents of the short-circuit releases are graded.

Zone-Selective Interlocking (ZSI) has been developed by SIEMENS for the SENTRON VL circuit breakers to prevent long, undesired release times when several circuit breakers are connected in series.

ZSI enables the tripping delay to be reduced to a maximum of 100 ms for the circuit breaker upstream from the location of the short-circuit.

When selecting a circuit breaker, the circuit breaker must be capable of dealing with the initial balanced short-circuit current I_K at the mounting point.

You can find more information in the manual "SENTRON WL and SENTRON VL (PROFIBUS) circuit breakers with communication capability (PROFIBUS)" (Order No. A5E01051347-01).

Installing/mounting 7

7.1 Installation methods

Installation overview

The SENTRON VL circuit breakers are available in fixed-mounted, plug-in or withdrawable versions, with three or four poles.

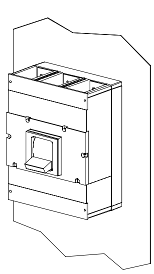
Table 7- 1 Overview of installation methods

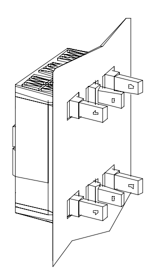
Circuit breaker type	Fixed	Plug-in	Withdrawable part
VL 160X	x	x	-
VL 160	x	x	x
VL 250	x	x	x
VL 400	x	x	x
VL 630	x	x	x
VL 800	x	-	x
VL 1250	x	-	x
VL 1600	x	-	x

Fixed mounting

Mounting plate connection

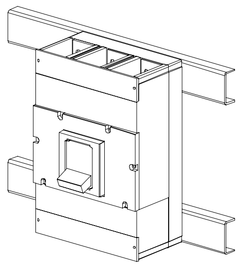
SENTRON VL circuit breakers can be installed direct onto the mounting plate. If busbars or terminals are used to connect the circuit breaker on the back of the mounting plate, the appropriate safety clearances must be observed (see technical overview)





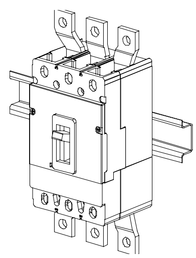
Mounting rail connection

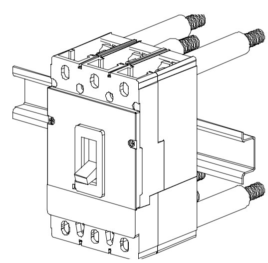
SENTRON VL circuit breakers from Siemens can be mounted direct onto the mounting rails supplied by the customer. The appropriate safety clearances must be observed.



Busbar connections

Busbars or cables can be connected direct to the front of busbar extensions or to bolts for connections on the back. If straight busbar extensions are used, terminal covers or phase barriers are recommended.

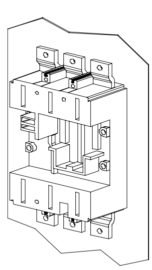


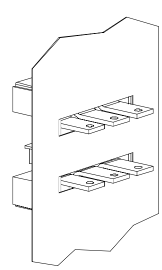


Plug-in version

Mounting plate connection

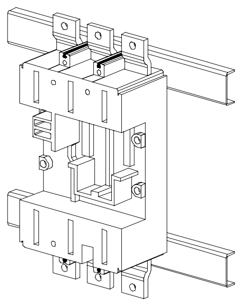
Plug-in sockets with pad-type terminals on the front or rear are available for direct connection of cables or busbars. The plug-in socket is attached direct to the mounting plate or mounting rail supplied by the customer.

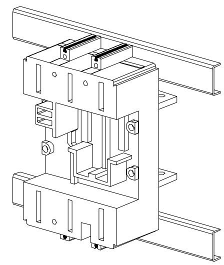




Mounting rail connection

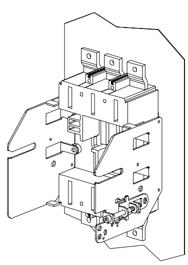
The appropriate safety clearances must be observed. Terminal covers or phase barriers are available for the front connecting bars. Circuit breakers cannot be removed from the plug-in socket in the "On" position. The circuit breaker will switch to the "tripped" position if attempts are made to remove it in the "ON" position.

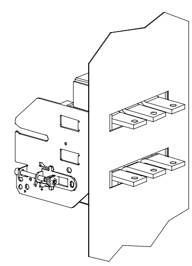




Withdrawable version

SENTRON VL circuit breakers can be used as withdrawable devices. They can be connected on either the front or the back. Safety covers are provided and are required for final installation.





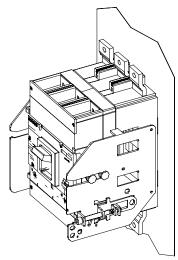
In the connected position, the circuit breaker is completely engaged, and all contacts - supply, outgoing and auxiliary contacts - are connected to the guide frame. The circuit breaker is ready for operation.

Note

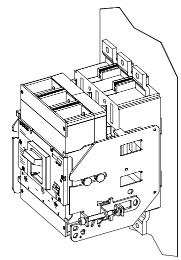
Safety interlock

A safety interlock prevents the circuit breaker from being removed when it is switched on. The safety interlock causes the circuit breaker to switch off so that the arc which occurs inside the circuit breaker when current flows can be extinguished.

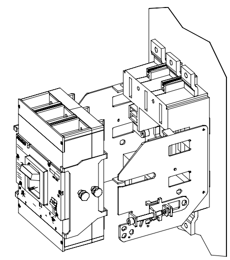
The circuit breaker can be installed in and removed from the guide frame when it is in the removable position.



Connected position



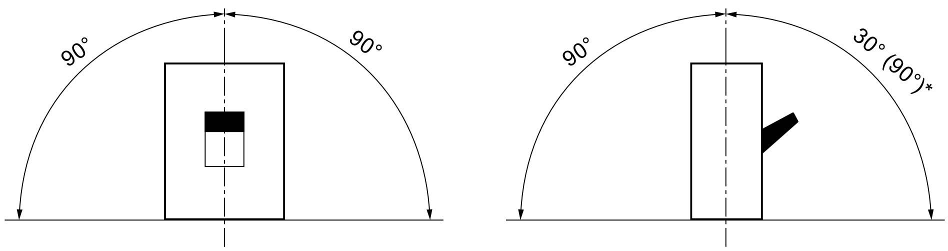
Disconnected position



Removable position

7.2 Mounting and safety clearances

All SENTRON VL circuit breakers can be mounted in the positions shown.



* Max. permissible current load factor 0.9; with internal accessories only.

Figure 7-1 Mounting/installation

Safety clearances

During a short-circuit interruption, high temperatures, ionized gases and high pressures occur in and above the arcing chambers of the circuit breaker.

Safety clearances are required to:

- allow the pressure to be distributed
- prevent fire or damage caused by any escaped ionized gases
- prevent a short circuit to grounded sections
- prevent arcing or short-circuit currents to live sections

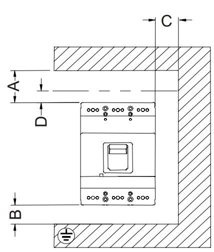


Figure 7-2 Safety clearances

Table 7- 2 Permissible safety clearances in accordance with IEC 60947

Circuit breaker type	Switching capacity	A ≤ 415 V	A > 415-690 V		B ≤ 690 V	C ≤ 690 V	D ≤ 690 V
		With or without covers	Without covers	With covers			
VL160X	Standard High	35 mm	70 mm	35 mm	25 mm	25 mm	35 mm
VL160	Standard High Very high	50 mm	100 mm	50 mm	25 mm	25 mm	35 mm
VL250	Standard High Very high	50 mm	100 mm	50 mm	25 mm	25 mm	35 mm
VL400	Standard High Very high	50 mm	100 mm	50 mm	25 mm	25 mm	35 mm
VL630	Standard High Very high	50 mm	100 mm	50 mm	25 mm	25 mm	35 mm
VL800	Standard High Very high	50 mm	100 mm	50 mm	25 mm	25 mm	35 mm
VL1250	Standard High Very high	70 mm	100 mm	70 mm	30 mm	30 mm	50 mm
VL1600	Standard High Very high	100 mm	100 mm	100 mm	100 mm	30 mm	100 mm

Definition of the permissible safety clearances in [mm] between

- Q: Circuit breaker and current paths (uninsulated and grounded metal)
- B: Circuit breaker phase terminal and lower panel
- C: Sides of the circuit breaker and side panels left/right (uninsulated and grounded metal)
- D: Circuit breaker and non-conductive parts with at least 3 mm thick insulation (insulator, insulated bar, painted plate)

If uninsulated conductors are connected to terminals 1, 3, 5 and 7, they must be insulated from each other independently of the direction of the mains supply (see Chapter 3.1.1.). This can be achieved using phase barriers or terminal covers.

Terminal covers must be used for the main terminals at voltages of ≥ 600 V AC or ≥ 500 V DC.

Note

We recommend you also insulate connections 2, 4, 6 and 8 from each other for additional safety.

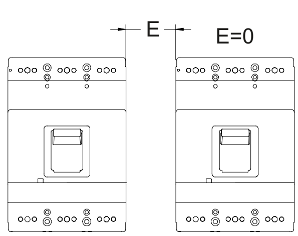


Figure 7-3 Minimum clearance between two horizontally or vertically installed circuit breakers

Minimum clearance between two horizontally or vertically installed circuit breakers.

Ensure the busbar or cable connection does not reduce the air insulation distance. The permissible clearance between two circuit breakers applies for both fixed-mounted and plug-in versions. Some accessories may increase the width of the circuit breaker.

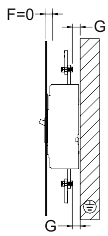


Figure 7-4 Minimum clearance between the circuit breaker and metal

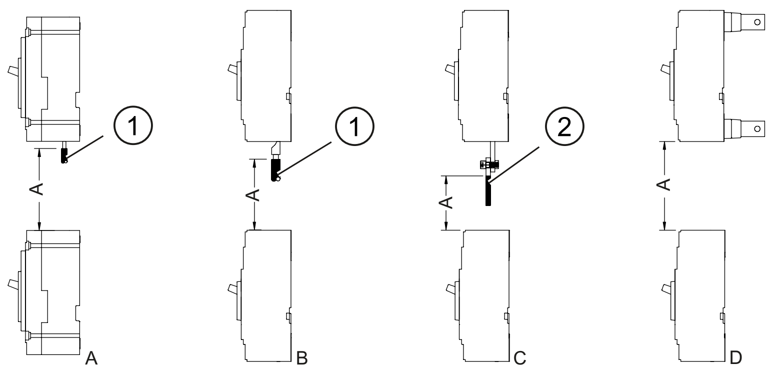
The clearance between the terminal and the grounded metal must be G ≥ 12 mm.

If the clearance to ground G is < 12 mm, live parts must be insulated or a suitable barrier must be installed.

CAUTION
Depending on the application, appropriate air and creepage distances must be observed, e.g. IEC 60439-1.

Safety clearances between circuit breakers

Minimum clearance between two circuit breakers installed above one another with different kinds of connections.



A Connection on the front with cable, direct

B Connection on the front with cable lug

C Connection on the front with flat connecting bar

D Connection on the back with plug-in socket or busbar terminals

① Insulation

② Insulation busbar

Figure 7-5 Table of different connection types

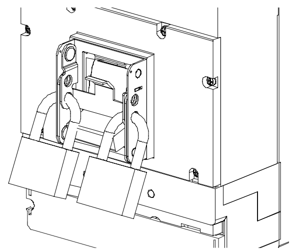
Table 7- 3 Safety clearances between circuit breakers

Circuit breaker type	VL160X	VL160	VL250	VL400	VL630	VL800	VL1250	VL1600
Switching capacity	LV HRC	NHL			NHL			
A ≤ 690 V	160 mm				200 mm			

The clearances given in the table are necessary to enable any ionized gases arising during a short-circuit to disperse

7.3 Locking devices

Locking device for the toggle handle



Locking device for the toggle handle

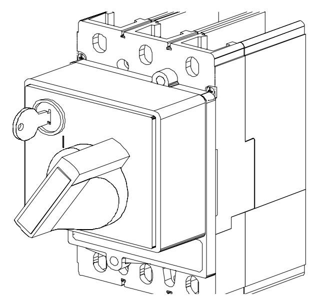
The locking device for the toggle handle is designed to be easily attached to the circuit breaker collar. This device allows the handle to be locked in the "OFF" position. The locking device for the toggle handle can be installed in 3-pole and 4-pole circuit breakers. Up to 3 padlocks with shackle diameters ranging from 5 to 8 mm may be used. (Not for the VL160X with RCD module)

Safety lock for the rotary operating mechanism and the motorized operating mechanism

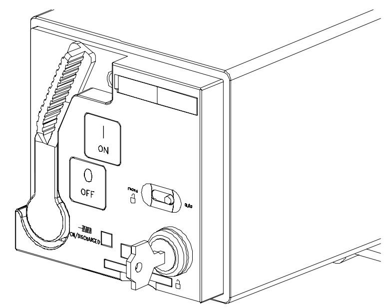
A safety lock can be used for both the rotary operating mechanism and the motorized operating mechanism.

The safety lock is used to lock the circuit breaker in the "OFF" position. The key can only be removed when the circuit breaker is in the "OFF" position. The key cannot be removed when the rotary operating mechanism or the motorized operating mechanism is in the "ON" position.

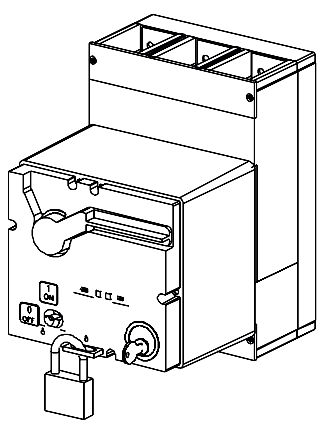
Each safety lock is supplied with its own locking system as standard.



Rotary operating mechanism on the front with key



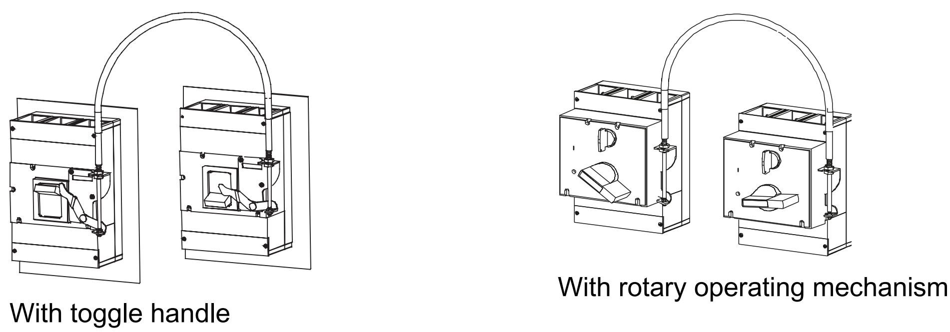
Stored-energy operator for the VL250 (key)

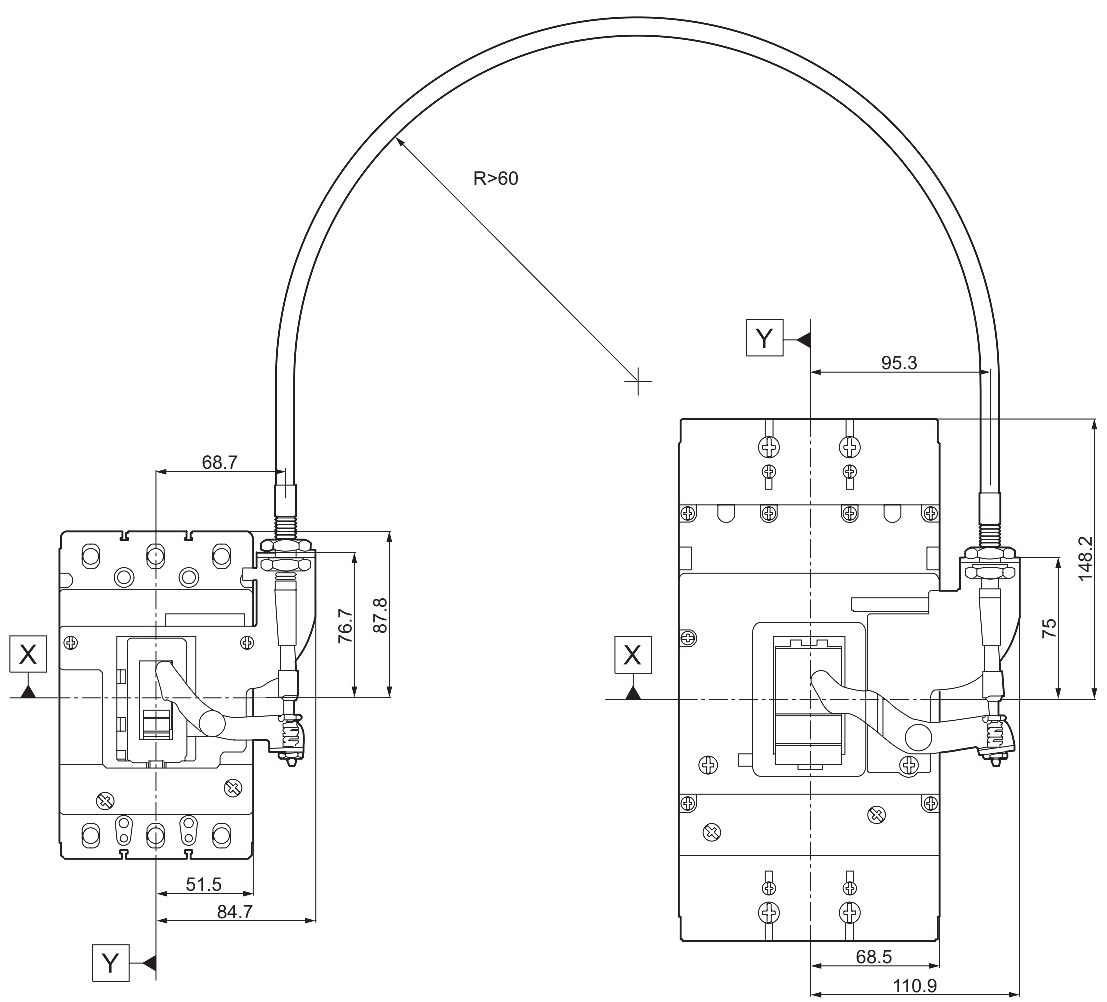


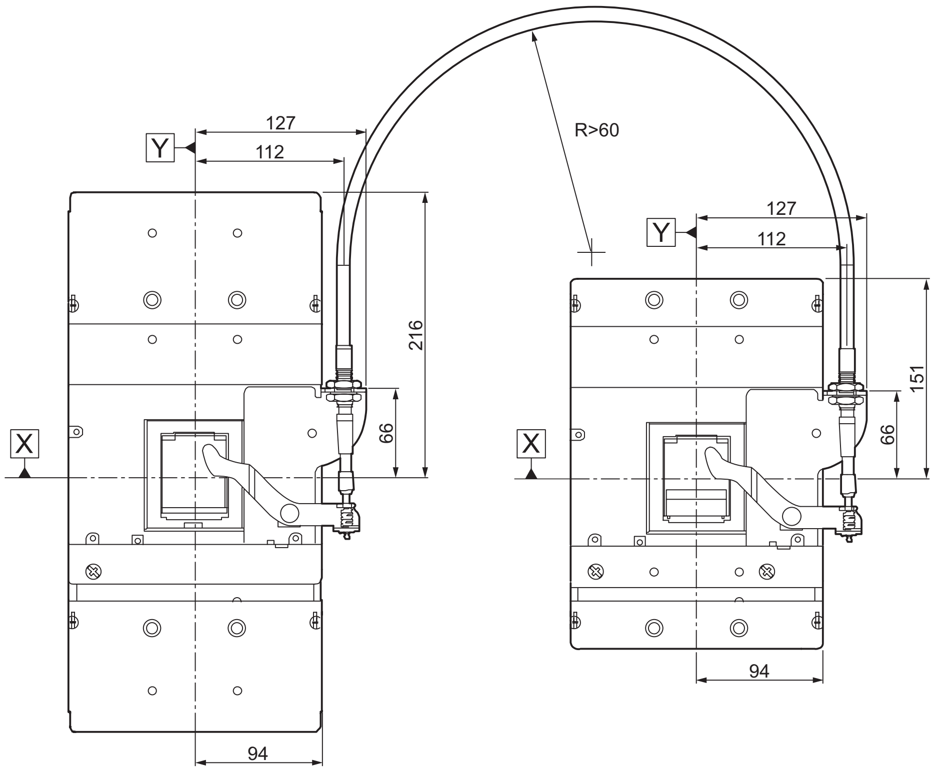
Stored-energy motorized operating mechanism for the VL630

Mutual interlocking of two circuit breakers (bowden wire) in the fixed-mounted, plug-in and withdrawable versions

Table 7- 4 Mounting options







The combination table below shows the mutual locking options of the individual circuit breaker breaker dimensions:

Table 7- 5 Locking with bowden wire

	3VL9 300-8LA00 for VL160X (3VL1), VL160 (3VL2) and VL250 (3VL3)	**3VL9 400-8LA00** for VL400 (3VL4)	**3VL9 600-8LA00** for VL630 (3VL5) and VL800 (3VL6)	**3VL9 800-8LA00** for VL1250 (3VL7) and VL1600 (3VL8)
3VL9 300-8LA00 for VL160X (3VL1), VL160 (3VL2) and VL250 (3VL3)	✓	-	-	-
3VL9 400-8LA00 for VL400 (3VL4)	-	✓	-	-
3VL9 600-8LA00 for VL630 (3VL5) and VL800 (3VL6)	-	-	✓	-
3VL9 800-8LA00 for VL1250 (3VL7) and VL1600 (3VL8)	-	-	-	✓

Two SENTRON VL circuit breakers can be mutually mechanically interlocked using a bowden cable and the locking modules.

Modules with the same dimensions or with the dimensions specified above (e.g. VL250 and VL400) can be locked together.

Use of this accessory kit means only one of the circuit breakers is in the "ON" position at any time.

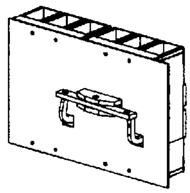
Fixed-mounted and plug-in circuit breakers use different interlocking modules. However, these are compatible with each other. This enables both to be used in locking circuits.

Two circuit breakers can be mounted side by side or one above the other. The distance between the two circuit breakers depends on the length of the bowden cable and its minimum bending radius. The cable comes in lengths of 0.5, 1.0 and 1.5 m. The minimum bending radius for each cable is 60 mm. The length of the bowden cable must not be altered by the customer. The bowden cable has a mechanical endurance of 10,000 operations. Each bowden cable must be ordered separately.

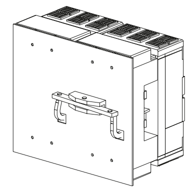
Note

Not possible in combination with the motorized operating mechanism.

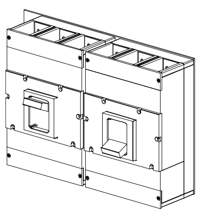
Mutual interlocking (rear interlocking module) of two circuit breakers in the fixed-mounted, plug-in and withdrawable versions



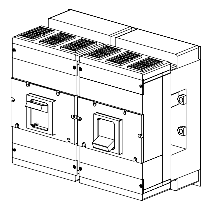
Fixed-mounted version (lock at rear)



Plug-in version (lock at rear)



Fixed-mounted version (lock at front)



Plug-in version (lock at front)

The rear interlocking module enables mutual mechanical interlocking of two SENTRON VL circuit breakers of the same frame size. The rear interlocking module is attached behind the circuit breakers to the mounting plate supplied by the customer.

A tappet on each end of the rocker automatically accesses the either of the breakers through an opening in the mounting plate and the base of the circuit breakers. The rear interlocking module prevents both circuit breakers from being in the "ON" position at the same time.

The rear interlocking module can be used with fixed-mounted, plug-in and withdrawable circuit breakers.

Cross wiring of internal accessories via the rear of the circuit breaker is not prevented.

This locking version is possible with all operating mechanism types (toggle handle, rotary operating mechansim, and motorized operating mechanism).

Connecting 8

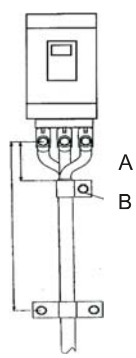
8.1 Cables and busbars

SENTRON VL molded-case circuit breakers can be connected using cables, flexible copper bars or busbars. Either copper or aluminum can be used.

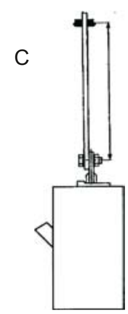
Thermal and electrodynamic loads affect these conductors if a short-circuit occurs. To avoid dangerous effects, it is necessary to size them properly and to ground them correctly.

The diagrams and tables below show the recommendeded maximum clearance between the circuit breaker and the first support.

Overview of cable and busbar mounting methods



Cable mounting



Busbar mounting

Table 8- 1 Recommended cable mounting clearances

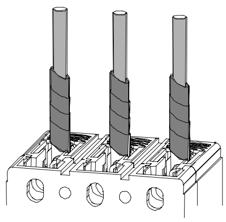
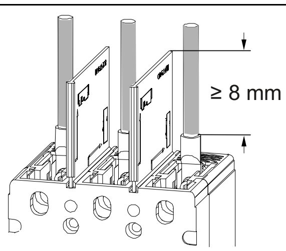
Support dimensions	VL160X	VL160	VL250	VL400	VL630	VL800	VL1250	VL1600
A cable mm	100	100	130	150	300			
B cable mm	400	400	400	400	600			
C bar mm	250							

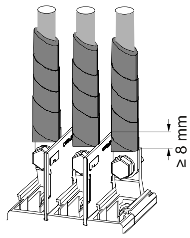
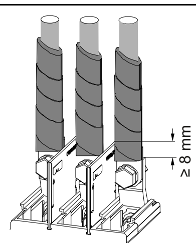
This table applies for all switching capacities

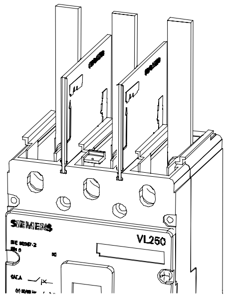
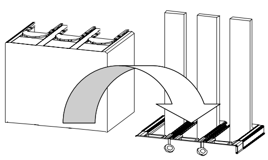
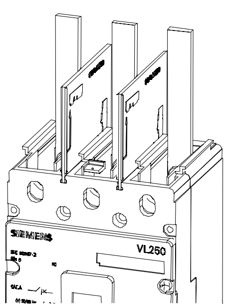
Rated operating voltage: V_e ≤ 600 V AC / 500 V DC

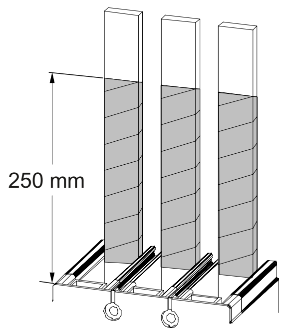
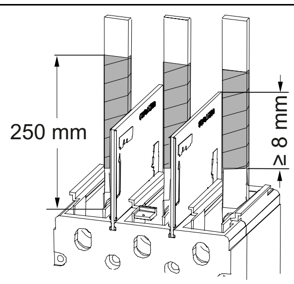
(data about switching capacity I_{cu} is based on 400/415 V AC)

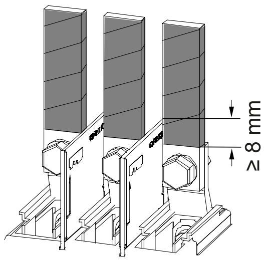
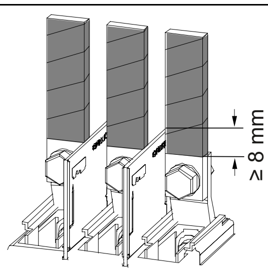
Table 8- 2 Connection methods (600 V AC/500 V DC)

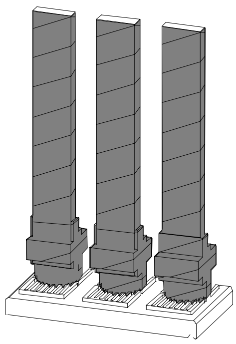
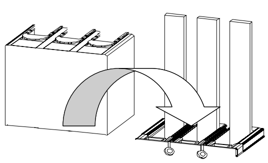
Circuit breaker dimensions	VL160X	VL160	VL250	VL400	VL630	VL800	VL1250	VL1600
	Permissible switching capacity class for V_e ≤ 600 V AC/500 V DC							
 • Cable mounted direct • Insulated up to the circuit breaker • Accessories: – None	N H	N H L	N H L	N H L	N H L	N H L	N H L	Not applicable
 • Cable with cable lug • Insulation 8 mm above phase barrier • Accessories: – Phase barriers – Weitkowitz cable lug – Connection with screw-type terminals	N H	N H L	N H L	N H L	N	N	N	Not applicable

Circuit breaker dimensions	VL160X	VL160	VL250	VL400	VL630	VL800	VL1250	VL1600
 • Cable with cable lug • Front connecting bars, standard • Insulation 8 mm above phase barrier • Accessories: – Phase barriers – Connection with screw-type terminals – Front connecting bars, standard	N H	N H L	N H L	N H L	N	N	N	Not applicabl e
 • Front flared • busbar extensions • Insulation 8 mm above phase barrier • Accessories: – Phase barriers – Connection with screw-type terminals – Front flared busbar extensions	N H	N H L	N H L	N H L	N	N	N	N

Circuit breaker dimensions	VL160X	VL160	VL250	VL400	VL630	VL800	VL1250	VL1600
 • Connecting bar, direct mounted • Without insulation • Accessories: – Phase barriers – Connection with screw-type terminals	N	N	N	N	N	N	N	N
 • Connecting bar, direct mounted • With extended terminal cover • Without insulation • Accessories: – Extended connection cover – Connection with screw-type terminals	N H	N H L	N H L	N H L	N H L	N H L	N H L	N H L
REVERSE  • Connecting bar, direct mounted • Incoming supply from overcurrent trip unit side • Without insulation • Accessories: – Phase barriers – Connection with screw-type terminals	N H	N H L	N H L	N H L	N H L	N H L	N H L	N H L

Circuit breaker dimensions	VL160X	VL160	VL250	VL400	VL630	VL800	VL1250	VL1600
 • Connecting bar, direct mounted • Insulation 250 mm from the circuit breaker • Accessories: – Connection with screw-type terminals	N H	N H L	N H L	N H L	N	N	N	N
 • Connecting bar, direct mounted • Insulation 8 mm above phase barrier and 250 mm from circuit breaker • Accessories: – Phase barriers – Connection with screw-type terminals	N H	N H L	N H L	N H L	N	N	N	N

Circuit breaker dimensions	VL160X	VL160	VL250	VL400	VL630	VL800	VL1250	VL1600
 • Connecting bar • Front connecting bars, standard • Insulation 8 mm above phase barrier and 250 mm from circuit breaker • Accessories: – Phase barriers – Connection with screw-type terminals – Front connecting bars, standard	N H	N H L	N H L	N H L	N	N	N	N
 • Connecting bar • Front flared • busbar extensions • Insulation 8 mm above phase barrier and 250 mm from circuit breaker • Accessories: – Phase barriers – Connection with screw-type terminals – Front flared busbar extensions	N H	N H L	N H L	N H L	N	N	N	N

Circuit breaker dimensions	VL160X	VL160	VL250	VL400	VL630	VL800	VL1250	VL1600
 • Connecting bar • Front connecting bars, standard • Insulation 250 mm from the circuit breaker • Accessories: – Connection with screw-type terminals – Front connecting bars, standard	N H	N H L	N H L	N H L	N H L	N H L	N H L	N H L
 • Connecting bar • Front connecting bars, standard • With extended connection cover • Without insulation • Accessories: – Extended connection cover – Connection with screw-type terminals – Front connecting bars, standard	N H	N H L	N H L	N H L	N H L	N H L	N H L	N H L

N: Low

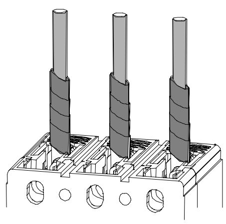
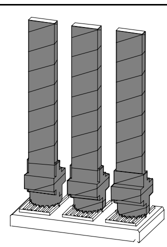
H: High

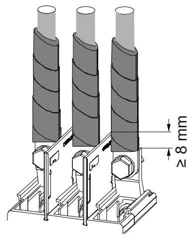
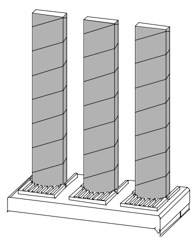
L: Very high

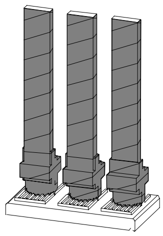
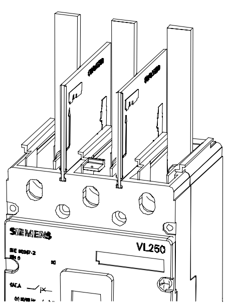
Rated operating voltage: V_e ≤ 690 V AC/600 V DC

(data about switching capacity I_{cu} is based on 690 V AC)

Table 8- 3 Connection methods (690 V AC/600 V DC)

Circuit breaker dimensions	VL160X	VL160	VL250	VL400	VL630	VL800	VL1250	VL1600
	Permissible switching capacity class for V_e ≤ 600 V AC/500 V DC							
 • Cable mounted direct • Insulated up to the circuit breaker • **Accessories:** – Standard terminal cover	N H	N H L	N H L	N H L	N H L	N H L	N H L	Not applicable
 • Cable with cable lug • Weitkowitz cable lug • Front connecting bars, standard • Insulated up to the circuit breaker • **Accessories:** – Standard terminal cover – Connection with screw-type terminals – Front connecting bars, standard	N H	N H L	N H L	N H L	N H L	N H L	N H L	Not applicable

Circuit breaker dimensions	VL160X	VL160	VL250	VL400	VL630	VL800	VL1250	VL1600
 • Cable with cable lug • With extended connection cover • Accessories: – Extended terminal cover – Connection with screw-type terminals	N H	N H L	N H L	N H L	Not applicable	Not applicable	Not applicable	Not applicable
 • Connecting bar, direct mounted • Insulation 250 mm from the circuit breaker • Accessories: • Standard terminal cover • Connection with screw-type terminals	N H	N H L	N H L	N H L	N H L	N H L	N H L	N H L

Circuit breaker dimensions	VL160X	VL160	VL250	VL400	VL630	VL800	VL1250	VL1600
 • Connecting bar • Front connecting bars, standard • Insulation 250 mm from the circuit breaker • Accessories: – Standard terminal cover – Connection with screw-type terminals – Front connecting bars, standard	N H	N H L	N H L	N H L	N H L	N H L	N H L	N H L
REVERSE  • Connecting bar, direct mounted • Incoming supply from overcurrent trip unit side • Without insulation • Accessories: – Phase barriers – Connection with screw-type terminals	N	N H L	N H L	N H L	N H L	N H L	N H L	N H L

N: Low

H: High

L: Very high

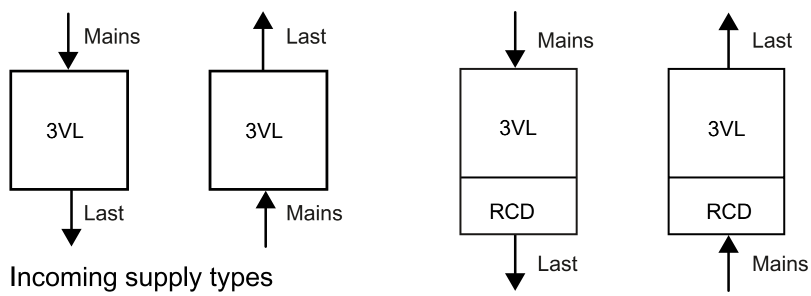
8.2 Main connection types for fixed mounting

Main conductor connection for SENTRON VL fixed-mounted version

There are different methods of connecting the circuit breaker main conductors for fixed mounting.

Network connection

The SENTRON VL circuit breakers can be supplied with power from above and below.

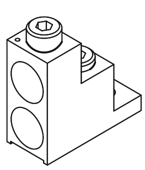


Incoming supply types

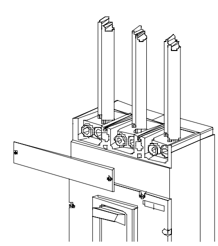
Network: Incoming supply
Load: Outgoing feeder

Multiple feed-in terminal for round cables (copper/aluminum)

The multiple feed-in terminals for incoming supply and outgoing feeders consist of an aluminum body with tin plating to prevent oxidation. Both aluminum and copper cables may be used. Only one conductor is permitted per terminal. Multiple feed-in terminals are available for the SENTRON VL 160X to VL 1250 circuit breakers. Additional screw-type terminals are required for the SENTRON VL 160X and VL 160 circuit breakers.



Multiple feed-in terminals

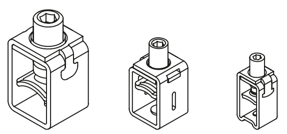


Use of multiple feed-in terminals

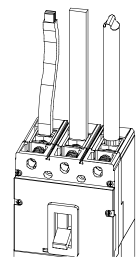
For additional information, refer to the technical data in Chapter Configuration of main connections (Page 150).

Box terminals (copper cables or bars)

The steel box terminal is supplied as standard for use with the SENTRON VL160X and VL160 circuit breakers. It is optional for VL250 to VL400. The terminal is designed to connect either a conductor or a solid/flexible copper bar.



Box terminals

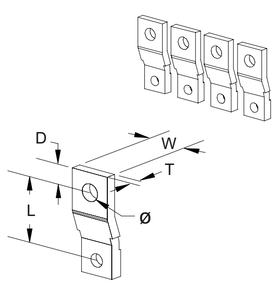


Box terminals with solid/flexible copper bars or cables

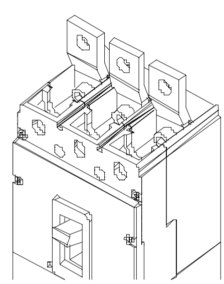
For additional information, refer to the technical data in Chapter Configuration of main connections (Page 150).

Front connecting bars

Connecting bars are used to connect the circuit breakers to busbars or cables in electrical systems. Front connecting bars are supplied with the SENTRON VL1600 as standard. Phase barriers are also included. Extended terminal covers can be fitted if necessary. Screw-type terminals with a metric thread are required for the SENTRON VL160X and 160 (see Connection with screw-type terminals).



Front connecting bar



Use of front connecting bars

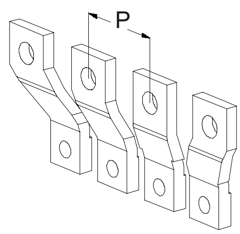
For additional information, refer to the technical data in Chapter Configuration of main connections (Page 150).

Front flared busbar extensions

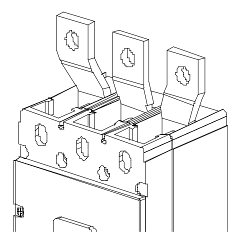
Front flared busbar extensions are used to establish busbar connections in switchboards or other electrical equipment. Normal use enables them to be matched to the next largest circuit breaker. Phase barriers are also included.

Note

Cannot be combined with extended terminal covers! Additional screw-type terminals are required for the SENTRON VL160 and VL160X.



Flared busbar extensions

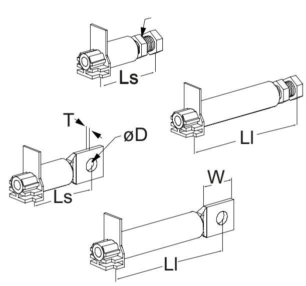


Use of flared busbar extensions

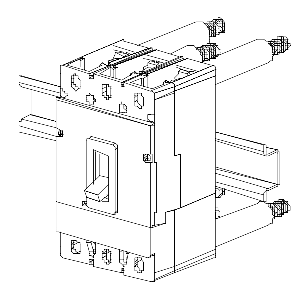
For additional information, refer to the technical data in Chapter Configuration of main connections (Page 150).

Rear terminals

Rear terminals are used to adapt the SENTRON VL circuit breakers to switchboards or other applications that require rear connection. They are bolted direct to a standard SENTRON VL circuit breaker without requiring any modification. Circuit breakers mounted in switchboards or other electrical equipment may be removed from the front by removing the fixing screw that connects the circuit breaker to the terminal



Round terminals

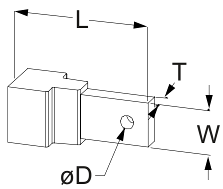


Use of the terminals

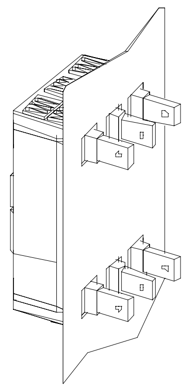
For additional information, refer to the technical data in Chapter Configuration of main connections (Page 150).

Rear flat busbar terminals

Rear flat busbar terminals are used to adapt SENTRON VL630 to VL1600 circuit breakers to switchboards or other applications that require rear connection. They are screwed direct to a standard SENTRON VL circuit breaker without requiring any modification. A vertical or horizontal connection is established, depending on the way the busbar terminals are mounted to the rear of the circuit breaker. Circuit breakers mounted in switchboards or other electrical equipment with the help of rear flat busbar terminals may be removed from the front by removing the fixing screw that connects the circuit breaker to the terminal



Bus bars

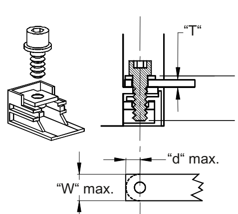


Bus bars

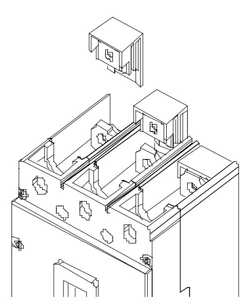
For additional information, refer to the technical data in Chapter Configuration of main connections (Page 150).

Connection with screw-type terminals

The screw-type terminal with metric thread slides onto the incoming and outgoing terminal of the SENTRON VL circuit breaker and acts as a threaded adapter for connecting busbars and cable lugs. The customer is responsible for providing screws and washers for the terminals and busbars if the size specified below is exceeded. Screw-type terminals are supplied for use wtih the SENTRON VL250 to VL1250 as standard



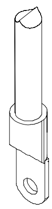
Connection with screw-type terminals

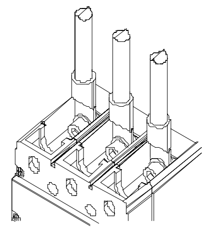


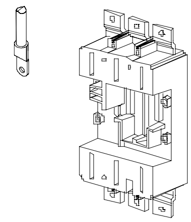
Establishing a connection with screw-type terminals

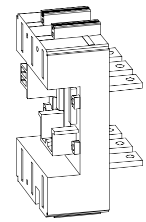
For additional information, refer to the technical data in Chapter Configuration of main connections (Page 150).

Connection with cable lugs


Cable lug


Use of cable lug No. 1


Use of cable lug No. 2


Use of cable lug No. 3

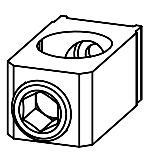
Cable lugs (ring cable lugs) are used to connect the cables to the terminals of the circuit breaker.

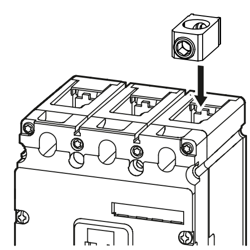
Cable lugs in accordance with DIN 46220 with a narrow flange are recommended (VL1 to VL4).

Connection terminal for round conductors (copper/aluminum)

Round conductor connection terminals for the incoming supply and outgoing feeders consist of an aluminum body with tin plating to prevent oxidation. Both aluminum and copper cables may be used. Only one conductor is permitted per terminal.

Round conductor connection terminals are available for the SENTRON VL 160X to VL 400 circuit breakers. Additional screw-type terminals are required for the SENTRON VL 160X and VL 160 circuit breakers.





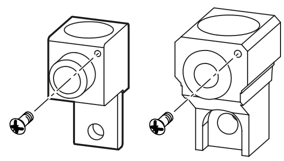
For additional information, refer to the technical data in Chapter Configuration of main connections (Page 150).

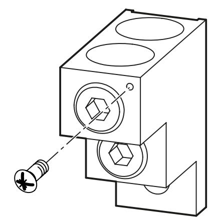
Auxiliary conductor connection terminal

3VL offers two methods of connecting auxiliary conductors

A) Connection with lug to round conductor connection terminal

The 3VL1-3VL7 round conductor terminals are provided with an M3 drill hole. Using the screw with contact washer provided, cable lugs up to 2.5 mm² can be connected.





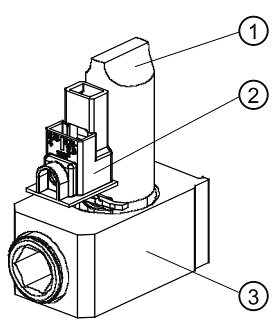
CAUTION
The maximum total ampacity of the round conductor connection terminals must not be exceeded.

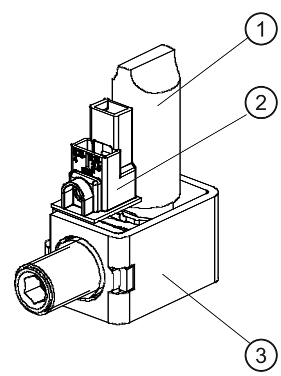
Example:

Round conductor connection terminal 3VL9 115-4TD30 I_{max} = 160 A without connected auxiliary conductor. Auxiliary conductor with 2.5 A load, means I_{max} = 157.5 A for the round conductor connection terminal.

B) Connection with auxiliary connection terminal in box or round conductor connection terminal

The auxiliary connection terminal is an additional component that is inserted into a round conductor connection terminal or steel box terminal additionally to the main conductor.





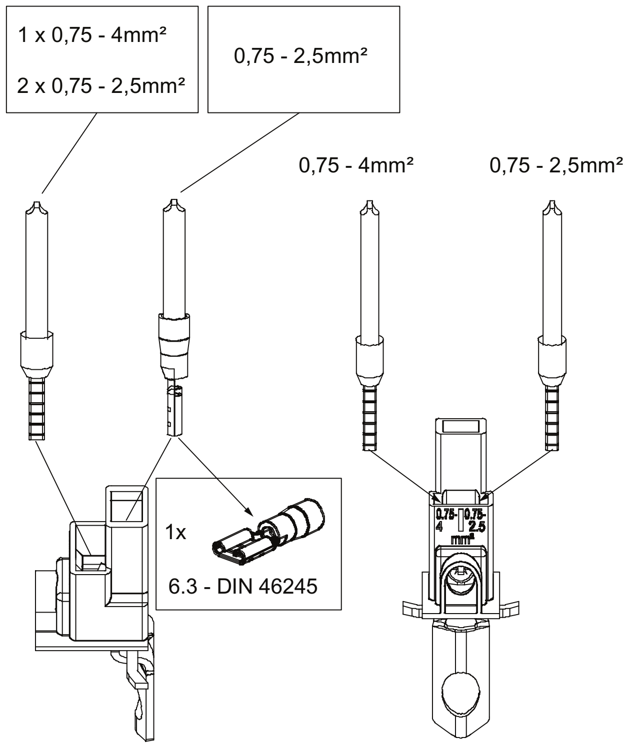
① Main conductor

② Auxiliary conductor connection terminal

③ Round conductor connection terminal

Several auxiliary conductors can be connected to the auxiliary connection terminal:

- 1 x stranded with core end sleeve max. 4 mm² + 1 x stranded with AMP connector 6.3
- 1 x stranded with core end sleeve max. 4 mm² + 1 x stranded with core end sleeve max. 2.5 mm²



The maximum ampacity I_{max} = 6 A must not be exceeded.

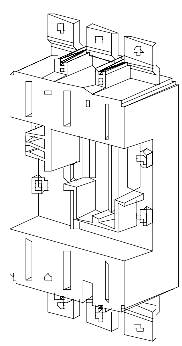
8.3 Main connection methods for plug-in and withdrawable version

Main conductor connection for plug-in and withdrawable version

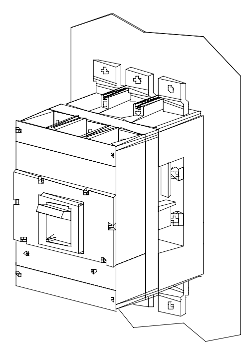
There are different methods of connecting the circuit breaker main conductors for the plug-in and withdrawable version.

Plug-in socket: Connection on the front with busbar connection pieces

Plug-in sockets simplify installation and removal of the SENTRON VL circuit breakers. The circuit breaker has been developed together with the plug-in socket in such a way as to prevent disconnection in the "ON" position. Busbars or cables can be connected on the front. A connection cover is supplied and is to be used both for the incoming and the outgoing side. An additional phase barrier for insulation between the connections is possible (see Connection covers/barriers and phase barriers). If the circuit breaker is in the connected position, the primary voltage is supplied via multiple clamping contacts in the guide frame



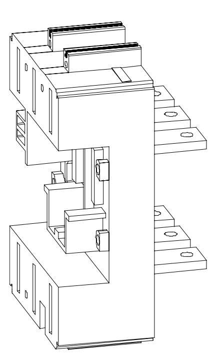
Plug-in socket (front)



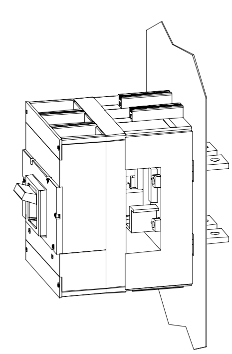
Plug-in sockets with busbar connection (busbar covers are not shown)

Plug-in socket: Connection on the back with flat busbar terminals

Busbars and cables can be connected on the back. Vertical and horizontal connections are possible depending on the configuration of the connecting bars.



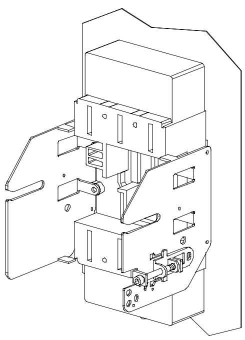
Plug-in socket (rear)



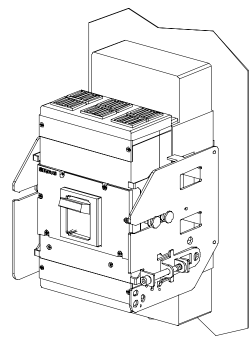
Plug-in socket with rear flat busbar terminals

Withdrawable version: Connection on the front with busbar connection pieces

The withdrawable version enables the insertion and removal of the SENTRON VL circuit breaker without requiring the disconnection of incoming or outgoing cables or busbars. A special operating mechanism, attached to the stationary assembly, is used to insert or remove the circuit breaker. A mechanical interlock prevents the circuit breaker from being moved from the connected position to the disconnected position when it is switched on. The circuit breaker will trip before the multiple clamping contacts between the circuit breaker and the guide frame open. A locking device with padlock is provided on the stationary arm of the withdrawable unit. The customer can lock the circuit breaker in either the disconnected or connected position



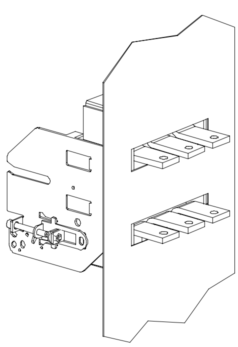
Withdrawable version with front busbar connections and terminal covers



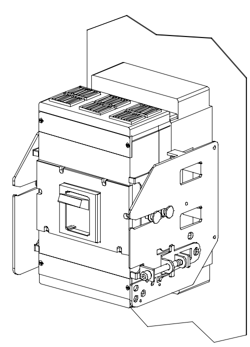
Withdrawable version with front busbar connections

Withdrawable version: Connection on the back with flat busbar terminals

It is possible to configure the busbars for horizontal connection when the withdrawable assembly with rear flat busbar terminals is used. A separate kit is available for vertical connection of circuit breakers up to and including VL250.



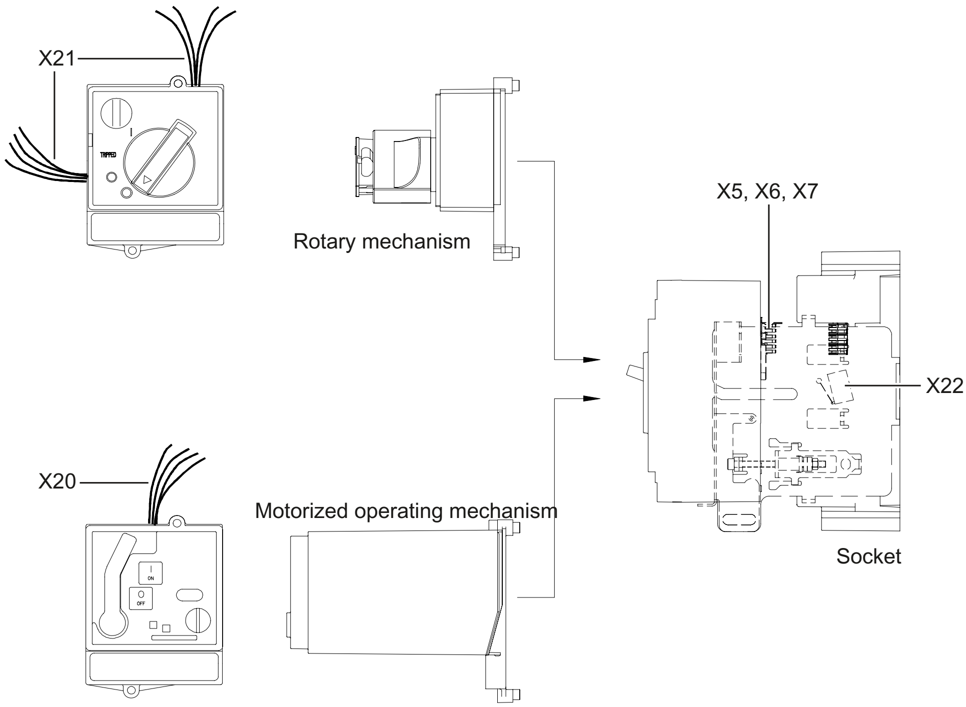
Withdrawable version with rear flat busbar connections (rear)

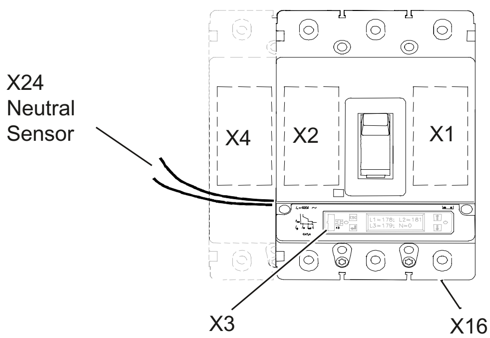


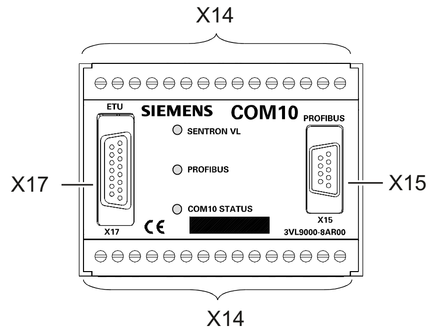
Withdrawable version with rear flat busbar connections (front)

8.4 Terminal assignments

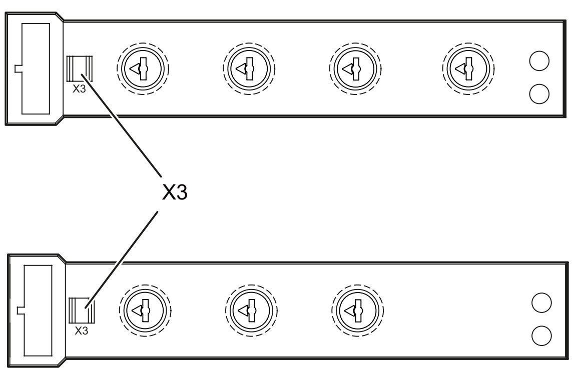
The figures below show the locations and positions of the terminals for the individual functions.

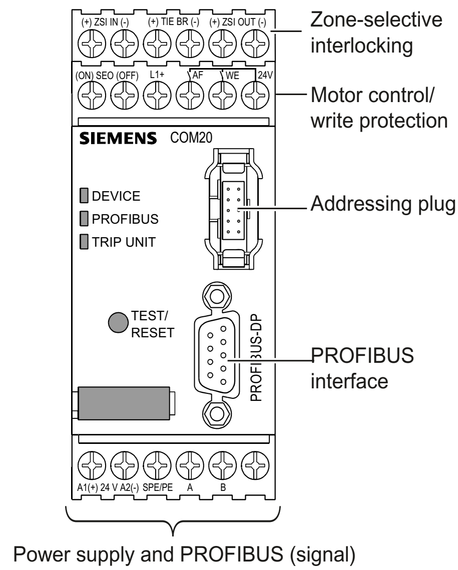




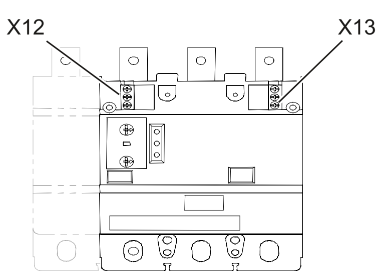


Electronic trip unit ETU

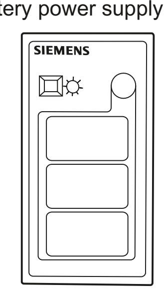




RCD module



Battery power supply device



8.5 Auxiliary switch designations

Connection designations for auxiliary switches (NC and NO)

If the circuit breakers are supplied from the factory with integral auxiliary switches, these are designated in accordance with the operating instructions.

The compartments (cutouts) in each circuit breaker (behind the front cover) for installing accessories are designated X1, X2 and X4. The terminals of the installed accessories are numbered consecutively. The contacts remain as designated in the catalog.

The contact designations on the auxiliary (HS) and alarm switches (AS) are replaced by the stickers supplied.

8.6 Description of the terminals

Description of the terminals

The exact positions and functions of the separate terminals are listed in this table for the **Description of the terminals**.

Table 8- 4 Overview of the secondary connections

Number	Where are the circuit breakers/accessories?	Description	
X1	Right-hand accessory compartment of the circuit breaker	Shunt release and undervoltage release, auxiliary and alarm switches VL160X to VL400 VL630 to VL1600	X1.1 + X1.2 X1.1 to X1.6 X1.1 to X1.8
X2	Left-hand accessory compartment of the circuit breaker	Auxiliary switches and alarm switches VL160X to VL400 VL630 to VL1600	X2.1 to X2.6 X2.1 to X2.8
X3	Connection socket to ETU LCD	I/O connection for portable tester or communication adapter	
X4	Left accessory compartment of the circuit breaker (4-pole only)	Auxiliary switches and alarm switches VL160X to VL400 VL630 to VL1600	X4.1 to X4.6 X4.1 to X4.8
X5	Auxiliary current plug-in connection for plug-in socket/guide frame	Motorized operating mechanism Remote tripping RCD module If no motorized operating mechanism is available: Remote tripping display RCD module	X5.1 to X5.5 X5.6 to X5.8 X5.1 to X5.3
X6	Auxiliary current plug-in connection for plug-in socket/guide frame	Shunt release or undervoltage release Auxiliary switches or alarm switches If motorized operating mechanism is available: Remote tripping display RCD module	X6.1 to X6.2 X6.3 to X6.8 X6.6 to X6.8

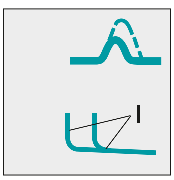
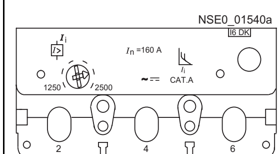
Number	Where are the circuit breakers/accessories?	Description	
X7	Auxiliary current plug-in connection for plug-in socket/guide frame	VL400 to VL1600 only Auxiliary & alarm switches	X7.1 to X7.8
X8		Reserved	
X9		Reserved	
X10 (plug-in)		Reserved	
X11 (plug-in)		Reserved	
X12	RCD module	VL160 to VL400 only Remote tripping display	X12.1 to X12.3
X13	RCD module	VL160 to VL400 only Remote control	X13.1 to X13.3
X14		COM 10 (Profibus module)	
X15		COM 10 (Profibus connection)	
X16		Connection for the communication module	
X17		COM 10 (circuit breaker connection)	
X18, X19	Hand-held tester for ETU/LCD-ETU	Reserved	
X20	Motor	X20.1 N/L voltage supply X20.2 ON (electr. ON) X20.3 OFF (electr. OFF) X20.4 L1 / L+ voltage supply X20.5 protective conductor	
X21	Rotary mechanism Leading auxiliary contacts (connection cables)	Leading NO contacts NC/NO X21.1 to X21.3 switch A X21.4 to X21.6 switch B Leading NC contacts NC/NO X21.7 to X21.9 switch A X21.10 to X21.12 switch B	
X22	Plug-in socket Withdrawable device Position switch	Position signaling contacts X22.1 to X22.3 switch A X22.4 to X22.6 switch B	
ZSI IN	COM20	Input Variable	
ZSI OUT	COM20	Output Variable	
TIE BR	COM20	Tie breaker, input	
WE	COM20	Profibus write protection, input	
SEO/MO	COM20	Motorized operating mechanism with/without stored energy, output	
A/B	COM20	PROFIBUS signal A / B	
SPE/PE	COM20	Shield / PE	

9 Displays and operator controls

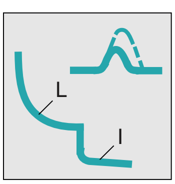
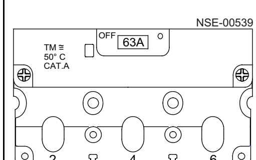
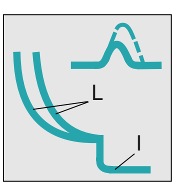
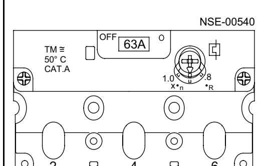
9.1 Overcurrent trip unit without LCD display

The different setting options of the individual overcurrent trip units without LCD display are explained using the examples listed:

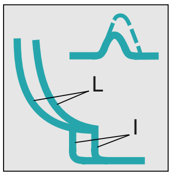
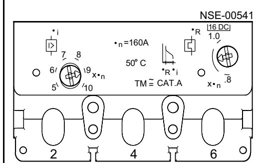
Magnetic overcurrent trip units M VL160-VL630

Characteristic curve	Application	View
	**Starter protection M, I function** Short-circuit protection, adjustable I_i = 7 to 15 x I_n, for VL160 to VL630 (frame-size-dependent)	

Thermomagnetic overcurrent trip units TM VL160X

Characteristic curve	Application	View
	**Line protection TM, LI/LIN function** Overload protection fixed, short-circuit protection fixed	
	**Line protection TM, LI/LIN function** Overload protection adjustable I_R = 0.8 to 1 x I_n Short-circuit protection fixed	

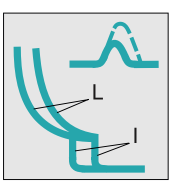
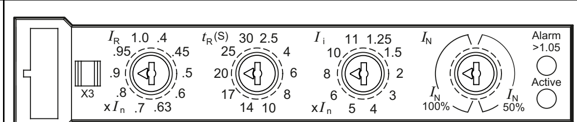
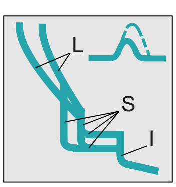
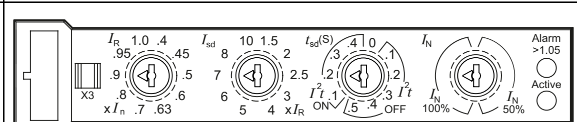
Thermomagnetic overcurrent trip units TM VL160-VL630

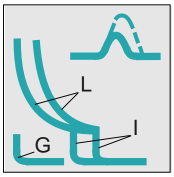
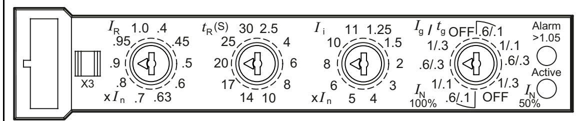
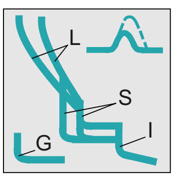
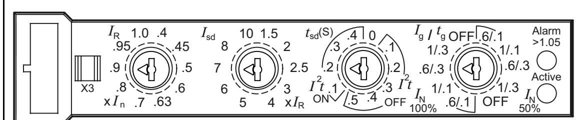
Characteristic curve	Application	View
	**Line protection TM, LI/LIN function** Overload protection adjustable I_R = 0.8 to 1 x I_n Short-circuit protection adjustable I_i = 5 to 10 x I_n for VL160 to VL630	

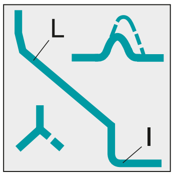
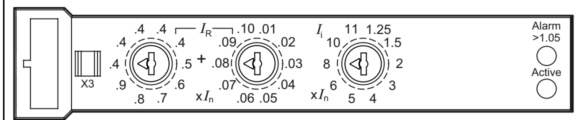
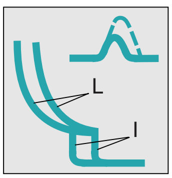
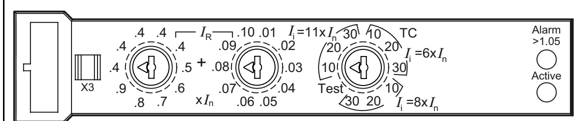
Electronic trip units ETU VL160-VL1600

The electronic trip units include the following operating features:

- No auxiliary voltage is necessary for the tripping system.
- All ETUs have a thermal memory
- A flashing green LED indicates correct operation of the microprocessor
- Overload status (I > 1.05 x I_R) is indicated by a permanently lit yellow LED (alarm)
- Integral self-test function
- Plug-in socket for tester
- Communication link to PROFIBUS DP and Modbus

Characteristic curve	Application	View
	**ETU10 for line protection, LI/LIN function** Overload protection I_R = 0.4; 0.45; 0.5 to 0.95; 1 x I_n, time-lag class t_R = 2.5 to 30 Short-circuit protection (instantaneous) I_i = 1.25 to 11 x I_n (frame-size-dependent) Neutral conductor protection I_n = 50%/100% x I_R, "TA" and "NA" versions	
	**ETU20 for line and generator protection, LSI / LSIN function** Overload protection I_R = 0.4; 0.45; 0.5 to 0.95; 1 x I_n Short-circuit protection (short-time delay) I_{sd} = 1.5 to 10 x I_R, t_{sd} = 0 to 0.5 s I^2t selectable on/off Short-circuit protection (instantaneous) I_i = 11 x I_n (fixed setting, frame-size-dependent) Neutral conductor protection I_n = 50%/100% x I_R, "TF" and "NF" versions	

Characteristic curve	Application	View
	**ETU12 for line protection, LIG/LING function** Overload protection I_R = 0.4; 0.45; 0.5 to 0.95; 1 x I_n time-lag class t_R = 2.5 to 30 Short-circuit protection (instantaneous) I_i = 1.25 to 11 x I_n (frame-size-dependent) On 4-pole circuit breakers: neutral conductor protection 50%/100% × I_R Ground fault protection: I_g = 0.6/1.0 I_n, t_g = 0.1/0.3 s measuring method No. 1: (G_R) vectorial summation current formation in the three phases and neutral conductor (4-conductor systems); $I_{\Delta n}$ = I_n, versions "SL", "SF", "ML", "MF", "TN", "NN"	
	**ETU22 for line and generator protection, LSIG/LSING function** Overload protection I_R = 0.4; 0.45; 0.5 to 0.95; 1 x I_n, Short-circuit protection (short-time delay) I_{sd} = 1.5 to 10 x I_R, t_{sd} = 0 to 0.5 s I^2t selectable on/off Short-circuit protection (instantaneous) I_i = 11 x I_n (fixed setting, frame-size-dependent) On 4-pole circuit breakers: neutral conductor protection 50%/100% × I_R Ground fault protection: I_g = 0.6/1.0 I_n, t_g = 0.1/0.3 s Measuring method No. 1: (G_R) vectorial summation current formation in the three phases and neutral conductor (4-conductor systems); $I_{\Delta n}$ = I_n, versions "SG", "MG", "SH", "MH", "TH", "NH"	

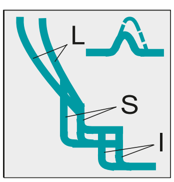
Characteristic curve	Application	View
	**ETU10M for line and generator protection, LI function** Finely adjustable overload protection I_R = 0.41; 0.42 to 0.98; 0.99; 1 x I_n, Trip class t_C = 10 (fixed setting) Thermal memory Short-circuit protection (instantaneous) I_i = 1.25 to 11 x I_n (frame-size-dependent) with phase failure sensitivity (40% I_R fixed setting)	
	**ETU30M for motor and generator protection, LI function** Finely adjustable overload protection I_R = 0.41; 0.42 to 0.98; 0.99; 1 x I_n, Trip class t_C = 10, 20, 30 Thermal memory Short-circuit protection (instantaneous) I_i = 6 to 11 x I_n with phase failure sensitivity (40% I_R fixed setting)	

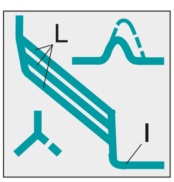
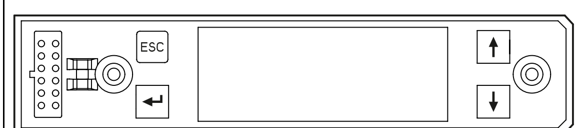
9.2 Overcurrent trip unit with LCD display

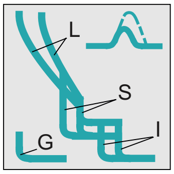
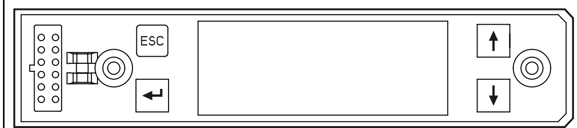
The electronic trip units with LCD display have the following operating features:

- No auxiliary voltage is necessary for the tripping system.
- Current display
- An illuminated green LCD display indicates correct operation of the microprocessor
- Overload status ($I > 1.05 \times I_R$) is indicated by "overload" on the LCD display
- Direct, user-friendly, menu-driven setting of the absolute values of the protection parameters in the Ampere values via keys
- Integral self-test function
- Plug-in socket for tester
- Communication link to PROFIBUS DP and Modbus

Electronic trip unit LCD ETU

<table>
<tr><th>Characteristic curve</th><th>Application</th><th>View</th></tr>
<tr>
<td>


</td>
<td>ETU40 for line protection, LI/LSI/LSIN function, ETU40M motor/generator protection, LI function
Overload protection $I_R = 0.4$ to $1 \times I_n$,
Trip class $t_C = 5$ to 30 at ETU40M
Time-lag class $t_R = 2.5$ to 30 at ETU40
Thermal memory selectable on/off,
with phase failure sensitivity with ETU40M
(5 to 50% I_R adjustable)
Short-circuit protection (short-time delay) on ETU40
$I_{sd} = 1.5$ to $10 \times I_R$, $t_{sd} = 0$ to 0.5 s
I^2t selectable on/off on ETU40
Short-circuit protection (instantaneous)
$I_i = 1.25$ to $11 \times I_n$ (frame-size-dependent)</td>
<td>
</td>
</tr>
</table>

Characteristic curve	Application	View
	**ETU42 for line protection,LSIG/LSING function** Overload protection I_R = 0.4 to 1 x I_n Time-lag class t_R = 2.5 to 30 Thermal memory selectable on/off Short-circuit protection (short-time delay) I_{sd} = 1.5 to 10 x I_R, t_{sd} = 0 to 0.5 s I^2t selectable on/off Short-circuit protection (instantaneous) I_i = 1.25 to 11 x I_n (frame-size-dependent) Ground fault protection: Measuring method No. 1: (G_R) vectorial summation of the currents in the three phases and neutral conductor (4-conductor systems); $I_{\Delta n}$ = 0.4 to 1 x I_n, versions "CL", "CM", "CN" Measuring method No. 2: (G_{GND}) direct measurement of the ground-fault current using a current converter, I_g = 0.4 to 1 x I_n, t_g = 0.1 to 0.5 s; "CM" version On 4-pole circuit breakers: neutral conductor protection N: 50 to 100% I_R selectable or adjustable.	

MENU on the LCD display of the overcurrent trip unit

The following languages are available:

- English (default)
- Spanish
- German
- French

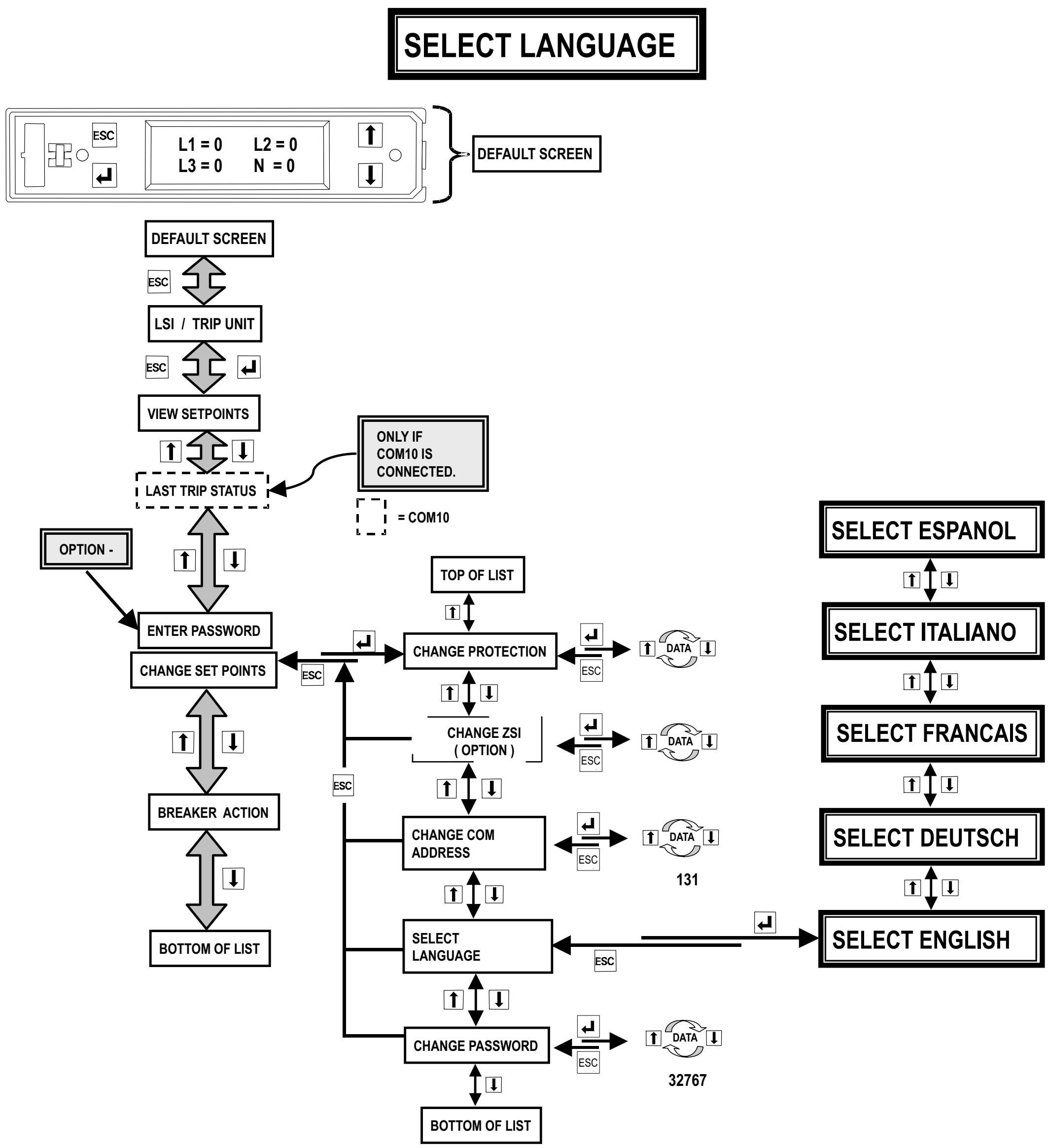


Figure 9-1 MENU on the LCD display of the overcurrent trip unit

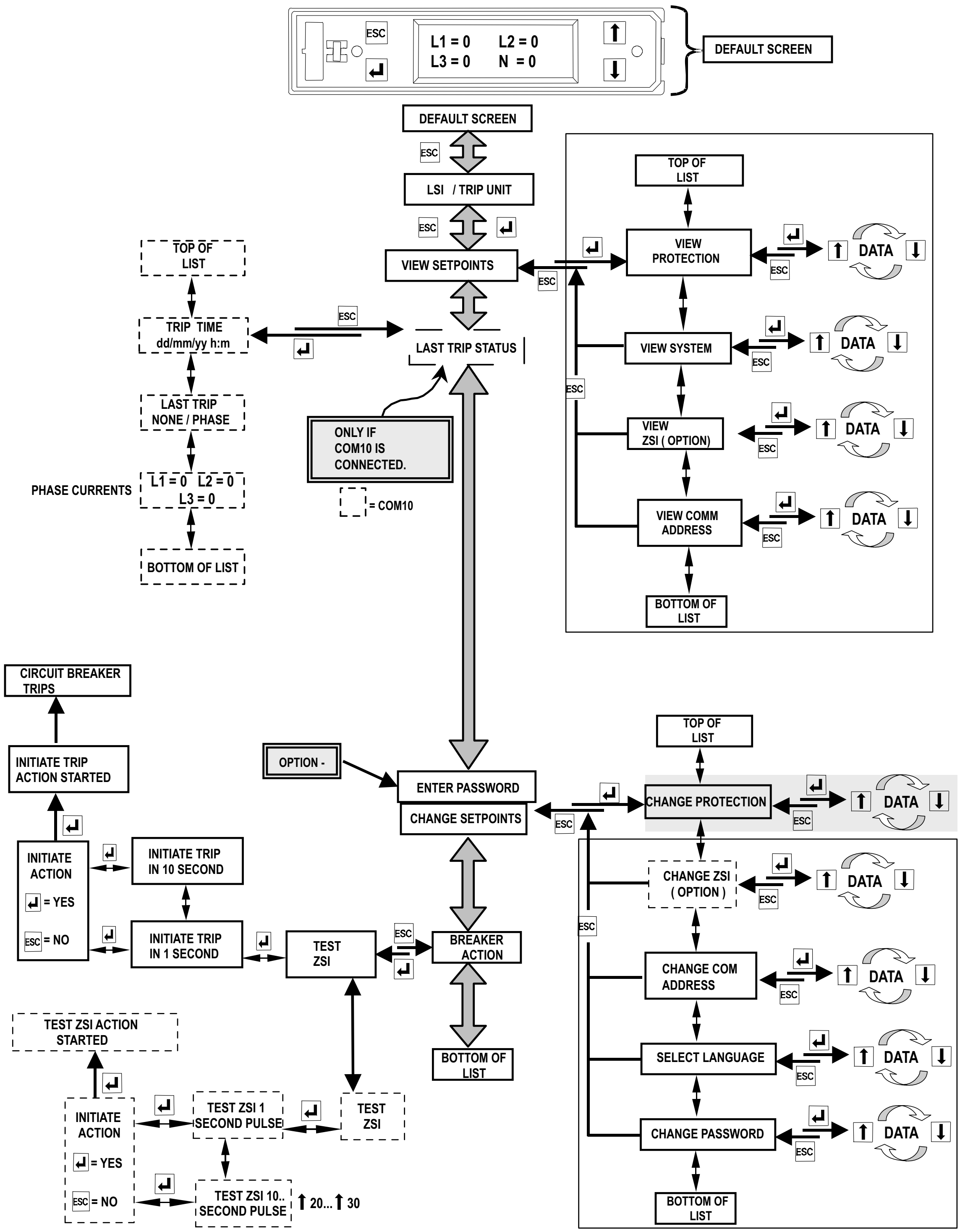


Figure 9-2 Detailed menu of the LCD-ETU 40 electronic trip unit

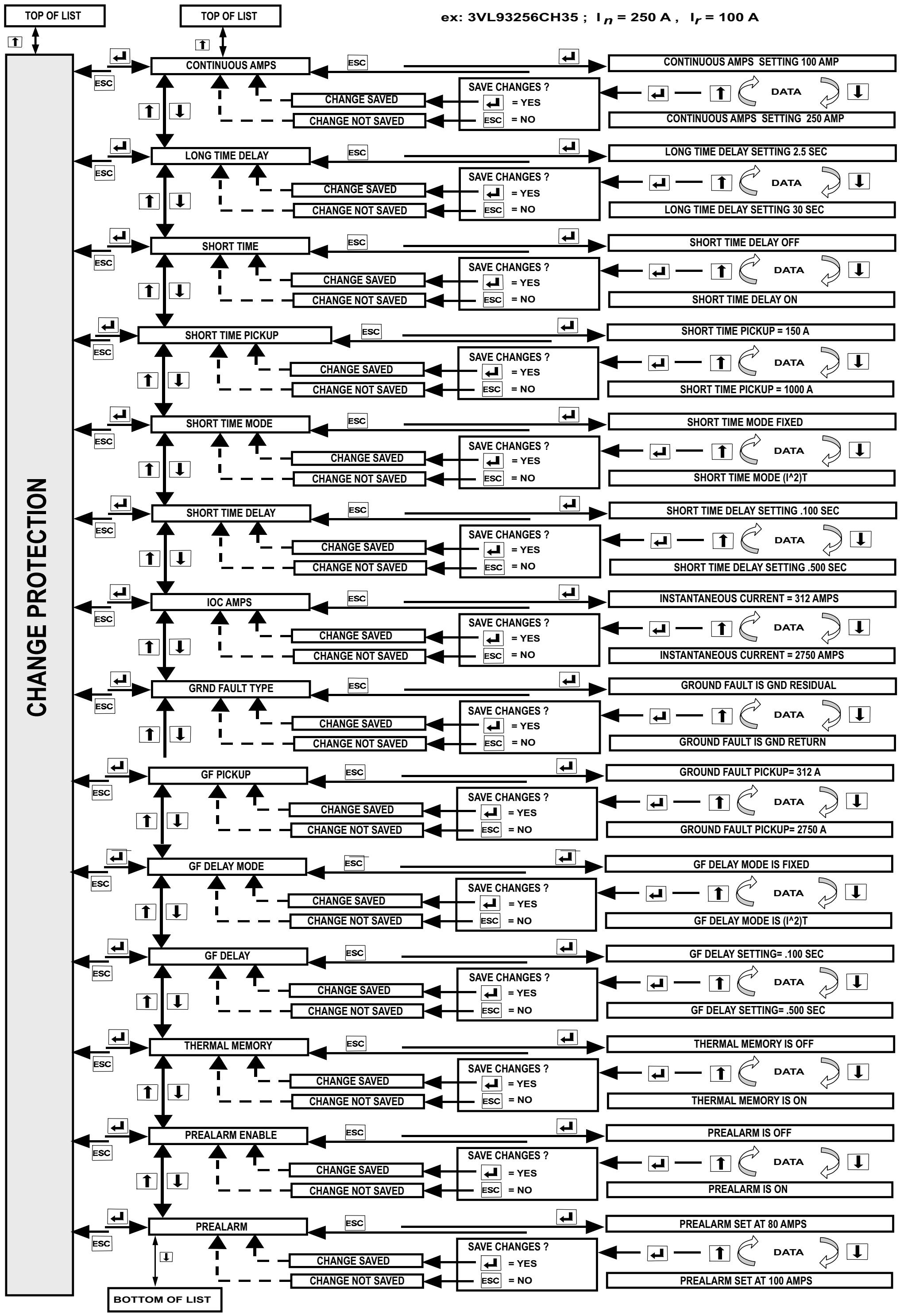


Figure 9-3 Example: Changing the type of the protection of the LCD-ETU 40

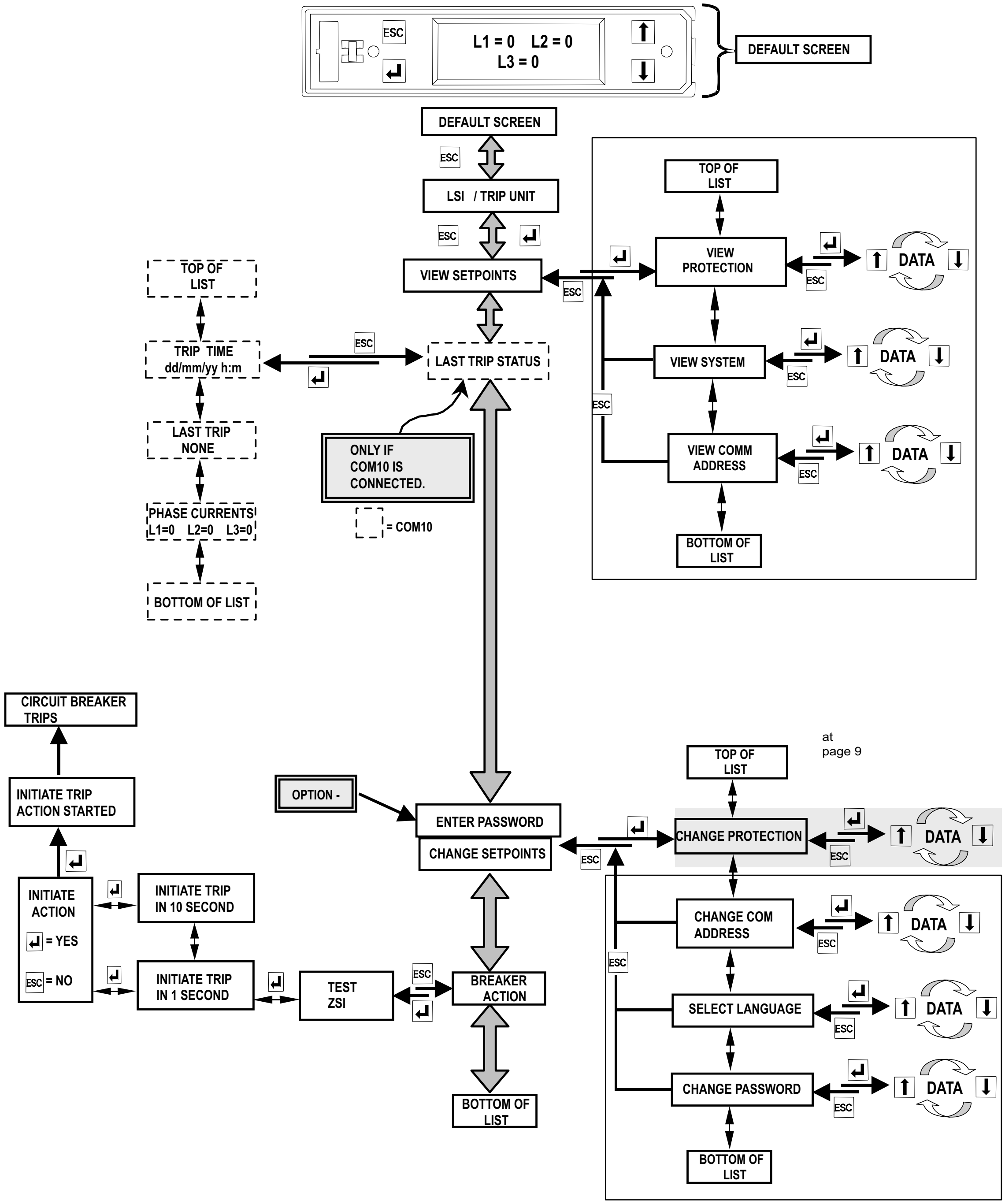


Figure 9-4 Detail of the menu of the LCD-ETU 40 electronic trip unit

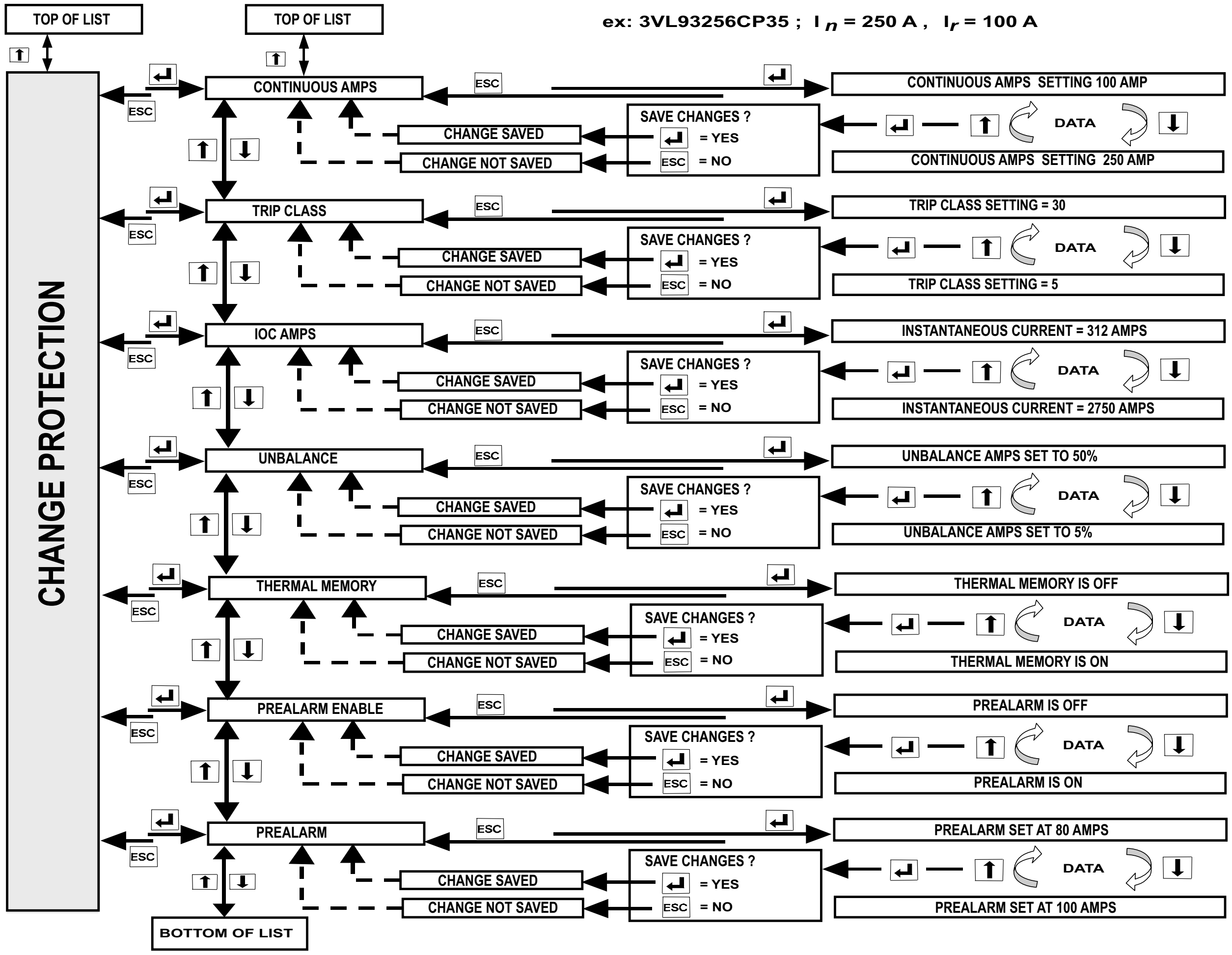


Figure 9-5 Example: Changing the type of the protection of the LCD-ETU 40 M

Commissioning

The overcurrent trip unit must be activated before it can be parameterized. A minimum load current of approximately 20% of the relevant rated current I_n of the circuit breaker is required.

The "LCD-ETU" trip unit is preset in the factory with the maximum settings for the overload release and the short-circuit release. This means activation, and hence parameterization, is possible when a load is connected with a minimum current of approximately 20% of the relevant rated current 'I_n'.

Changing the parameters for the overload and short-circuit releases during operation to a value under the present operating value causes instantaneous tripping.

If this minimum load current is not available, the required auxiliary power can be supplied using the 3VL9000-8AP00 hand-held tester. In circuit breakers with communication capability, the trip unit is supplied with power by the COM10.

Note:

The hand-held tester can be hired from the Instrument Center (SIRENT) in Erlangen, Germany:

Address of SIRENT Rentals, Sales and Service. Rental and sales of tools, and measuring and test devices:

SIEMENS AG

SIRENT Service Center

I IS IN OLM LC ITM OP

Günther-Scharowsky-Str. 2

91058 Erlangen, Germany

Tel. 09131-7-33310

Fax. 09131-7-33320

sirent.industry@siemens.com

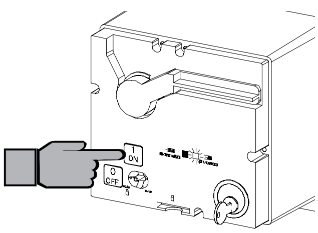
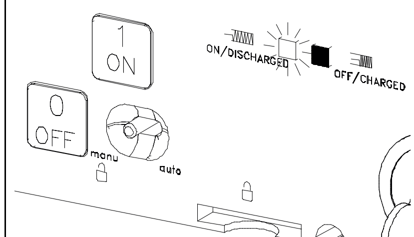
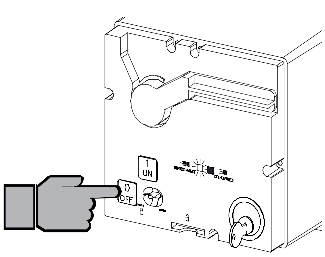
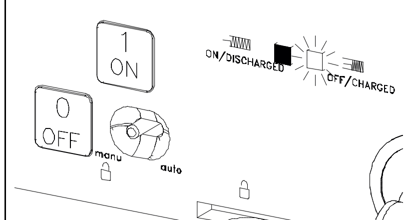
http://intranet.siemens.de/sirent

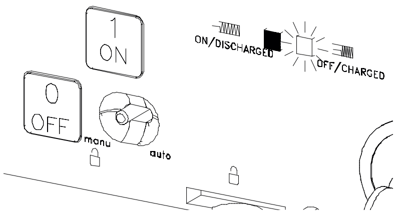
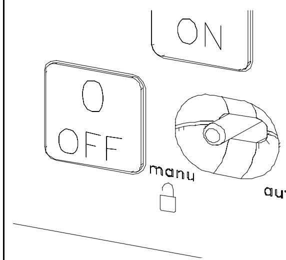
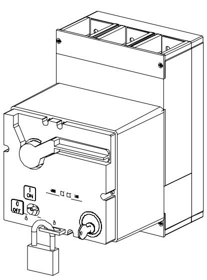
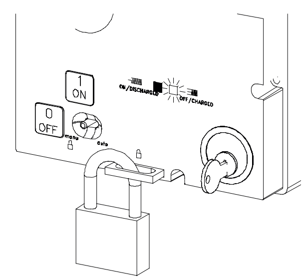
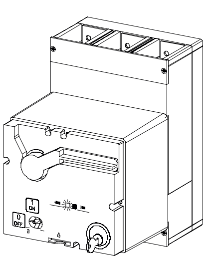
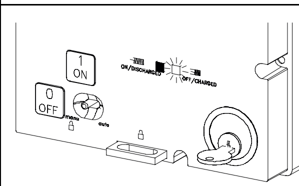
Hiring conditions can be viewed by entering the device number "S7P460" of the Instrument Center.

9.3 Stored-energy motorized operating mechanism

Function description for stored-energy motorized operating mechanism

Requirement: Supply voltage is applied

Status	Operation	Display
The stored-energy spring mechanism is charged The toggle handle of the SENTRON VL is in the "OFF/RESET" position.  The stored-energy motorized operating mechanism is charged. SENTRON VL toggle handle is in the "OFF/RESET" position.	**Local operation:** Power ON: Press the "ON" button. **Remote operation:** Power ON: "ON" signal The stored-energy spring mechanism discharges when actuated, and switches the SENTRON VL to the "ON" position.	"ON/Discharged"  Display: Stored-energy spring mechanism discharged
The stored-energy spring mechanism is discharged. The toggle handle of the SENTRON VL is in the tripped position.  The stored energy operator is discharged SENTRON VL toggle handle is in the "ON" or tripped position	**Local operation:** To switch off: Press the "OFF" button **Remote operation:** To switch off: "OFF" signal The toggle handle of the SENTRON VL moves to the "OFF" position. The motor charges the stored-energy spring mechanism.	"OFF/Charged"  Display: Stored-energy spring mechanism compressed

Status	Operation	Display
 Stored-energy motorized operating mechanism Automatic (remote)/manual (local) changeover switch	Only remote operation is possible in Auto mode. The local operating controls are deactivated. The manual clamping handle works when the operating mechanism is in the "ON/Discharged" position. Only local operation is possible in manual mode. Remote signals are blocked. The "ON" button operates mechanically and releases the stored-energy spring mechanism. The "OFF" button operates the motor that charges the stored-energy spring mechanism. Using a mechanical locking device, the "OFF" button can be configured such that the SENTRON VL will trip when the button is pressed. This makes it possible to immediately trip the circuit breaker. When this occurs, the toggle handle initially goes to the "tripped" position, and then the motor movement takes it to the "OFF/RESET" position.	 Local/remote changeover switch
 Locking slide with padlock	The auto/manual changeover switch must be set to manual mode to be able to locally lock the circuit breaker in the "OFF" position. Between 1 and 3 padlocks with shackle diameters ranging between 4 and 8 mm can be accommodated on the locking slide. The operating mechanism cover cannot be removed. Compatible with locking using the safety lock feature.	 Locking slide with padlock (enlarged)
 Mechanical locking with safety lock	The auto/manual changeover switch must be set to manual mode to be able to locally lock the circuit breaker in the "OFF" position. The locking device with safety key prevents local and remote operation. The key can only be removed in the locked switch position ("OFF"). The locking slide protrudes out of the operating mechanism cover to indicate that the operating mechanism is locked. The operating mechanism cover cannot be removed when it is locked. Compatible with the padlock feature.	 Mechanical locking with safety lock

Parameter assignment/addressing 10

10.1 Setting the parameters

Settings on the ETU

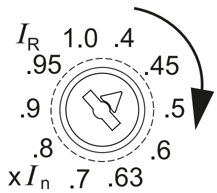
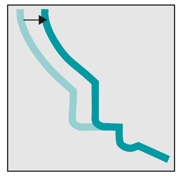
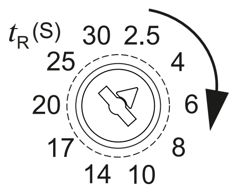
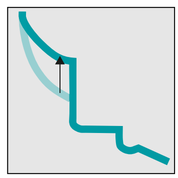
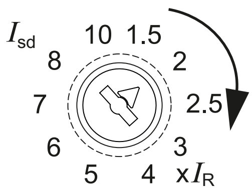
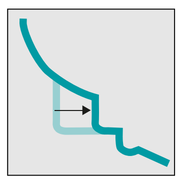
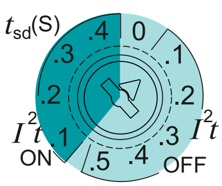
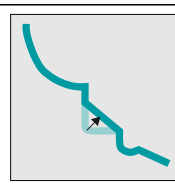
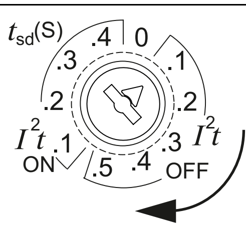
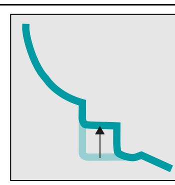
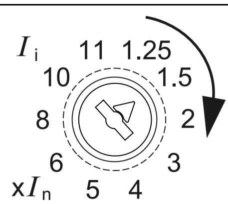
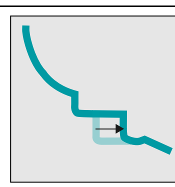
The values to be set on the electronic trip unit of the circuit breaker depend on the technical environment (switching station, cables), the network configuration, and the type of equipment to be protected. There is no rule of thumb for protection settings. These values can be calculated by the relevant electrical planning engineer.

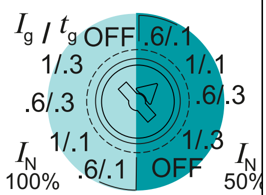
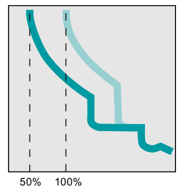
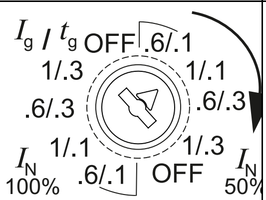
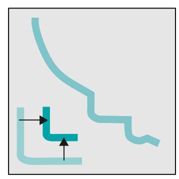
The Siemens software tool SIMARIS Design (www.siemens.com/simaris) offers a simple, quick and safe solution for dimensioning switching and protecting devices.

Tripping characteristic curve and settings parameters

The current-time curve of a trip unit offers the best method for detecting the tripping characteristics of a trip unit. The tripping characteristic curve reflects the behavior of the circuit breaker in the event of a fault, e.g. overload or short-circuit. The time required to trip is defined at a specific current. The tripping characteristic curve is divided into different sections. Each section reflects the tripping behavior of the circuit breaker at a specific current level Depending on the type, the trip units can be supplied with or without the S, N, or G functions (L, S, I, N, G designations in accordance with IEC60947-2, Annex K).

- L long time delay = overload protection with current-dependent long time delay and current-dependent tripping curve (I^2t = constant)
- L short time delay = short-circuit protection with current-dependent or current-independent short time delay and current-dependent tripping curve (I^2t_{sd} = constant)
- I Instantaneous = short-circuit protection with instantaneous adjustable tripping.
- N Neutral protection = protection of the neutral conductor with adjustable, current-dependent tripping curve.
- G Ground fault = ground fault protection with current-independent short time delay

Parameter		Setting buttons	Effect on characteristic curve	Brief description	Reason
L	I_R			Tripping current of the overload protection I_R = 0.4 to 1 x I_n	Limitation of the overload range by setting to the operating current of the circuit to be protected
	t_R			Delay (or time-lag class) in the overload range. The set time is the tripping time at 6 x I_R. t_R = 2.5 to 30s	Improved selectivity in the overload range in switching stations with several grade levels when the rated currents differ only slightly
S	I_{sd}			Tripping current of the short-time delay short-circuit protection I_{sd} = 1.5 to 10 x I_R	Limitation of the short-circuit range in which the current has to be shut off faster than in the overload range, but with a shorter time delay to achieve time selectivity to downstream switchgear
	I^2t_{sd}			Switchover from a constant time delay to a I^2t characteristic curve in the short-circuit range I^2t_{sd} = ON or OFF	Improved selectivity with downstream switchgear, e.g. LVHRC fuses
	t_{sd}			Delay time of the short-circuit protection. Please note: The selection between t_{sd} = constant and I^2t characteristic t_{sd} = 0 to 0.5 s with the position of the rotary encoding switch	Improved selectivity of the short-circuit protection in switching stations with several grade levels
I	I_i			Tripping current of the instantaneous short-circuit release I_i = 1.25 to 11 x I_n	Limitation of the short-circuit range in which the impermissibly high current has to be switched off as quickly as possible. This also takes place for the self protection of the circuit breaker

Parameter		Setting buttons	Effect on characteristic curve	Brief description	Reason
N	I_N			Tripping current of the neutral conductor protection I_N = 0.5 or 1 x I_R	Monitoring of a possible overload of a neutral conductor or protection of a conductor with reduced cross-section
G	I_g/t_g			Tripping current and time delay of the ground fault protection I_g = Off, 1 or 0.6 x I_n t_g = Off, 0.1 s or 0.3 s	Monitoring of a ground fault

Setting of the protection parameter for line and generator protection

The settings vary depending on the trip unit (ETU10, ETU12, ETU20, ETU22, LCD-ETU40 and LCD-ETU42). The following parameters can be set depending on the version:

L **overload release** I_R:

The overload release I_R is set to the operating current I_B of the circuit to be protected. This takes place with the help of the left rotary encoding switch I_R that is set to the factor I_B/I_n (example: IB = 250 A, In = 315 A => setting factor 250 / 315 = 0.79 corresponds to 0.8 on the rotary encoding switch).

Delay time t_r:

The delay time (or time-lag class) t_r can be set using another rotary encoding switch. The set time is the tripping time at 6 x I_r. In this way, selectivity can be achieved in the overload range, for example, when the rated current range scarcely differs.

S **short-time delay short-circuit protection** I_{sd}:

The short-time delay short-circuit protection can be set with regard to the tripping value of the current I_{sd} and the delay time t_{sd}. I_{sd} refers to the tripping value of the overload release I_R and can be set between 1.5 to 10 x I_R (depending on the circuit breaker).

Delay time t_{sd}:

Depending on the requirements and on the trip unit, selectivity in the relevant circuit can be achieved with appropriate selection of the delay time t_{sd}. If the rotary encoding switch is in the "ON" range, this means the delay time is current-dependent. The I^2t value is constant. For example, the higher the current, the faster the circuit breaker will trip (equivalent to the overload release I_R). In contrast, the delay time in the "OFF" position is current-independent, that is, constant. If the current reaches the set value I_{sd}, the circuit breaker trips after the set time t_{sd}. The degree to which the current exceeds the value I_{sd} is not important.

I^2t waveform:

A I^2t waveform of the characteristic curve can be switched in (depending on the ETU), the delay time t_{sd} is based on the reference point 8 x I_R. Two different procedures are used to form the characteristic curve. As well as a fixed time delay for all currents in the characteristic curve section, the I^2t characteristic can also be used. The tripping time falls continuously as the current increases, and the product of squared current and time remains constant.

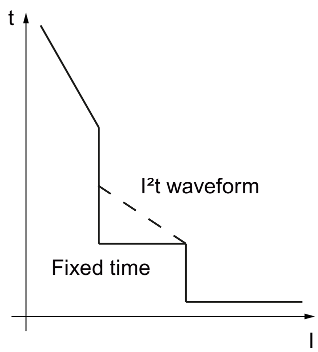


Figure 10-1 I^2t

I instantaneous short-circuit protection I_i:

On some trip units, the instantaneous short-circuit release I_i can also be set. This refers to the rated current I_n of the circuit breaker. It must always be noted that either the instantaneous short-circuit release (I_{sd}) or the delayed short-circuit release (I_i) handles the protection of personnel. The tripping current of the short-circuit release of the circuit breaker is set to a value that is at least 20% (tolerance of the trip unit) lower than the lowest short-circuit current at the installation location and simultaneously higher than the maximum operating current +20%. This guarantees that the circuit breaker will trip within the required time even with the smallest short-circuit current, and that correct currents will not result in unwanted trips.

G ground fault protection I_g:

The tripping value of the of the ground fault release I_g is fixed to the rated breaker current on the ETU12 and ETU22. The tripping current of the ground fault release can be set to between 0.6 and 1 x I_n, and the delay time t_g can be set between 0.3 s and 0.6 s. The measuring methods for the ground fault protection are specified on the representation of the trip unit. On the ETU42, the tripping current of the ground fault release can be set to between 0.4 and 1 x In, and the delay time can be set between 0.1 s and 0.5 s.

Note

Ground-fault protection

It must be noted that the ground fault protection is not an r.c.b. circuit breaker (FI or RCD in the building installation). Fault currents to ground therefore cannot be detected, only ground "short"-circuits. However, an RCD module can be used for 3VL.

10.2 Setting the protection parameters for motor protection (ETU10M, ETU30M and LCD-ETU 40M)

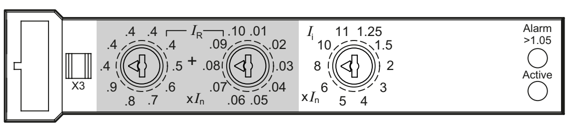
The selection of the circuit breaker is oriented around the rated operating current of the motor; the releases are specially designed for overload protection of 3-phase motors.

Overload release I_R:

The overload release I_R is set to the rated current of the motor, similarly to protection parameters for line and generator protection. The overload protection is finely adjustable with the left rotary encoding switch (first decimal place) and the center rotary encoding switch (second decimal place) in the range between I_R = 0.41; 0.42 to 0.98; 0.99; 1 x I_n (I_n = rated breaker current).

Example

Adjusting to the motor current 360 A is carried out for the rotary encoding switch left and center (ETU10M and ETU30M) (rated breaker current I_n = 500 A) as follows:



Overload protection setting

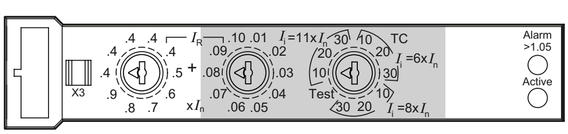
Setting I_R / rated breaker current I_n = 360 A / 500 A = 0.72

1. Setting the rotary encoding switch left factor 0.7
2. Setting the rotary encoding switch center factor 0.02

Short-circuit release I_i

Furthermore, instantaneous short-circuit release I_i can also be set depending on the trip unit. This setting value refers to the rated current I_n of the circuit breaker. As with line and generator protection, the minimum short-circuit must be taken into account when selecting the setting.

With the ETU30M version, you must note that the setting of the short-circuit release is selected in combination with the time-lag class. The rotary encoding switch is divided into three areas here, corresponding to the values 6, 8 or 11 x I_n. The desired time-lag class can be selected within these ranges.



Setting the time-lag class/trip class

The SENTRON VL circuit breaker offers the option of selecting from various time-lag classes or trip classes for different motor applications.

One version (ETU10M) contains a thermal memory and phase failure sensitivity based on a fixed trip class 10.

With the *ETU30M* overcurrent trip unit, both the time-lag class TC and the tripping current of the short-circuit release are set in combination with the right rotary encoding switch.

Another version (*ETU40M*) with an LCD trip unit permits step by step setting from class 5 to 30. The setting in accordance with CLASS 5 is only used on motors with an extremely low overload capacity. In CLASS 30, by contrast, the motor must be suitable for starting under a heavy load. That is, the trip class must be adapted to the start-up time of the motor.

Definition of the trip class

The trip class specifies the start-up times during the motor start in accordance with IEC 60947-4-1. The trip class is defined by the tripping time at 7.2-times the set current level (in the cold state). Combinations with CLASS 10 are generally used.

The tripping times are as follows:

- CLASS 10 A between 2 and 10 secs,
- CLASS 10 between 4 and 10 secs,
- CLASS 20 between 6 and 20 secs,
- CLASS 30 between 9 and 30 secs.

Applications such as fans, require longer start-up times.

Phase failure sensitivity

The "phase failure sensitivity" function is also integrated into the releases for motor protection ETU10M, ETU30M and ETU40M. This ensures that the motor is reliably protected against overheating if a phase interruption or a large fluctuation occurs. The phase failure sensitivity protects 3-phase AC motors against overheating while only 2 phases are active. The specified operating current IR is automatically reduced to 80% of the set value if the r.m.s. values of the operating currents in the three phases in the case of the ETU10M and ETU30M trip units differ by more than 40%. If an adjustable phase unbalance of 5 to 50% is set on the ETU40M trip unit, the set operating current I_R is automatically reduced to 87% of the set value.

Thermal memory

The trip units for motor protection ETU10M, ETU30M and ETU40M have a "thermal memory" that takes account of the pre-loading of the motor (tripping in the case of overload). The function of the fixed "thermal memory" cannot be switched off (except on the ETU40M). Following an overload trip of the circuit breaker, the tripping time is reduced by the thermal pre-loading of the circuit breaker in such a way that further overloads cannot harm the motor windings.

After an overload trip, the tripping times are reduced in accordance with the tripping characteristic curves so that even the inrush current can result in a trip. A cooling time dependent on the size of the motor is required before the motor can be switched on again.

11 Service and maintenance

11.1 Preventive measures

Maintenance

> **⚠ DANGER**
>
> **Qualified personnel**
>
> Functionality tests and maintenance tasks must only be carried out by qualified personnel due to the dangers associated with electrical equipment.

The following inspection intervals must be defined by the operator (customer) depending on the conditions of use of the relevant SENTRON 3VL molded-case circuit breaker:

- At least 1 x per year
- After serious high-power shutdowns
- After trips caused by the electronic overcurrent trip unit
- Additional testing of downstream circuit breakers.

Inspection

Checks are required within the scope of the inspection(s) and/or after 1,000 rated current shutdowns. Please proceed as follows:

- External circuit breaker housing
  - Examine all visible surfaces for oxidation, residues or other adverse effects.
  - Remove residues with a lint-free, dry and clean cloth. (Never use chemical cleaners or water)

> **CAUTION**
>
> **Damage to the circuit breaker**
>
> Never carry out repairs to the plastic casing or the interior of the circuit breaker! Molded-case circuit breakers contain only maintenance-free components.

- Electrical and mechanical functions of the circuit breaker
  - Test the operating lever to check the mechanical functioning of the circuit breaker contacts
- Function of the mechanical on and off switch
  - Operate the trip button, if available. Return the circuit breaker to the starting position after each operation.
- Main circuits and control circuits, function.

- Check connections are tight
  - Check the terminal screws for proper torque values
  - Spot checking of input and output cables
  - Spot checking of terminal accessories
  - Replace damaged terminal accessories after cleaning the terminal area
- Check, and, if necessary, correct, the settings of the electronic overcurrent trip unit in accordance with the system conditions
  - Electronic circuit breaker releases must only be tested with a device especially supplied for this purpose.

The operator (customer) must arrange for the disposal of the molded-case circuit breaker or the replaced parts at the end of their service life in accordance with the currently applicable legal requirements and guidelines.

11.2 Troubleshooting

Notes on troubleshooting

Table 11- 1 Troubleshooting

Circuit breaker status	Causes of faults	Corrective action
Overload causes circuit breaker to trip:	Excessive current	The circuit breaker is functioning correctly and switches off an overload that occurs. Check to see if the operating current has exceeded the thermal tripping limit.
		Carry out a visual inspection of the terminals. Discoloration indicates the terminals are loose. The proper torque values for the terminals are listed in the operating manual supplied with every circuit breaker. See also the figures in the Chapters Multiple feed-in terminal for round cables, and Box terminals
	Connecting cable not correctly connected to the circuit breaker	Carry out a visual inspection of the terminals for discoloration. Cables can become loose during service due to various reasons such as vibration (machine tool applications) and cold flow (for aluminum cables)
	Ambient temperature too high	This can be a problem on hot summer days or in areas subject to extreme heat. Although all SENTRON VL circuit breakers are calibrated for application at an ambient temperature of 50 °C, the temperatures in the housings can exceed this level. It may be necessary to consider derating the I_n or I_R values. See the Chapters Use in harsh environments, and Derating factors
	Overcurrent trip unit not correctly connected to the circuit breaker.	If none of the above suggestions apply, the overcurrent trip unit must be removed from the circuit breaker and inspected for discoloration. The tightening torque values are listed in the operating manual supplied with every circuit breaker.
Short-circuit causes circuit breaker to trip:	Excessive making current, e.g. motor	Adjust the magnetic trip rating to the next highest setting or until the circuit breaker does not trip when the motor is started.
	High current peaks, e.g. when changing from star to delta in star-delta starters.	A current peak of up to 20 times the nominal current of the motor can occur when changing from star to delta. In this case, the short-circuit release "I" must be set to a higher value. However, this may result in the loss of the desired higher motor protection function.

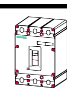
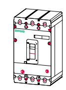
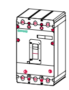
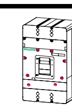
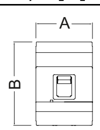
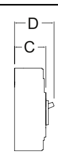
Circuit breaker status	Causes of faults	Corrective action
Mechanical and electrical functions:	High humidity	The circuit breakers must not be used in environments with high humidity since this can cause dielectric and insulation problems. In such environments, appropriate measures need to be taken, such as placing the circuit breaker in an enclosure.
	Corrosion	The circuit breakers are not designed to be used in aggressive environments. In such environments, the circuit breaker should be installed in a housing.
	Function of the internal accessories	Determine what type of internal accessories are installed. Remove the circuit breaker cover and determine the type of accessories using the circuit breaker order number. Then check for correct functioning. • Undervoltage release: Ensure the correct voltage is connected to the undervoltage release since otherwise, the circuit breaker cannot be tripped. • Shunt release: Ensure the voltage is not applied to the shunt release since this can also prevent the circuit breaker from tripping. • Auxiliary and alarm switches: The auxiliary and alarm switches do not have any effect on the functioning of the circuit breaker.

Technical data 12

12.1 Technical overview

The technical overview lists all the operating data and dimensions as well as the possible overcurrent tripping methods and the switching capacities of the SENTRON VL circuit breakers. The RCD blocks overview contains the relevant operating data.

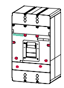
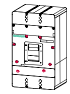
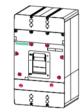
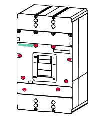
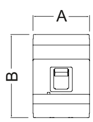
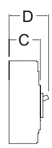
Table 12- 1 Technical overview VL160X, VL160 to VL400

			VL160X		VL160		VL250		VL400	
										
Rated current at 50 °C ambient temperature			16 to 160 A		26 to 160 A		80 to 250 A		125 to 400 A	
Rated operating voltage V_e[1)] (AC) 50-60 Hz [V]			690		690		690		690	
Number of poles			3	4	3	4	3	4	3	4
(DC)[2)] [V]			500	500	600	600	600	600	600	600
		mm A	105	139	105	139	105	139	139	183
		mm B	157	157	175	175	175	175	279	279
		mm C	87	87	87	87	87	87	106	106
		mm D	107	107	107	107	107	107	138	138
Overcurrent trip unit										
Thermomagnetic TM			X	X	X	X	X	X	X	X
Electronic trip unit ETU LCD			-- --	-- --	X	X	X	X	X	X
Interchangeability			--	--	X	X	X	X	X	X

1) Rated insulation voltage of the main conducting paths V_i = 800 V AC

2) Rated direct current data apply only for thermomagnetic overcurrent trip units. The values apply for at least 3 conductor paths in series and extremely high switching capacity "L". For switching direct current, the maximum permissible direct voltage per conductor path must be observed; see also "Suggested circuits for DC networks" in the Chapter "Application planning".

Table 12- 2 Technical overview VL630 to VL1600

	VL630		VL800		VL1250		VL1600	
								
Rated current at 50 °C ambient temperature	252 to 630 A		320 to 800 A		400 to 1250 A		640 to 1600 A	
Rated operating voltage V_e[1)] (AC) 50-60 Hz [V]	690		690		690		690	
Number of poles	3	4	3	4	3	4	3	4
(DC)[2)] [V]	600	600	--	--	--	--	--	--
  mm A	190	253	190	253	229	305	229	305
mm B	279	279	406	406	406	406	406	406
mm C	106	106	119	119	157	157	15	152
mm D	138	138	151	151	207	207	207	207
Overcurrent trip unit								
Thermomagnetic TM	X	X	--	--	--	--	--	--
Electronic trip unit ETU LCD	X	X	X	X	X	X	X	X
Interchangeability	X	X	X	X	X	X	X	X

1) Rated insulation voltage of the main conducting paths V_i = 800 V AC

2) Rated direct current data apply only for thermomagnetic overcurrent trip units. The values apply for at least 3 conductor paths in series and extremely high switching capacity "L". For switching direct current, the maximum permissible direct voltage per conductor path must be observed; see also "Suggested circuits for DC networks" in the Chapter "Application planning".

Table 12- 3 Standard switching capacity VL160X, VL160 to VL400

SENTRON VL - N rated breaking current (kA) balanced (standard switching capacity)						
Type	SENTRON		Vl160X	VL160	VL250	VL400
			I_{cu}/I_{cs}	I_{cu}/I_{cs}	I_{cu}/I_{cs}	I_{cu}/I_{cs}
IEC 60947-2	4)	Up to 240 V AC	65/65	65/65	65/65	65/65
		415 V AC	40/40	40/40	40/40	45/45
		440 V AC	25/20	25/20	25/20	32/26
		500 V AC	18/14	25/20	25/20	25/20
		690 V AC	8/4 3)	12/6	12/6	15/8
	5)	Up to 250 V DC	30/30	32/32	32/32	32/32
		500 V DC	--	--	--	--
		600 V DC	--	--	--	--

3) For rated currents above 25 A. The VL160X is not available with a rated voltage of 690 V AC for rated currents of 16 A and 20 A.

4) At 240 V AC, 415 V AC, and 525 V AC max. 5% overvoltage; at 440 V AC, 500 V AC and 690 V AC max. 10% overvoltage.

5) At 250 V DC, 500 V DC and 600 V DC max. 5% overvoltage.

The voltage values apply for at least 3 conductor paths in series, see also "Suggested circuits for DC networks" in the Chapter "Application planning".

Table 12- 4 Standard switching capacity VL630 to VL1600

SENTRON VL - N rated breaking current (kA) balanced (standard switching capacity)						
Type	SENTRON		VL630	VL800	VL1250	VL1600
			I_{cu}/I_{cs}	I_{cu}/I_{cs}	I_{cu}/I_{cs}	I_{cu}/I_{cs}
IEC 60947-2	4)	Up to 240 V AC	65/65	65/65	65/35	65/35
		415 V AC	45/45	50/50	50/25	50/25
		440 V AC	35/26	35/26	35/26	35/26
		525 V AC	25/20	25/20	25/20	25/20
		690 V AC	20/10	20/10	20/10	20/10
	5)	Up to 250 V DC	32/32	--	--	--
		500 V DC	--	--	--	--
		600 V DC	--	--	--	--

Table 12- 5 High switching capacity VL160X, VL160 to VL400

SENTRON VL - H rated breaking current (kA) balanced (high switching capacity)						
Type	SENTRON		VL160X	VL160	VL250	VL400
			I_{cu}/I_{cs}	I_{cu}/I_{cs}	I_{cu}/I_{cs}	I_{cu}/I_{cs}
IEC 60947-2	4)	Up to 240 V AC	100/75	100/75	100/75	100/75
		415 V AC	70/70	70/70	70/70	70/70
		440 V AC	42/32	50/38	50/38	50/38
		525 V AC	30/23	40/30	40/30	40/30
		690 V AC	12/6 3)	12/6	12/6	15/8
	5)	Up to 250 V DC	30/30	32/32	32/32	32/32
		500 V DC	30/30	32/32	32/32	32/32
		600 V DC	--	--	--	--

3) For rated currents above 25 A. The VL160X is not available with a rated voltage of 690 V AC for rated currents of 16 A and 20 A.

4) At 240 V AC, 415 V AC, and 525 V AC max. 5% overvoltage; at 440 V AC, 500 V AC and 690 V AC max. 10% overvoltage.

5) At 250 V DC, 500 V DC and 600 V DC max. 5% overvoltage.

The voltage values apply for at least 3 conductor paths in series, see also "Suggested circuits for DC networks" in the Chapter "Application planning".

Table 12- 6 High switching capacity VL630 to VL1600

SENTRON VL - H rated breaking current (kA) balanced (high switching capacity)						
Type	SENTRON		VL630	VL800	VL1250	VL1600
			I_{cu}/I_{cs}	I_{cu}/I_{cs}	I_{cu}/I_{cs}	I_{cu}/I_{cs}
IEC 60947-2	4)	Up to 240 V AC	100/75	100/75	100/50	100/50
		415 V AC	70/70	70/70	70/35	70/35
		440 V AC	50/38	50/38	50/38	50/38
		525 V AC	40/30	40/30	40/30	40/30
		690 V AC	30/15	30/15	30/15	30/15
	5)	Up to 250 V DC	32/32	--	--	--
		500 V DC	32/32	--	--	--
		600 V DC	--	--	--	--

Table 12- 7 Very high switching capacity VL160X, VL160 to VL400

SENTRON VL - L rated breaking current (kA) balanced (very high switching capacity)						
Type	SENTRON		VL160X	VL160	VL250	VL400
			I_{cu}/I_{cs}	I_{cu}/I_{cs}	I_{cu}/I_{cs}	I_{cu}/I_{cs}
IEC 60947-2	4)	Up to 240 V AC	--	200/150	200/150	200/150
		415 V AC	--	100/75	100/75	100/75
		440 V AC	--	75/50	75/50	75/50
		525 V AC	--	50/38	50/38	50/38
		690 V AC	--	12/6	12/6	15/8
	5)	Up to 250 V DC	--	32/32	32/32	32/32
		500 V DC	--	32/32	32/32	32/32
		600 V DC	--	32/32	32/32	30/32

Table 12- 8 Very high switching capacity VL630 to VL1600

SENTRON VL - L rated breaking current (kA) balanced (very high switching capacity)						
Type	SENTRON		VL630	VL800	VL1250	VL1600
			I_{cu}/I_{cs}	I_{cu}/I_{cs}	I_{cu}/I_{cs}	I_{cu}/I_{cs}
IEC 60947-2	4)	Up to 240 V AC	200/150	200/150	200/100	200/100
		415 V AC	100/75	100/75	100/50	100/50
		440 V AC	75/50	75/50	75/50	75/50
		525 V AC	50/38	50/38	50/38	50/38
		690 V AC	35/17	35/17	35/17	35/17
	5)	Up to 250 V DC	32/32	--	--	--
		500 V DC	32/32	--	--	--
		600 V DC	32/32	--	--	--

4) At 240 V AC, 415 V AC, and 525 V AC max. 5% overvoltage; at 440 V AC, 500 V AC and 690 V AC max. 10% overvoltage.

5) At 250 V DC, 500 V DC and 600 V DC max. 5% overvoltage.

The voltage values apply for at least 3 conductor paths in series, see also "Suggested circuits for DC networks" in the Chapter "Application planning".

12.2 Configuration of main connections

Main conductor connection for SENTRON VL fixed-mounted version

Terminals for cable

Table 12- 9 Terminals

<table>
<tr><th></th><th></th><th colspan="2">VL160X/
VL160</th><th colspan="2">VL250</th><th colspan="2">VL400</th><th>VL400</th><th>VL630</th><th>VL800</th><th>VL1250</th></tr>
<tr><td>Conductor cross section
Cu / Al (mm^2)</td><td>multi-core
Finely-stranded</td><td colspan="2">16-70
16-50</td><td colspan="2">25-185
25-120</td><td colspan="2">50-300
50-240</td><td>50-120
50-95</td><td>50-240
50-185</td><td>50-185
50-150</td><td>50-185
50-150</td></tr>
<tr><td>Cable connection option</td><td></td><td colspan="2">1</td><td colspan="2">1</td><td colspan="2">1</td><td>2</td><td>2</td><td>3</td><td>4</td></tr>
<tr><td>Tightening torque for securing conductor</td><td>Nm</td><td>16
25-35
50-70
[mm^2]</td><td>6
9
14</td><td>25-35
50-185
[mm^2]</td><td>14
31</td><td>50-120
150-300
[mm^2]</td><td>31
56</td><td>31</td><td>34</td><td>42</td><td>42</td></tr>
<tr><td>Tool (hex wrench)</td><td></td><td colspan="2">4</td><td colspan="2">8</td><td colspan="2">12</td><td>8</td><td>8</td><td>8</td><td>8</td></tr>
<tr><td>Tightening torque
Securing terminals</td><td>Nm</td><td colspan="2">-</td><td colspan="2">14</td><td colspan="2">15</td><td>15</td><td>15</td><td>15</td><td>24</td></tr>
<tr><td>Tool
(Allen key)[1)]</td><td></td><td colspan="2">-</td><td colspan="2">4</td><td colspan="2">6</td><td>6</td><td>6</td><td>8</td><td>8</td></tr>
</table>

1) For the fixing screws for the connection pieces

Box terminals

Table 12- 10 Box terminals

Type of cable		VL160X/VL160	VL250	VL400
Solid/stranded	mm^2	2,5-70	25-185	50-300
Finely stranded with end sleeve	mm^2	2,5-50	25-120	50-240
Busbar dimensions W x H x D	mm	12 x 10 x 19	17 x 10 x 24	25 x 10 x 28
Tightening torque	Nm	4 (2,5 ... 10 mm^2) 8 (16 ... 70 mm^2)	12	25
Tool (Allen key)		4	5	8

Front connecting bars

Table 12- 11 Front connecting bars

Dimensions (mm)	VL160X/ VL160	VL250	VL400	VL630	VL800	VL1250 / VL1600
Width (W)	20	22	30,5	42	51	60
Length (L)	44,5	44,5	81,75	69,75	91,5	102,25
Clearance (D)	10	13	15	15	15	20
Thickness (T)	6,5	6,5	9,5	9,5	9,5	16
inside (∅)	7	11	11	11	13	13

Front flared busbar extensions

Table 12- 12 Front flared busbar extensions

Dimensions (mm)	VL160X/ VL160	VL250	VL400	VL630	VL800
Pole clearance (P)	44,5	44,5	63,5	76	76

Rear terminals

Table 12- 13 Rear terminals

Thread round terminal	VL160X/ VL160	VL250	VL400
Short length (Ls) mm	66	66	73
Long length (Ll) mm	123	123	131
Thread	M12	M12	M12
Pad-type terminal	VL160X/VL160	VL250	VL400
Short length (Ls) mm	51,5	51,5	98
Long length (Ll) mm	108,5	108,5	157
Bore hole Ø	11	11	11
W / W / T	25 / 25 / 4	25 / 25 / 4	28 / 28 / 8

Rear flat busbar terminals

Table 12- 14 Rear flat busbar terminals

mm	VL630	VL800	VL1250	VL1600
Width(W)	32	50	50	60
Length(L)	66,5	142	142	178
inside(∅ D)	11	13 (2x)	13 (2x)	13 (2x)
Allen key/hex wrench opening	6 / -	6 / -	6 / -	- / 18
Tightening torque for fixing screw	15	15	15	30

Connection with screw-type terminals

Table 12- 15 Connection with screw-type terminals

Circuit breaker		VL160X	VL160	VL250	VL400	VL630	VL800	VL1250
Screw customer busbar T	mm	M6 x 20 1-7	M6 x 20 1-7	M8 x 20 1-7	M8 x 25 3-10	M6 x 40 (2x) 5-10	M8 x 40 (2x) 10-15	M8 x 50 (2x) 15-20
Max. torque	Nm	6	6	10	15	15	24	24
Busbar d_{max} W_{max}	mm mm	6 19	9 24	9 24	10 32	10 42	13 50	13 50

12.3 Switching capacity overview

The dimensioning of the circuit breakers for the individual application can be seen from the overview tables of the switching capacity of the SENTRON VL module, as well as the table for deviating network frequencies.

Application case: System protection

3- and 4-pole circuit breakers

Table 12- 16 Overview of switching capacity for line protection for VL160X, VL160 to VL1600

Rated current I_n [A]	VL160X	VL160	VL250	VL400	VL630	VL800	VL1250	VL1600
16	X							
20	X							
25	X	X						
32	X	X						
40	X	X						
50	X	X						
63	X	X						
80	X	X	X					
100	X	X	X					
125	X	X	X	X				
160	X	X	X	X				
200			X	X				
250			X	X	X			
315				X	X	X		
400				X	X	X	X	
600					X	X	X	
630					X	X	X	X
800						X	X	X
100							X	X
1250								X
1600								X

Application case: Motor Protection

3-pole circuit breakers

Table 12- 17 Overview of switching capacity for motor protection for VL160 to VL630

Rated current I_n [A]	VL160	VL250	VL400	VL630
16				
20	X			
25	X			
32	X			
40	X			
50	X			
63	X			
80	X	X		
100	X	X		
125	X	X		
160	X	X	X	
200		X	X	
250		X	x	X
315			X	X
400				X
500				X

Application case: Starter combinations

3-pole circuit breakers

Table 12- 18 Overview of switching capacity of starter combination for VL160 to 630

Rated current I_n [A]	Vl160	VL250	VL400	VL630
16				
20				
25				
32				
40				
50				
63	X			
80	X			
100	X			
125	X	X		
160	X	X		
200		X	X	
250		X	X	
315			X	X
400				X
500				X

Application case: Non-automatic circuit breakers

3- and 4-pole circuit breakers

Table 12- 19 Overview of switching capacity for non-automatic circuit breakers for VL160X, VL160 to VL1600

Rated current I_n [A]	Vl160X	Vl160	VL250	VL400	VL630	VL800	VL1250	VL1600
16	X	X						
20	X	X						
25	X	X						
32	X	X						
40	X	X						
50	X	X						
63	X	X						
80	X	X						
100	X	X						
125	X	X						
160	X	X	X					
200			X					
250			X	X				
315				X				
400				X	X			
500					X			
630					X	X		
800						X	X	
1000							X	
1250							X	X
1600								X

12.4 Switching capacity overview

The individual switching capacities and those of SENTRON VL circuit breakers used in different network frequencies are listed in the table below:

Switching capacity in different applications

Application case	Type	Standard switching capacity N (40, 45, 50 kA/415 V AC)	High switching capacity H (70 kA/415 V AC)	Very high switching capacity L (100 kA/415 V AC)
For line protection 3- and 4-pole circuit breakers	VL160X	X	X	
	VL160	X	X	X
	VL250	X	X	X
	VL400	X	X	X
	VL630	X	X	X
	VL800	X	X	X
	VL1250	X	X	X
	VL1600	X	X	X
For motor protection 3-pole circuit breakers	VL160	X	X	X
	VL250	X	X	X
	VL400	X	X	X
	VL630	X	X	X
For starter combination 3-pole circuit breakers	VL160	X	X	X
	VL250	X	X	X
	VL400	X	X	X
	VL630	X	X	X
Non-automatic circuit breakers 3- and 4-pole circuit breakers	VL160X	X	X	
	VL160	X	X	X
	VL250	X	X	X
	VL400	X	X	X
	VL630	X	X	X
	VL800	X	X	X
	VL1250	X	X	X
	VL1600	X	X	X

Use in deviating network frequencies

Version	Type	Use of VL in networks with			
		16 2/3 Hz	50 / 60 Hz	400 Hz	DC
VL160X	TM	Yes	Yes	On request	Yes
VL160	ETU / LCD	No	Yes	No	No
	TM	Yes	Yes	On request	Yes
VL250	ETU / LCD	No	Yes	No	No
	TM	Yes	Yes	On request	Yes
VL400	ETU / LCD	No	Yes	No	No
	TM	Yes	Yes	On request	Yes
VL630	ETU / LCD	No	Yes	No	No
	TM	Yes	Yes	On request	Yes
VL800	ETU / LCD	No	Yes	No	No
VL1250	ETU / LCD	No	Yes	No	No
VL1600	ETU / LCD	No	Yes	No	No

12.5 Derating factors

The tables for derating factors apply for SENTRON VL used under difficult operating conditions in the following areas:

Use at altitudes above 2000 meters

Table 12- 20 Derating factors for high altitudes

Circuit breaker	Characteristic values	Altitude (m)						
		2000	3000	4000	5000	6000	7000	8000
All	Switching capacity I_{CU}/I_{CS}	1,0	0,9	0,8	0,7	0,6	0,5	0,4
	Operating voltage V_{max}	1,0	0,9	0,8	0,7	0,6	0,5	0,4
	Operating current I_{max} [1)]	1,00	0,96	0,92	0,88	0,84	0,80	0,76
	Set current I_r [2)]	1,00	1,02	1,04	1,06	1,08	1,10	1,12

1) At max. ambient temperature 50 °C

2) Thermomagnetic trip units only

Thermomagnetic overcurrent trip unit:

Fixed mounting:

Table 12- 21 Derating factors of thermomagnetic overcurrent trip unit

Circuit breaker	I_n At 50 °C [A]	Cross-section Cu [mm²] min.	Cross-section Al [mm²]min	Max. rated uninterrupted current according to the ambient temperature x I_n			
				40 °C	50 °C	60 °C	70 °C
VL160X	16	2,5	4	1	1	0,93	0,86
	20	2,5	4	1	1	0,93	0,86
	25	4	6	1	1	0,93	0,86
	32	6	10	1	1	0,93	0,86
	40	10	10	1	1	0,93	0,86
	50	10	16	1	1	0,93	0,86
	63	16	25	1	1	0,93	0,86
	80	25	35	1	1	0,93	0,86
	100	35	50	1	1	0,93	0,86
	125	50	70	1	1	0,93	0,86
	160	70	95	1	1	0,93	0,86
VL160	50	10	16	1	1	0,93	0,86
	63	16	25	1	1	0,93	0,86
	80	25	35	1	1	0,93	0,86
	100	35	50	1	1	0,93	0,86
	125	50	70	1	1	0,93	0,86
	160	70	95	1	1	0,93	0,86
VL250	200	95	120	1	1	0,93	0,86
	250	120	185	1	1	0,93	0,86
VL400	200	95	120	1	1	0,93	0,86
	250	120	185	1	1	0,93	0,86
	315	185	2x120	1	1	0,93	0,86
	400	240	2x150	1	1	0,93	0,86
VL630	315	185	2x120	1	1	0,93	0,86
	400	240	2x150	1	1	0,93	0,86
	500	2x150	2x185	1	1	0,93	0,86
	630	2x185	2x240	1	1	0,93	0,86

Plug-in or withdrawable version:

Table 12- 22 Derating factors Thermomagnetic overcurrent trip units (plug-in or withdrawable version

Circuit breaker	Trip unit Thermomagnetic TM		Coefficient at			
	From [A]	To [A]	40 °C	50 °C	60 °C	70 °C
VL160X	16	40	1	1	1	1
VL160 & VL160X	50	100	1	1	1	1
	125	160	1	0,9	0,9	0,9
VL250	200	250	1	0,9	0,9	0,9
VL400	200	250	1	1	1	1
	315	400	1	0,9	0,9	0,9
VL630	315	400	1	1	1	1
	500	630	1	0,85	0,85	0,85

Example for VL250:

- I_n = 200 A at 50 °C
- Ambient temperature = 60 °C

I_n = 200 x 0.93 = 186 A for fixed-mounted version

I_n = 200 x 0,93 x 0.9 = 167 A for plug-in version

Thermomagnetic overcurrent trip unit + RCD module

Fixed mounting:

Table 12- 23 Derating factors for thermomagnetic overcurrent trip unit + RCD module (fixed mounting)

Circuit breaker	I_n at 50 °C [A]	Cross-section Cu [mm²]min	Cross-section Al [mm²]min	Max. rated uninterrupted current according to the ambient temperature x I_n			
				40 °C	50 °C	60 °C	70 °C
VL160X	16	2,5	4	1	1	0,93	0,80
	20	2,5	4	1	1	0,93	0,80
	25	4	6	1	1	0,93	0,80
	32	6	10	1	1	0,93	0,80
	40	10	10	1	1	0,93	0,80
	50	10	16	1	1	0,93	0,80
	63	16	25	1	1	0,93	0,80
	80	25	35	1	1	0,93	0,80
	100	35	50	1	1	0,93	0,80
	120	50	70	1	1	0,93	0,80
	160	70	95	1	1	0,93	0,80
VL160	50	10	16	1	1	0,93	0,80
	63	16	25	1	1	0,93	0,80
	80	25	35	1	1	0,93	0,80
	100	35	50	1	1	0,93	0,80
	125	50	70	1	1	0,93	0,80
	160	70	95	1	1	0,93	0,80
VL250	200	95	120	1	1	0,86	0,80
	250	120	185	1	1	0,86	0,80
VL400	200	95	120	1	1	0,86	0,80
	250	120	185	1	1	0,86	0,80
	315	185	2x120	1	1	0,86	0,80
	400	240	2x150	1	1	0,86	0,80

Plug-in or withdrawable version:

Table 12- 24 Derating factors for thermomagnetic overcurrent trip unit + RCD module (plug-in or withdrawable version)

Circuit breaker	Trip unit Thermomagnetic TM		Coefficient at			
	From [A]	To [A]	40 °C	50 °C	60 °C	70 °C
VL160X	16	40	1	1	1	1
VL160 & VL160X	50	100	1	0,97	0,97	0,97
	125	160	1	0,88	0,88	0,88
VL250	200	250	1	0,85	0,85	0,85
VL400	200	250	1	0,97	0,97	0,97
	315	400	1	0,85	0,85	0,85

Electronic trip unit

Fixed mounting:

Table 12- 25 Derating factors for electronic trip unit (fixed mounting)

Circuit breaker	I_n At 50 °C [A]	Cross-section Cu [mm²]min	Cross-section Al [mm²]min	Max. rated uninterrupted current according to the ambient temperature x I_n			
				40 °C	50 °C	60 °C	70 °C
VL160	63	16	25	1	1	1	0,80
	100	35	50	1	1	1	0,80
	160	70	95	1	1	1	0,80
VL250	200	95	120	1	1	1	0,80
	250	120	185	1	1	0,95	0,80
VL400	315	185	2x120	1	1	1	0,80
	400	240	2x150	1	1	0,95	0,80
VL630	630	2x185	2x240	1	1	0,95	0,80
VL800	800	2x50x5		1	1	0,95	0,80
VL1250	1000	2x60x5		1	1	1	0,80
	1250	2x80x5		1	1	0,95	0,80
VL1600	1600	2x100x5		1	1	0,95	0,80

Note

The electronic trip units with the ordering data Sx, Mx, Tx and Nx have a thermal self-protection feature that trips the breaker if the electronics components reach 100°C.

Plug-in or withdrawable version:

Table 12- 26 Derating factors for electronic trip units (plug-in or withdrawable version)

Circuit breaker	Trip unit Thermomagnetic TM		Coefficient at			
	From [A]	To [A]	40 °C	50 °C	60 °C	70 °C
VL160	63	100	1	1	1	1
	125	160	1	0,9	0,9	0,9
VL250	200	250	1	0,9	0,9	0,9
VL400	315	400	1	0,9	0,9	0,9
VL630		630	1	0,85	0,85	0,85
VL800		800	1	0,9	0,9	0,9
VL1250	1000	1250	1	0,95	0,95	0,95
VL1600		1600	1	0,8	0,8	0,8

Example for VL250:

- I_n = 250 A at 50 °C
- Ambient temperature = 60 °C

- I_n = 250 x 0.95 = 237 A for fixed-mounted version
- I_n = 250 x 0.95 x 0.9 = 213 A for plug-in version

- Set I_R to the next possible value

- I_R = 0.95 In for fixed-mounted version
- I_R = 0.8 In for plug-in version

Thermomagnetic overcurrent trip unit, setting values Ir (thermal)

Table 12- 27 Derating factors for low setting values

Circuit breaker	At 0 °C	At 10 °C	At 20 °C	At 30 °C	At 40 °C	At 50 °C	At 60 °C	At 70 °C
VL160X	0,80	0,84	0,88	0,92	0,96	1	1,04	1,08
VL160	0,80	0,84	0,88	0,92	0,96	1	1,04	1,08
VL250	0,80	0,84	0,88	0,92	0,96	1	1,04	1,08
VL400	0,80	0,84	0,88	0,92	0,96	1	1,04	1,08
VL630	0,80	0,84	0,88	0,92	0,96	1	1,04	1,08

Table 12- 28 Derating factors for high setting values

Circuit breaker	At 0 °C	At 10 °C	At 20 °C	At 30 °C	At 40 °C	At 50 °C	At 60 °C	At 70 °C
VL160X	0,65	0,72	0,79	0,86	0,93	1	1,07	1,14
VL160	0,65	0,72	0,79	0,86	0,93	1	1,07	1,14
VL250	0,65	0,72	0,79	0,86	0,93	1	1,07	1,14
VL400	0,65	0,72	0,79	0,86	0,93	1	1,07	1,14
VL630	0,65	0,72	0,79	0,86	0,93	1	1,07	1,14

Example for VL250:

- I_n = 250 A at 50 °C
- Setting the thermal overcurrent trip unit: 250 A

- Ambient temperature = 20 °C
- Corrected setting = 250 x 0.87 = 217 A

12.6 Power loss

Power loss for fixed-mounted circuit breakers

Thermomagnetic overcurrent trip units (TM)

The table below shows the **power loss and the current path resistance for thermomagnetic overcurrent trip units (TM)**. The power loss applies for I_n with 3-phase balanced load. The specified power loss is the sum of all current paths. The current path resistance is only a guide value and can fluctuate.

Table 12- 29 Power loss for thermomagnetic overcurrent trip units (TM)

Type	Rated current [A]	Power loss [W]	Path resistance [mΩ]
VL160X	16	11	14
	20	17	14
	25	7	3,7
	32	11	3,6
	40	16	3,3
	50	15	2,0
	63	18	1,5
	80	24	1,3
	100	22	0.73
	125	31	0,66
	160	41	0,53
VL160	50	16	2,1
	63	21	1,8
	80	27	1,4
	100	27	0,90
	125	36	0,77
	160	55	0,63
VL250	200	60	0,47
	250	71	0,38
VL400	200	60	0,50
	250	84	0,45
	315	120	0,40
	400	175	0,36
VL630	315	85	0,29
	400	120	0,25
	500	170	0,23
	630	230	0,19

Electronic trip units (ETU / LCD-ETU)

The table below shows the **power loss for electronic trip units (ETU / LCD-ETU)**. The power loss applies for I_n with 3-phase balanced load. The specified power loss is the sum of all current paths.

The current path resistance is only a guide value and can fluctuate.

Table 12- 30 Power loss for electronic overload releases (ETU / LCD-ETU)

Type	Rated current [A]	Power loss [W]	Path resistance [mΩ]
VL160	63	7	0,59
	100	16	0,53
	160	40	0,52
VL250	200	42	0,35
	250	60	0,32
VL400	315	60	0,2
	400	90	0,19
VL630	630	160	0,13
VL800	800	250	0,13
VL1250	1000	135	0,045
	1250	210	0,045
VL1600	1600	260	0,034

Starter combinations

The table below shows the **power loss and the current path resistance for starter combinations**. The power loss applies for I_n with 3-phase balanced load. The specified power loss is the sum of all current paths. The current path resistance is only a guide value and can fluctuate.

Table 12- 31 Power loss for starter combinations

Type	Rated current [A]	Power loss [W]	Path resistance [mΩ]
VL160	63	7	0,59
	100	16	0,53
	160	40	0,52
VL250	250	60	0,32
VL400	200	30	0,25
	250	42	0,22
	315	60	0,20
VL630	315	59	0,20
	500	118	0,16

Molded-case non-automatic circuit breakers

The table below shows the **power loss and the current path resistance for molded-case non-automatic circuit breakers.** The power loss applies for I_n with 3-phase balanced load. The specified power loss is the sum of all current paths. The current path resistance is only a guide value and can fluctuate.

Table 12- 32 Power loss for molded-case circuit breakers

Type	Rated current [A]	Power loss [W]	Path resistance [mΩ]
VL160X	100	13	0,43
	160	34	0,44
VL160	100	16	0,53
	160	40	0,52
VL250	250	60	0,32
VL400	400	90	0,19
VL630	630	160	0,13
VL800	800	250	0,13
VL1250	1250	210	0,045
VL1600	1600	260	0,034

12.7 Mechanical operating mechanisms

The following technical data apply for the mechanical operating mechanisms of the SENTRON VL circuit breakers:

- Door-coupling rotary operating mechanisms

Table 12- 33 Overview of accessories for door-coupling rotary operating mechanisms

Type	Rated current	Operating mechanism without knob	Extension shaft	Standard operating mechanism	Emergency-stop mechanism
				Order No.	Order No.
VL160X	16 ... 160	3VL9300-3HE00	6 x 6 mm	8UC7111-6BD15	8UC7121-8BD15
VL160	50 ... 160	3VL9300-3HE00	6 x 6 mm		
VL250	200 ... 250	3VL9300-3HE00	6 x 6 mm		
VL400	200 ... 400	3VL9400-3HE00	8 x 8 mm	8UC7262-6BD26	8UC7262-8BD26
VL630	315 ... 600	3VL9600-3HE00	8 x 8 mm		
VL800	320 ... 800	3VL9600-3HE00	8 x 8 mm		
VL1250	400 ... 1250	3VL9800-3HE00	12 x 12 mm	8UC7314-6BD44	8UC7324-8BD44
VL1600	640 ... 1600	3VL9800-3HE00	12 x 12 mm		

The knob can be locked with a padlock; with cover frame and display plate, actuator plate for shafts, extension shaft (300 mm) or coupling for extension shaft available.

- Leading auxiliary switch for rotary operating mechanisms

Table 12- 34 Leading auxiliary switches

Technical data	VL160X- VL1600
Thermal rated current I_{th} [A]	2
Rated switching capacity [A]	
cosφ = 1 (resistive)	2
cosφ = 0.7 (inductive)	0.5
Rated operating voltage [V]	230
Rated operating current [A]	
cosφ = 1 (resistive)	2
cosφ = 0.7 (inductive)	0.5
Backup fuse [A]	2

12.8 Motorized operating mechanisms

The following specifications apply for the motorized operating mechanism with and without stored energy (model-dependent and size-dependent) for the SENTRON VL circuit breaker:

Table 12- 35 Stored-energy motorized operating mechanism for VL160x, VL160 to VL400 or without stored energy for VL160X, VL160 and VL250 (deviating values in brackets)

Type			VL160X	VL160	VL250	VL400
Synchronizable			X	X	X	X
Operating range		[V]	0,85 ... 1.1 U_S	0,85 ... 1.1 U_S	0,85 ... 1.1 U_S	0,85 ... 1.1 U_S
Minimum command duration at V_S		[ms]	50	50	50	50
Max. command duration 1)		[ms]	Jog/pushbutton or continuous command			
Total closing time		[ms]	<100 (3000)	<100 (3000)	<100 (3000)	<100
Break time		[s]	<5 (3)	<5 (3)	<5 (3)	<5
Pause between ON and OFF command		[s]	>1 (3)	>1 (3)	>1 (3)	>1
Pause between OFF and ON command		[s]	>5 (3)	>5 (3)	>5 (3)	>5
Max. permissible switching frequency		1/h	120	120	120	120
Electrical data						
Power consumption		[VA]	100			200
Rated control supply voltage Vs	50 .. 60 Hz AC	[V]	48, 60, 110/127, 230/250			
	DC	[V]	24, 48, 60, 110/127, 230/250			
Fuse (time-lag "T"), DIAZED		[A]	4 at 48V AC, 60V AC; 2 at 110/127V, 230/250V)			2
Circuit breaker, C characteristic		[A]	4 at 48V AC, 60V AC; 2 at 110/127V, 230/250V)			2

1) Circuit-dependent; changeover contact also permissible but note the pause times between ON and OFF.

Table 12- 36 Stored-energy motorized operating mechanism for VL630 and VL800 or without stored energy for VL1250 and VL1600

Type			VL630	VL800	VL1250	VL1600
Synchronizable			X	X	--	--
Operating range		[V]	0,85 ... 1.1 U_S	0,85 ... 1.1 U_S	0,85 ... 1.1 U_S	0,85 ... 1.1 U_S
Minimum command duration at V_S		[ms]	50	50	50	50
Max. command duration [1)]		[ms]	Jog or pushbutton command			
Total closing time		[ms]	<100	<100	<5000	<5000
Break time		[s]	<5	<5	<5	<5
Pause between ON and OFF command		[s]	>1	>1	>1	>1
Pause between OFF and ON command		[s]	>5	>5	>5	>5
Max. permissible switching frequency		1/h	60	60	30	30
Electrical data						
Power consumption		[VA]	250			
Rated control supply voltage Vs	50 .. 60 Hz AC	[V]	48, 60, 110/127, 230/250			
	DC	[V]	24, 48, 60, 110/127, 230/250			
Fuse (time-lag "T"), DIAZED		[A]	4 at 48V AC, 60V AC; 2 at 110/127V, 230/250V			
Circuit breaker, C characteristic		[A]	4 at 48V AC, 60V AC; 2 at 110/127V, 230/250V			

12.9 Capacitor banks

Selection of the circuit breaker for protecting and switching capacitors

This table takes account of only a few typical applications and combinations. The appropriate selection must be made for all other applications.

Table 12- 37 Selection examples for capacitor protection circuits

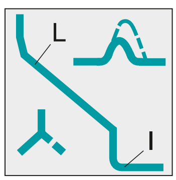
Rated voltage [50 Hz]	Q_c capacitor bank power [kvar]	Capacitor rated current x 1.5 = I_R of the SENTRON VL [A]	Upstream SENTRON VL circuit breaker		
			Type	I_R [A]	I_i [A]
230 V	15	56	VL160	50-63	600
	30	113	VL160	100-125	1000
400 V	25	54	VL160	50-63	600
	50	108	VL160	100-125	1000
	100	216	VL250	200-250	2000
415 V	20	42	VL160	40-50	600
	40	84	VL160	80-100	1000
525 V	25	42	VL160	40-50	600
	50	84	VL160	80-100	1000

12.10 Motor Protection

The following characteristic values in the relevant tables apply for the SENTRON VL circuit breakers in motor protection with different trip classes:

- Trip class ETU10M fixed
- Trip class ETU30M adjustable
- Trip class ETU40M adjustable

Circuit breakers for motor protection with fixed trip class ETU10M



Characteristic value of circuit breakers for motor protection with fixed trip class ETU10M

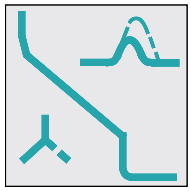
These circuit breakers possess an adjustable overload and short-circuit release and a fixed trip class.

They are current-limiting and have a phase failure sensitivity feature

Table 12- 38 Circuit breakers for motor protection with fixed trip class ETU 10M

Circuit breaker	Rated current I_n [A]	Max. rated power of the motor at 50 Hz AC [kW]		Adjustable range of the overload protection I_R [A]	Adjustable range of the short-circuit protection I_I [A]	Trip class TC [s]
		380 / 415 V	500 V			
VL160	63	30	37	0.41-1.0 x I_n	1.25-11 x I_n	10
	100	37. 45	55	0.41-1.0 x I_n	1.25-11 x I_n	10
	160	55. 75	75. 90	0.41-1.0 x I_n	1.25-11 x I_n	10
VL250	200	90, 110	110, 132	0.41-1.0 x I_n	1.25-11 x I_n	10
	250	132	160	0.41-1.0 x I_n	1.25-11 x I_n	10
VL400	315	160	200	0.41-1.0 x I_n	1.25-11 x I_n	10
	315	200	250	0.41-1.0 x I_n	1.25-11 x I_n	10
VL630	500	250	355	0.41-1.0 x I_n	1. 5-12.5 x I_n	10

Circuit breakers for motor protection with adjustable trip class ETU30M



Characteristic curve of circuit breakers for motor protection with adjustable trip class ETU30M

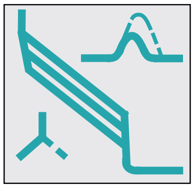
These circuit breakers possess an adjustable overload and short-circuit release and an adjustable trip class.

They are current-limiting and have a phase failure sensitivity feature

Table 12- 39 Circuit breakers for motor protection with adjustable trip class ETU30M

Circuit breaker	Rated current I_n [A]	Max. rated power of the motor at 50 Hz AC [kW]		Adjustable range of the overload protection I_R [A]	Adjustable range of the short-circuit protection I_I [A]	Trip class TC [s]
		380 / 415 V	500 V			
VL160	63	30	37	0.41-1.0 x I_n	6/8/1 x I_n	10/20/30
	100	37. 45	55	0.41-1.0 x I_n	6/8/1 x I_n	10/20/30
	160	55, 75	75, 90	0.41-1.0 x I_n	6/8/1 x I_n	10/20/30
VL250	200	90, 110	110, 132	0.41-1.0 x I_n	6/8/1 x I_n	10/20/30
	250	132	160	0.41-1.0 x I_n	6/8/1 x I_n	10/20/30
VL400	315	160	200	0.41-1.0 x I_n	6/8/1 x I_n	10/20/30
	315	200	250	0.41-1.0 x I_n	6/8/1 x I_n	10/20/30
VL630	500	250	355	0.41-1.0 x I_n	6/8/12.5 x I_n	10/20/30

Circuit breakers for motor protection with adjustable trip class ETU 40M



Characteristic curve of circuit breakers for motor protection with adjustable trip class ETU40M

These circuit breakers possess an adjustable overload and short-circuit release and an adjustable trip class. They are current-limiting and have a phase failure sensitivity feature. They are also equipped with an LCD display for indicating the current and for parameterization.

Communication via Profibus is possible.

Table 12- 40 Circuit breakers for motor protection with adjustable trip class ETU40M

Circuit breaker	Rated current I_n [A]	Max. rated power of the motor at 50 Hz AC [kW]		Adjustable range of the overload protection I_R [A]	Adjustable range of the short-circuit protection I_I [A]	Trip class TC [s]
		380 / 415 V	500 V			
VL160	63	30	37	25-63	1.25-11 x I_n	5/10/15/20/30
	100	37, 45	55	40-100	1.25-11 x I_n	5/10/15/20/30
	160	55, 75	75, 90	63-160	1.25-11 x I_n	5/10/15/20/30
VL250	200	90, 110	110, 132	80-200	1.25-11 x I_n	5/10/15/20/30
	250	132	160	100-250	1.25-11 x I_n	5/10/15/20/30
VL400	315	160	200	126-315	1.25-11 x I_n	5/10/15/20/30
	315	200	250	126-315	1.25-11 x I_n	5/10/15/20/30
VL630	500	250	355	200-500	1.25-12.5 x I_n	5/10/15/20/30

12.11 RCD modules

The RCD modules have the following technical data for their line protection function:

Table 12- 41 Overview of RCD modules

RCD module Circuit breaker for line protection 3- and 4-pole	Rated current I_n [A]	Differential currents $I_{\Delta n}$ Adjustable [A]	Delay time t_d Adjustable [s]	Rated operating voltage V_e [V AC]
VL160X (installed from below) (installed from the left)	160	0.03 0.10 0.30 0.50 1.00 3.00	Instantaneous 0.06 0.10 0.25 0.50 1.00	127-480
VL160	160			127-480 230-690
VL250	250			127-480 230-690
VL400	400			127-480 230-690

12.12 Undervoltage release

The undervoltage releases of the SENTRON VL circuit breakers have the following technical data:

Table 12- 42 Undervoltage releases for VL160X, VL160 to VL400

		VL160X	VL160	VL250	VL400
Operating voltage [V]					
Drop (circuit breaker trips)		0.35-0.70 V_s	0.35-0.70 V_s	0.35-0.70 V_s	0.35-0.70 V_s
Pick-up (circuit breaker can be switched on)		0.85-1.10 V_s	0.85-1.10 V_s	0.85-1.10 V_s	0.85-1.10 V_s
Power consumption					
AC 50 / 60 Hz [VA]	110-127 V	1.5	1.5	1.5	1.5
	220-250 V	1.5	1.5	1.5	1.5
	208 V	1.8	1.8	1.8	1.8
	277 V	2.1	2.1	2.1	2.1
	380-415 V	1.6	1.6	1.6	1.6
	440-480 V	1.8	1.8	1.8	1.8
	500-525 V	2.05	2.05	2.05	2.05
	600 V	2.4	2.4	2.4	2.4
DC [W]	12 V	0.75	0.75	0.75	0.75
	24 V	0.8	0.8	0.8	0.8
	48 V	0.8	0.8	0.8	0.8
	60 V	0.8	0.8	0.8	0.8
	110-127 V	0.8	0.8	0.8	0.8
	220-250 V	0.8	0.8	0.8	0.8
Max. opening (release) time [ms]		50	50	50	50

Table 12- 43 Undervoltage release for VL630 to VL1600

		VL630	VL800	VL1250	VL1600
Operating voltage [V]					
Drop (circuit breaker trips)		0.35-0.70 V_s	0.35-0.70 V_s	0.35-0.70 V_s	0.35-0.70 V_s
Pick-up (circuit breaker can be switched on)		0.85-1.10 V_s	0.85-1.10 V_s	0.85-1.10 V_s	0.85-1.10 V_s
Power consumption					
AC 50 / 60 Hz [VA]	110-127 V	1.1	1.1	1.1	1.1
	220-250 V	2.1	2.1	2.1	2.1
	208 V	2.2	2.2	2.2	2.2
	277 V	1.6	1.6	1.6	1.6
	380-415 V	2.0	2.0	2.0	2.0
	440-480 V	2.3	2.3	2.3	2.3
	500-525 V	2.9	2.9	2.9	2.9
DC [W]	12 V	1.2	1.2	1.2	1.2
	24 V	1.4	1.4	1.4	1.4
	48 V	1.5	1.5	1.5	1.5
	60 V	1.6	1.6	1.6	1.6
	110-127 V	1.2	1.2	1.2	1.2
	220-250 V	1.5	1.5	1.5	1.5
Max. opening (release) time [ms]		80	80	80	80

12.13 Undervoltage release connection data

Maximum connection lengths of the undervoltage releases depending on V_n

The table below lists the maximum lengths for the connecting cables of the undervoltage release family 1 (VL160X to VL400) and/or 2 (VL630 to VL1600). The values in the table refer to 100% V_n in the case of the undervoltage releases. A cable loss of 15% V_n has been taken into account.

Table 12- 44 Undervoltage release family 1

MLFB	Rated voltage in V From-to			A = 0.5 mm² l_{max} in m	A = 1 mm² l_{max} in m	A = 1.5 mm² l_{max} in m
3VL9400-1UN00	12	12	DC	0.93	1.85	2.78
3VL9400-1UP00	24	24	DC	3.66	7.32	10.98
3VL9400-1UU00	48	48	DC	11.79	23.57	35.36
3VL9400-1UV00	60	60	DC	21.98	43.97	65.95
3VL9400-1UR00	110	127	DC	99.59	199.17	298.76
3VL9400-1US00	220	250	DC	362.13	724.26	1086.40
3VL9400-1UD00	24	24	AC	1.78	3.57	5.35
3VL9400-1UG00	110	127	AC	41.00	82.01	123.01
3VL9400-1UM00	208	208	AC	97.18	194.35	291.53
3VL9400-1UH00	220	250	AC	153.40	306.80	460.20
3VL9400-1UQ00	277	277	AC	176.75	353.51	530.26
3VL9400-1UJ00	380	415	AC	438.14	876.28	1314.42
3VL9400-1UK00	440	480	AC	559.36	1118.72	1678.08
3VL9400-1UL00	525	550	AC	607.60	1215.21	1822.81

The table below shows the values of the undervoltage releases of Family 2:

Table 12- 45 Undervoltage release family 2

MLFB	Rated voltage in V From-to			A=0.5 mm² I_{max} in m	A=1 mm² I_{max} in m	A=1.5 mm² I_{max} in m
3VL9400-1UN00	12	12	DC	0.50	1.00	1.50
3VL9400-1UP00	24	24	DC	2.00	4.00	5.99
3VL9400-1UU00	48	48	DC	7.38	14.76	22.13
3VL9400-1UV00	60	60	DC	11.46	22.92	34.37
3VL9400-1UR00	110	127	DC	48.59	97.18	145.77
3VL9400-1US00	220	250	DC	197.82	395.65	593.47
3VL9400-1UD00	24	24	AC	2.42	4.85	7.27
3VL9400-1UG00	110	127	AC	57.22	114.43	171.65
3VL9400-1UM00	208	208	AC	90.29	180.57	270.86
3VL9400-1UH00	220	250	AC	193.66	387.32	580.98
3VL9400-1UQ00	277	277	AC	228.91	457.82	686.73
3VL9400-1UJ00	380	415	AC	406.90	813.81	1220.71
3VL9400-1UK00	440	480	AC	510.87	1021.74	1532.62

12.14 Shunt release

The shunt releases of the SENTRON VL circuit breakers have the following technical data:

Table 12- 46 Shunt releases for VL160X, VL160 to VL400

		Group 1			
		VL160X	VL160	VL250	VL400
Response voltage: Pick-up (circuit breaker trips) [V]		0.7-1.10 V_s	0.7-1.10 V_s	0.7-1.10 V_s	0.7-1.10 V_s
Power consumption					
AC 50 / 60 Hz [VA]	48-60 V	158-200	158-200	158-200	158-200
	110-127 V	136-158	136-158	136-158	136-158
	208-277 V	274-350	274-350	274-350	274-350
	380-600 V	158-237	158-237	158-237	158-237
DC [W]	12 V	110	110	110	110
	24 V	110	110	110	110
	48-60 V	110-172	110-172	110-172	110-172
	110-127 V	220-254	220-254	220-254	220-254
	220-250 V	97-110	97-110	97-110	97-110
Max. in-service period [s]		Interrupts automatically			
Max. opening (release) time [ms]		50	50	50	50
Fuse (time-lag) [A]		4 (AC 48-60, 110-127 V,208-277 V) 2 (all others)			
Circuit breaker, [A] C characteristic		5			

Table 12- 47 Shunt release for VL630 to VL1600

		Group 2			
		VL630	VL800	VL1200	VL1600
Response voltage: Pick-up (circuit breaker trips) [V]		0.7-1.10 V_s	0.7-1.10 V_s	0.7-1.10 V_s	0.7-1.10 V_s
Power consumption					
AC 50 / 60 Hz [VA]	48-60 V	300-480	300-480	300-480	300-480
	110-127 V	302-353	302-353	302-353	302-353
	208-277 V	330-439	330-439	330-439	330-439
	380-600 V	243-384	243-384	243-384	243-384
DC [W]	12 V	50	50	50	50
	24 V	360	360	360	360
	48-60 V	50-820	50-820	50-820	50-820
	110-127 V	302-353	302-353	302-353	302-353
	220-250 V	348-397	348-397	348-397	348-397
Max. in-service period [s]		Interrupts automatically			
Max. opening (release) time [ms]		50	50	50	50
Fuse (time-lag) [A]		4 (AC 48-60, 110-127 V,208-277 V) 2 (all others)			
Circuit breaker, [A] C characteristic		5			

12.15 Shunt release connection data

Maximum connection lengths of the shunt releases depending on Vn

The table below lists the maximum lengths for the connecting cables of the shunt releases of family 1 (VL160X to VL400) and/or 2 (VL630 to VL1600). The values in the table refer to 100% V_n in the case of the shunt releases. A cable loss of 30 % V_n has been taken into account.

Table 12- 48 Shunt release family 1

MLFB	Rated voltage in V From-to			A = 0.5 mm² l_{max} in m	A = 1 mm² l_{max} in m	A = 1.5 mm² l_{max} in m
3VL9400-1SC00	24	24	DC	10.71	21.43	32.14
3VL9400-1SJ00	48	60	DC	52.97	105.94	158.91
3VL9400-1SK00	110	127	DC	216.69	433.39	650.08
3VL9400-1SQ00	220	250	DC	999.20	1998.39	2997.59
3VL9400-1SM00	48	60	AC	52.97	105.94	158.91
3VL9400-1SR00	110	127	AC	216.69	433.39	650.08
3VL9400-1ST00	208	277	AC	626.00	1252.00	1878.01
3VL9400-1SV00	380	600	AC	6982.33	13964.66	20946.99

Table 12- 49 Shunt release family 2

MLFB	Rated voltage in V From-to			A = 0.5 mm² l_{max} in m	A = 1 mm² l_{max} in m	A = 1.5 mm² l_{max} in m
3VL9800-1SC00	24	24	DC	10.11	20.22	30.34
3VL9800-1SJ00	48	60	DC	38.52	77.05	115.57
3VL9800-1SK00	110	127	DC	240.77	481.54	722.31
3VL9800-1SQ00	220	250	DC	770.46	1540.93	2311.39
3VL9800-1SM00	48	60	AC	38.52	77.05	115.57
3VL9800-1SR00	110	127	AC	240.77	481.54	722.31
3VL9800-1ST00	208	277	AC	770.46	1540.93	2311.39
3VL9800-1SV00	380	600	AC	4755.21	9510.42	14265.62

12.16 Auxiliary switches and alarm switches

The auxiliary and alarm switches of the SENTRON VL circuit breakers have the following technical data:

Table 12- 50 Auxiliary switches and alarm switches

Technical data		
Rated insulation voltage V_i		
with degree of pollution in accordance with IEC 60947-1	Class 3	
Elements with screw-type terminal	400 V	
Rated impulse withstand voltage V_{imp}		
• Screw-type terminals, spring-loaded terminals	6 kV	
Conventional thermal current I_{th}	10 A	
Rated operating current I_e		
Rated operating voltage V_e		
• Alternating current 50 / 60 Hz, AC-12	at V_e	I_e
– Screw-type terminals	24 V	10 A
	48 V	10 A
	110 V	10 A
	230 V	10 A
	400 V	10 A
	600 V	10 A
• Alternating current 50 / 60 Hz, AC-15	at V_e	I_e
– Screw-type terminals	24 V	6 A
	48 V	6 A
	110 V	6 A
	230 V	6 A
	400 V	3 A
	600 V	1 A
• Direct current, DC-12	at V_e	I_e
– Screw-type terminals	24 V	10 A
	48 V	5 A
	110 V	2.5 A
	230 V	1 A
• Direct current, DC-13	at V_e	I_e
– Screw-type terminals	24 V	3 A
	48 V	1.5 A
	110 V	0.7 A
	230 V	0.3 A

Technical data	
Contact reliability Test voltage/test current	5 V/1 mA
Short-circuit protection weld-free in accordance with IEC 60947-5-1 • DIAZED fuse links, utilization category gL/gG • Miniature circuit breaker with C characteristic in accordance with IEC 60898 (VDE 0641)	10 A TDz, 16 A D 10 A
Connection cross-sections • Screw-type terminals – Stranded, with end sleeves in accordance with DIN 46228 – Solid – Solid, with end sleeves in accordance with DIN 46228 – Single- or multi-core	2 × (0.5 to 1.5) mm² 2 × (1 to 2.5) mm² 2 × (0.5 to 0.75) mm² 2 × AWG 18 to 14
Tightening torques • Connection screws	0.8 Nm
Rated voltage • Switching devices	300 V AC
Continuous current	10 A
Switching capacity	A 300, R 300, A 600 same polarity

12.17 Position signaling switch

The position signaling switch of the SENTRON VL circuit breakers have the following technical data:

Table 12- 51 Position signaling switch

Technical data		
Connection cross-sections Screw-type terminal	Standard cross-sections (DIN 46228)	
Tightening torques Screws for cable connection	0.5 Nm	
Rated operating temperature	–40 °C to +85 °C	
Data in accordance with IEC/EN 61058		
Rated operating current I_e with rated operating voltage V_e Standard operation	At V_e 250 V AC/400 V AC	I_e 16 A/10 A
Rated making capacity	At 250 V AC 16 A	At 400 V AC 10 A
Rated thermal current I_{th}	16 A	
Rated operating voltage	250 V AC	400 V AC
Rated breaking capacity cosφ = 1 (resistive) cosφ = 0.7 (inductive)	At 250 V AC 16 A 4 A	At 400 V AC 10 A 4 A
Short-circuit fuse (quick-response)	At 250 V AC 16 A	At 400 V AC 10 A
Data according to UL 1054		
Rated operating current I_e with rated operating voltage V_e Alternating current Standard operation	With V_e, power, [horsepower] 125/250 V AC, 1HP	I_e 16 A
Flammability class	UL94V-0	

12.18 Ground fault protection classes

There are different ground fault protection classes for the individual overcurrent trip units:

Table 12- 52 Overview of ground fault protection classes

Trip unit	Ordering data	Ground fault protection class
ETU22	SG, MG[1)]	Vectorial summation current formation (3-conductor system)
ETU22	SH, NH[1)]	Vectorial summation current formation (4-conductor system)
ETU22	TH, NH[1)]	Vectorial summation current formation (4-conductor system)
LCD-ETU42	CL	Vectorial summation current formation (3-conductor system)
LCD-ETU42	CM	Vectorial summation current formation (3-conductor system)/direct recording of the ground-fault current in the neutral point of the transformer
LCD-ETU42	CN	Vectorial summation current formation (4-conductor system)

1 With communication

12.19 IP degrees of protection

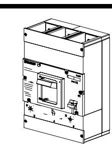
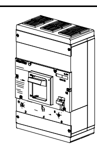
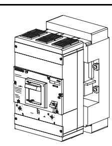
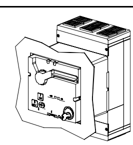
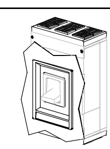
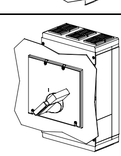
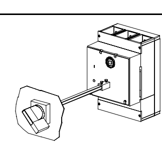
All SENTRON VL molded-case circuit breakers are constructed with degree of protection IP20 regardless of size and version.

A wide range of additional accessories is also available for the basic version of the SENTRON VL circuit breaker in IP20.

The accessories listed below are designed to provide a higher degree of protection:

The degree of protection in accordance with IEC 60529 is listed in the table below:

Table 12- 53 Overview of degrees of protection

Circuit breaker	Protection	Degree of protection
	**Circuit breaker** Finger-proof Protected against solid foreign bodies with a diameter of 12.5 mm or larger.	IP20
	**Circuit breaker with terminal cover** Protected against access to live parts with a tool. Protected against solid foreign bodies with a diameter of 2.5 mm or larger.	IP30
	**Plug-in circuit breaker** Finger-proof Protected against solid foreign bodies with a diameter of 12.5 mm or larger.	IP20 IP30[1]
	**Circuit breaker with cover frame and motorized operating mechanism** Protected against access to live parts with a wire. Protected against solid foreign bodies with a diameter of 1.0 mm or larger.	IP40[2]
	**Circuit breaker with cover frame for door cutout** Protected against access to live parts with a wire. Protected against solid foreign bodies with a diameter of 1.0 mm or larger.	IP40[2]
	**Circuit breaker with cover frame and rotary operating mechanism on front** Protected against access to live parts with a wire. Protected against solid foreign bodies with a diameter of 1.0 mm or larger.	IP40[2]
	**Circuit breaker with door coupling rotary operating mechanism** Protected against ingress of dust and water jets from any direction.	IP65[2]

1) If the circuit breaker is installed and the supplied covers are mounted.

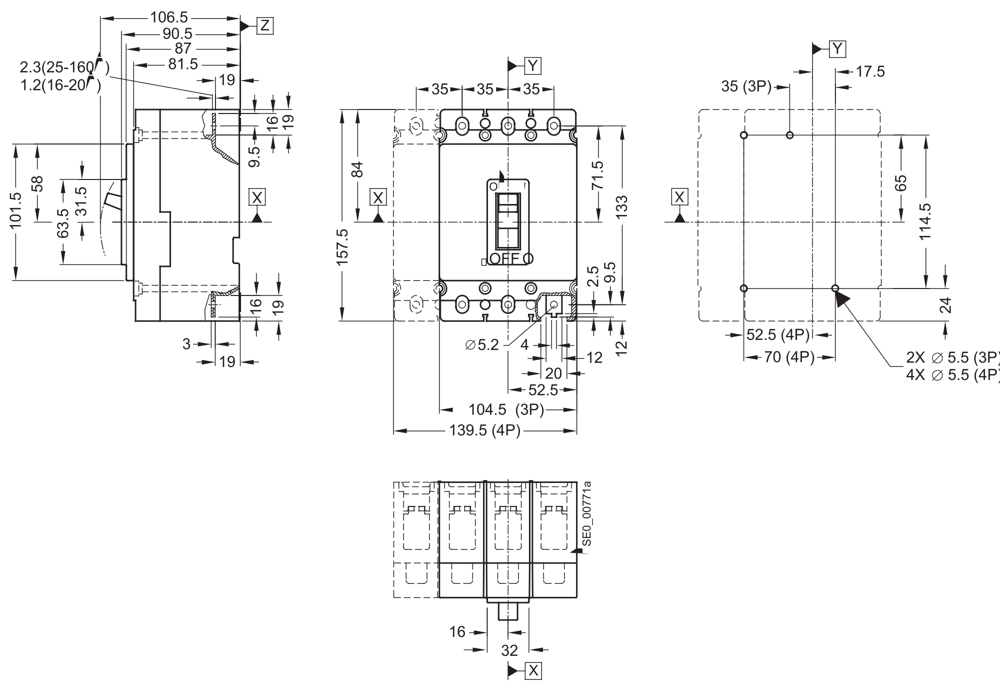
2) Depending on the degree of protection of the housing

Dimensional drawings 13

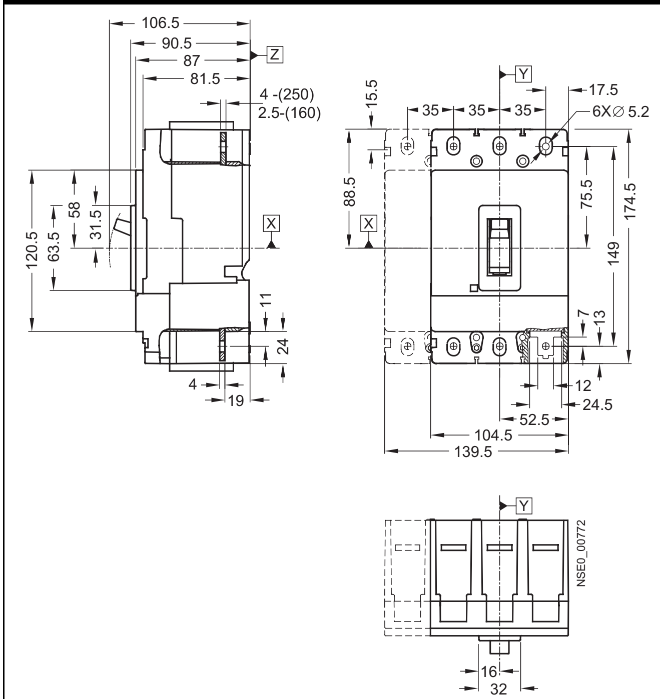
13.1 VL160X (3VL1), VL160 (3VL2), and VL250 (3VL3), 3- and 4-pole, to 250 A

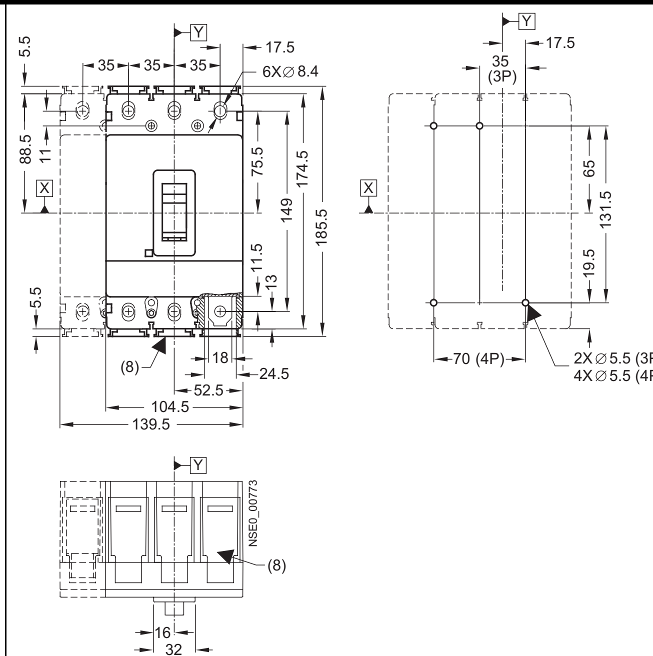
13.1.1 Circuit breakers

SENTRON VL160X (3VL1) circuit breaker and mounting instructions



SENTRON VL160/VL250 (3VL2/3VL3) circuit breakers	SENTRON VL250 (3VL3) circuit breaker	SENTRON VL250 (3VL3) circuit breaker	SENTRON SENTRON VL160 and VL250 (3VL2 and 3VL3) circuit breakers mounting instructions





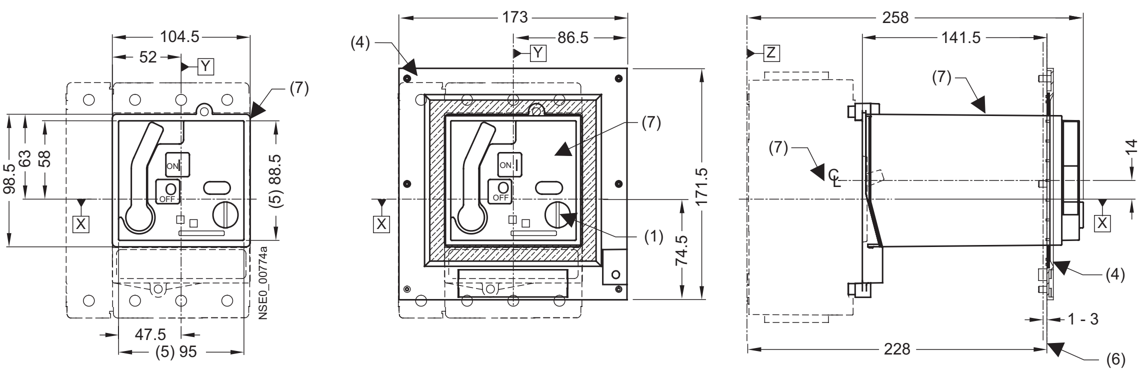
Note

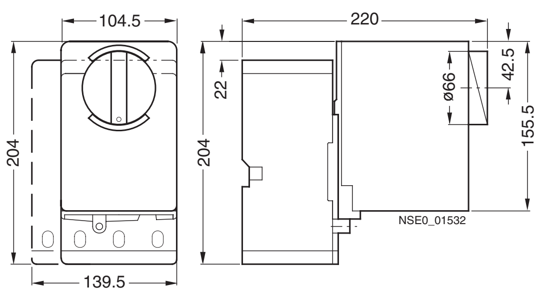
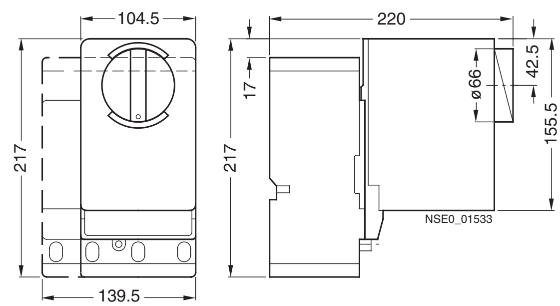
Note:

The 5.5-mm extension at each end of the SENTRON VL250 (3VL3) circuit breaker is only to be observed when using box terminals or round conductor terminals (8).

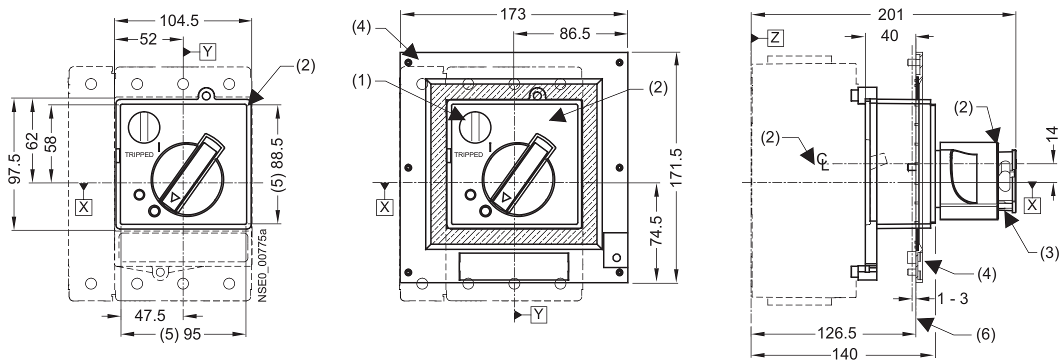
13.1.2 Operating mechanisms

Stored-energy motorized operating mechanism



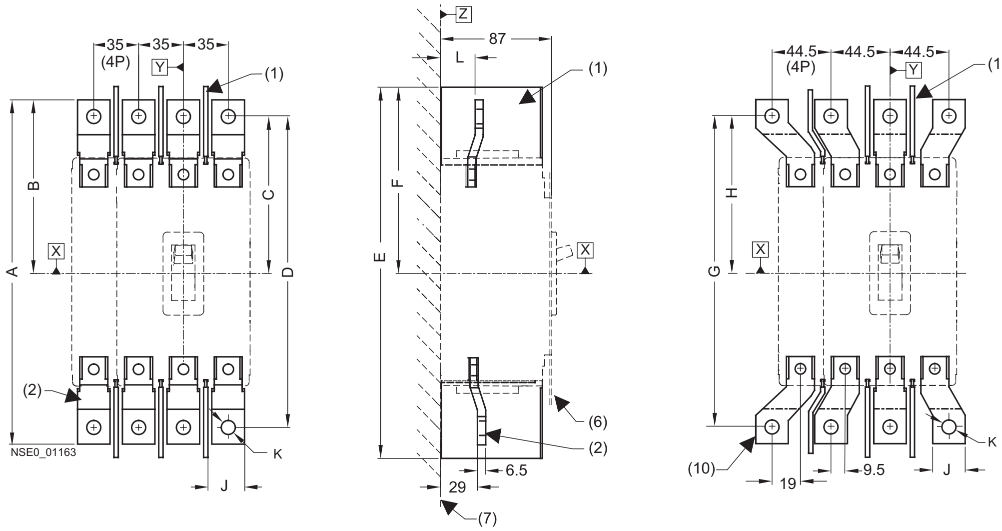
Motorized operating mechanism for VL160X (3VL1)	Motorized operating mechanism for VL160 (3VL2) and VL250 (3VL3)
	

Front rotary operating mechanism



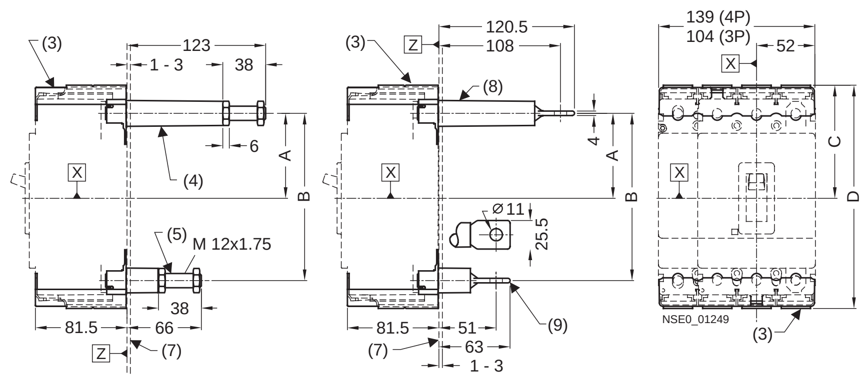
(1) Safety locks
(2) Front rotary operating mechanism
(3) Padlock barrier
(4) Cover frame for door cutout (for circuit breakers with operating mechanism)
(5) Grading for cover
(6) External surface of cabinet door
(7) Stored-energy motorized operating mechanism
(8) Terminal insulation

13.1.3 Connections and phase barriers



Type	A	B	C	D	E	F	G	H	J	K	L
VL160X (3VL1)	242	126	116	222	266,5	138.5	222	116	20	7	27
VL160 (3VL2)	258	130	120	238	283,5	143	238	120	20	7	27
VL250 (3VL3)	263,5	133	120	238	283,5	143	238	120	22	11	29

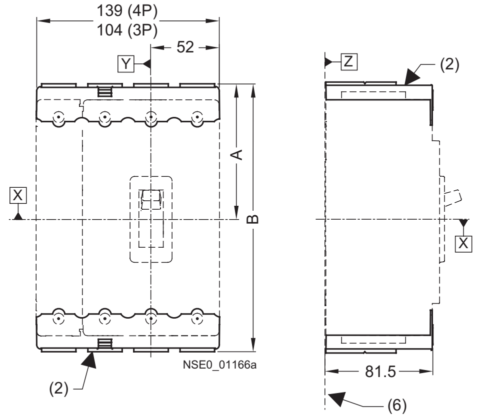
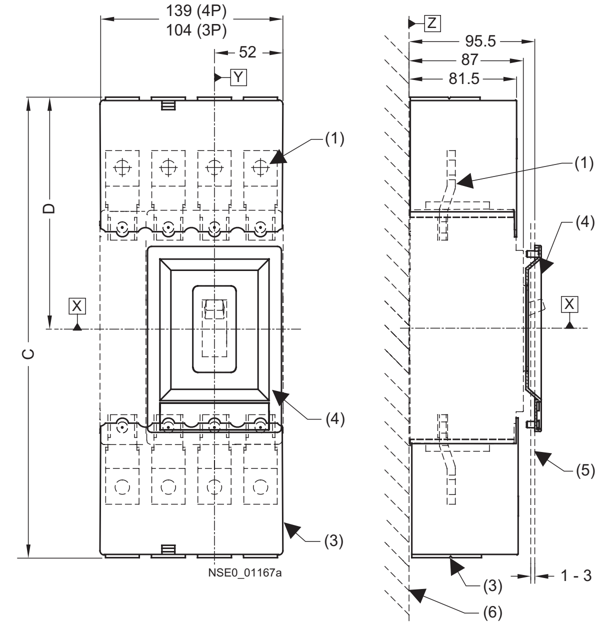
Circuit breaker with rear connections – long and short



Type	A	B	C	D
VL160X (3VL1)	71,5	133	96	182
VL160 (3VL2)	75,5	149	101	199
VL250 (3VL3)	75,5	149	101	199

(1)	Interphase barrier
(2)	Front connecting bars
(3)	Terminal covers (standard)
(4)	Rear connection threaded bolt (long)
(5)	Rear connection threaded bolt (short)
(6)	External surface of cabinet door
(7)	Mounting level
(8)	Rear pad-type terminals (long)
(9)	Rear pad-type terminals (short)
(10)	Flared busbar extensions

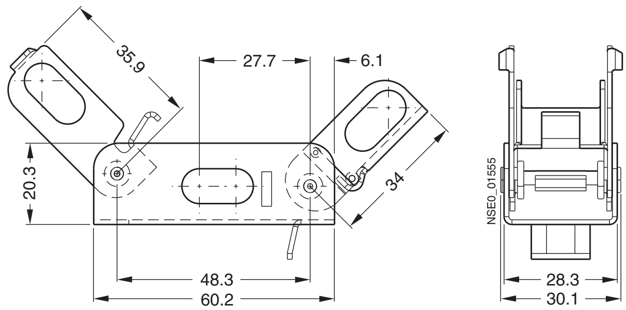
13.1.4 Terminal covers

Terminal covers, standard	Extended terminal covers
	

Type	A	B	C	D
VL160X (3VL1)	96	182	326.5	168.5
VL160 (3VL2)	101	199	343	173
VL250 (3VL3)	101	199	343	173

(1) Front connecting bars
(2) Terminal covers (standard)
(3) Terminal covers (extended)
(4) Cover frame for door cutout (for circuit breakers with toggle handle)
(5) External surface of cabinet door
(6) Mounting level

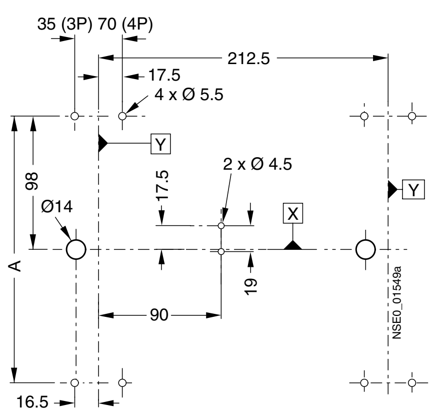
13.1.5 Locking device for the toggle handle



13.1.6 Rear locking module

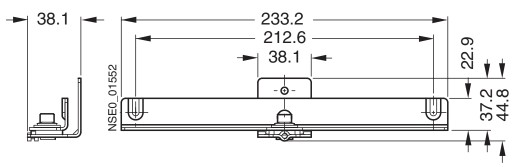
Rear interlocking module for plug-in/withdrawable circuit breakers, with front connection, with/without RCD module (withdrawable version only without RCD module)

For other detailed dimension drawings, please refer to the mounting instructions for the rear interlocking module.

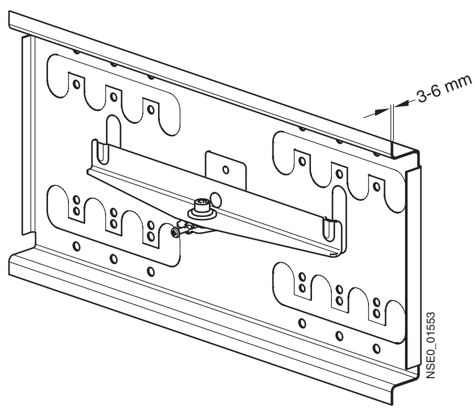


Type		A
Without RCD module	VL160X (3VL1), VL160 (3VL2), VL250 (3VL3)	194
With RCD module – "plug-in version" only	VL160X (3VL1), VL160 (3VL2), VL250 (3VL3)	315

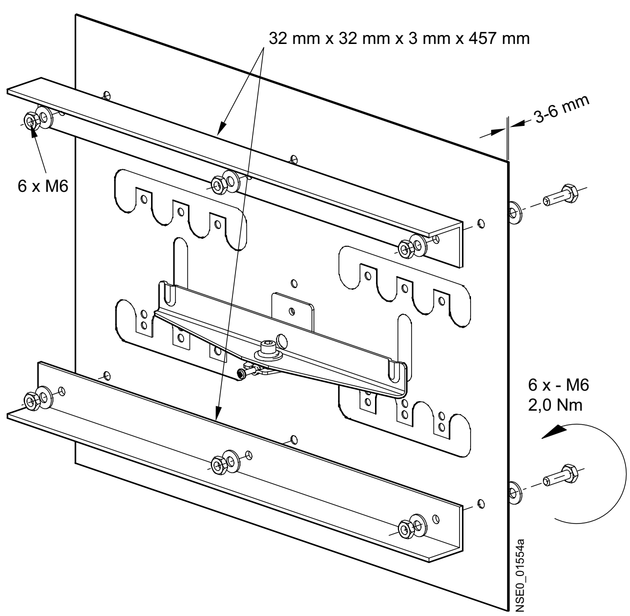
Rear interlocking module



Mounting plate, example 1, not included in the scope of supply

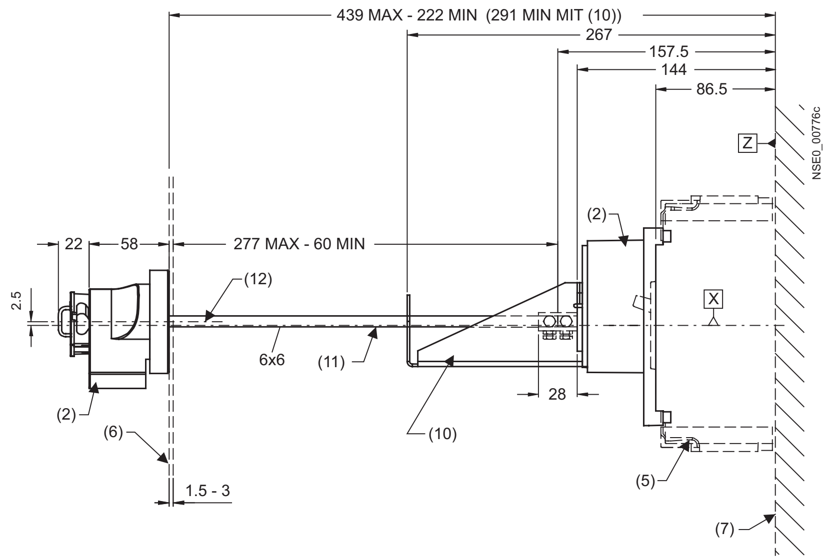


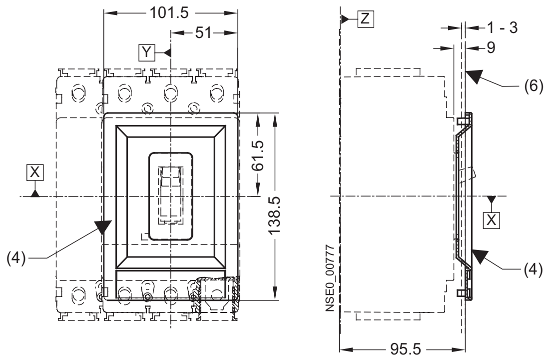
Mounting plate, example 2, not included in the scope of supply



13.1.7 Accessories

Circuit breakers with door coupling rotary operating mechanism



<table>
<tr><th>Cover frame for door cutout for circuit breakers with toggle handle</th><th>Cover frame for door cutout for circuit breakers with operating mechanism</th></tr>
<tr><td>


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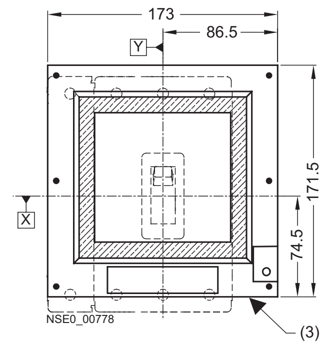
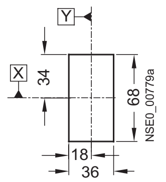
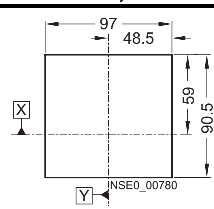
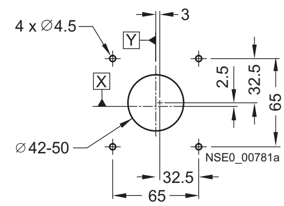
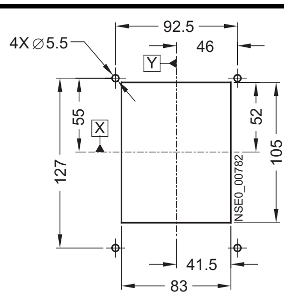
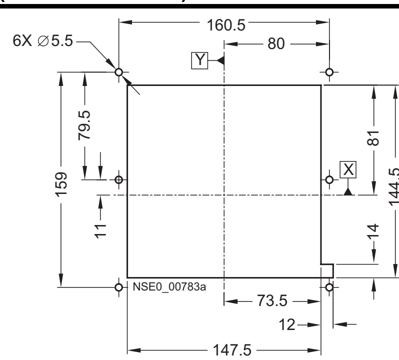
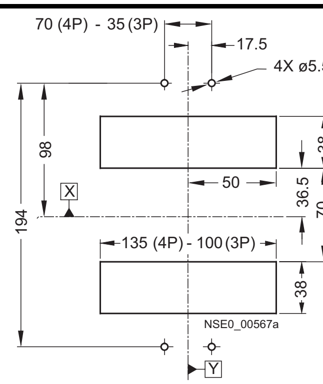


</td></tr>
</table>

(2)	Door-coupling rotary operating mechanism
(3)	Cover frame for door cutout (for circuit breakers with operating mechanism)
(4)	Cover frame for door cutout (for circuit breakers with toggle handle)
(5)	Terminal covers
(6)	External surface of cabinet door
(7)	Mounting level
(10)	Supporting bracket
(11)	Extension
(12)	Center line of operating mechanism shaft

13.1.8 Door cutouts

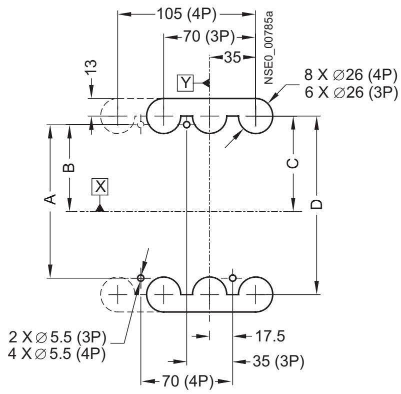
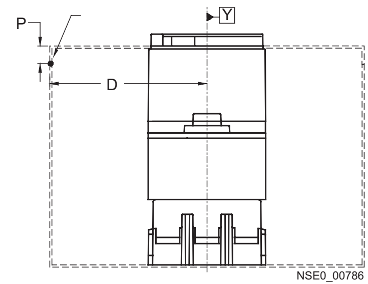
Door cutout Toggle handle (without cover frame)	Door cutout Front rotary operating mechanism and stored-energy motorized operating mechanism (without cover frame)	Door cutout Door coupling rotary operating mechanism
		

Door cutout toggle handle (with cover frame)	Door cutout front rotary operating mechanism, motorized operating mechanism with stored-energy spring mechanism and extension collar (with cover frame)	Drilling template and cutout for plug-in socket with connecting bars on rear
		

Note

Note:

Door cutouts require a minimum clearance between reference point Y and the door hinge.

<table>
<tr><th>Drilling template and cutout for rear connection</th><th>Door hinge point (see arrow)</th></tr>
<tr><td>
</td><td>

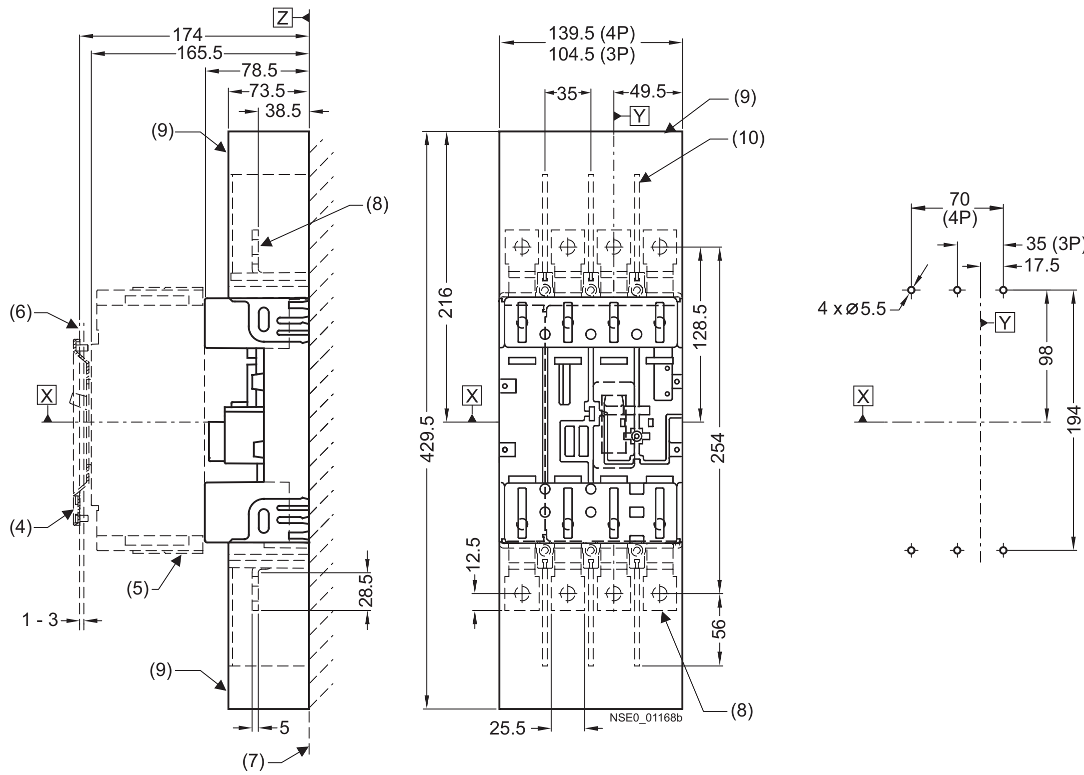
D > A from table + (P x 5)</td></tr>
</table>

Type	A	B	C	D
VL160X (3VL1)	114.5	65	71.5	133
VL160 (3VL2)	131.5	65	75.5	149
VL250 (3VL3)	131.5	65	75.5	149

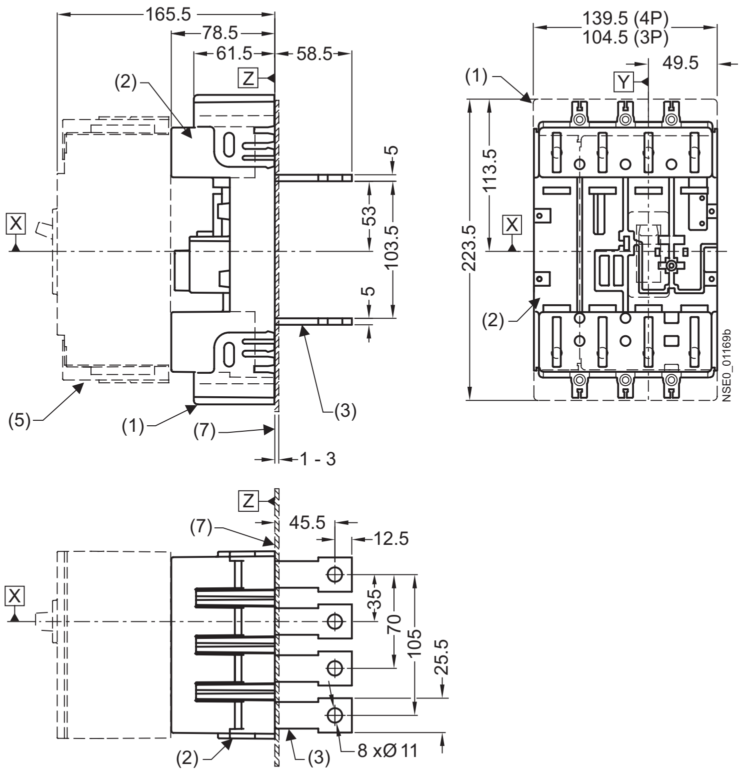
Combination	A
Circuit breaker only	100
Circuit breaker + plug-in socket + stored-energy motorized operating mechanism	100
Circuit breaker + plug-in socket + front rotary operating mechanism	200
Circuit breaker + withdrawable version	200

13.1.9 Plug-in socket and accessories

Plug-in socket with front connecting bars and drilling template for plug-in socket with front connecting bars



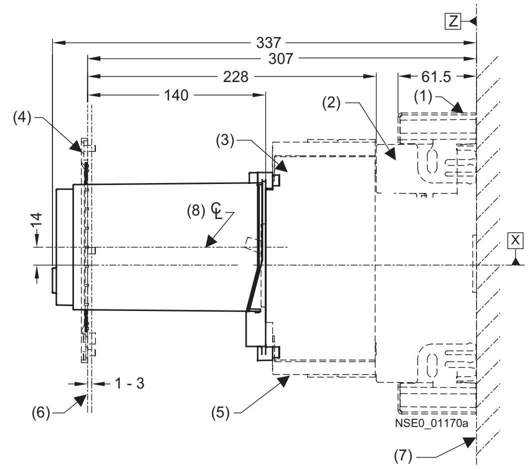
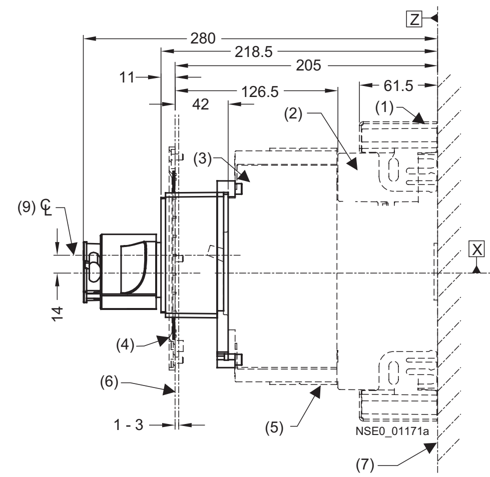
Plug-in socket with rear flat busbar terminals



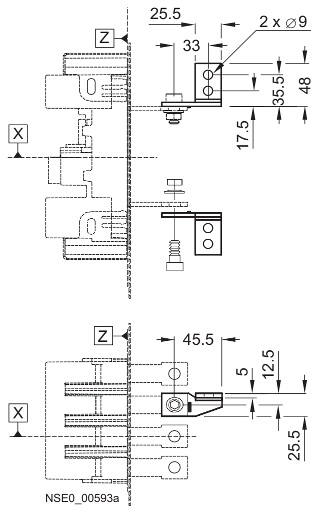
(1)	Plug-in socket with rear terminal covers
(2)	Socket
(3)	Plug-in socket with rear flat busbar terminals
(4)	Cover frame for door cutout (for circuit breakers with toggle handle)
(5)	Terminal covers (standard)
(6)	External surface of cabinet door
(7)	Mounting level
(8)	Plug-in socket with front connecting bars
(9)	Plug-in socket with terminal covers on the front
(10)	Phase barriers

13.1.10 VL160X (3VL1), 3- and 4-pole, up to 160 A

13.1.10.1 Plug-in socket and accessories

SENTRON VL160X (3VL1) circuit breaker with stored-energy motorized operating mechanism, mounted on plug-in socket	SENTRON VL160X (3VL1) circuit breaker with front rotary operating mechanism mounted on plug-in socket
	

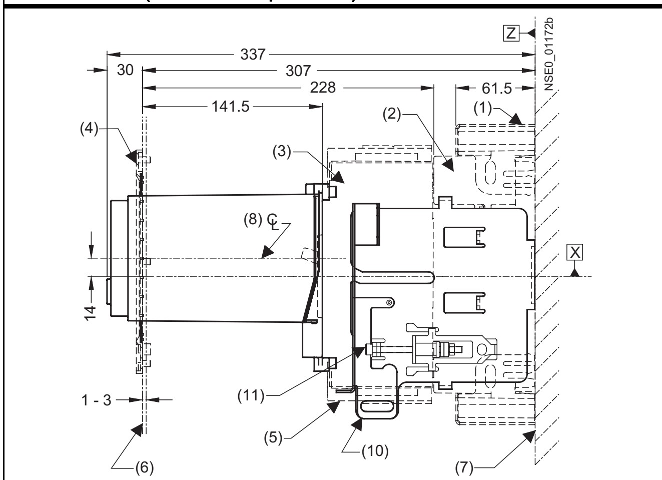
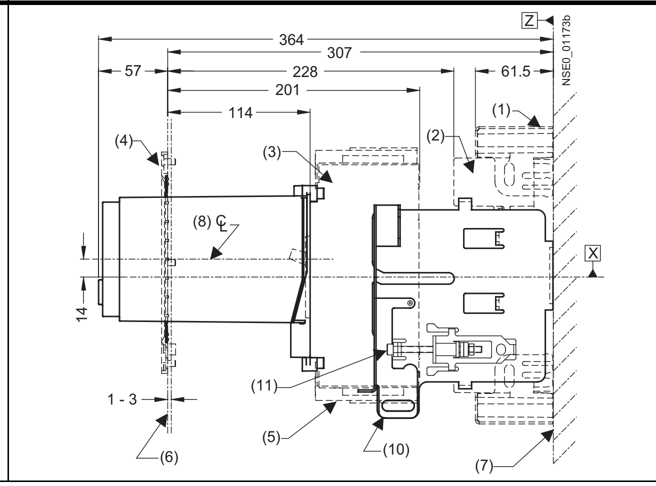
Connection adapter 90° angle

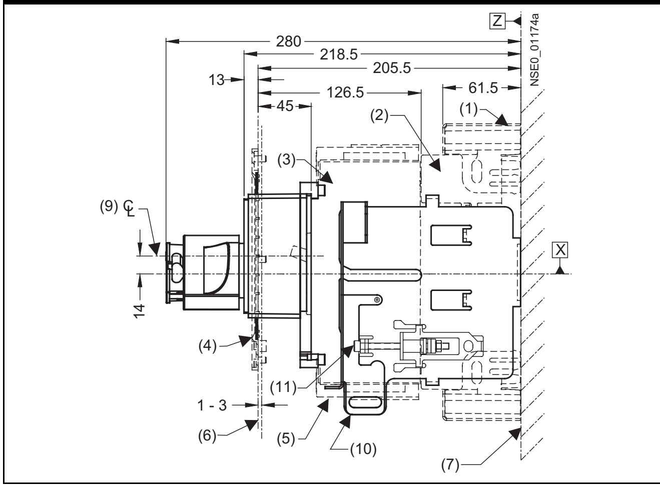


(1)	Plug-in socket with terminal covers
(2)	Socket
(3)	Circuit breaker
(4)	Cover frame for door cutout (for circuit breakers with operating mechanism)
(5)	Terminal covers (standard)
(6)	External surface of cabinet door
(7)	Mounting level
(8)	Stored-energy motorized operating mechanism
(9)	Front rotary operating mechanism

13.1.11 VL160 (3VL) and VL250 (3VL3), 3- and 4-pole, up to 250 A

13.1.11.1 Withdrawable version and accessories

<table>
<tr><th>SENTRON VL160 (3VL2) and VL250 (3VL3) circuit breakers with stored-energy motorized operating mechanism (connected position)</th><th>SENTRON VL160 (3VL2) and VL250 (3VL3) circuit breakers with stored-energy motorized operating mechanism (connected position)</th></tr>
<tr><td>
</td><td>
</td></tr>
</table>

<table>
<tr>
<th>SENTRON VL160 (3VL2) and VL250 (3VL3) circuit breakers with front rotary operating mechanism (connected position)</th>
<th>SENTRON VL160 (3VL2) and VL250 (3VL3) circuit breakers with front rotary operating mechanism (disconnected position)</th>
</tr>
<tr>
<td>


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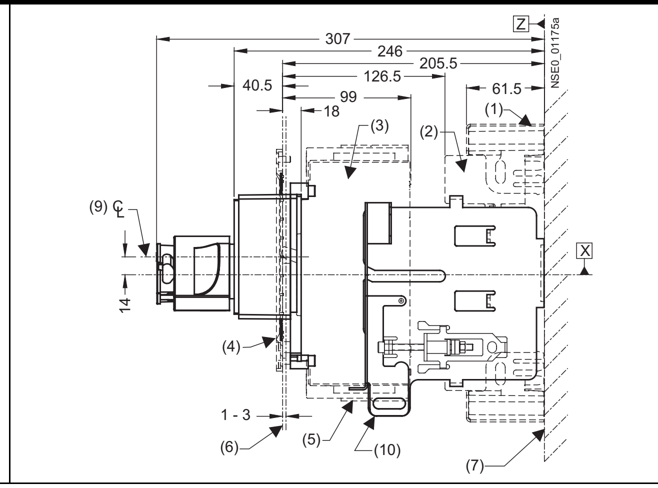
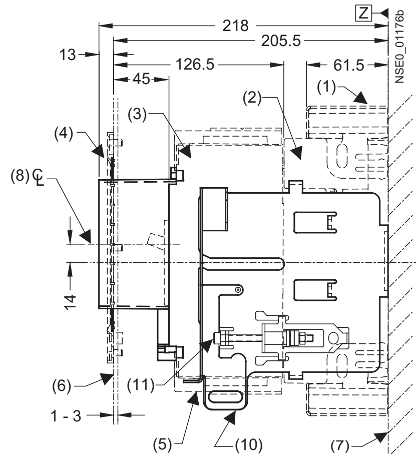
<td>


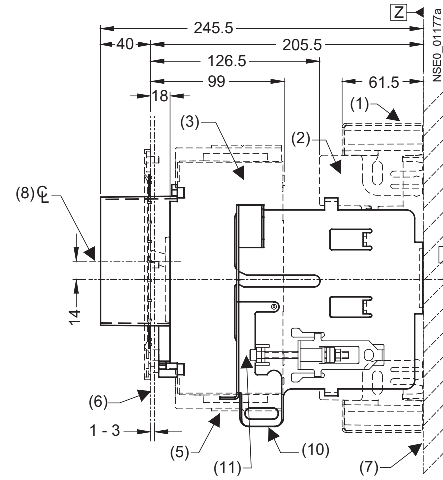
</td>
</tr>
</table>

(1)	Plug-in socket with terminal covers
(2)	Socket
(3)	Circuit breaker
(4)	Cover frame for door cutout (for circuit breakers with operating mechanism)
(5)	Terminal covers (standard)
(6)	External surface of cabinet door
(7)	Mounting level
(8)	Stored-energy motorized operating mechanism
(9)	Front rotary operating mechanism
(10)	Locking device for the racking mechanism
(11)	Racking mechanism

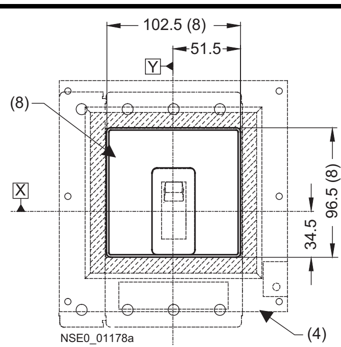
SENTRON VL160 (3VL2) and VL250 (3VL3) circuit breakers with extension collar (connected position)



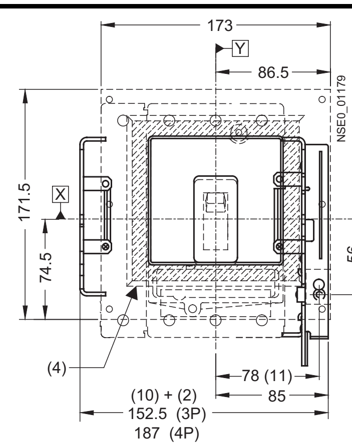
SENTRON VL160 (3VL2) and VL250 (3VL3) circuit breakers with extension collar (disconnected position)



Extension collar installation dimensions



Withdrawable version installation dimensions

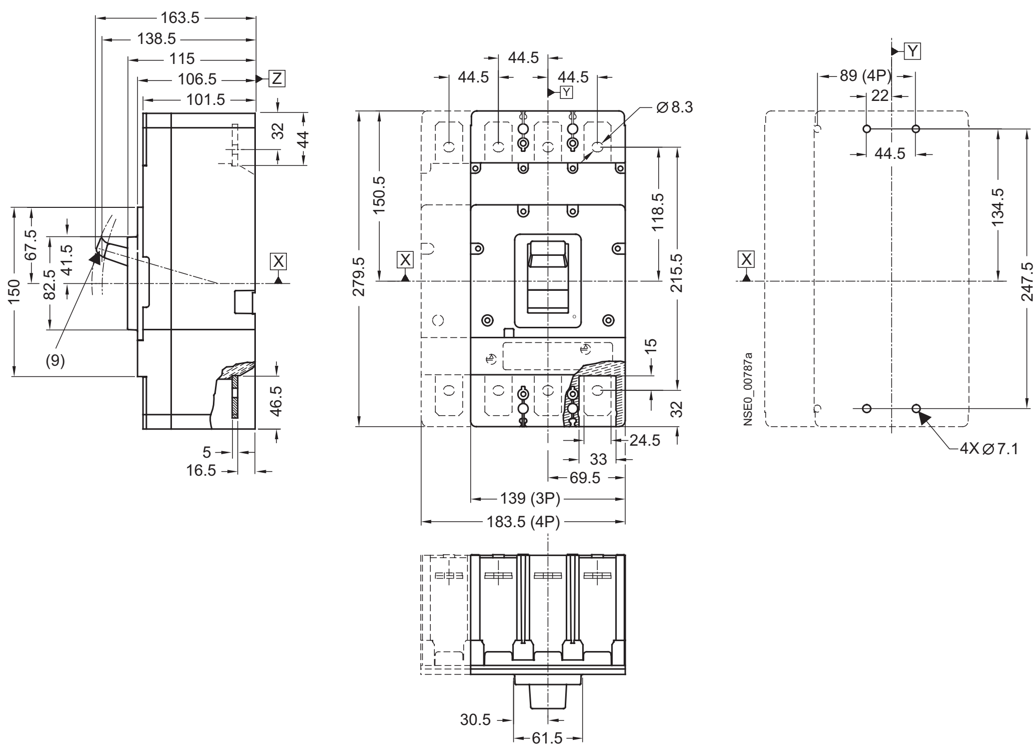


(1)	Plug-in socket with terminal covers
(2)	Socket
(3)	Circuit breaker
(4)	Cover frame for door cutout (for circuit breakers with operating mechanism)
(5)	Terminal covers (standard)
(6)	External surface of cabinet door
(7)	Mounting level
(8)	Extension collar
(10)	Locking device for the racking mechanism
(11)	Racking mechanism

13.2 VL400 (3VL4), 3- and 4-pole, up to 400 A

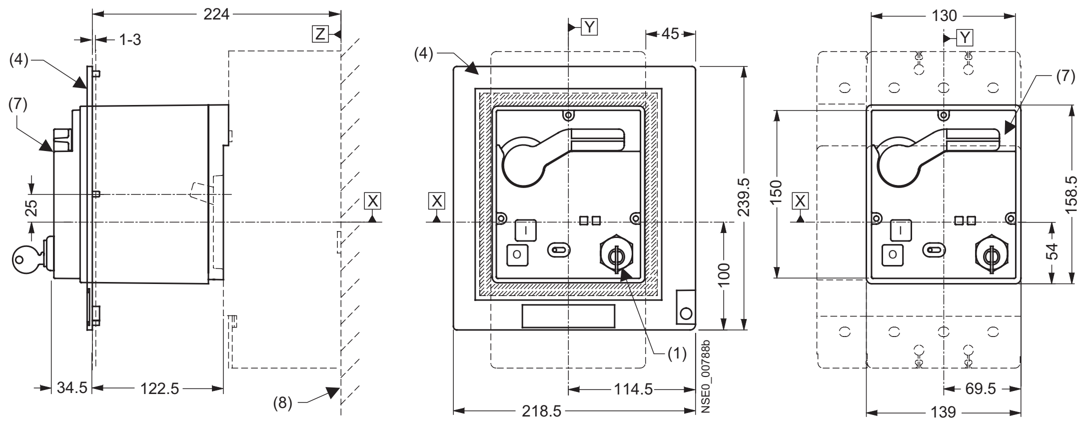
13.2.1 Circuit breaker

SENTRON VL400 (3VL4) circuit breaker and mounting instructions

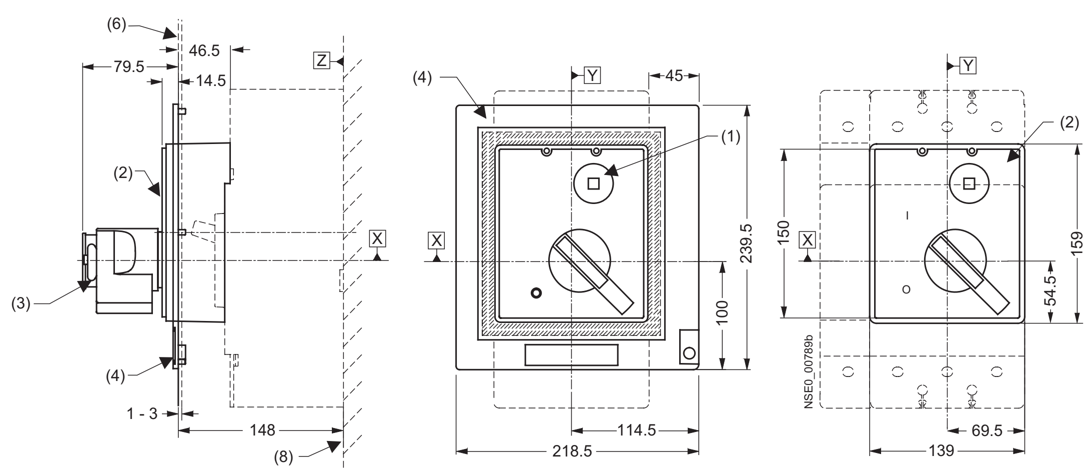


13.2.2 Operating mechanisms

Stored-energy motorized operating mechanism

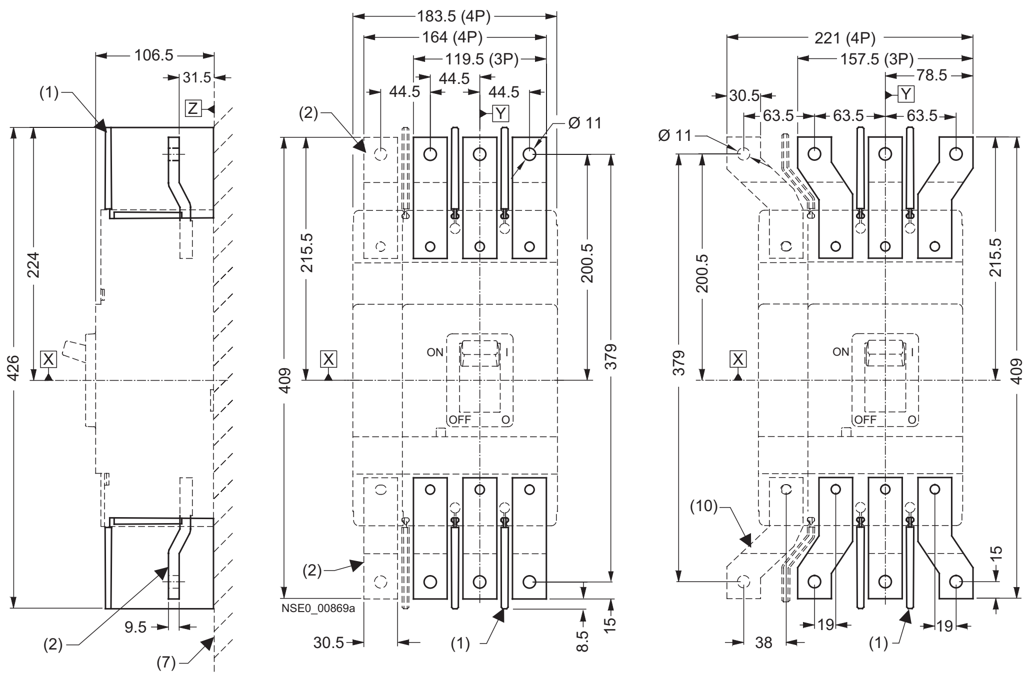


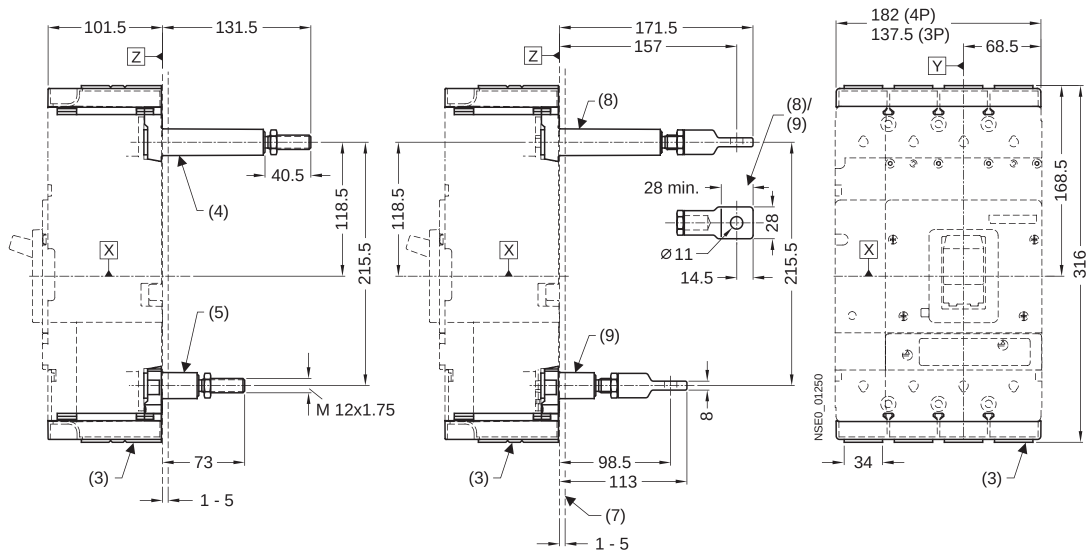
Front rotary operating mechanism



(1)	Safety lock
(2)	Front rotary operating mechanism
(3)	Padlock barrier
(4)	Cover frame for door cutout (for circuit breakers with operating mechanism)
(6)	External surface of cabinet door
(7)	Stored-energy motorized operating mechanism
(8)	Mounting level
(9)	Toggle handle extension

13.2.3 Connections and phase barriers

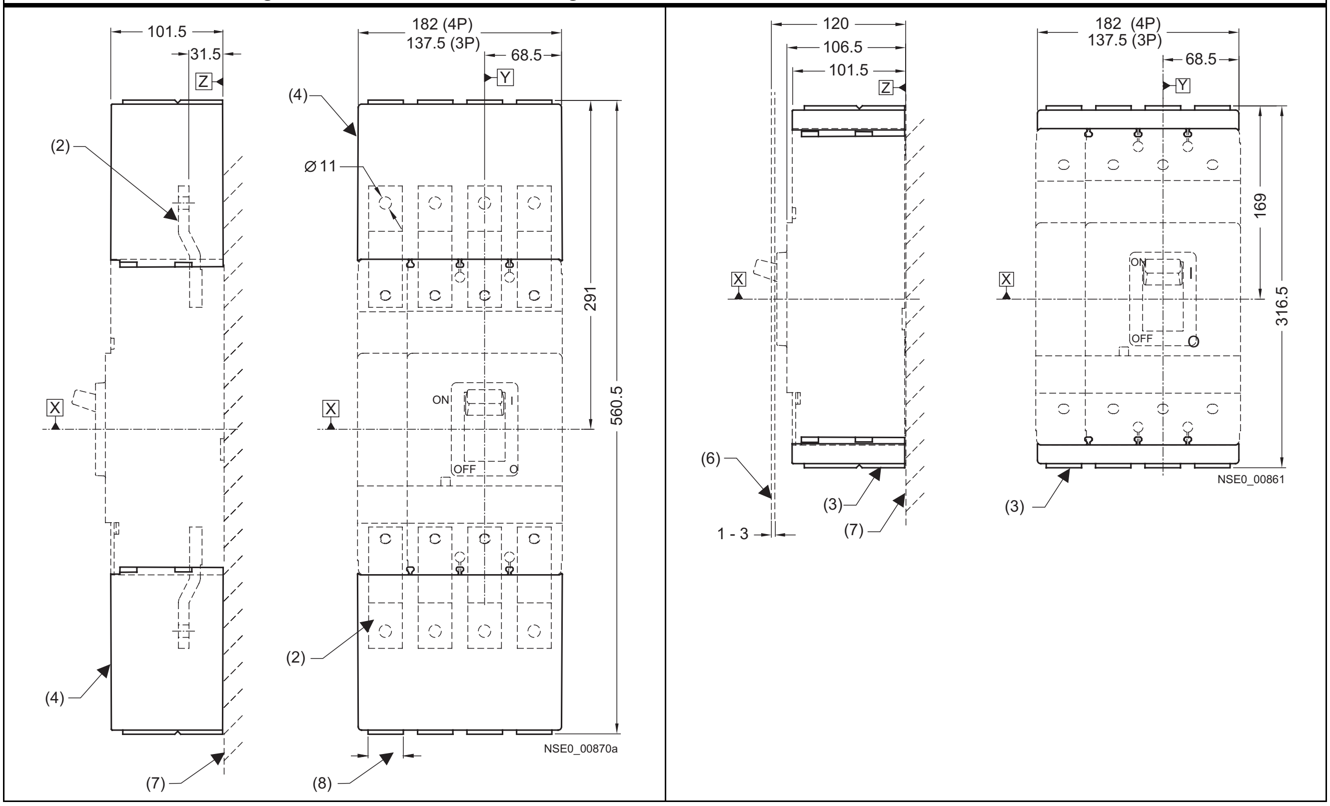




(1)	Interphase barrier
(2)	Front connecting bars
(3)	Terminal covers (standard)
(4)	Rear connection (long)
(5)	Rear connection (short)
(7)	Mounting level
(8)	Rear pad-type terminals (long)
(9)	Rear pad-type terminals (short)
(10)	Flared busbar extensions

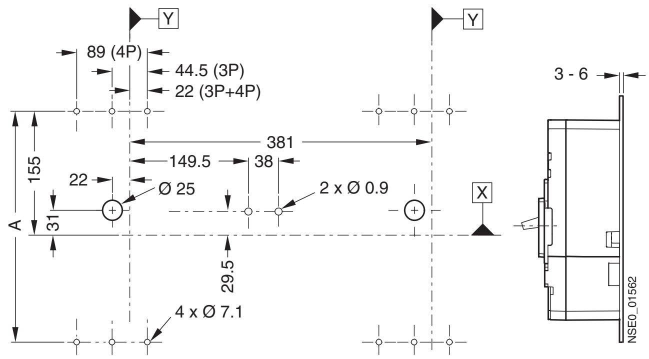
13.2.4 Terminal covers

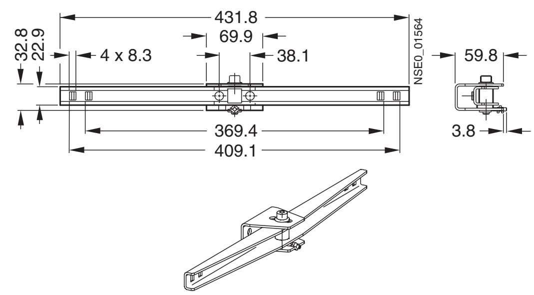
Circuit breaker mounting instructions front connecting bars



(2)	Front connecting bars
(3)	Terminal covers (standard)
(4)	Terminal covers (extended)
(6)	External surface of cabinet door
(7)	Mounting level
(8)	Cutout

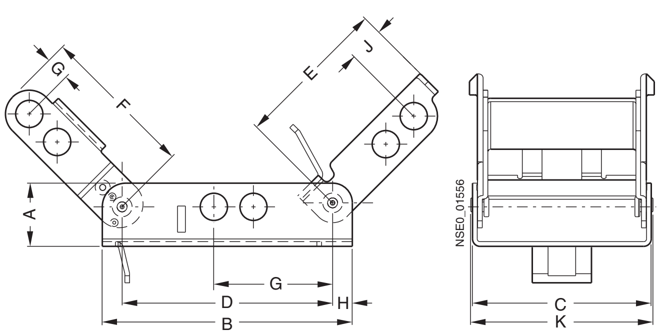
13.2.5 Rear interlocking module

<table>
<tr><th>Rear interlocking module for plug-in/withdrawable circuit breakers for front connection, with/without RCD module</th><th>Rear interlocking module</th></tr>
<tr><td>

For other detailed dimension drawings, please refer to the mounting instructions for the rear interlocking module.</td><td>
</td></tr>
</table>

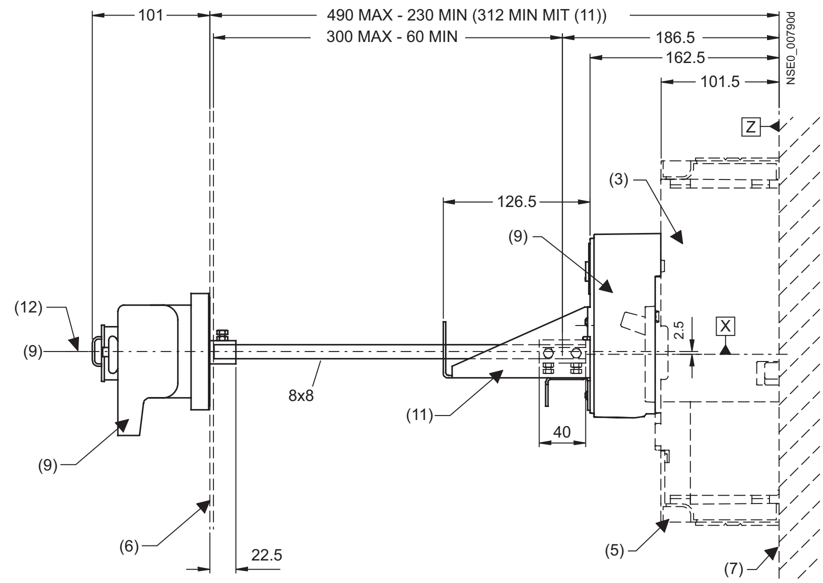
Type	A
Without RCD module VL400 (3VL4)	289
With RCD module VL400 (3VL4)	449

13.2.6 Locking devices, locking device for toggle handle and accessories

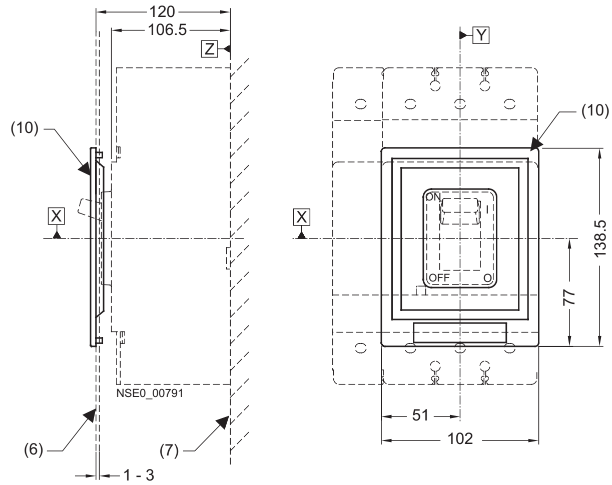


Type	a	b	c	d	e	f	g	h	i	k
3VL9 4	20.3	80.3	57.4	52.8	49.3	49.8	6.35	6.3	11.2	58.5
3VL9 6	21.6	79.8	71.1	62.0	50.4	46.5	12.9	8.9	8.6	72.2
3VL9 8	21.6	110.5	88.9	96.5	77.2	69.1	11.7	5.1	24.8	90.0

Plug-in socket door coupling rotary operating mechanism

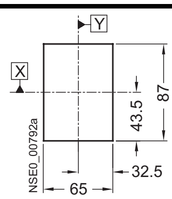
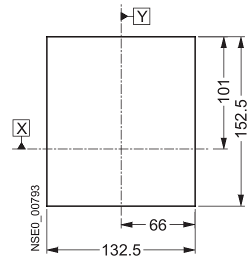
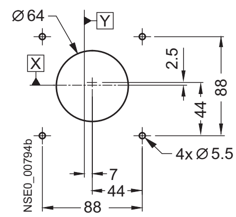


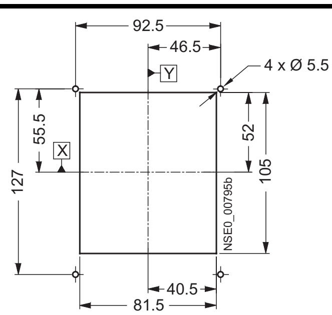
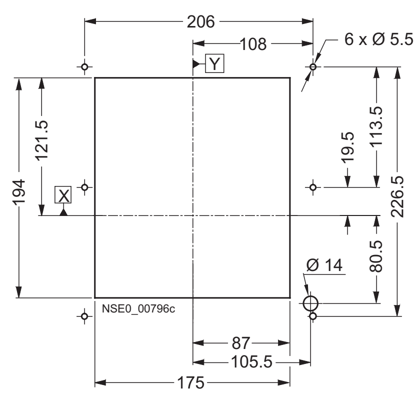
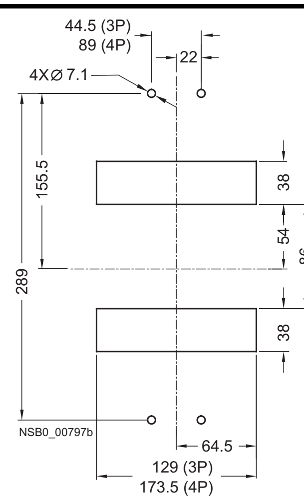
Cover frame for door cutout for circuit breakers with toggle handle



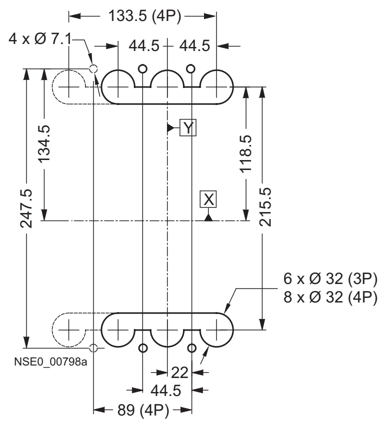
(3) Circuit breaker
(5) Terminal covers (standard)
(6) External surface of cabinet door
(7) Mounting level
(9) Door-coupling rotary operating mechanism
(10) Cover frame for door cutout
(for circuit breakers with toggle handle)
(11) Supporting bracket
(12) Center line of operating mechanism shaft

13.2.7 Door cutouts

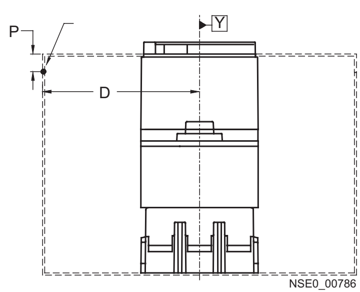
Door cutout toggle handle operating mechanism (without cover frame)	Door cutout front rotary operating mechanism and stored-energy motorized operating mechanism (without cover frame)	Door cutout Door coupling rotary operating mechanism
		

Door cutout toggle handle operating mechanism (with cover frame)	Door cutout front rotary operating mechanism, stored-energy motorized operating mechanism and extension collar (with cover frame)	Drilling template and cutout for plug-in socket with flat connecting bars on rear
		

Drilling template and cutout for rear connection



Door hinge point (see arrow)



Note

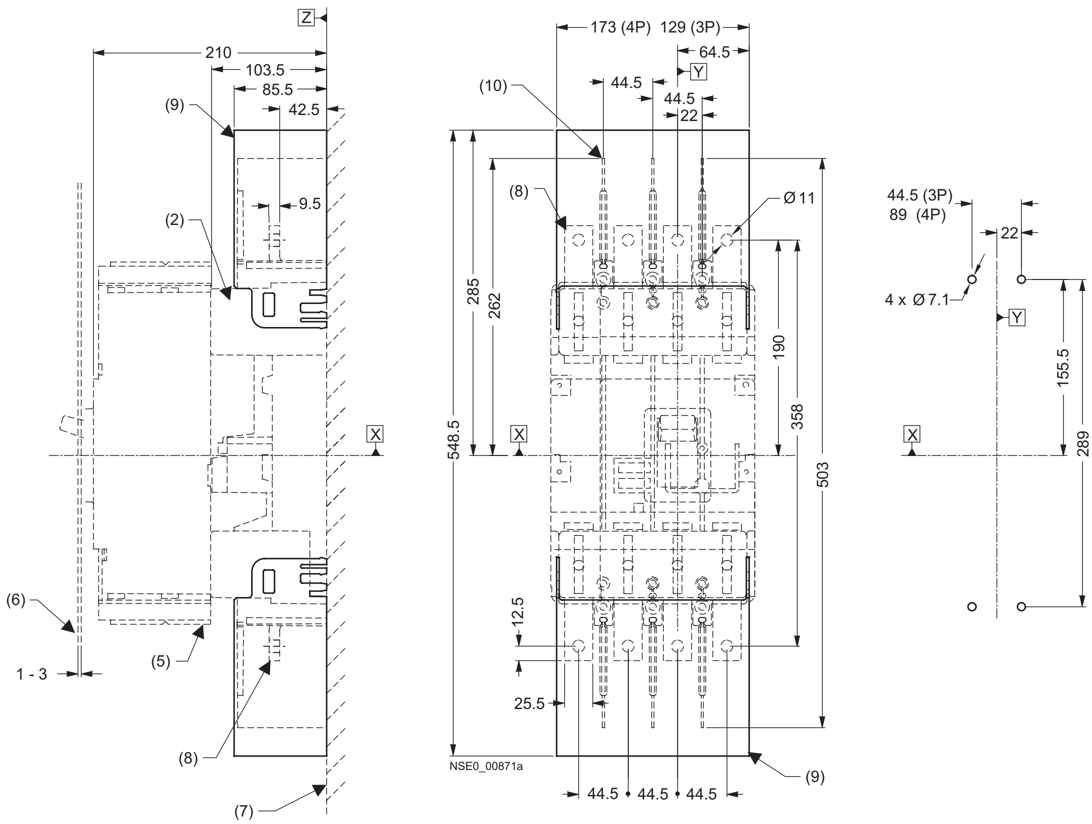
Note:

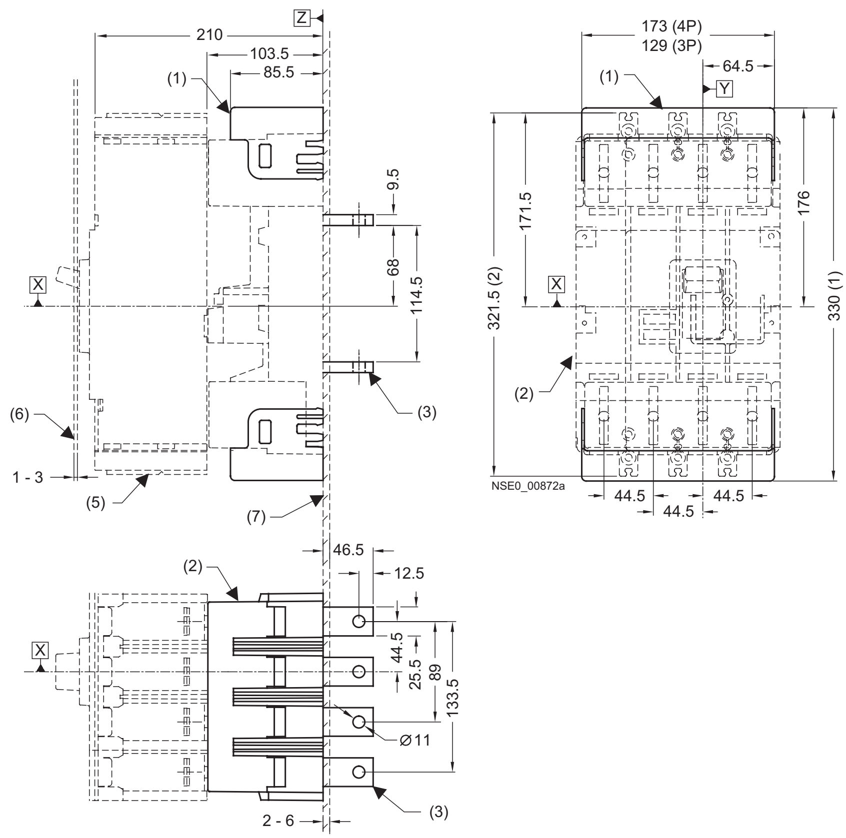
Door cutouts require a minimum clearance between reference point Y and the door hinge.

Combination	A
Circuit breaker only	150
Circuit breaker + plug-in socket + stored-energy motorized operating mechanism	150
Circuit breaker + plug-in socket + front rotary operating mechanism	200
Circuit breaker + withdrawable version	200

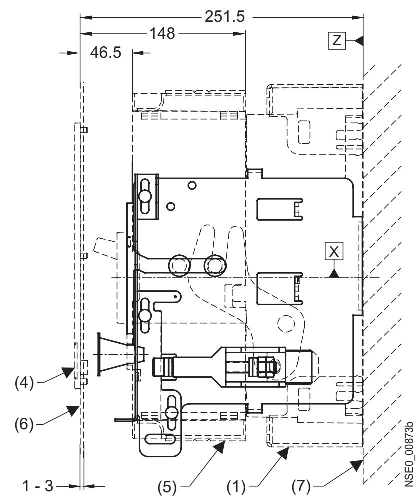
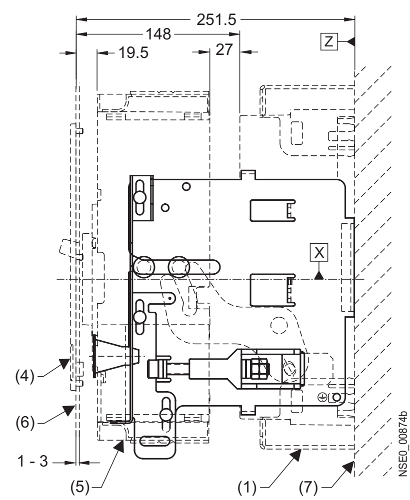
13.2.8 Plug-in socket and accessories

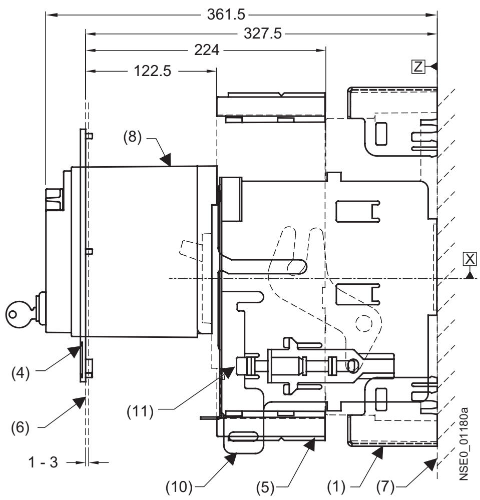
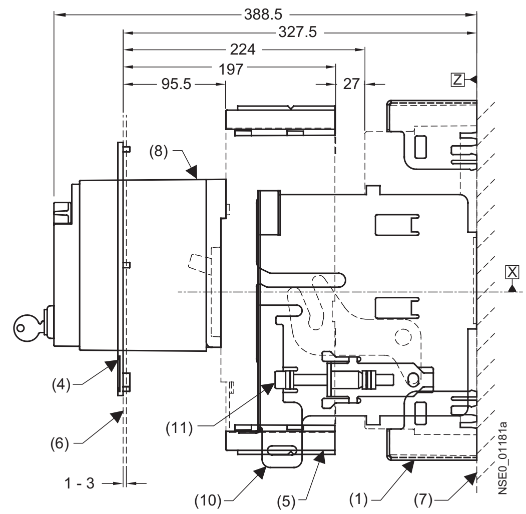
Plug-in socket and drilling template Plug-in socket with front connecting bars



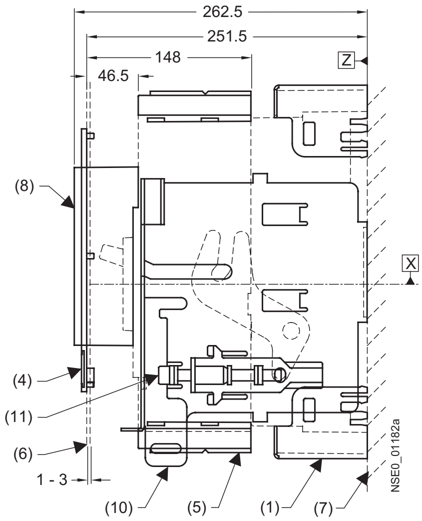
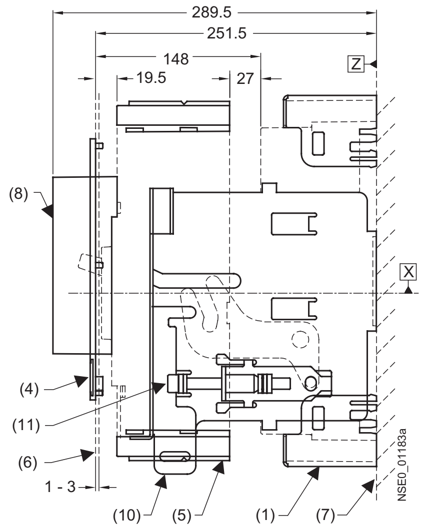


(1) Plug-in socket with rear terminal covers
(2) Socket
(3) Plug-in socket with rear flat connecting bars
(5) Terminal covers (standard)
(6) External surface of cabinet door
(7) Mounting level
(8) Plug-in socket with front connecting bars
(9) Plug-in socket with terminal covers on the front
(10) Interphase barrier

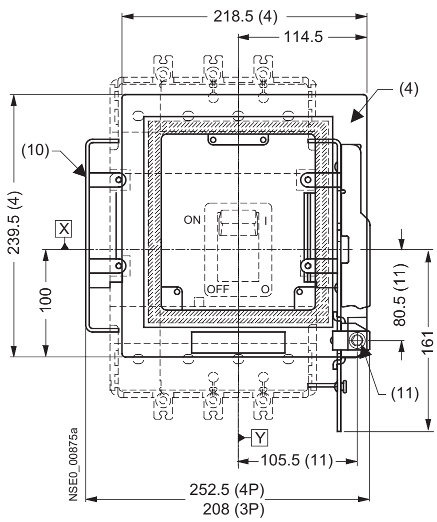
Plug-in socket front rotary operating mechanism (connected position)	Plug-in socket front rotary operating mechanism (disconnected position)
	

Plug-in socket stored-energy motorized operating mechanism (connected position)	Plug-in socket stored-energy motorized operating mechanism (disconnected position)
	

(1)	Plug-in socket with terminal covers
(4)	Cover frame for door cutout (for circuit breakers with operating mechanism)
(5)	Terminal covers (standard)
(6)	External surface of cabinet door
(7)	Mounting level
(8)	Stored-energy motorized operating mechanism
(9)	Front rotary operating mechanism
(10)	Locking device for the racking mechanism
(11)	Racking mechanism

Plug-in socket extension collar (connected position)	Plug-in socket extension collar (disconnected position)
	

Extension collar mounted on guide rail

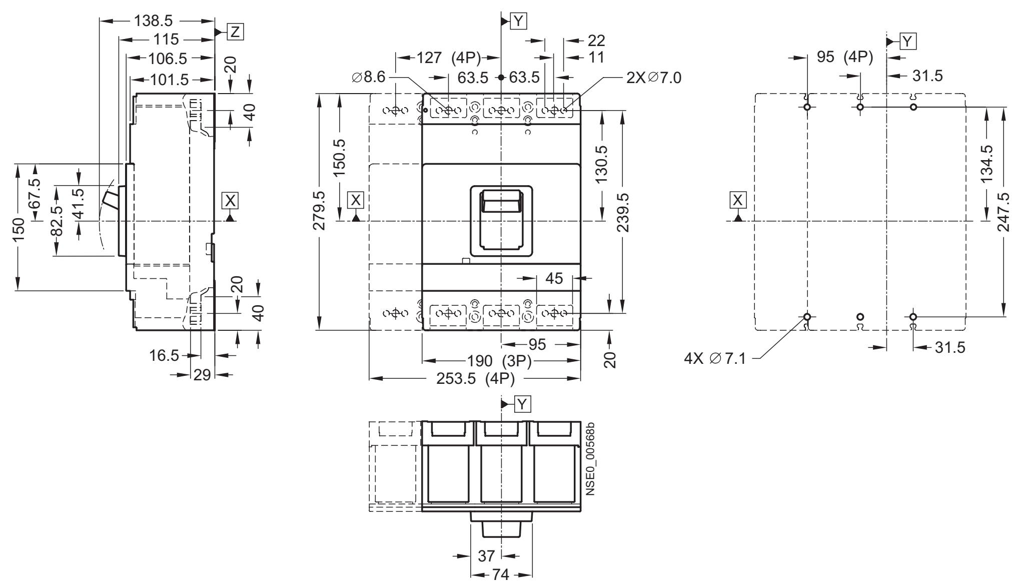


(1) Plug-in socket with terminal covers
(4) Cover frame for door cutout
(for circuit breakers with operating mechanism)
(5) Terminal covers (standard)
(6) External surface of cabinet door
(7) Mounting level
(8) Extension collar
(10) Locking device for the racking mechanism
(11) Racking mechanism

13.3 VL630 (3VL5), 3- and 4-pole, up to 630 A

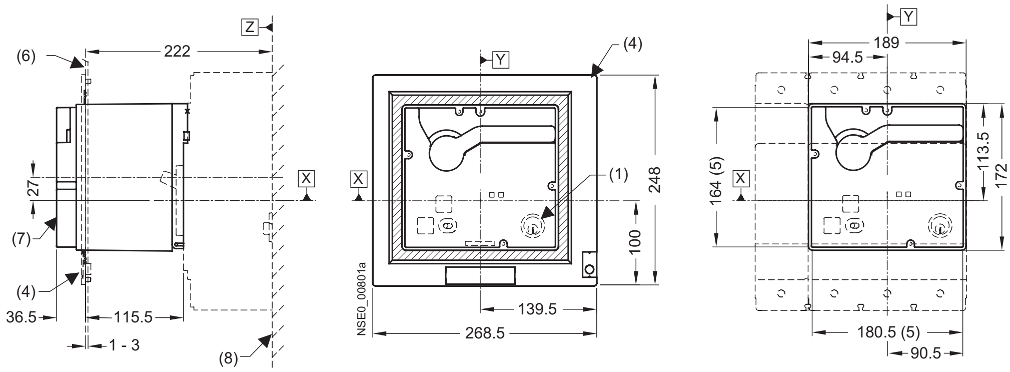
13.3.1 Circuit breaker

SENTRON VL630 (3VL5) circuit breaker and mounting instructions

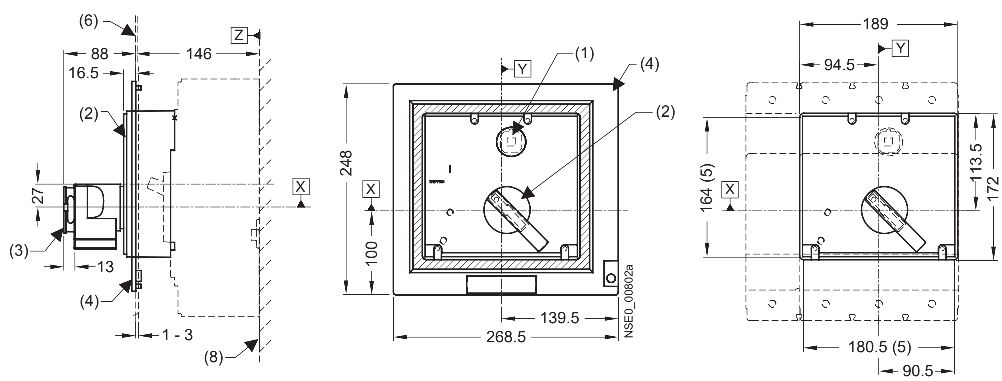


13.3.2 Operating mechanisms

Stored-energy motorized operating mechanism

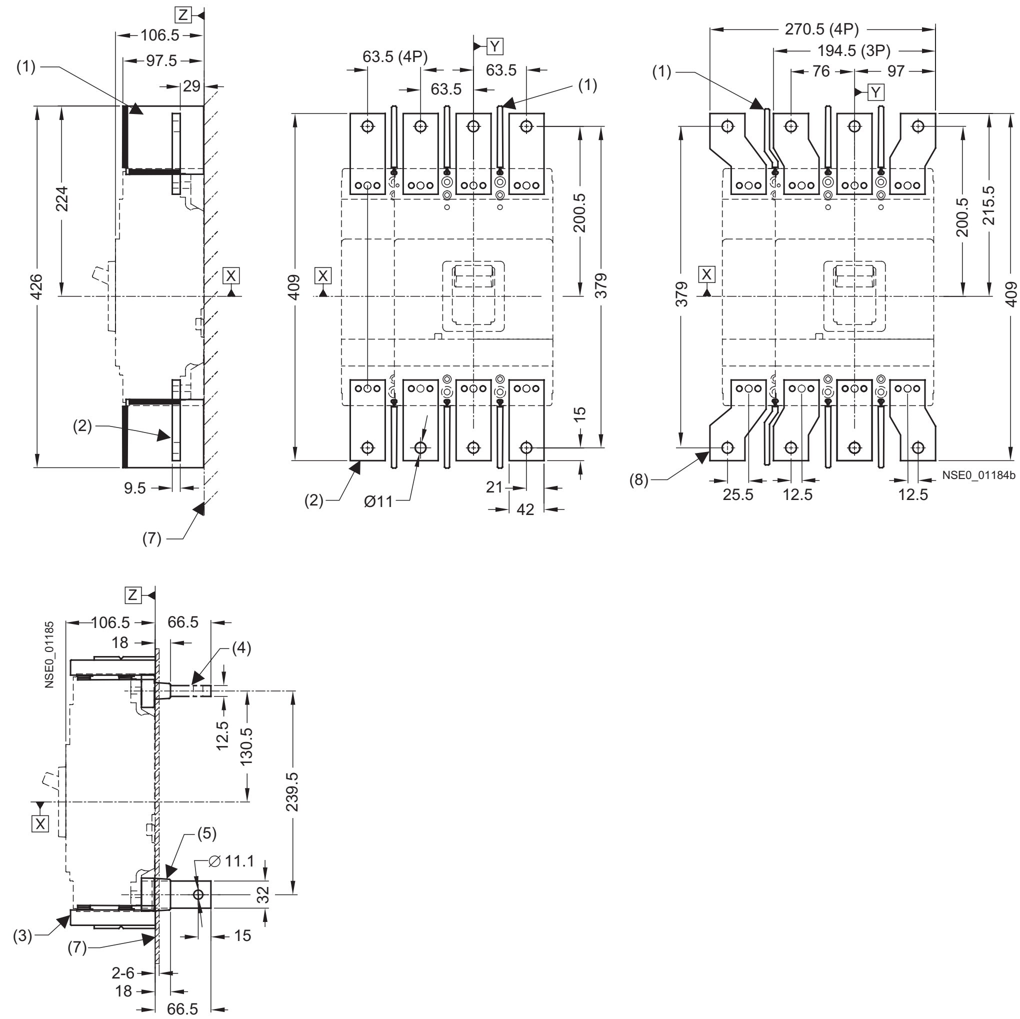


Front rotary operating mechanism



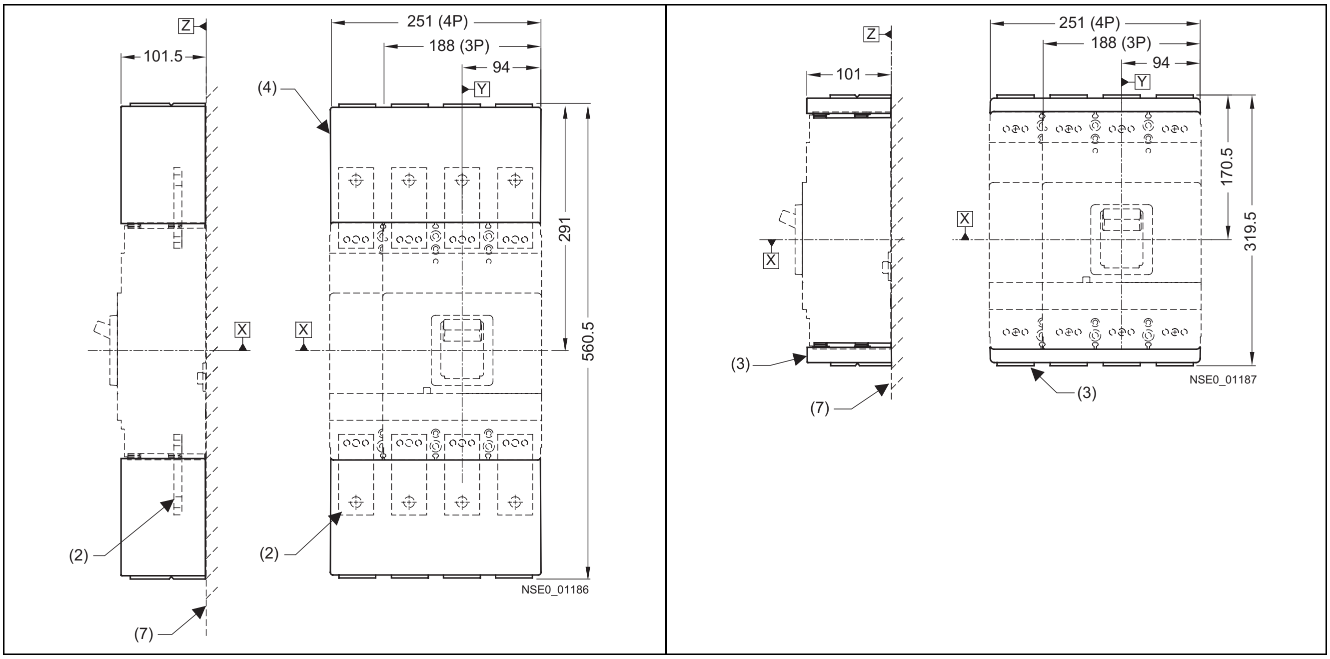
(1)	Safety lock
(2)	Front rotary operating mechanism
(3)	Padlock barrier
(4)	Cover frame for door cutout (for circuit breakers with operating mechanism)
(5)	Grading for cover
(6)	External surface of cabinet door
(7)	Stored-energy motorized operating mechanism
(8)	Mounting level

13.3.3 Connections and phase barriers



(1) Interphase barrier
(2) Front connecting bars
(3) Terminal covers (standard)
(4) Rear connection (horizontal connection)
(5) Rear connection (vertical connection)
(7) Mounting level
(8) Flared busbar extensions

13.3.4 Terminal covers

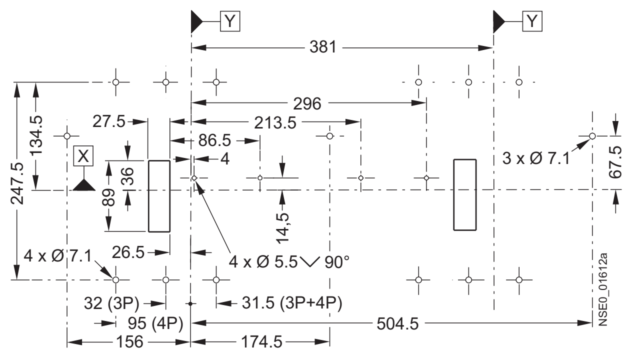


(2)	Front connecting bars
(3)	Terminal covers (standard)
(4)	Terminal covers (extended)
(7)	Mounting level

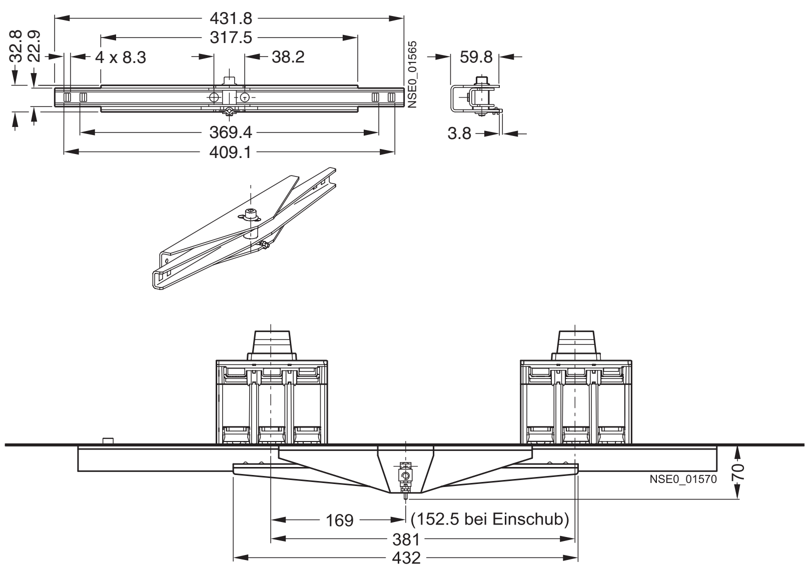
13.3.5 Rear interlocking module

Rear interlocking module for plug-in/withdrawable circuit breakers for front connection.

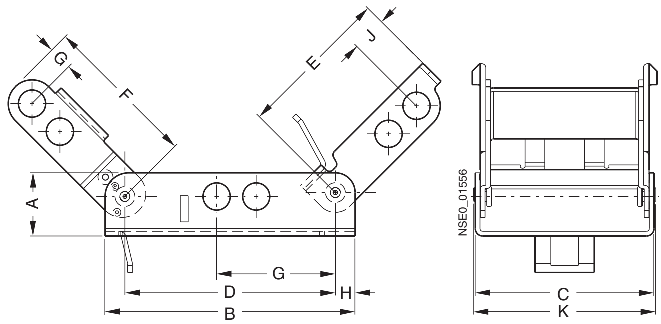
Rear interlocking module for plug-in/withdrawable circuit breakers for front connection.



Rear interlocking module



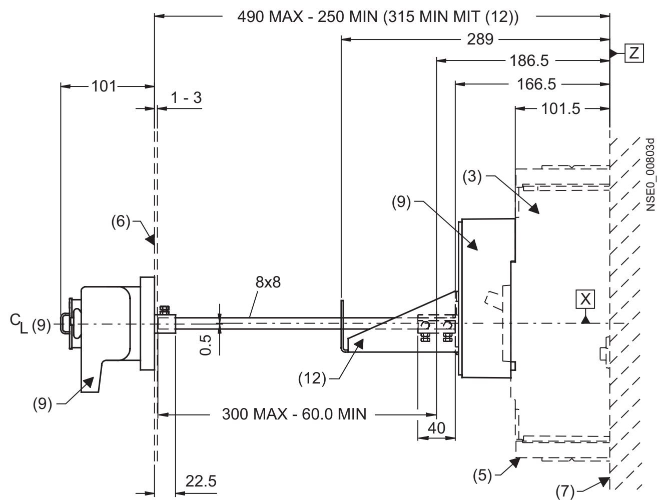
13.3.6 Locking and locking device for toggle handle

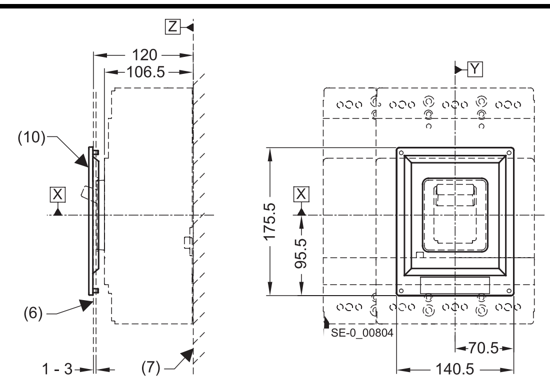
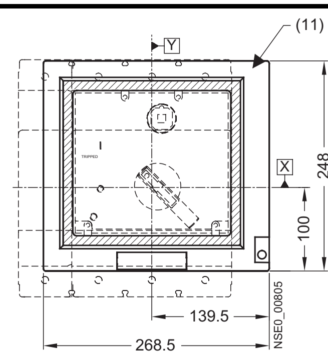


Type	a	b	c	d	e	f	g	h	i	k
3VL9 4	20.3	80.3	57.4	52.8	49.3	49.8	6.35	6.3	11.2	58.5
3VL9 6	21.6	79.8	71.1	62.0	50.4	46.5	12.9	8.9	8.6	72.2
3VL9 8	21.6	110.5	88.9	96.5	77.2	69.1	11.7	5.1	24.8	90.0

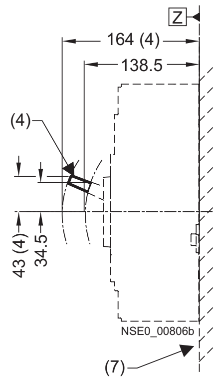
13.3.7 Accessories

Door-coupling rotary operating mechanism



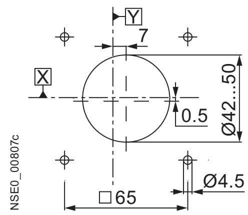
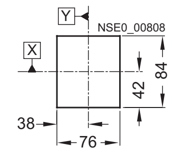
Cover frame for door cutout for circuit breakers with toggle handle	Cover frame for door cutout for circuit breakers with operating mechanism
	

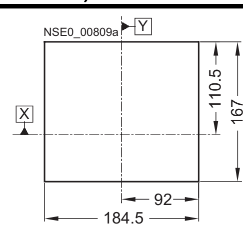
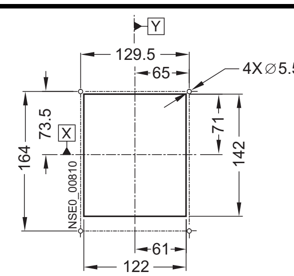
Toggle handle extension



(3)	Circuit breaker
(4)	Toggle handle extension
(5)	Terminal covers (standard)
(6)	External surface of cabinet door
(7)	Mounting level
(9)	Door-coupling rotary operating mechanism
(10)	Cover frame for door cutout (for circuit breakers with toggle handle)
(11)	Cover frame for door cutout (for circuit breakers with operating mechanism)
(12)	Supporting bracket

13.3.8 Door cutouts

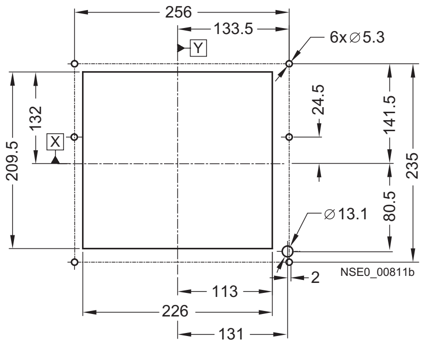
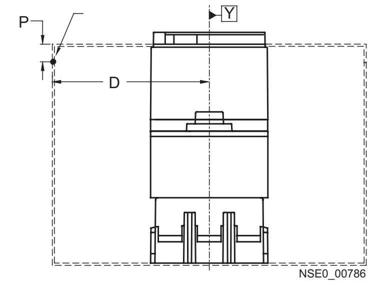
<table>
<tr><th>Door cutout door coupling rotary operating mechanism</th><th>Door cutout toggle handle operating mechanism (without cover frame)</th></tr>
<tr><td>
</td><td>
</td></tr>
</table>

<table>
<tr><th>Door cutout front rotary operating mechanism, stored-energy motorized operating mechanism and extension collar
(without cover frame)</th><th>Door cutout toggle handle operating mechanism (with cover frame)</th></tr>
<tr><td>
</td><td>
</td></tr>
</table>

Note

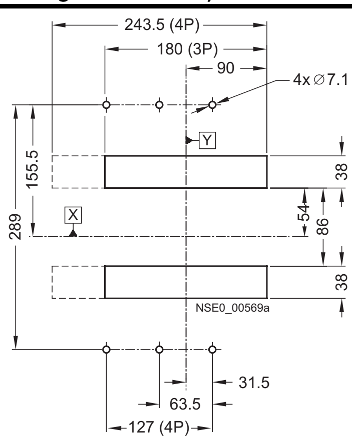
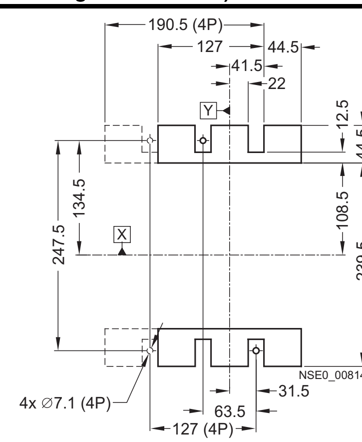
Note:

Door cutouts require a minimum clearance between reference point Y and the door hinge.

<table>
<tr><th>Door cutout front rotary operating mechanism, stored-energy motorized operating mechanism and extension collar (with cover frame)</th><th>Door hinge point (see arrow)</th></tr>
<tr><td>
</td><td>

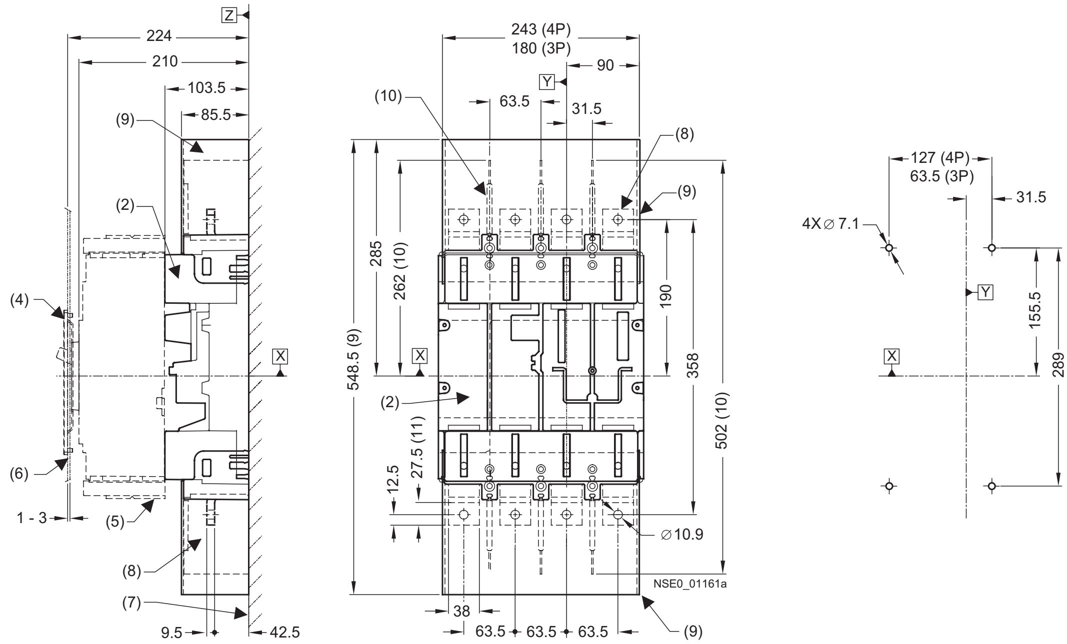
D > A from table + (P x 5)</td></tr>
</table>

Combination	A
Circuit breaker only	150
Circuit breaker + plug-in socket + stored-energy motorized operating mechanism	150
Circuit breaker + plug-in socket + front rotary operating mechanism	200
Circuit breaker + withdrawable version	200

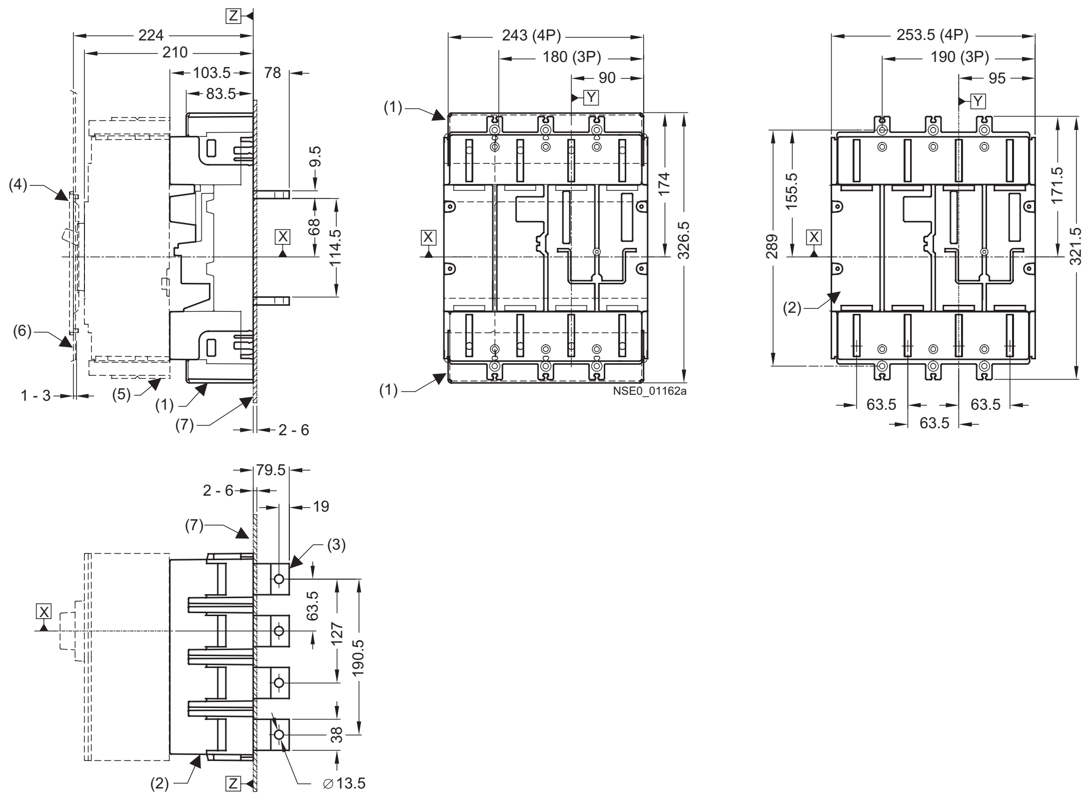
<table>
<tr><th>Drilling template and cutout for plug-in socket (with flat connecting bars on rear)</th><th>Drilling template and cutout for circuit breaker (with flat connecting bars on rear)</th></tr>
<tr><td>
</td><td>
</td></tr>
</table>

13.3.9 Plug-in socket and accessories

Plug-in socket with terminal covers on the front and drilling template for plug-in socket with front connecting bars

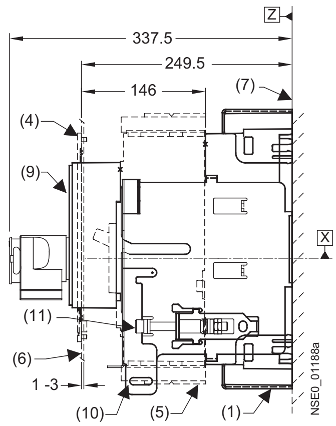
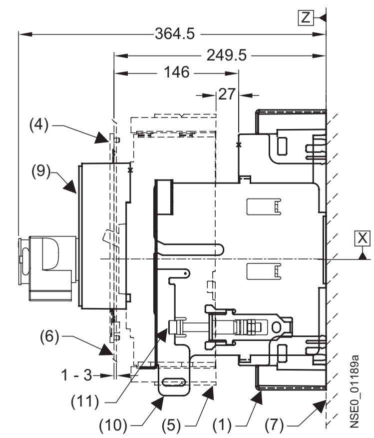


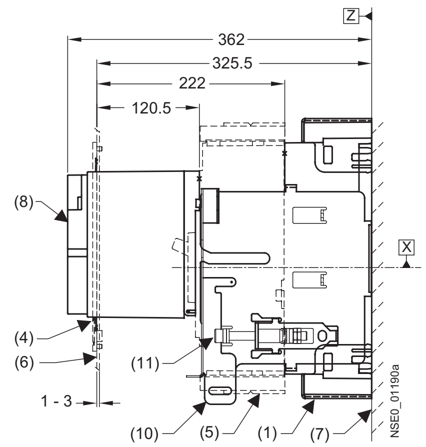
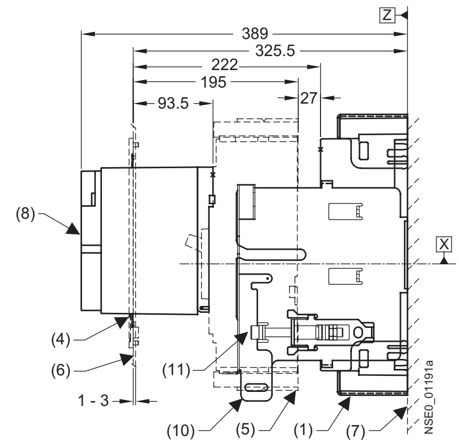
Plug-in socket, with terminal covers, rear flat connecting bars on plug-in socket



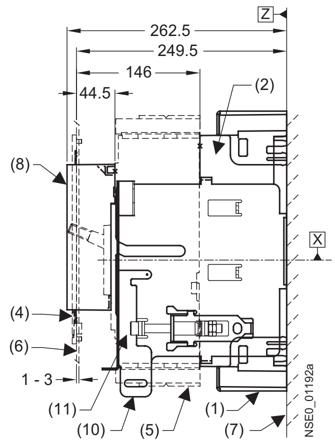
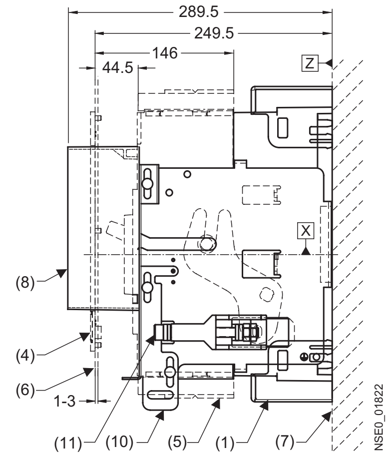
(1)	Plug-in socket with rear terminal covers
(2)	Socket
(3)	Plug-in socket with rear flat connecting bars
(4)	Cover frame for door cutout (for circuit breakers with toggle handle)
(5)	Terminal covers (standard)
(6)	External surface of cabinet door
(7)	Mounting level
(8)	Plug-in socket with front connecting bars
(9)	Plug-in socket with terminal covers on the front
(10)	Interphase barrier
(11)	Connection surface

13.3.10 Withdrawable version and accessories

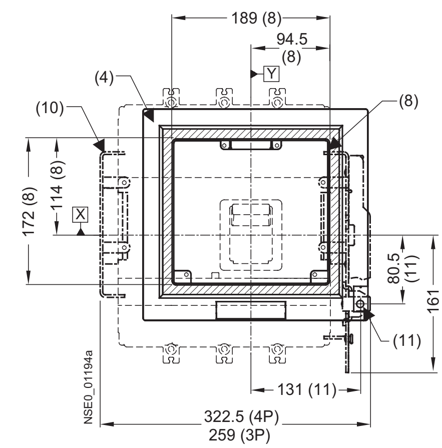
SENTRON VL630 (3VL5) circuit breaker with rotary operating mechanism in withdrawable design (connected position)	SENTRON VL630 (3VL5) circuit breaker with rotary operating mechanism in withdrawable design (disconnected position)
	

<table>
<tr><th>SENTRON VL630 (3VL5) circuit breaker with stored-energy motorized operating mechanism in withdrawable design (connected position)</th><th>SENTRON VL630 (3VL5) circuit breaker with stored-energy motorized operating mechanism in withdrawable design (disconnected position)</th></tr>
<tr><td>
</td><td>
</td></tr>
</table>

(1)	Plug-in socket with terminal covers
(4)	Cover frame for door cutout (for circuit breakers with operating mechanism)
(5)	Terminal covers (standard)
(6)	External surface of cabinet door
(7)	Mounting level
(8)	Stored-energy motorized operating mechanism
(9)	Front rotary operating mechanism
(10)	Locking device for the racking mechanism
(11)	Racking mechanism

<table>
<tr><th>SENTRON VL630 (3VL5) circuit breaker with extension collar in withdrawable design (connected position)</th><th>SENTRON VL630 (3VL5) circuit breaker with extension collar in withdrawable design (disconnected position)</th></tr>
<tr><td>
</td><td>
</td></tr>
</table>

SENTRON VL630 (3VL5) circuit breaker with extension collar in withdrawable design

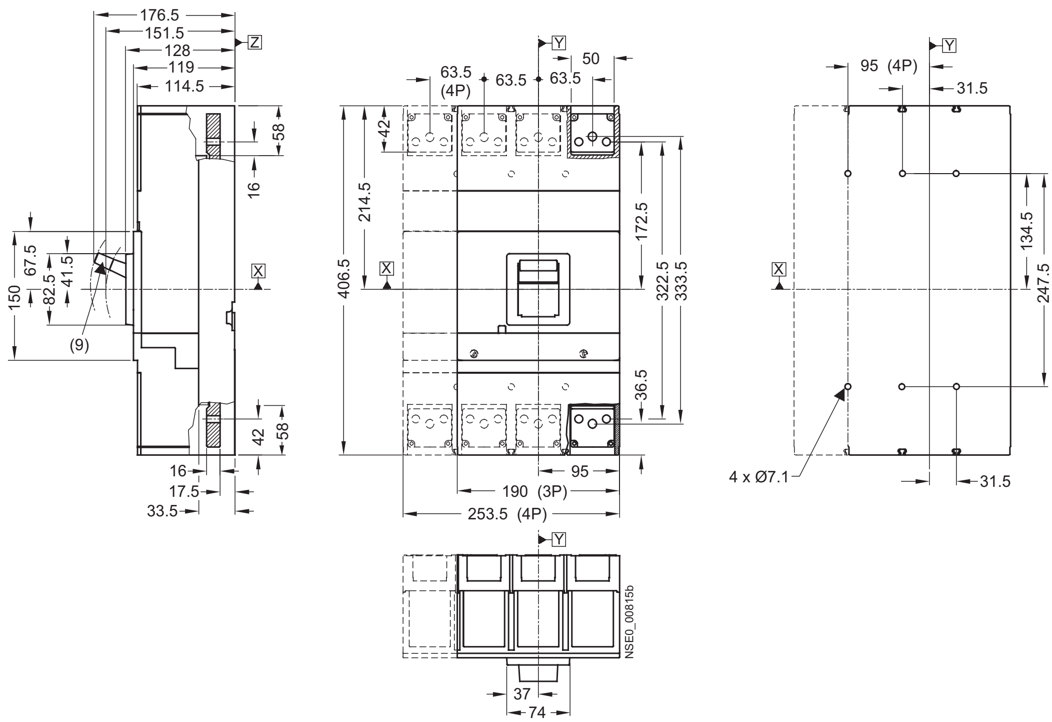


(1)	Plug-in socket with terminal covers
(2)	Socket
(4)	Cover frame for door cutout (for circuit breakers with operating mechanism)
(5)	Terminal covers (standard)
(6)	External surface of cabinet door
(7)	Mounting level
(8)	Extension collar
(10)	Locking device for the racking mechanism
(11)	Racking mechanism

13.4 VL800 (3VL6), 3- and 4-pole, up to 800 A

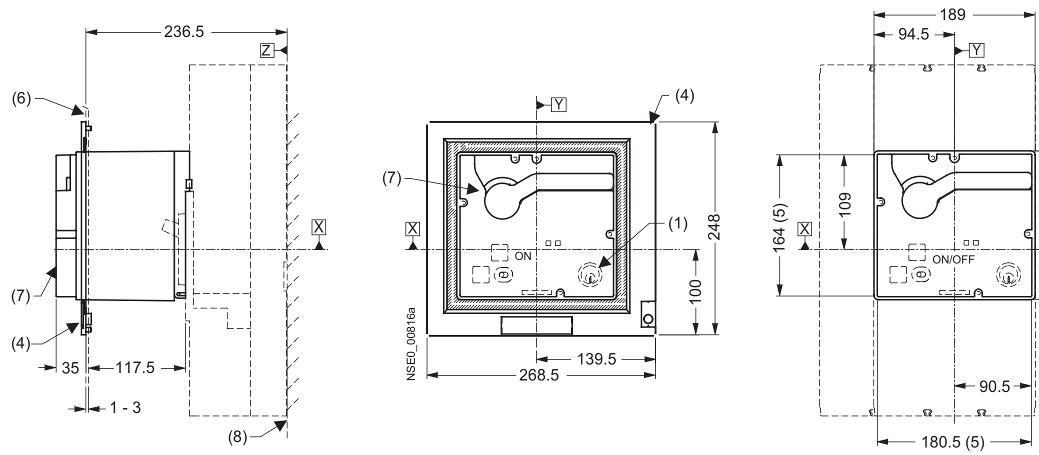
13.4.1 Circuit breaker

SENTRON VL800 (3VL6) circuit breaker and mounting instructions

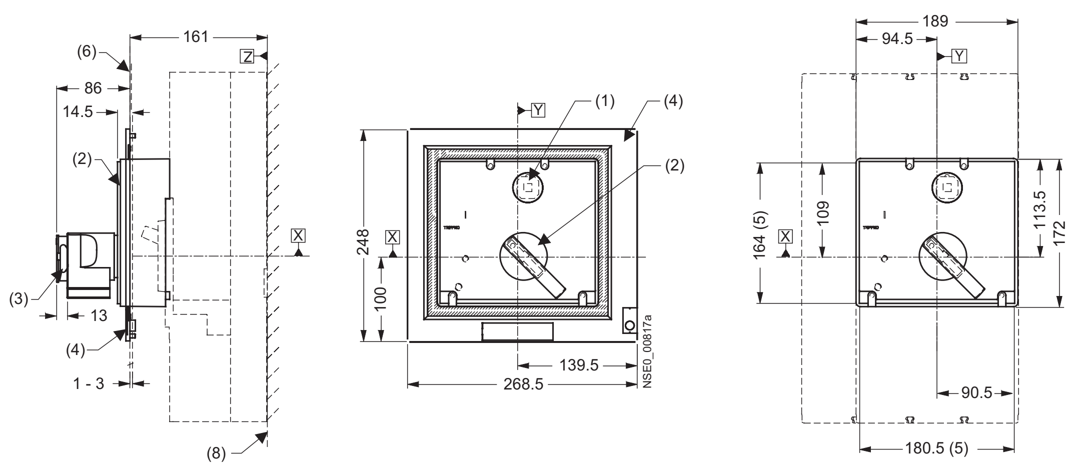


13.4.2 Operating mechanisms

Stored-energy motorized operating mechanism



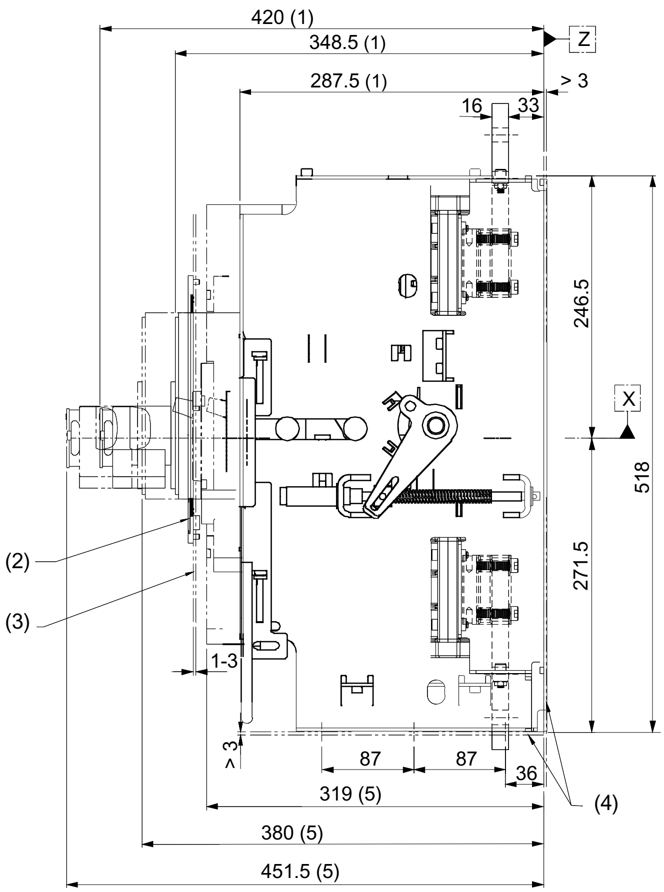
Front rotary operating mechanism



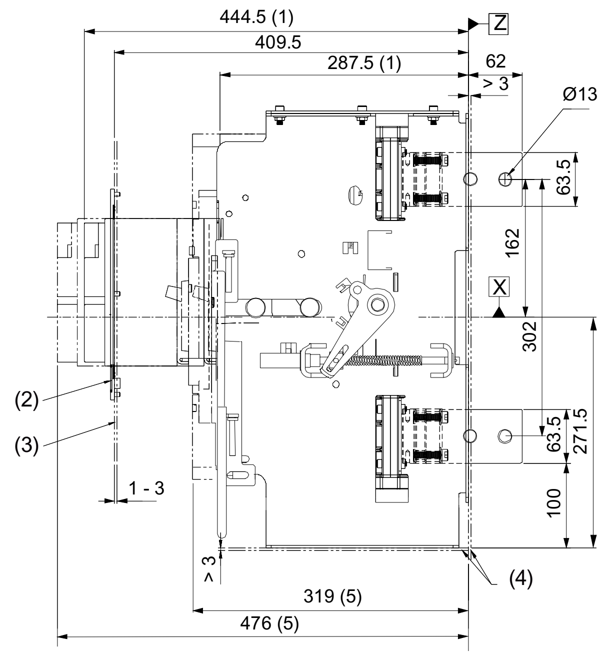
(1) Safety lock
(2) Front rotary operating mechanism
(3) Padlock barrier
(4) Cover frame for door cutout (for circuit breakers with operating mechanism)
(5) Grading for cover
(6) External surface of cabinet door
(7) Stored-energy motorized operating mechanism
(8) Mounting level
(9) Toggle handle extension

13.4.3 Withdrawable version

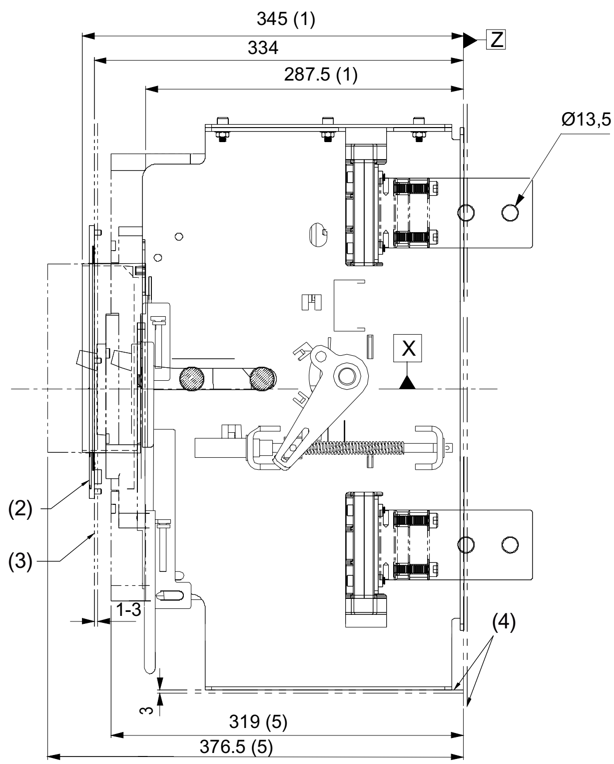
Withdrawable version with front rotary operating mechanism, insert position and remove position



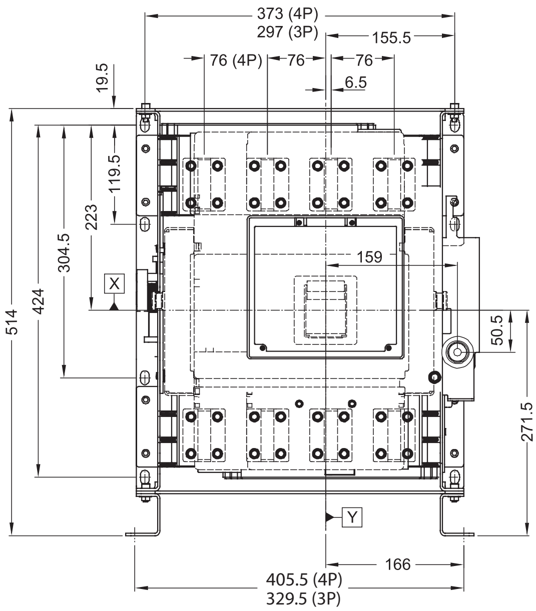
Withdrawable version with stored-energy motorized operating mechanism, insert position and remove position

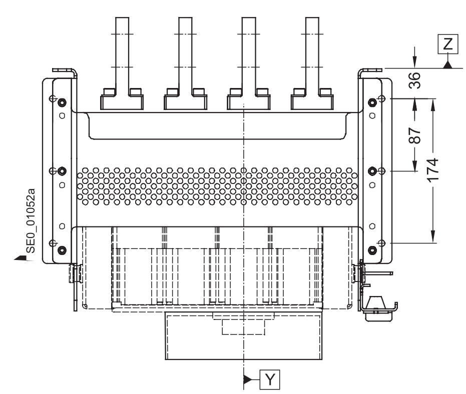


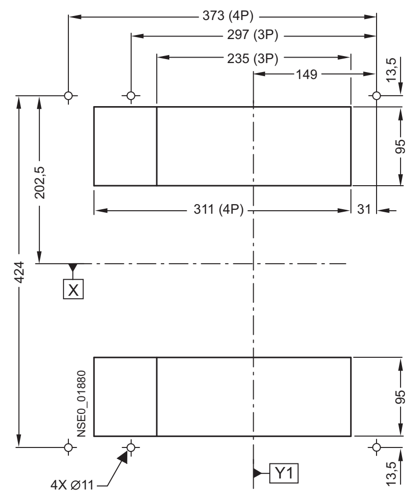
Withdrawable version with extension collar (without cover frame), insert position and remove position



(1)	Connected position
(2)	Cover frame for door cutout
(3)	External surface of cabinet door
(4)	Mounting level
(5)	Disconnected position

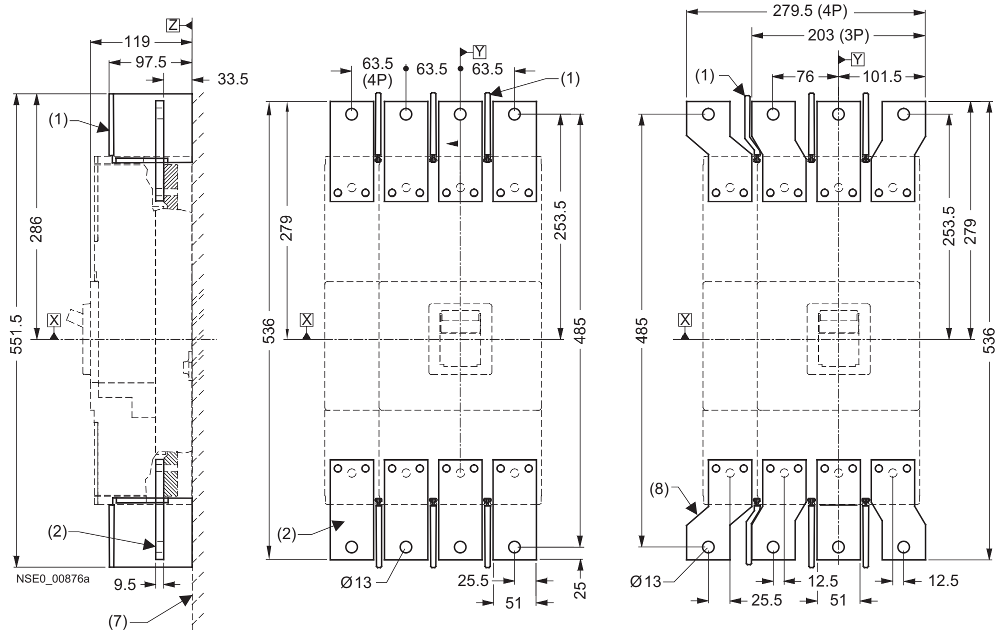
<table>
<tr><th>Withdrawable version</th><th>Drilling template and cutout for withdrawable modules with flat connecting bars on rear</th></tr>
<tr><td>




</td><td>


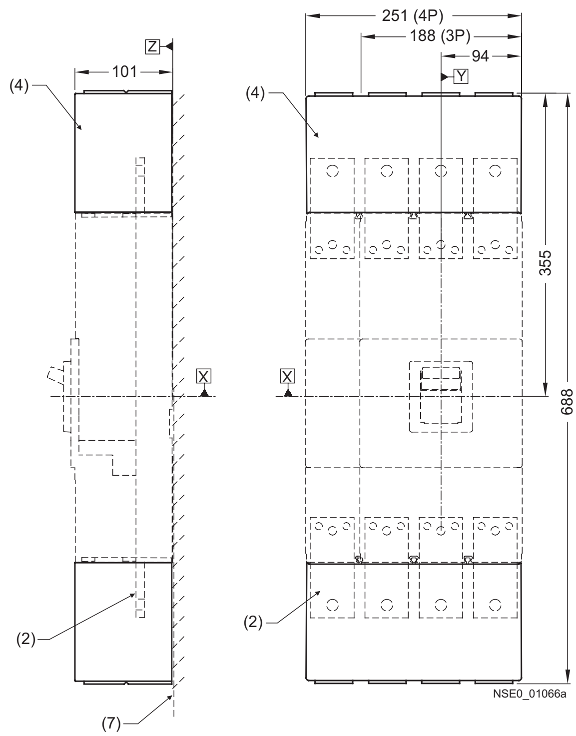
</td></tr>
</table>

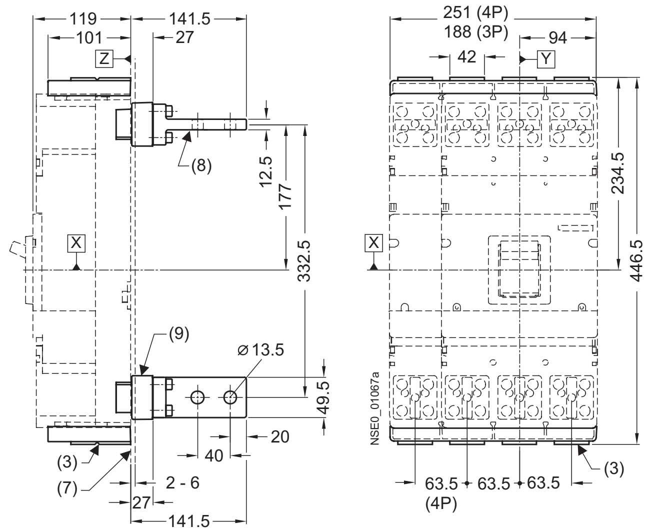
13.4.4 Connections and phase barriers

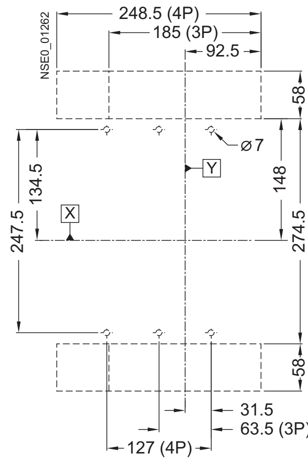


(1) Interphase barrier
(2) Front connecting bars
(7) Mounting level
(8) Flared busbar extensions

13.4.5 Terminal covers

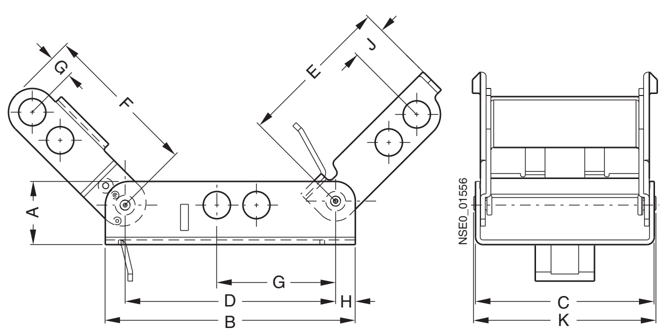






(2) Front connecting bars
(3) Terminal covers (standard)
(4) Terminal covers (extended)
(7) Mounting level
(8) Rear connection (horizontal mounting)
(9) Rear connection (vertical mounting)

13.4.6 Locking and locking device for toggle handle

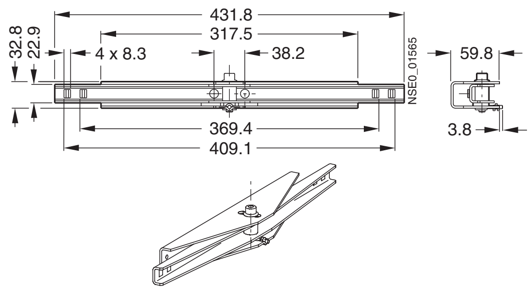


Type	a	b	c	d	e	f	g	h	i	k
3VL9 4	20.3	80.3	57.4	52.8	49.3	49.8	6.35	6.3	11.2	58.5
3VL9 6	21.6	79.8	71.1	62.0	50.4	46.5	12.9	8.9	8.6	72.2
3VL9 8	21.6	110.5	88.9	96.5	77.2	69.1	11.7	5.1	24.8	90.0

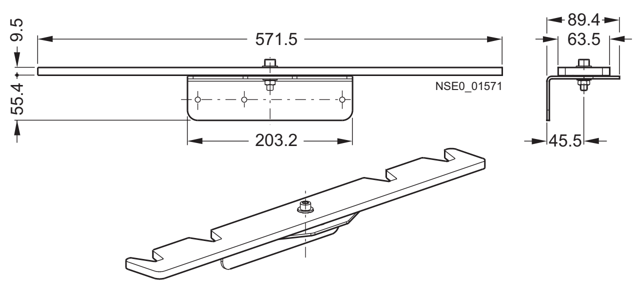
13.4.7 Rear interlocking module

Rear interlocking module 3-pole circuit breaker

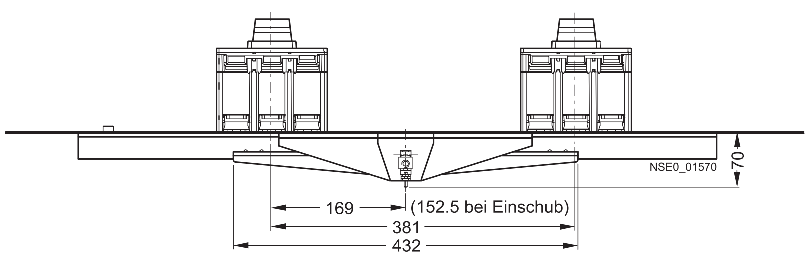
For other detailed dimension drawings, please refer to the mounting instructions for the rear interlocking module.



Rear interlocking module 4-pole circuit breaker

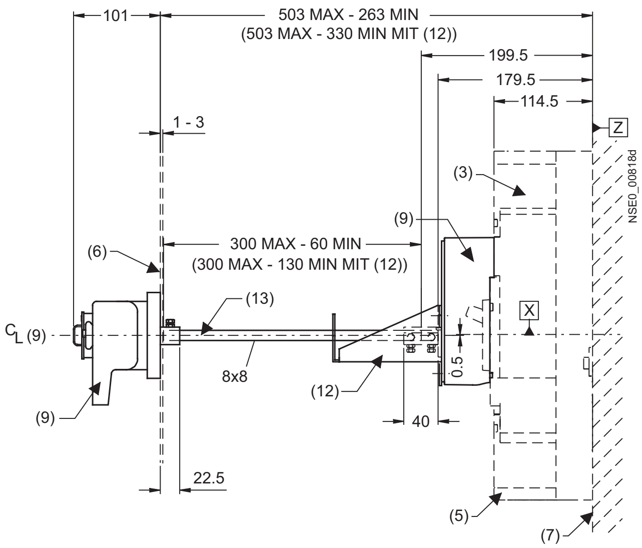


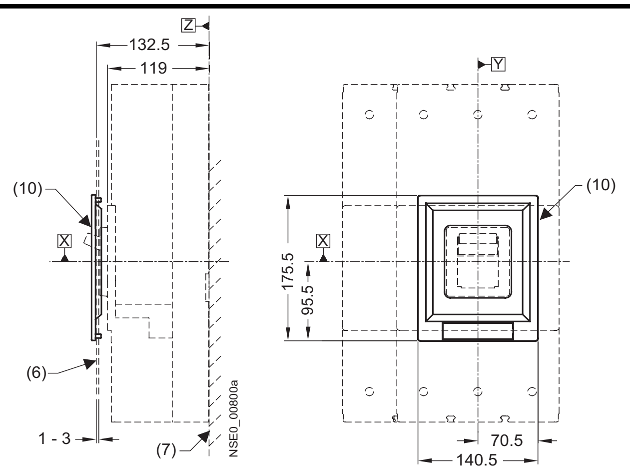
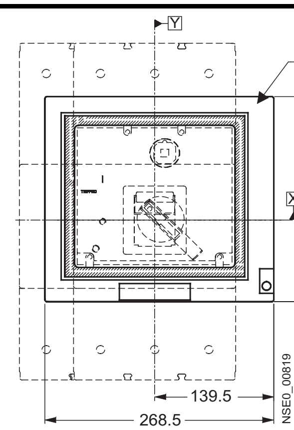
Rear interlocking module



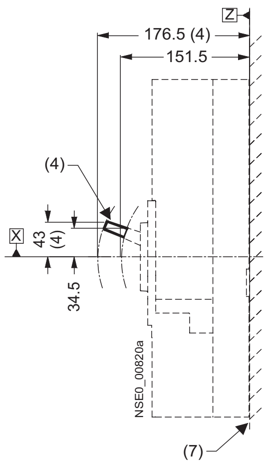
13.4.8 Accessories

Door-coupling rotary operating mechanism



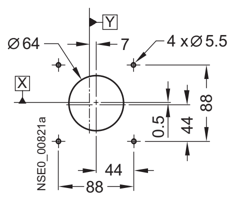
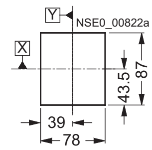
Cover frame for door cutout for circuit breakers with toggle handle	Cover frame for door cutout for circuit breakers with operating mechanism
	

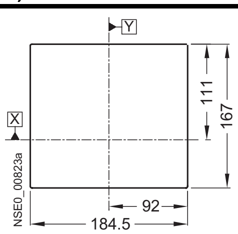
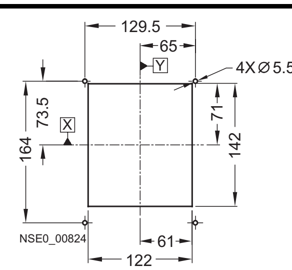
Toggle handle extension



(3)	Circuit breaker
(4)	Toggle handle extension
(5)	Terminal covers (standard)
(6)	External surface of cabinet door
(7)	Mounting level
(9)	Door-coupling rotary operating mechanism
(10)	Cover frame for door cutout (for circuit breakers with toggle handle)
(11)	Cover frame for door cutout (for circuit breakers with operating mechanism)
(12)	Supporting bracket
(13)	Center line of operating mechanism shaft

13.4.9 Door cutouts

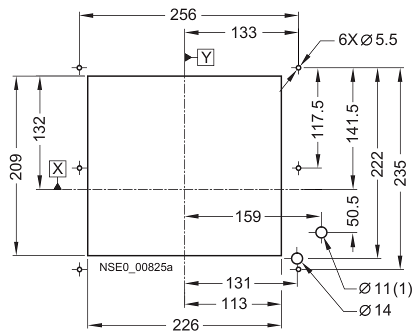
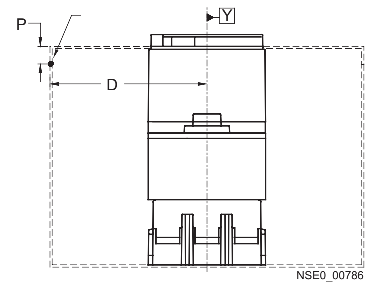
Door cutout Door coupling rotary operating mechanism	Door cutout toggle handle (without cover frame)
	

Door cutout front rotary operating mechanism, stored-energy motorized operating mechanism and extension collar (without cover frame)	Door cutout toggle handle (with cover frame)
	

Note

Note:

Door cutouts require a minimum clearance between reference point Y and the door hinge.

Door cutout front rotary operating mechanism, stored-energy motorized operating mechanism and extension collar (with cover frame)	Door hinge point (see arrow)
	 D > A from table + (P x 5)

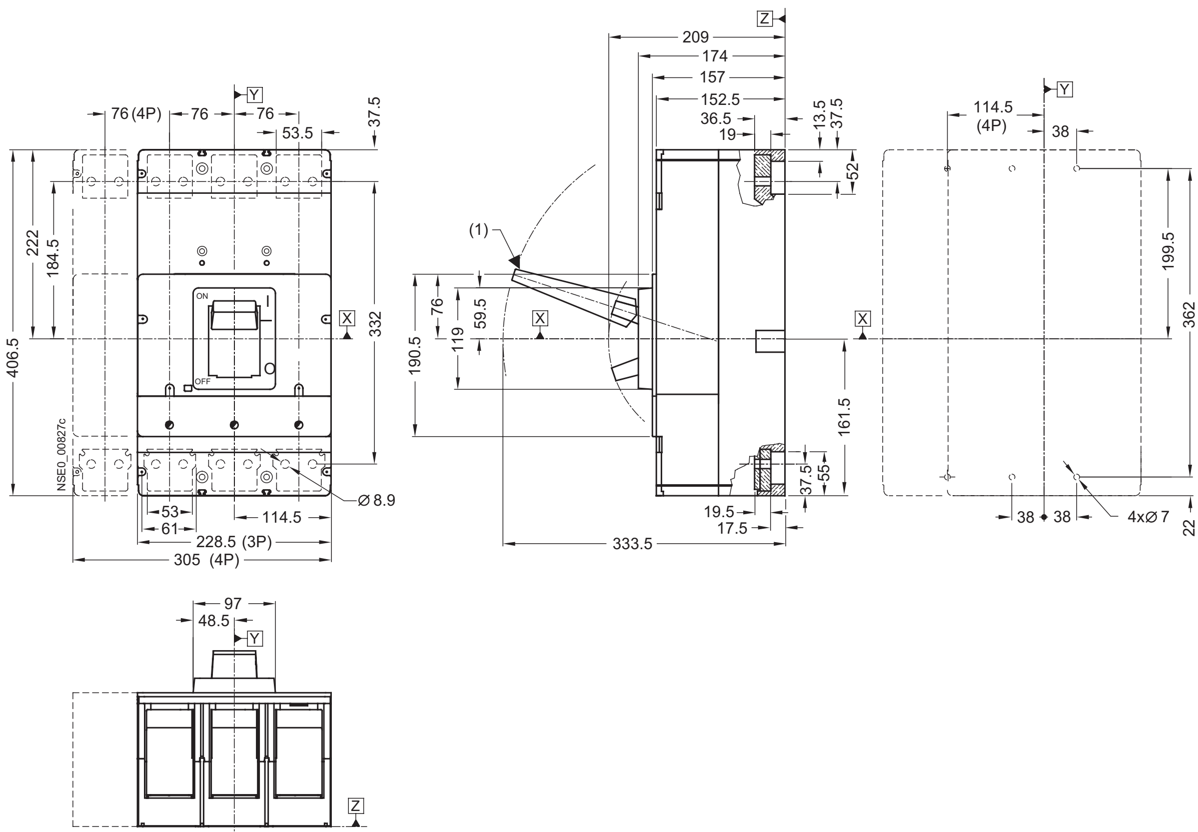
(1) Withdrawable version only

Combination	A
Circuit breaker only	150
Circuit breaker + plug-in socket + stored-energy motorized operating mechanism	150
Circuit breaker + plug-in socket + front rotary operating mechanism	200
Circuit breaker + withdrawable version	200

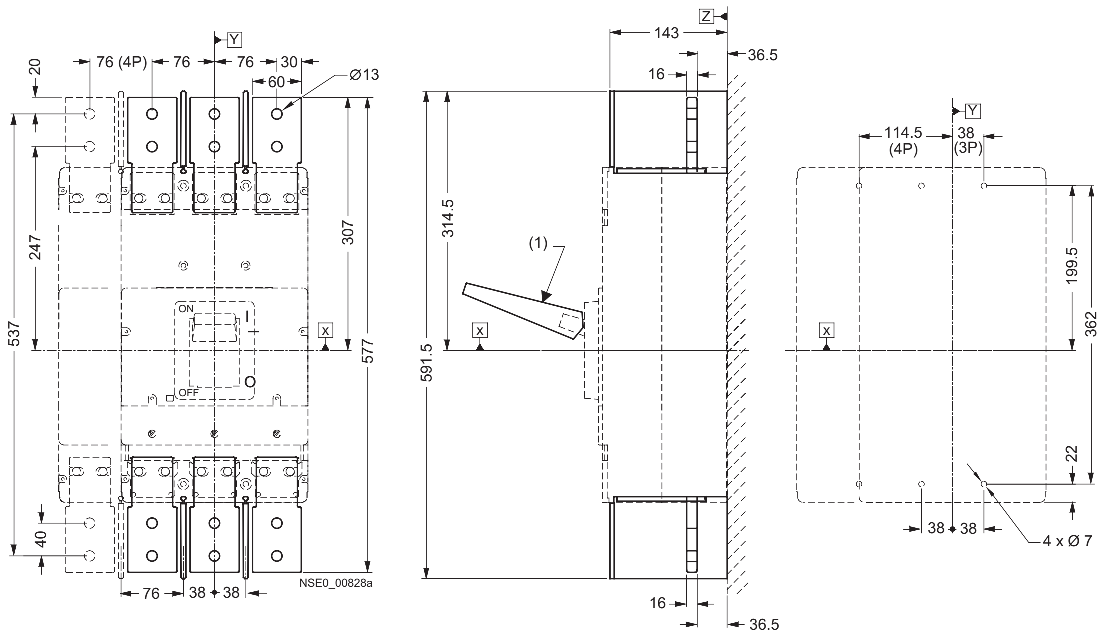
13.5 VL1250 (3VL7) and VL 1600 (3VL8), 3- and 4-pole, up to 1600 A

13.5.1 Circuit breaker

SENTRON VL1250 (3VL7) circuit breaker and mounting instructions



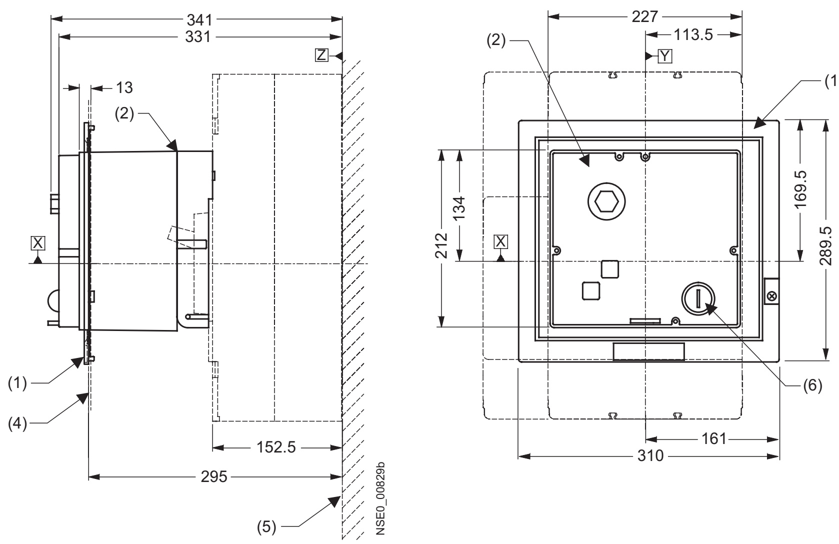
SENTRON VL1600 (3VL8) circuit breaker and mounting instructions



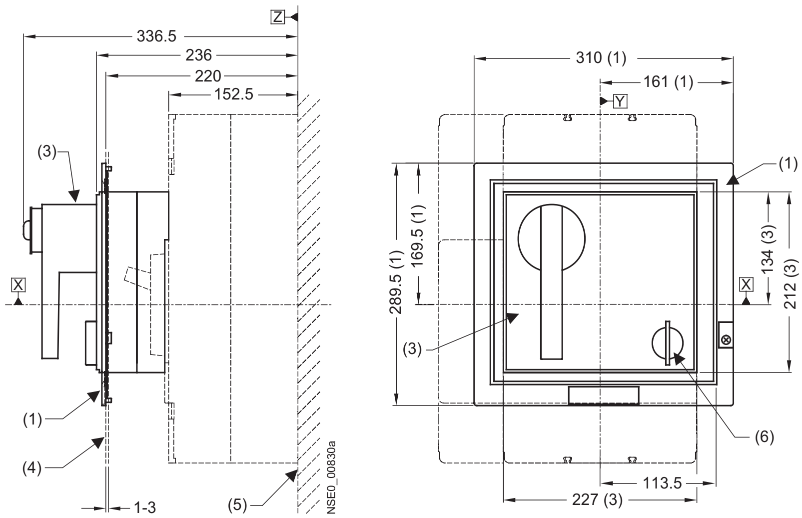
(1) Toggle handle extension

13.5.2 Operating mechanisms

Motorized operating mechanism



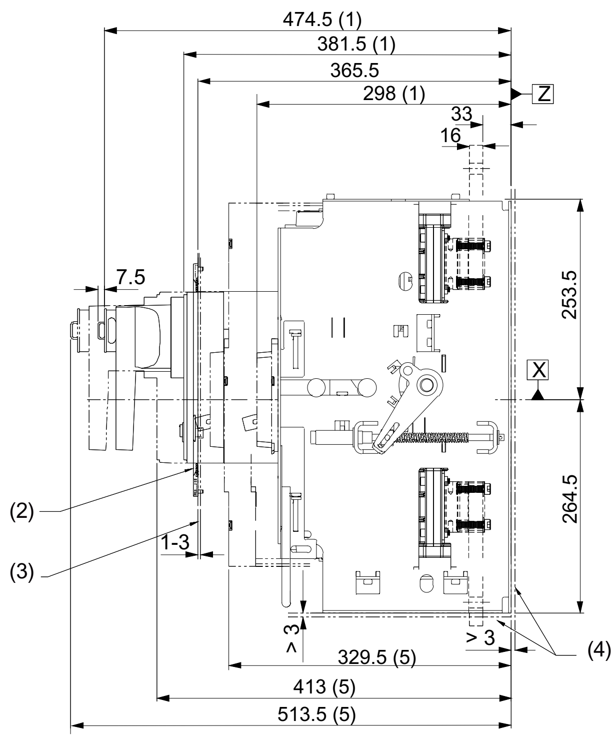
Front rotary operating mechanism



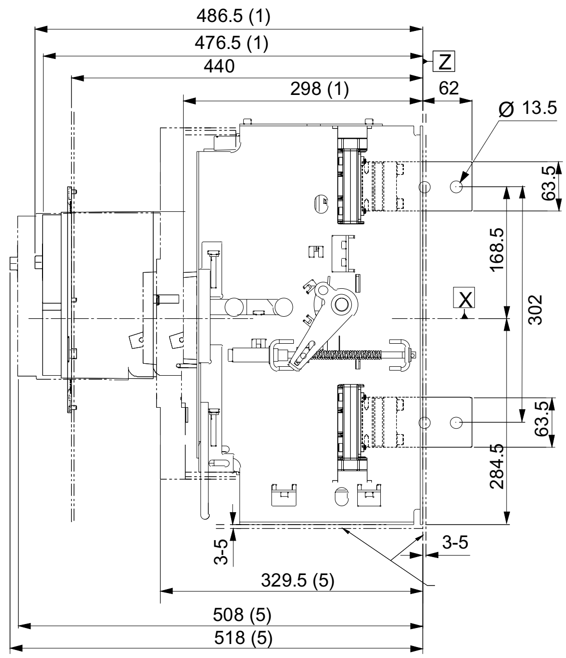
(1)	Cover frame for door cutout (for circuit breakers with operating mechanism)
(2)	Motorized operating mechanism
(3)	Front rotary operating mechanism
(4)	External surface of cabinet door
(5)	Mounting level
(6)	Safety lock

13.5.3 Withdrawable version

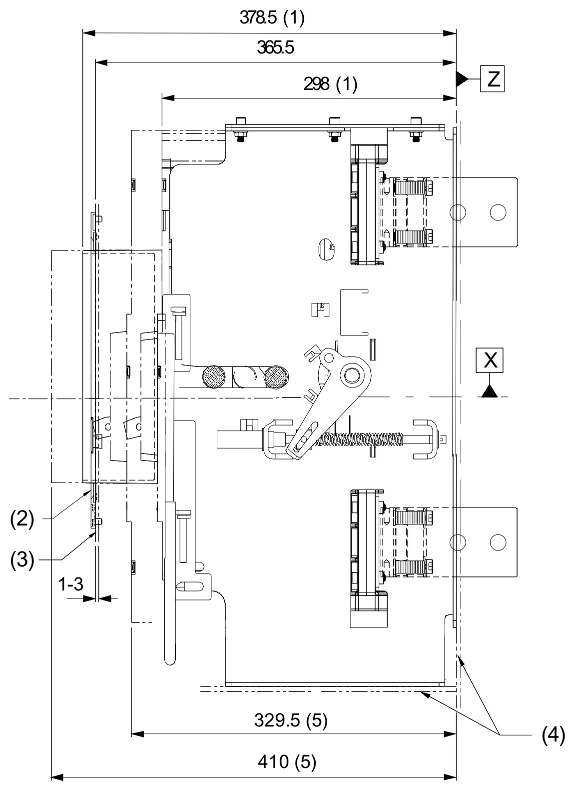
Withdrawable version with front rotary operating mechanism, insert position and remove position



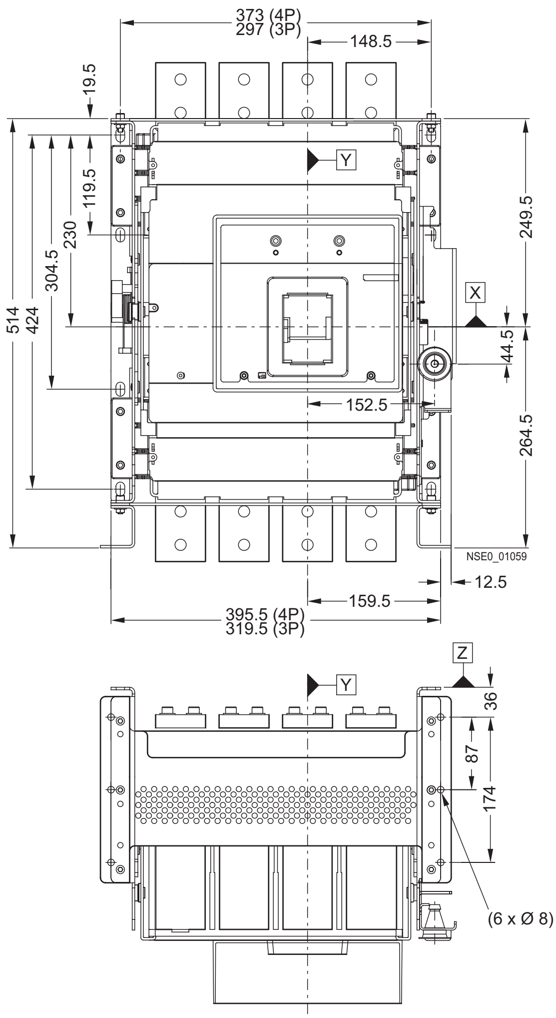
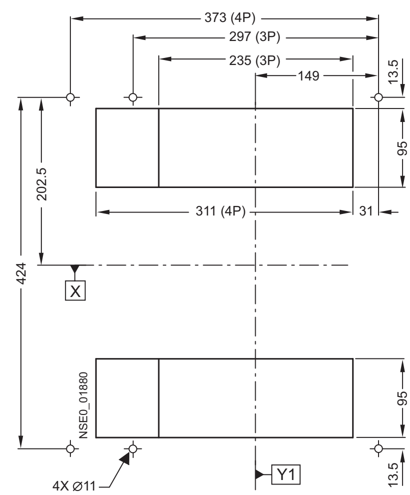
Withdrawable version with stored-energy motorized operating mechanism, insert position and remove position



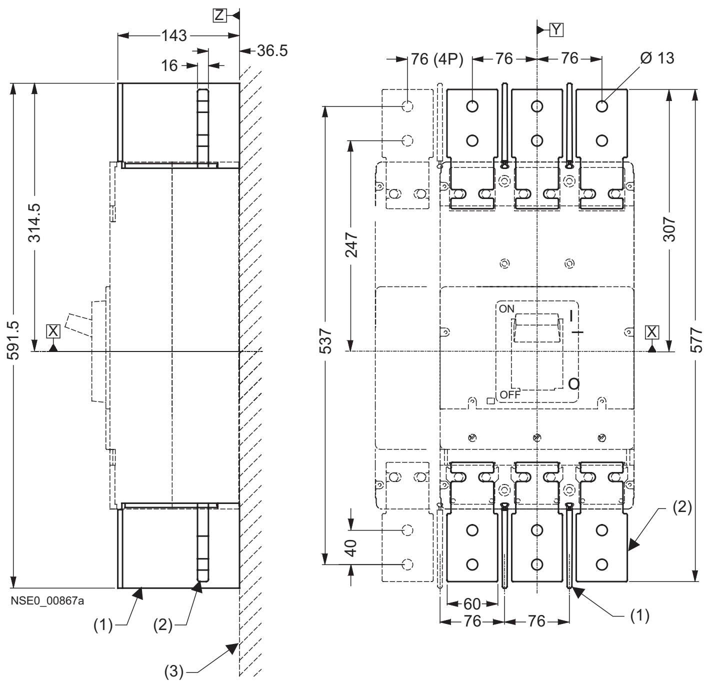
Withdrawable version with extension collar (without cover frame), insert position and remove position



(1)	Connected position
(2)	Cover frame for door cutout
(3)	External surface of cabinet door
(4)	Mounting level
(5)	Disconnected position

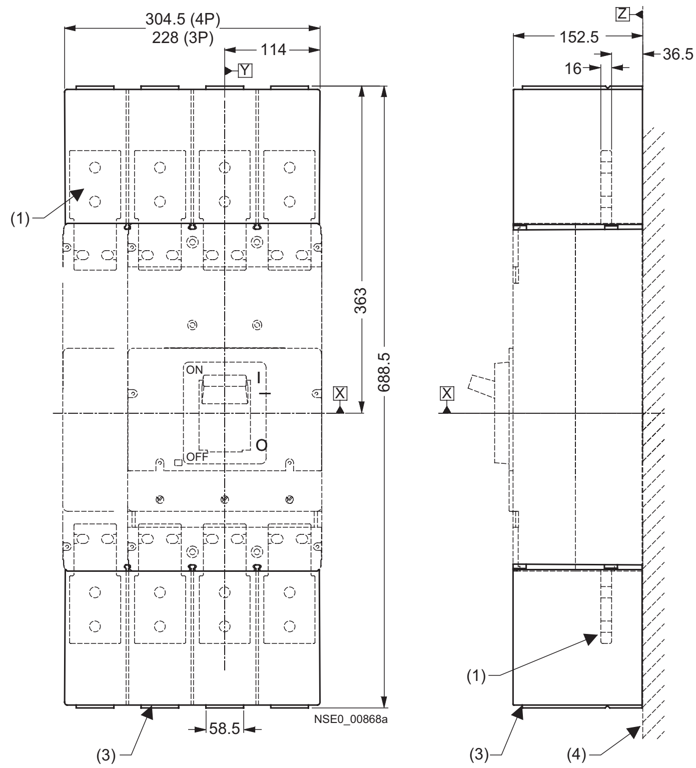
<table>
<tr><th>Withdrawable version</th><th>Drilling template and cutout for withdrawable modules with flat connecting bars on rear</th></tr>
<tr><td>
</td><td>
</td></tr>
</table>

13.5.4 Connections and phase barriers



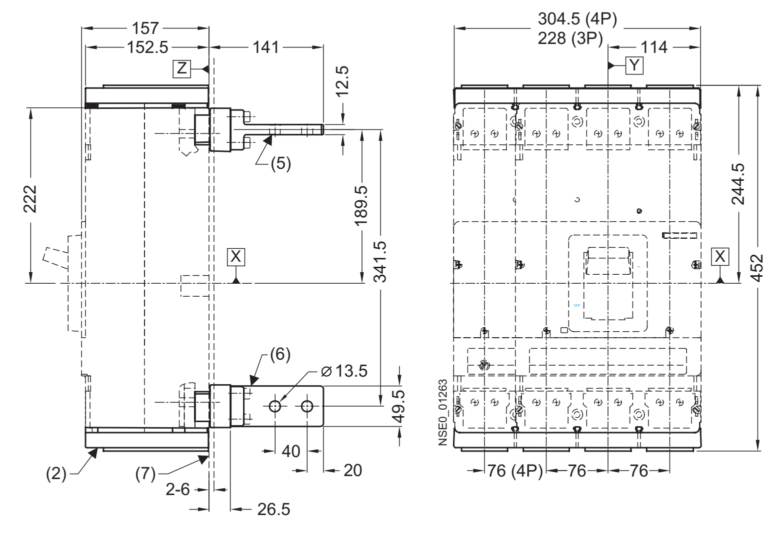
(1)	Interphase barrier
(2)	Front connecting bars
(3)	Mounting level

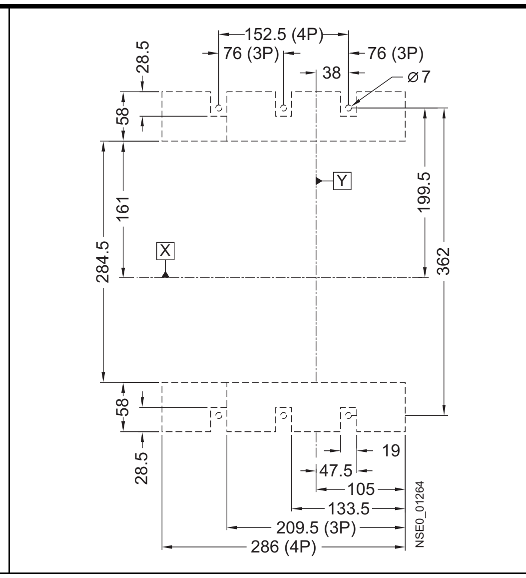
13.5.5 Terminal covers



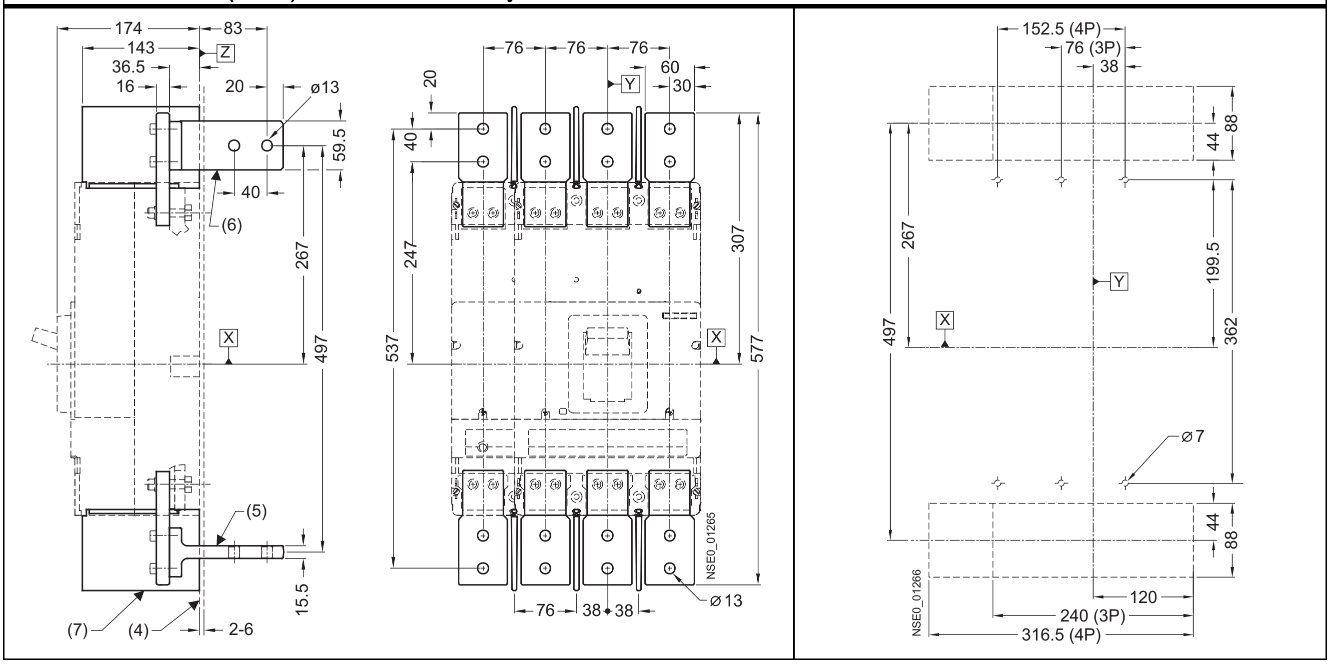
(1)	Front connecting bars
(2)	Terminal covers (short) – for SENTRON VL1250 (3VL7) circuit breakers only
(3)	Terminal covers (extended)
(4)	Mounting level
(5)	Rear connection (horizontal mounting)
(6)	Rear connection (vertical mounting)
(7)	Phase barriers

SENTRON VL1250 (3VL7) circuit breakers only





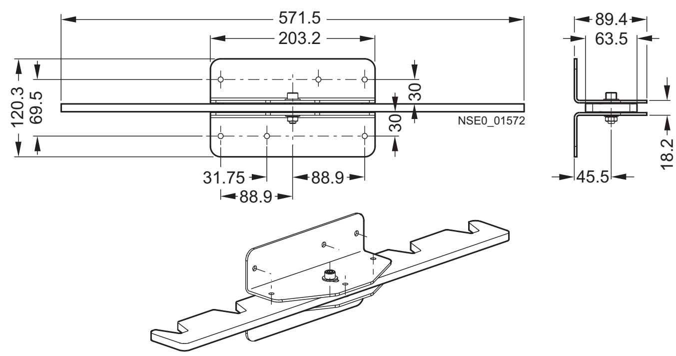
SENTRON VL1600 (3VL8) circuit breakers only



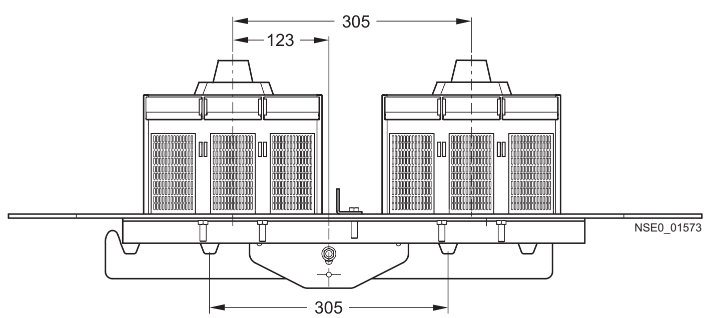
(1) Front connecting bars
(2) Terminal covers (short) – for SENTRON VL1250 (3VL7) circuit breakers only
(3) Terminal covers (extended)
(4) Mounting level
(5) Rear connection (horizontal mounting)
(6) Rear connection (vertical mounting)
(7) Phase barriers

13.5.6 Rear interlocking module

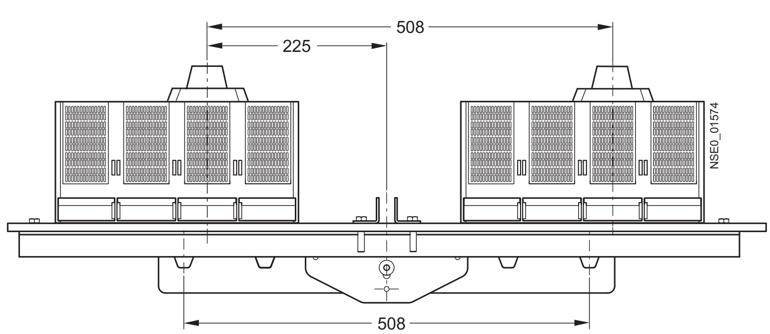
For other detailed dimension drawings, please refer to the mounting instructions for the rear interlocking module.



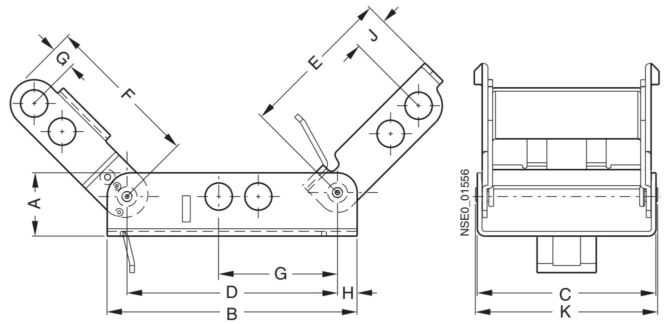
3-pole version



4-pole version



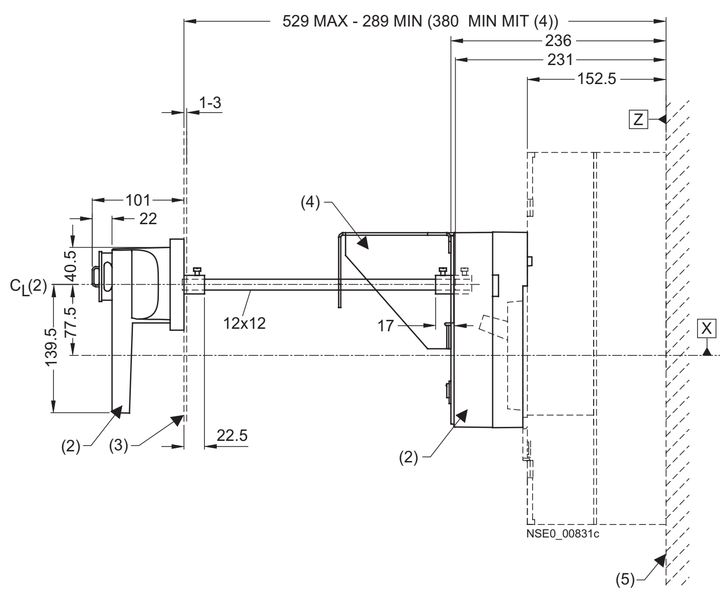
13.5.7 Locking and locking device for toggle handle



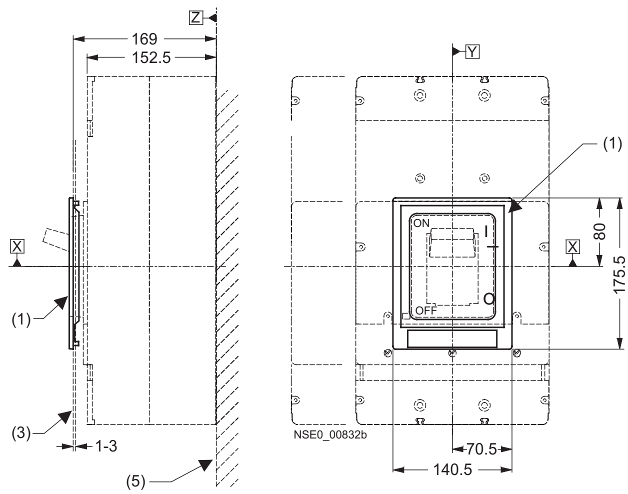
Type	a	b	c	d	e	f	g	h	i	k
3VL9 4	20.3	80.3	57.4	52.8	49.3	49.8	6.35	6.3	11.2	58.5
3VL9 6	21.6	79.8	71.1	62.0	50.4	46.5	12.9	8.9	8.6	72.2
3VL9 8	21.6	110.5	88.9	96.5	77.2	69.1	11.7	5.1	24.8	90.0

13.5.8 Accessories

Door-coupling rotary operating mechanism

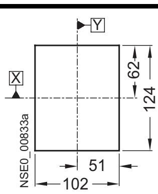
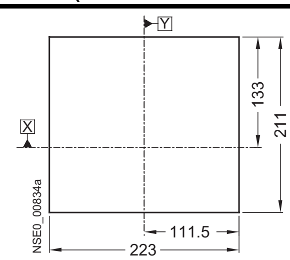
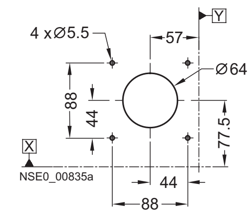


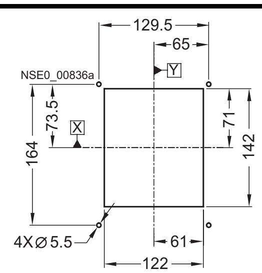
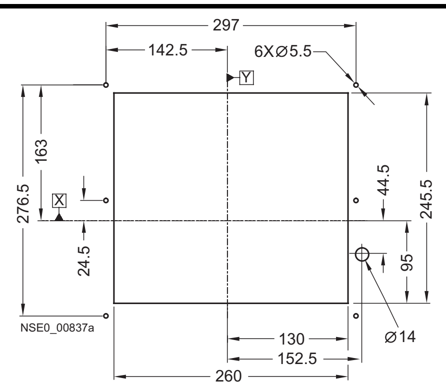
Cover frame for door cutout for circuit breakers with toggle handle



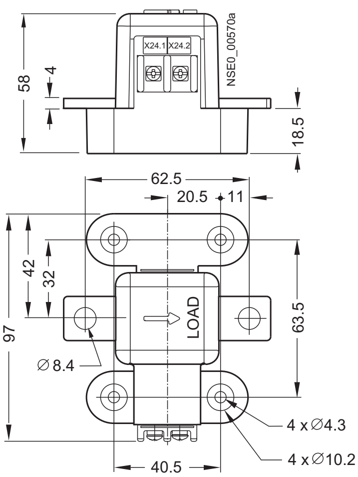
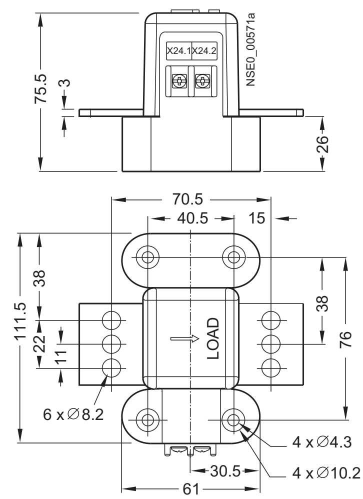
(1)	Cover frame for door cutout (for circuit breakers with toggle handle)
(2)	Door-coupling rotary operating mechanism
(3)	External surface of cabinet door
(4)	Supporting bracket
(5)	Mounting level

13.5.9 Door cutouts

<table>
<tr><th>Door cutout toggle handle
(without cover frame)</th><th>Door cutout front rotary operating mechanism and motorized operating mechanism (without cover frame)</th><th>Door cutout
Door coupling rotary operating mechanism</th></tr>
<tr><td>
</td><td>
</td><td>
</td></tr>
</table>

<table>
<tr><th>Door cutout toggle handle
(with cover frame)</th><th>Door cutout front rotary operating mechanism, motorized operating mechanism and extension collar (with cover frame)</th></tr>
<tr><td>
</td><td>
</td></tr>
</table>

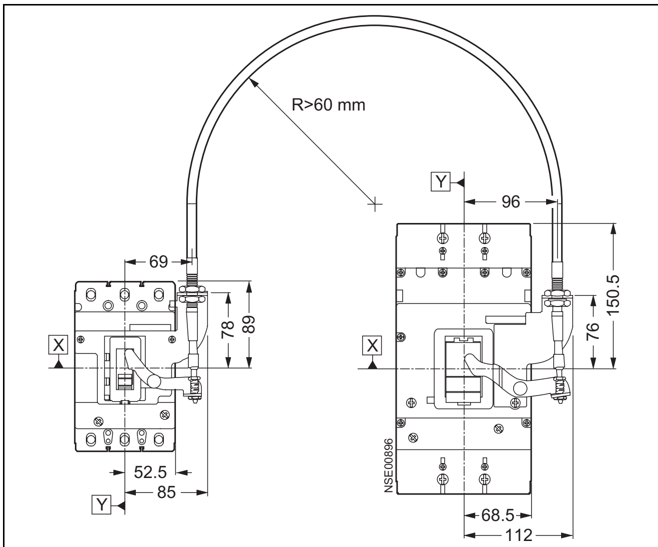
13.5.10 Current transformer

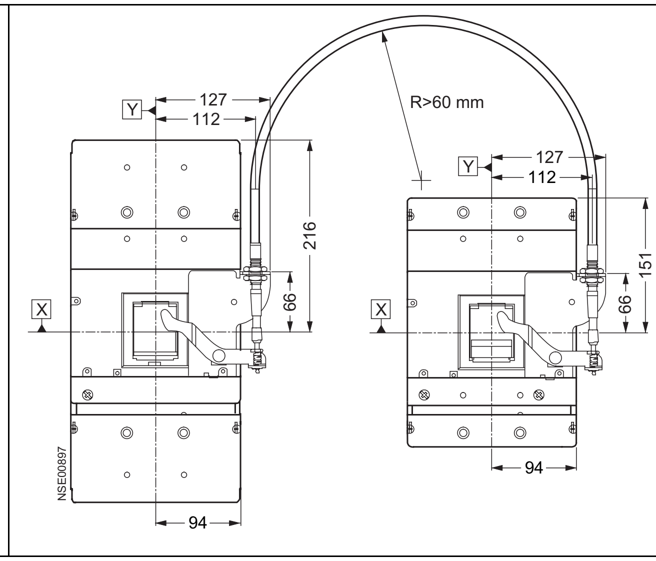
<table>
<tr><th>Current transformer for neutral conductor for ground fault protection in 4-conductor three-phase systems for SENTRON VL160 (3VL2)/VL250 (3VL3) circuit breakers</th><th>Current transformer for neutral conductor for ground fault protection in 4-conductor three-phase systems for SENTRON VL630 (3VL5)/VL800 (3VL6) circuit breakers</th></tr>
<tr><td>
</td><td>
</td></tr>
</table>

For other dimension drawings (for current transformers for 3VL4, 3VL7, 3VL8), please refer to the mounting instructions for current transformers.

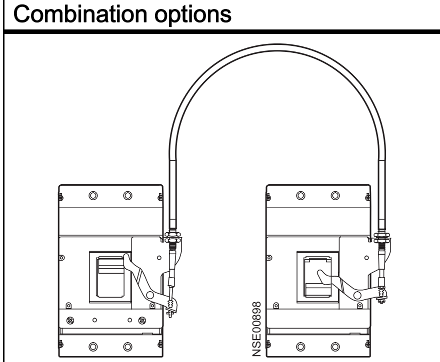
13.6 VL160X (3VL1) up to VL800 (3VL6), 3- and 4-pole, up to 800 A

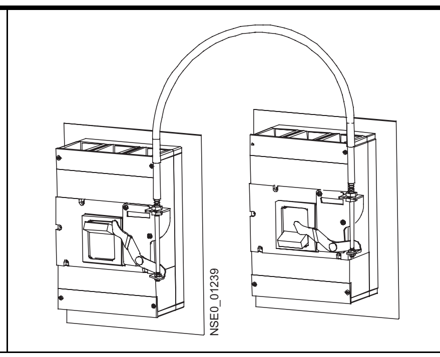
13.6.1 Locking with bowden wire

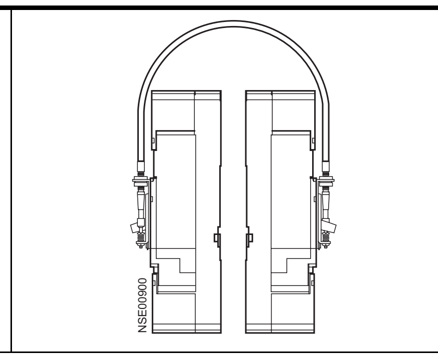




Combination options



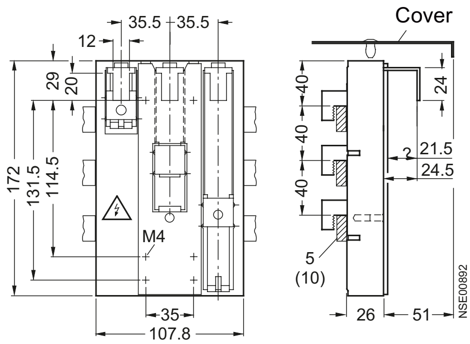
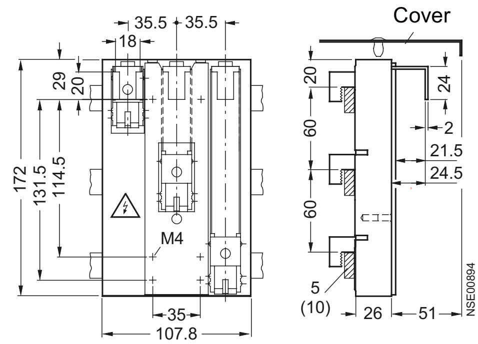


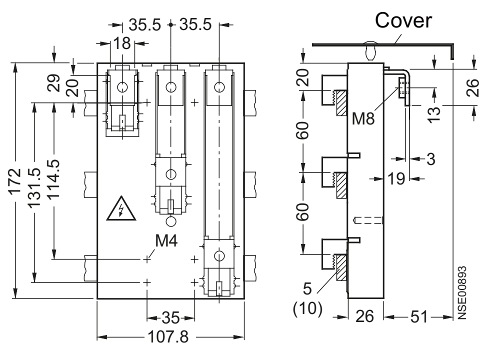
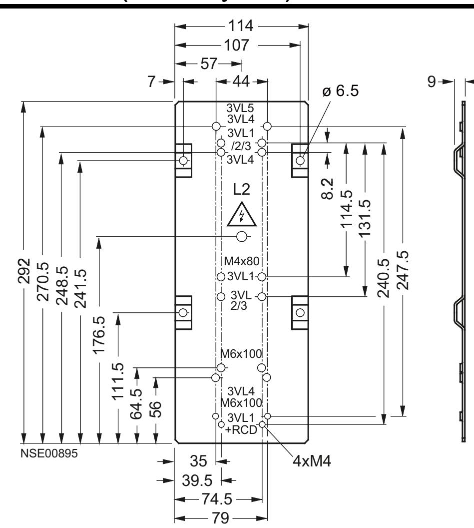


	3VL9 300-8LA00 for VL160X (3VL1), VL160 (3VL2) and VL250 (3VL3)	3VL9 400-8LA00 for VL400 (3VL4)	3VL9 600-8LA00 for VL630 (3VL5) and VL800 (3VL6)	3VL9 800-8LA00 for VL1250 (3VL7) and VL1600 (3VL8)
Locking with bowden wire				
3VL9 300-8LA00 for VL160X (3VL1), VL160 (3VL2) and VL250 (3VL3)	✓	-	-	-
3VL9 400-8LA00 for VL400 (3VL4)	-	✓	-	-
3VL9 600-8LA00 for VL630 (3VL5) and VL800 (3VL6)	-	-	✓	
3VL9 800-8LA00 for VL1250 (3VL7) and VL1600 (3VL8)	-	-	-	✓

✓ Combination possible

13.6.2 Busbar adapter system 8US1

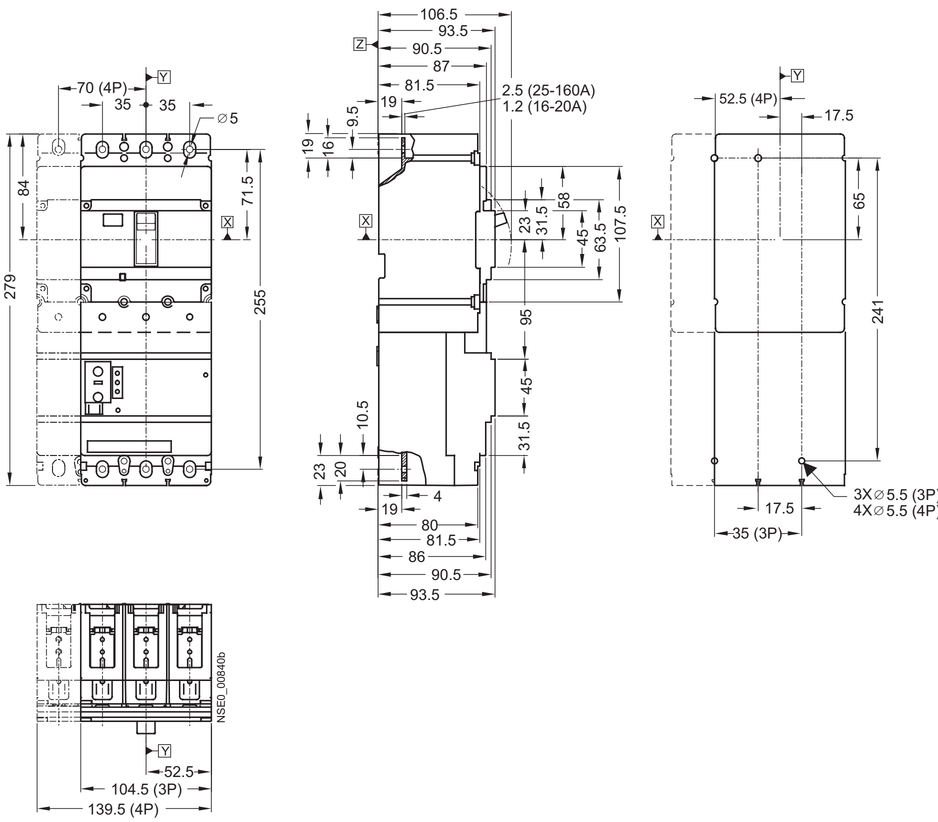
8US10 11-4SL01 (40-mm system)	8US12 11-4SL01 (60-mm system)
	

8US12 11-4SL00 (60-mm system)	8US19 27-4AF01 (60-mm system)
	

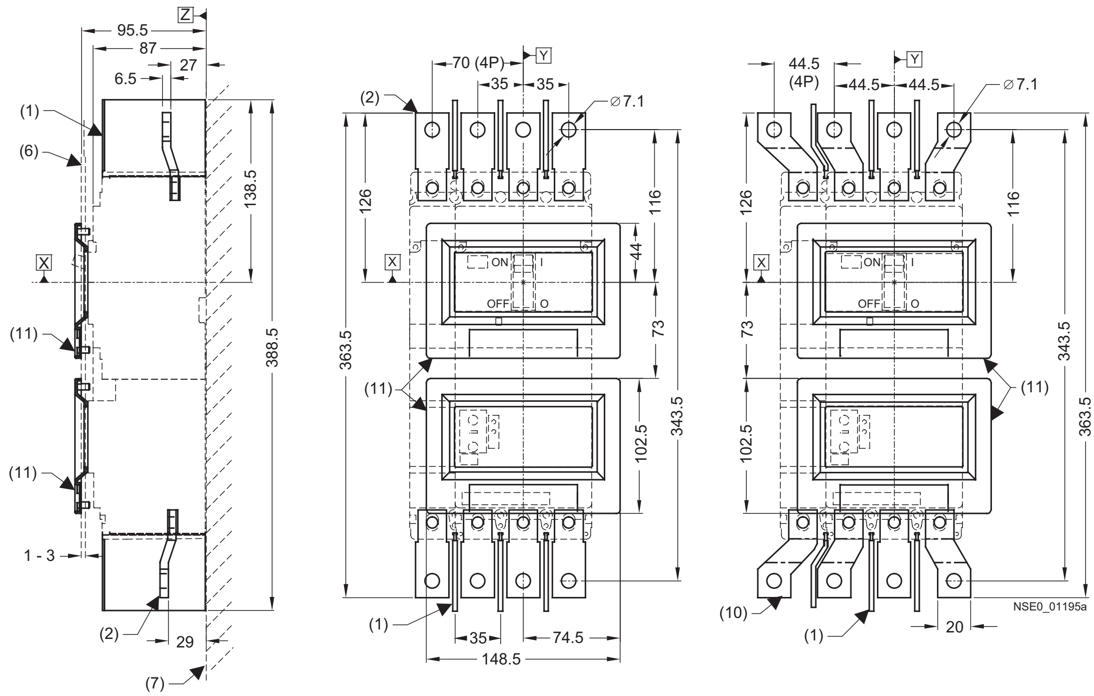
13.7 VL160X (3VL1) with RCD block, 3- and 4-pole, up to 160 A

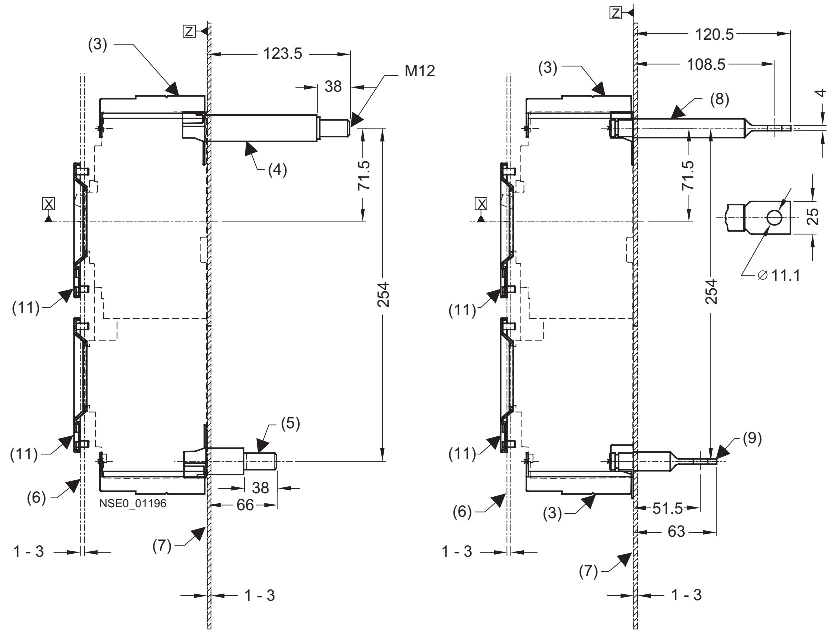
13.7.1 Circuit breakers

SENTRON VL160X (3VL1) circuit breaker with RCD module and mounting instructions



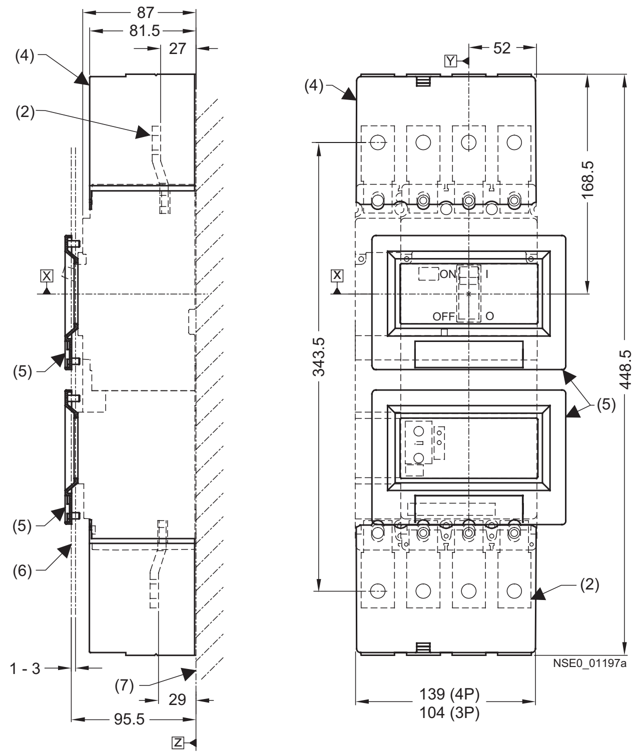
13.7.2 Connections and phase barriers

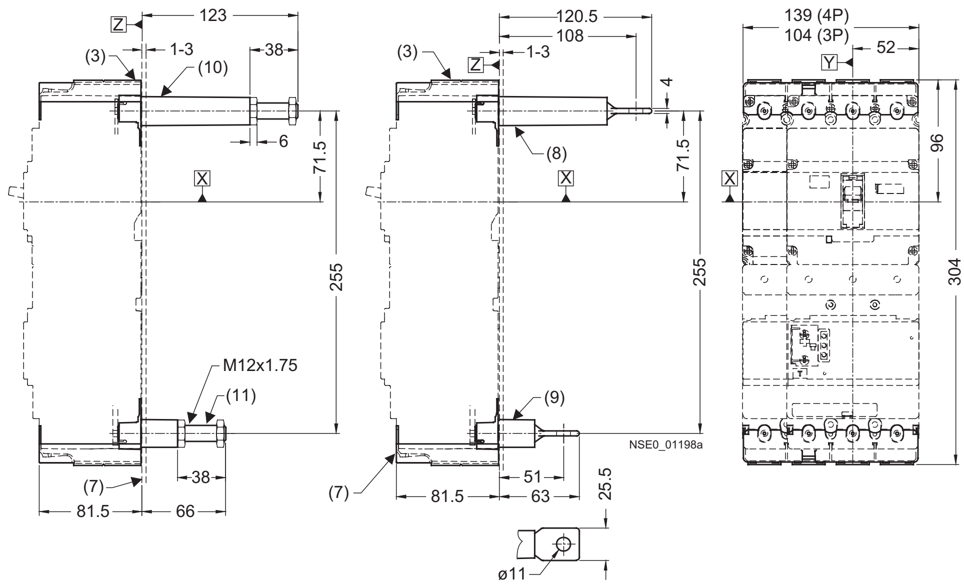




(1)	Interphase barrier
(2)	Front connecting bars
(3)	Terminal covers (standard)
(4)	Rear connection threaded bolt (long)
(5)	Rear connection threaded bolt (short)
(6)	External surface of cabinet door
(7)	Mounting level
(8)	Rear connection, long pad-type terminals
(9)	Rear connection, short pad-type terminals
(10)	Flared busbar extensions
(11)	Cover frame for door cutout (for circuit breakers with RCD module)

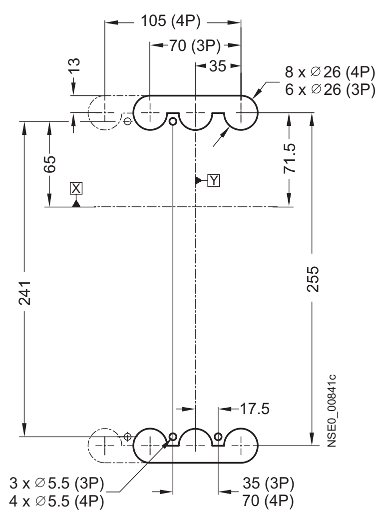
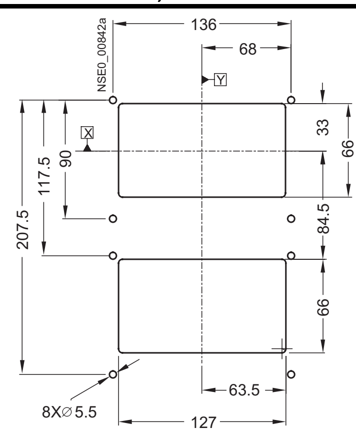
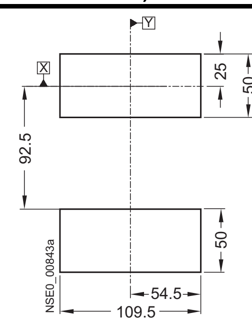
13.7.3 Terminal covers



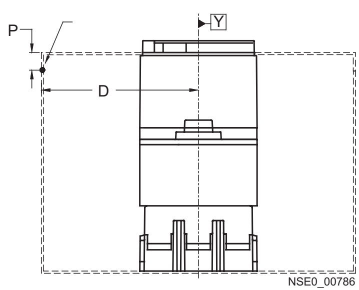


(2) Front connecting bars
(3) Terminal covers (standard)
(4) Terminal covers (extended)
(5) Cover frame for door cutout
(for circuit breakers with RCD module)
(6) External surface of cabinet door
(7) Mounting level
(8) Rear connection, long pad-type terminals
(9) Rear connection, short pad-type terminals
(10) Rear connection, long
(11) Rear connection, short

13.7.4 Door cutouts

Drilling template for rear connection	Door cutout toggle handle (with cover frame)	Door cutout toggle handle (without cover frame)
		

Door hinge point (see arrow)



D > A from table + (P × 5)

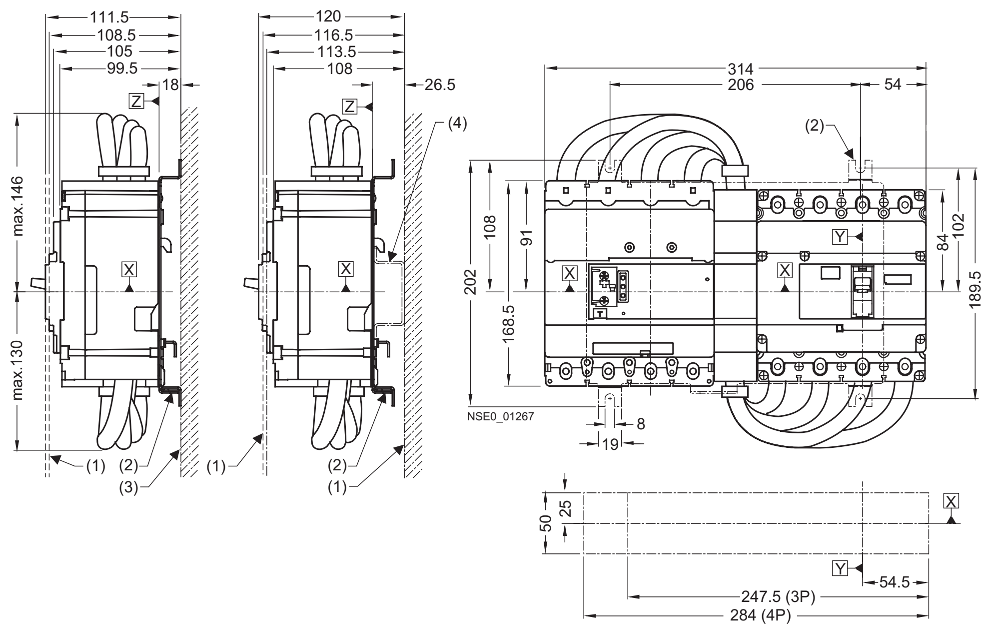
Note

Note:

Door cutouts require a minimum clearance between reference point Y and the door hinge.

Combination	A
Circuit breaker only	100
Circuit breaker + plug-in socket + stored-energy motorized operating mechanism	100
Circuit breaker + plug-in socket + front rotary operating mechanism	200

Circuit breaker with RCD module mounted on side



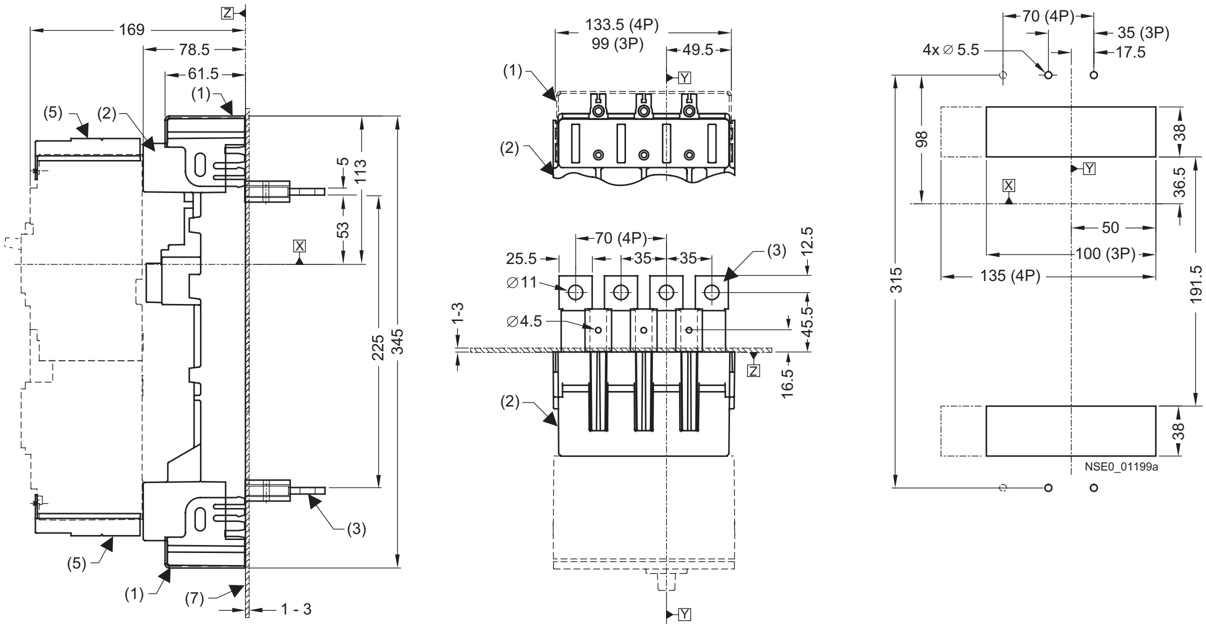
(1) External surface of cabinet door

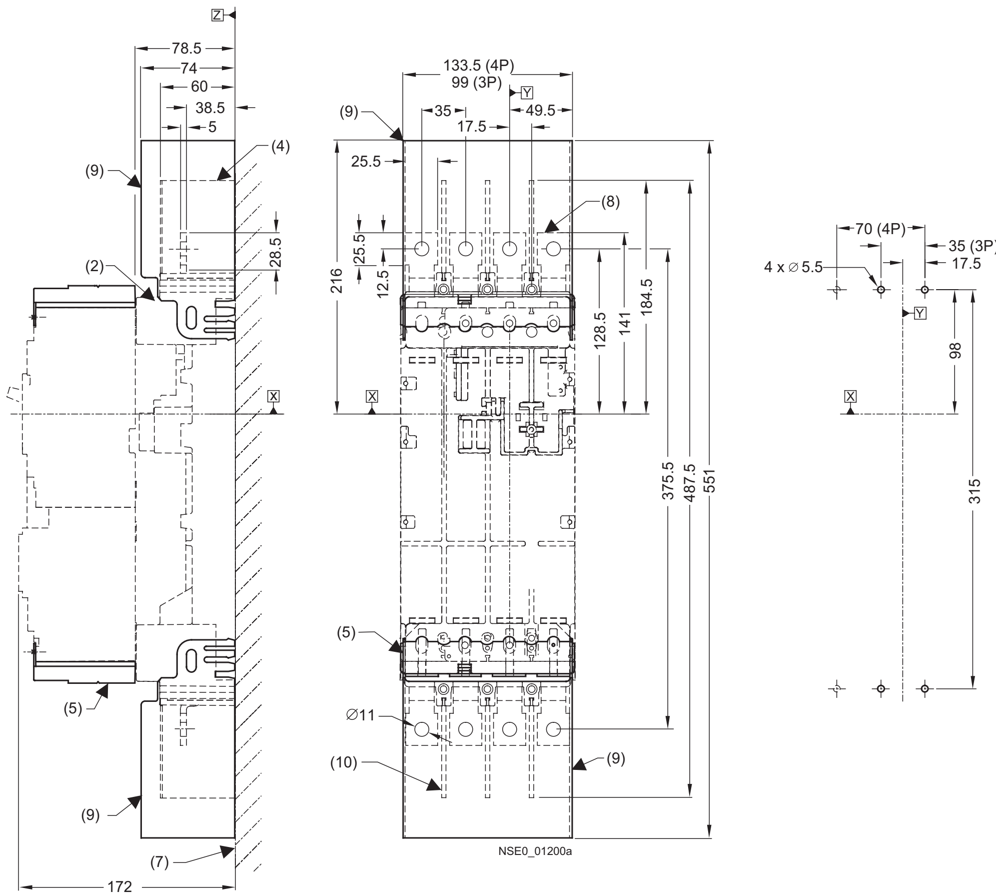
(2) Fastening bracket

(3) Mounting level

(4) Mounting rail TH 75 in accordance with DIN EN 60715 (to be provided by the customer)

13.7.5 Plug-in socket and accessories



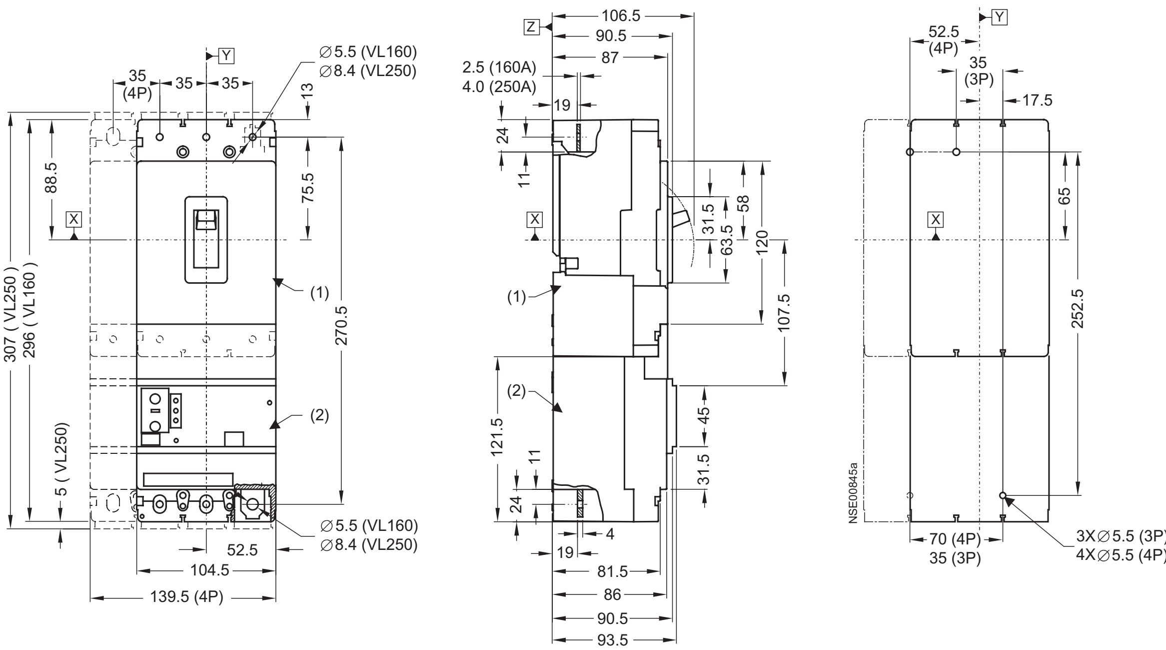


(1) Plug-in socket with rear terminal covers
(2) Plug-in socket for circuit breaker with RCD module
(3) Plug-in socket with rear flat busbar terminals
(4) Cover frame for door cutout
(for circuit breakers with RCD module)
(5) Terminal cover (standard)
(6) External surface of cabinet door
(7) Mounting level
(8) Plug-in socket with front connecting bars
(9) Plug-in socket with terminal covers on the front
(10) Interphase barrier

13.8 VL160 (3VL2) and VL250 (3VL3) with RCD module, 3- and 4-pole, to 250 A

13.8.1 Circuit breakers

SENTRON VL160 (3VL2) and VL250 (3VL3) circuit breakers with RCD module and mounting instructions



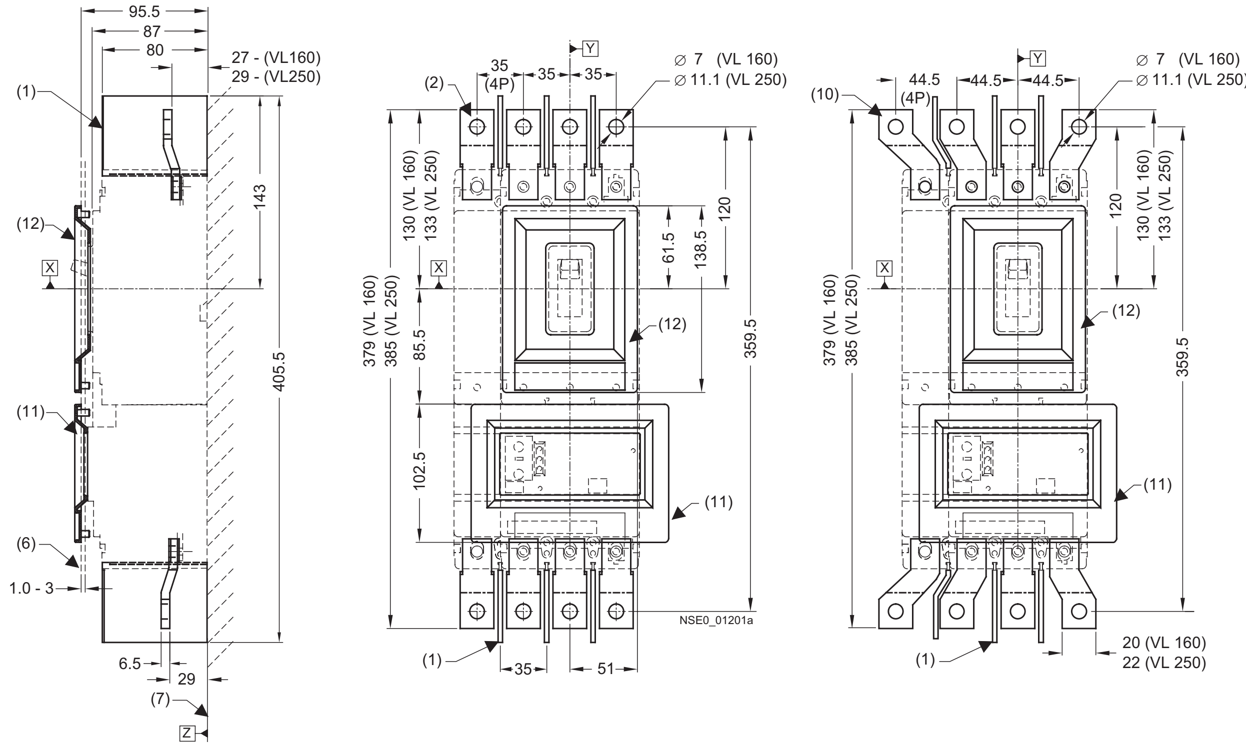
(1) Circuit breakers
(2) RCD module

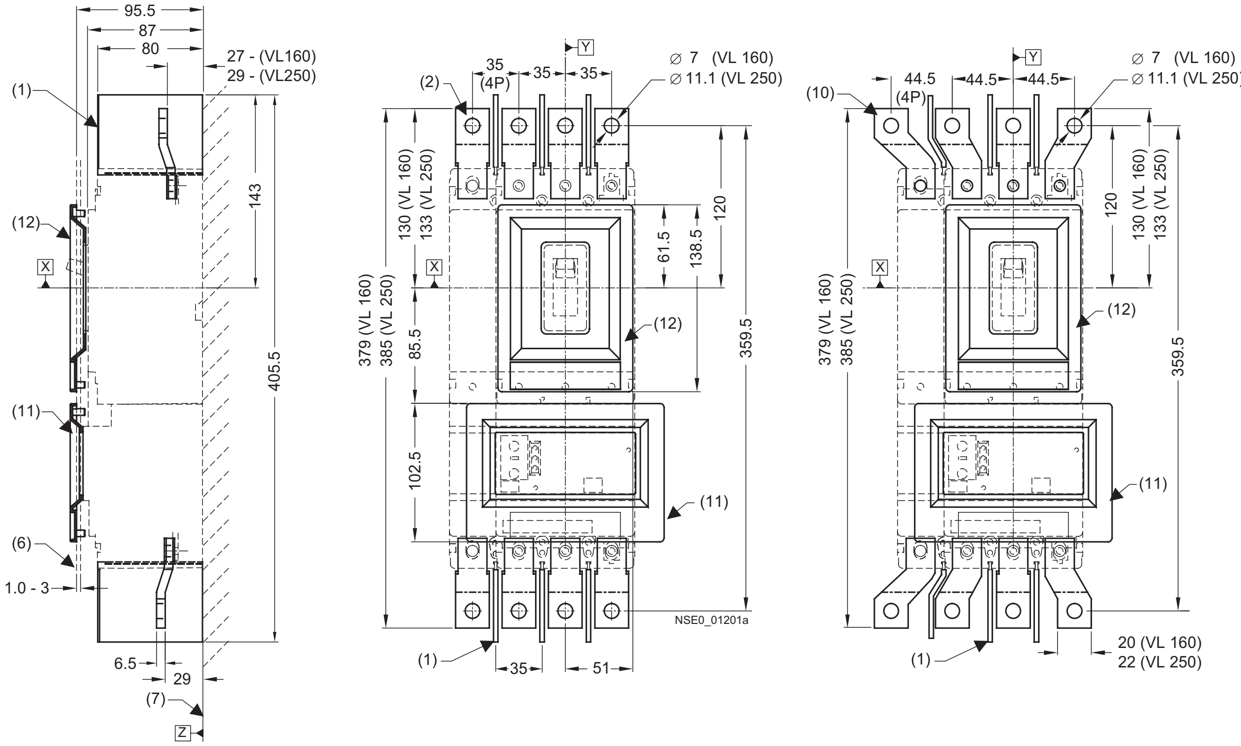
Note

Note for SENTRON VL250 (3VL3) circuit breakers:

The 5-mm extension (total height 307 mm) at each end is only significant when box terminals and round conductor terminals are used.

13.8.2 Connections and phase barriers

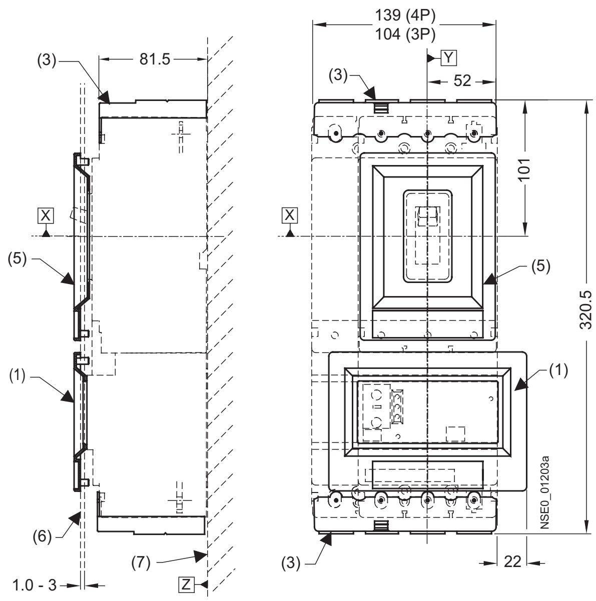


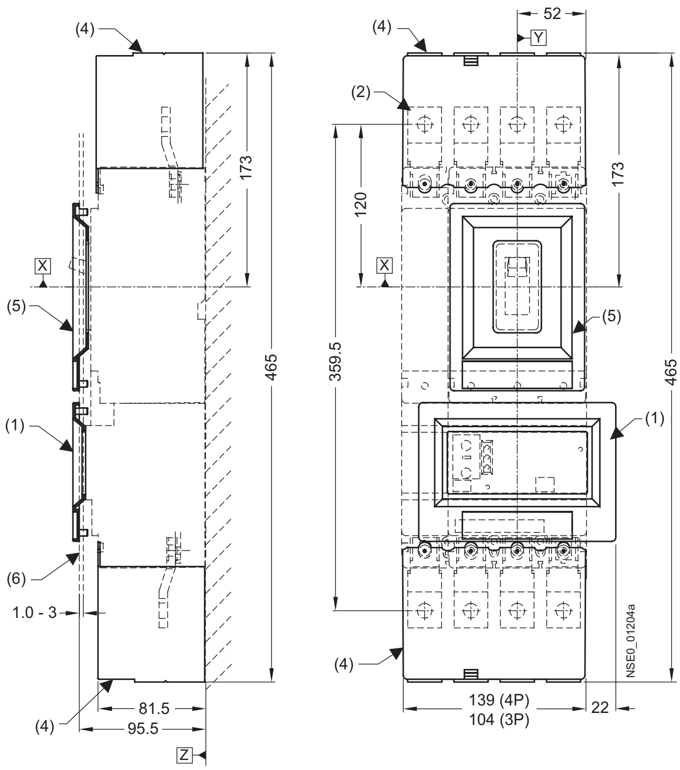


(1) Interphase barrier
(2) Front connecting bars
(3) Terminal covers (standard)
(4) Rear connections (long)
(5) Rear connections (short)
(6) External surface of cabinet door
(7) Mounting level
(8) Rear pad-type terminals (long)
(9) Rear pad-type terminals (short)
(10) Flared busbar extensions
(11) Cover frame for door cutout (for circuit breakers with RCD module)
(12) Cover frame for door cutout (for circuit breakers with toggle handle)

13.8.3 Terminal covers

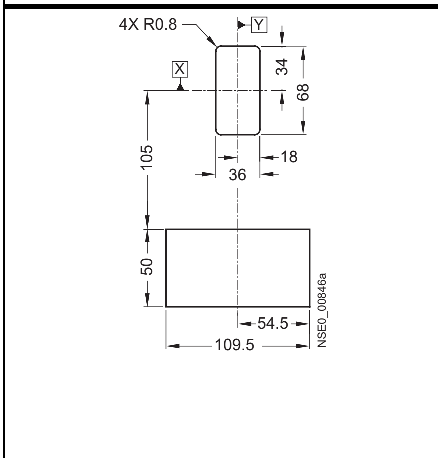
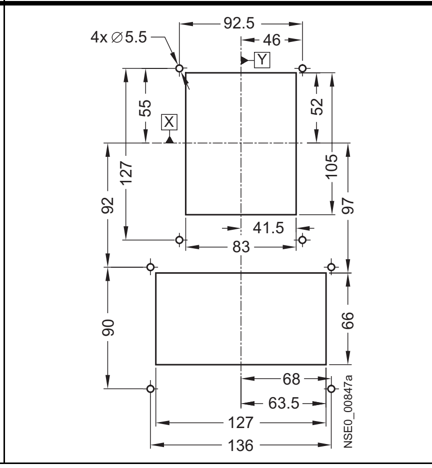
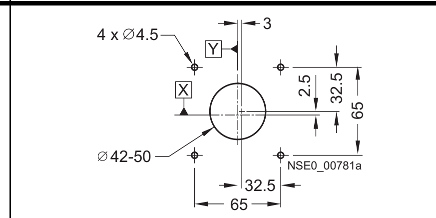
Dimensions of lower cover frame "VL160X (3VL1) with RCD block, 3- and 4-pole, up to 160 A", Terminal covers (Page 279).

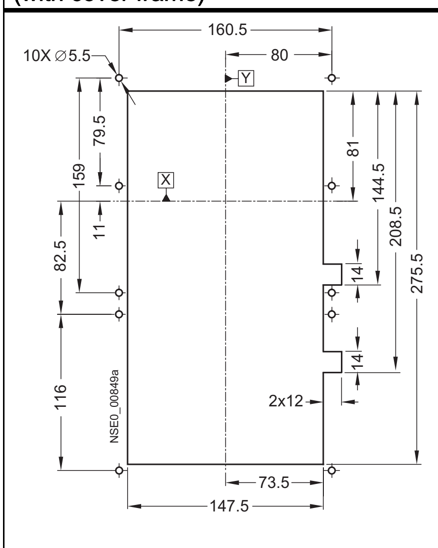
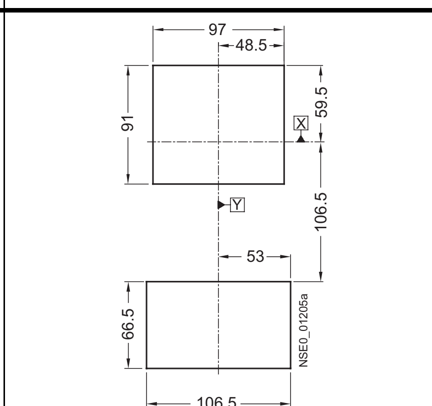
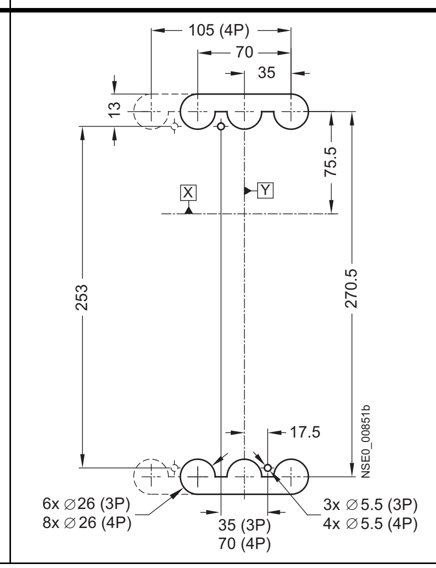




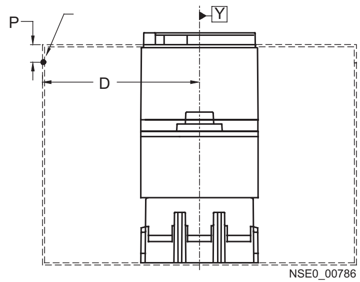
(1) Cover frame for door cutout (for circuit breakers with RCD module)
(2) Front connecting bars
(3) Terminal covers (standard)
(4) Terminal covers (extended)
(5) Cover frame for door cutout (for circuit breakers with toggle handle)
(6) External surface of cabinet door
(7) Mounting level

13.8.4 Door cutouts

Door cutout toggle handle (without cover frame)	Door cutout toggle handle (with cover frame)	Door cutout Door coupling rotary operating mechanism
		

Door cutout front rotary operating mechanism and stored-energy motorized operating mechanism (with cover frame)	Door cutout front rotary operating mechanism (without cover frame)	Drilling template for cutout rear connection bolts
		

Door hinge point (see arrow)



D > A from table + (P × 5)

Note

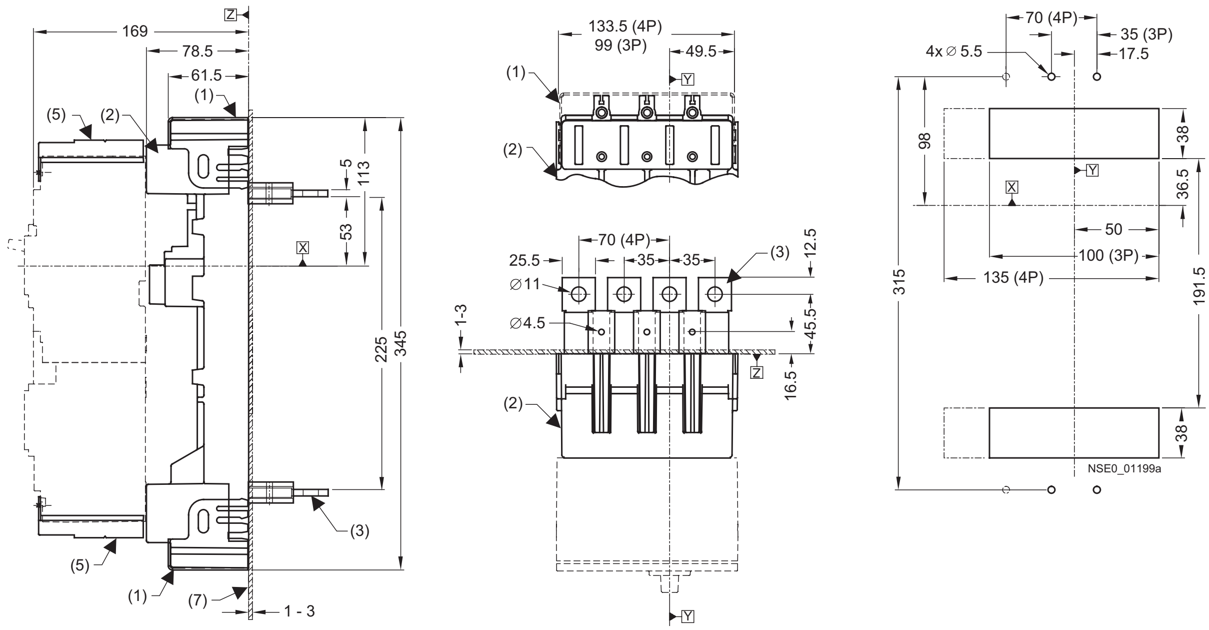
Note:

Door cutouts require a minimum clearance between reference point Y and the door hinge.

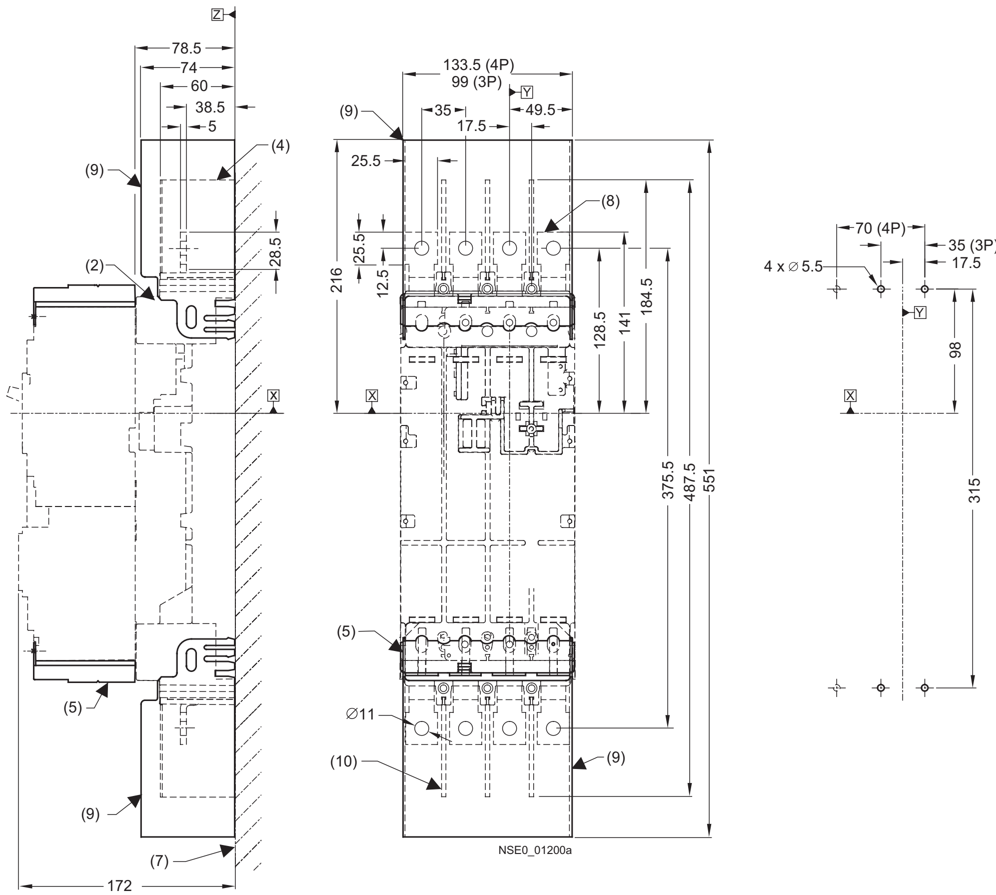
Combination	A
Circuit breaker only	100
Circuit breaker + plug-in socket + stored-energy motorized operating mechanism	100
Circuit breaker + plug-in socket + front rotary operating mechanism	200
Circuit breaker + withdrawable version	200

13.8.5 Plug-in socket and accessories

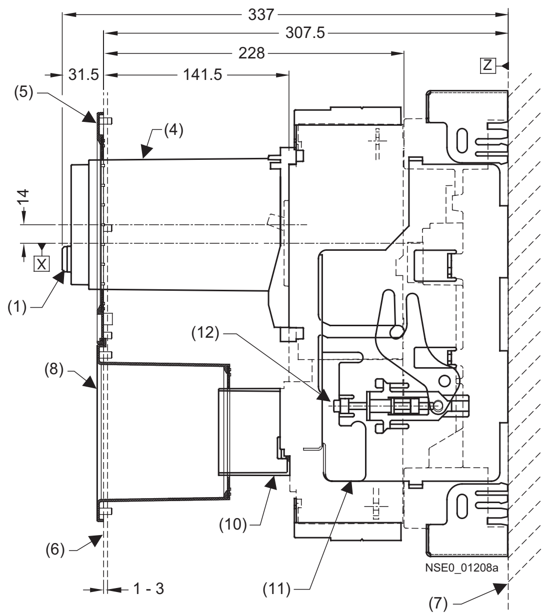
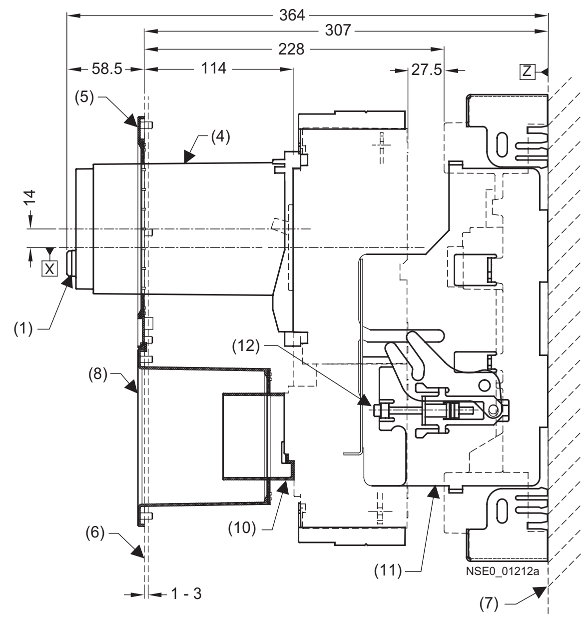
Plug-in socket and accessories with drilling template and cutout for plug-in socket with flat connecting bars on rear

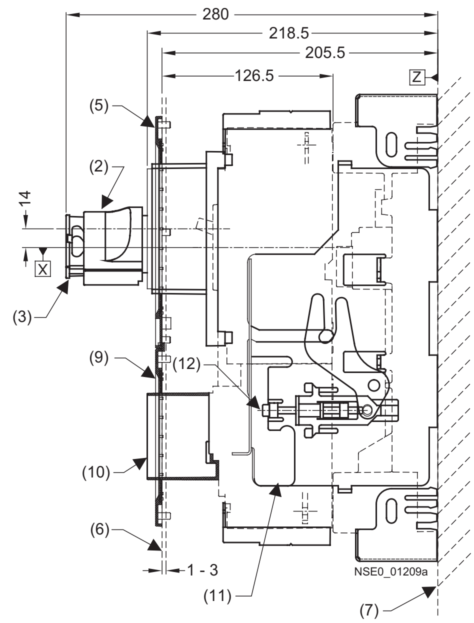
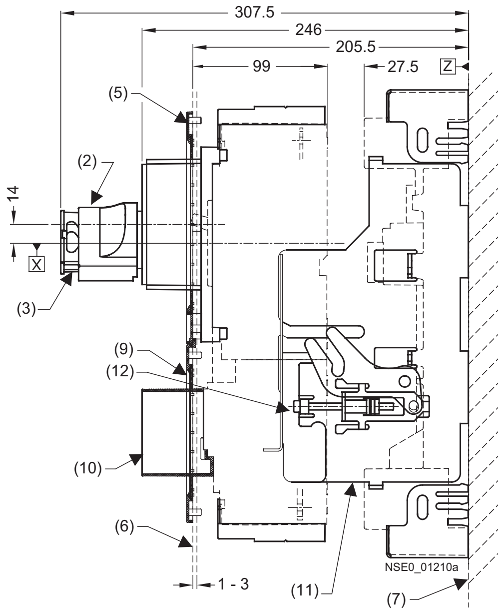


Drilling template and cutout for plug-in socket with flat connecting bars on rear with plug-in socket and accessories

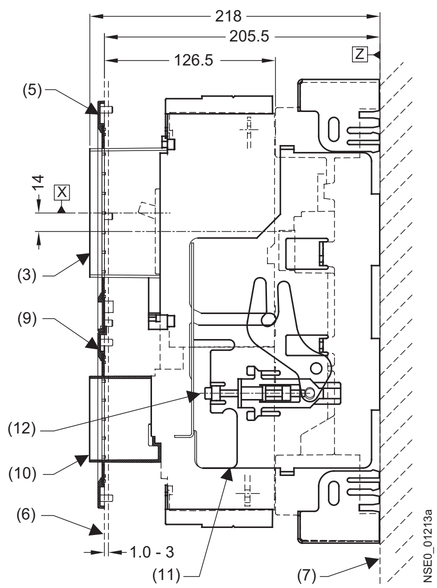
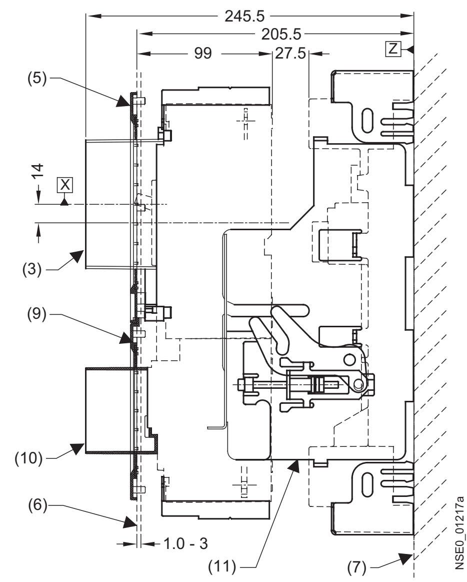


(1)	Plug-in socket with rear terminal covers
(2)	Plug-in socket for circuit breaker with RCD module
(3)	Plug-in socket with rear flat busbar terminals
(4)	Cover frame for door cutout (for circuit breakers with RCD module)
(5)	Terminal cover (standard)
(6)	External surface of cabinet door
(7)	Mounting level
(8)	Plug-in socket with front connecting bars
(9)	Plug-in socket with terminal covers on the front
(10)	Interphase barrier

SENTRON VL160 (3VL2) and VL250 (3VL3) circuit breakers with RCD module and stored-energy motorized operating mechanism (connected position)	SENTRON VL160 (3VL2) and VL250 (3VL3) circuit breakers with RCD module and stored-energy motorized operating mechanism (disconnected position)
	

<table>
<tr><th>SENTRON VL160 (3VL2) and VL250 (3VL3) circuit breakers with RCD module and front rotary operating mechanism (connected position)</th><th>SENTRON VL160 (3VL2) and VL250 (3VL3) circuit breakers with RCD module and front rotary operating mechanism (disconnected position)</th></tr>
<tr><td>
</td><td>
</td></tr>
</table>

(1)	Safety lock
(2)	Front rotary operating mechanism
(3)	Padlock barrier
(4)	Stored-energy motorized operating mechanism
(5)	Cover frame for door cutout (for circuit breakers with operating mechanism)
(6)	External surface of cabinet door
(7)	Mounting level
(8)	Cover frame for door cutout (for circuit breakers with RCD module, motorized operating mechanism)
(9)	Cover frame for door cutout (for circuit breakers with RCD module, toggle handle/rotary operating mechanism)
(10)	RCD extension collar
(11)	Locking device for the racking mechanism
(12)	Racking mechanism

SENTRON VL160 (3VL2) and VL250 (3VL3) circuit breakers with RCD module and extension collar (connected position)	SENTRON VL160 (3VL2) and VL250 (3VL3) circuit breakers with RCD module and extension collar (disconnected position)
	

<table>
<tr><th>Withdrawable version</th><th>Withdrawable version</th><th>Withdrawable version</th></tr>
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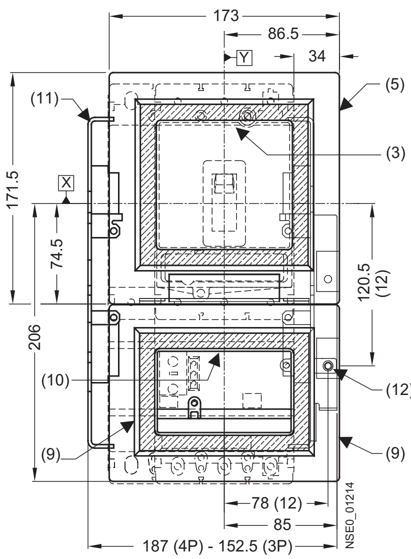

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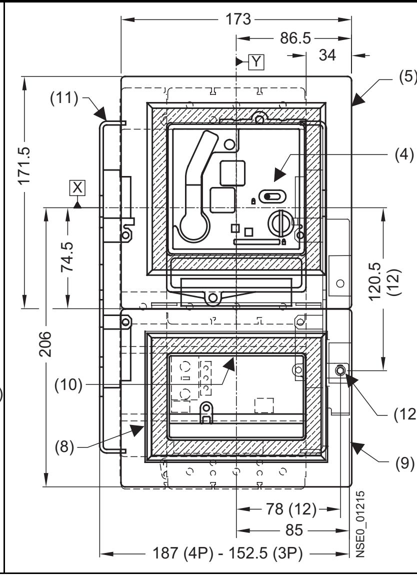

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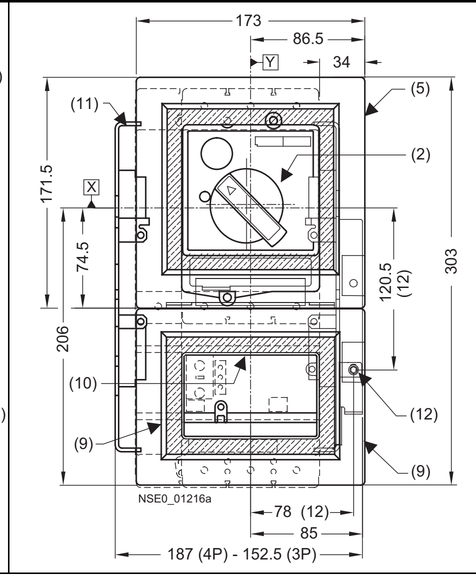


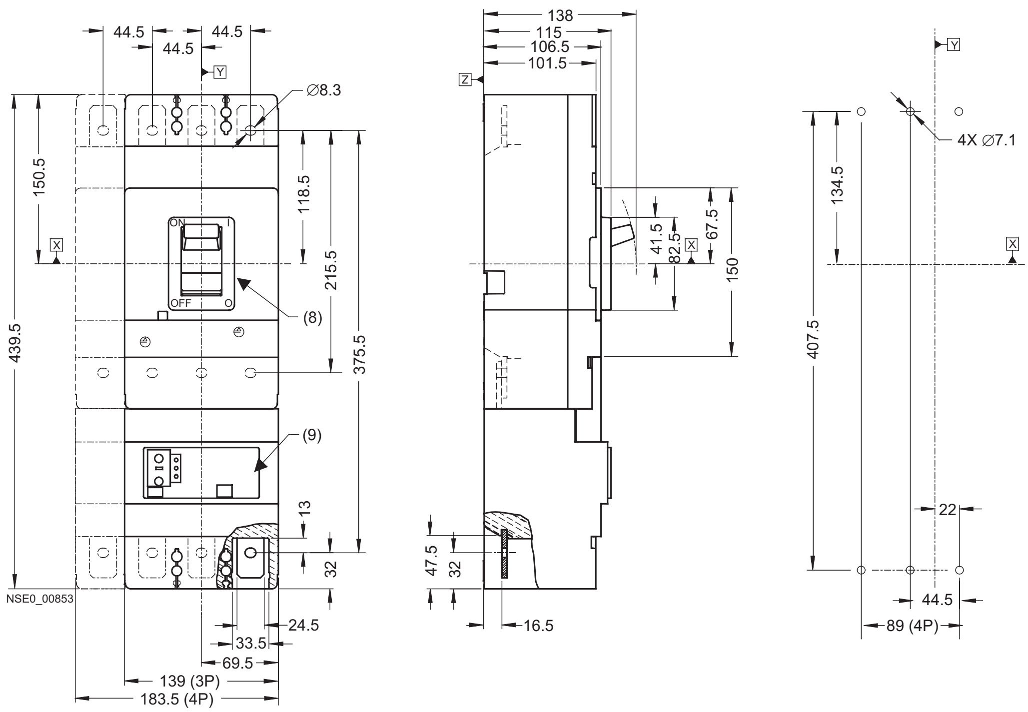
</td></tr>
</table>

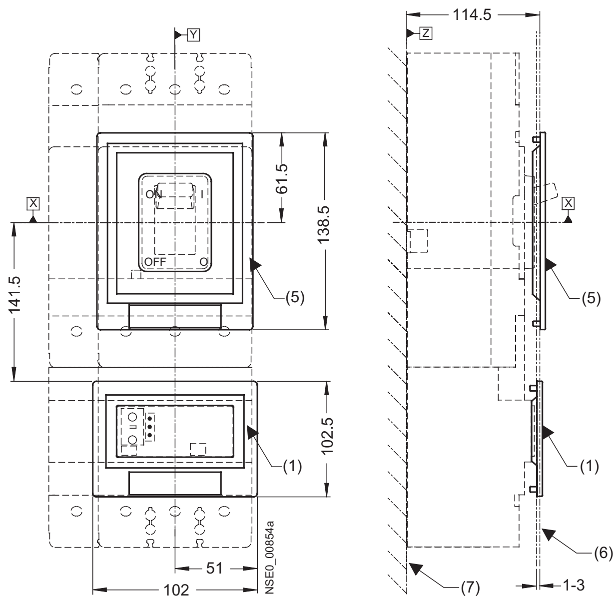
(2)	Front rotary operating mechanism
(3)	Circuit breaker extension collar
(4)	Stored-energy motorized operating mechanism
(5)	Cover frame for door cutout (for circuit breakers with operating mechanism)
(6)	External surface of cabinet door
(7)	Mounting level
(8)	Cover frame for door cutout (for circuit breakers with RCD module, motorized operating mechanism)
(9)	Cover frame for door cutout (for circuit breakers with RCD module, toggle handle/rotary operating mechanism)
(10)	RCD extension collar
(11)	Locking device for the racking mechanism
(12)	Racking mechanism

13.9 VL400 (3VL4) with RCD module, 3- and 4-pole, up to 400 A

13.9.1 Circuit breakers

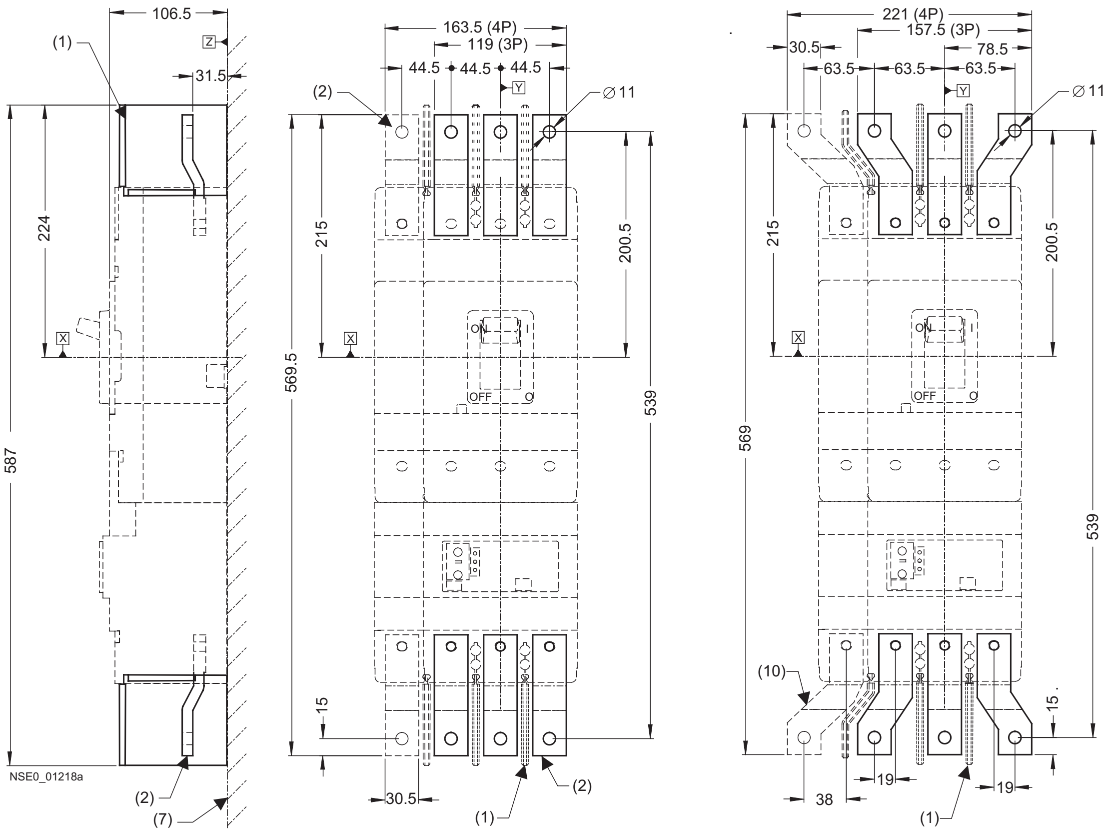
SENTRON VL400 (3VL4) circuit breaker with RCD module and drilling template for mounting the SENTRON VL400 (3VL4) circuit breaker with RCD front connecting bar

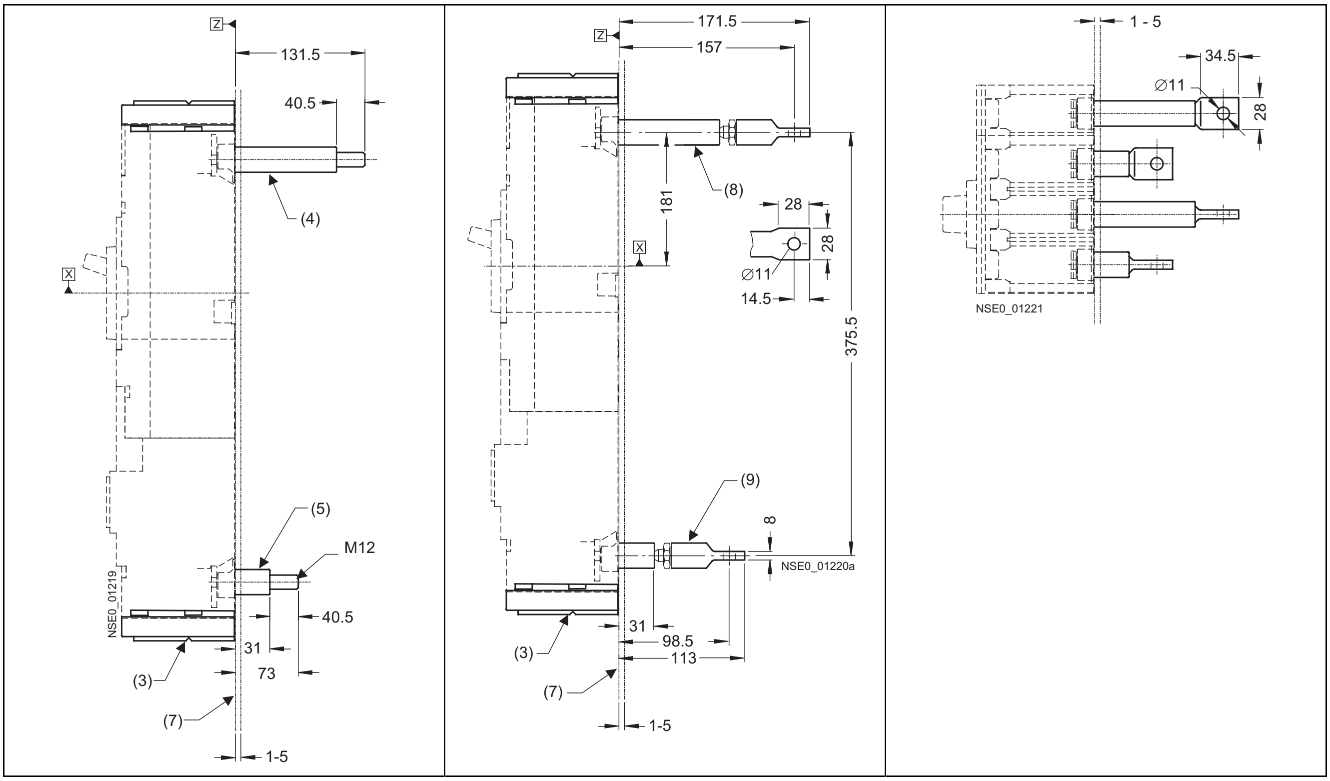




(1)	Cover frame for door cutout (for circuit breakers with RCD module)
(5)	Cover frame for door cutout (for circuit breakers with toggle handle)
(6)	External surface of cabinet door
(7)	Mounting level
(8)	Circuit breakers
(9)	RCD module

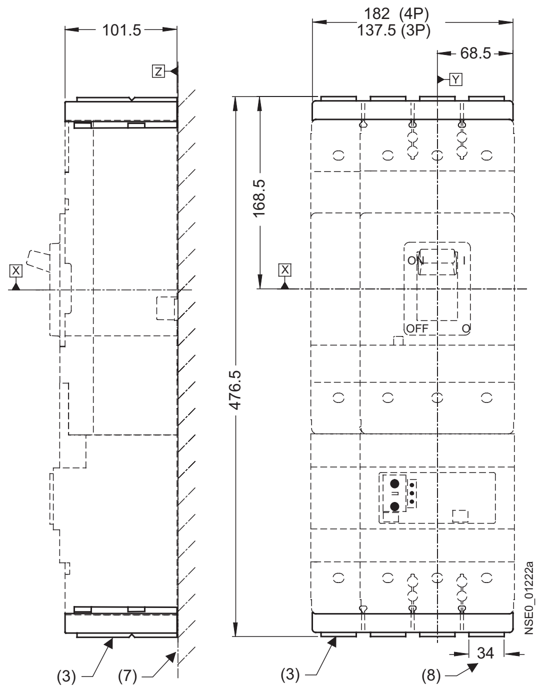
13.9.2 Connections and phase barriers

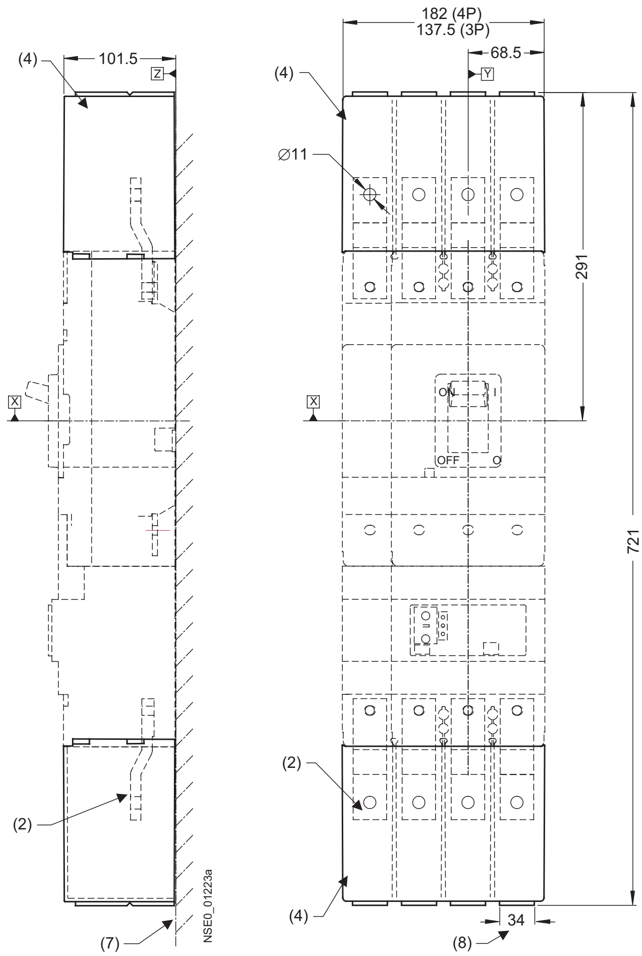




(1) Interphase barrier
(2) Front connecting bars
(3) Terminal covers (standard)
(4) Rear connections (long)
(5) Rear connections (short)
(7) Mounting level
(8) Rear pad-type terminals (long)
(9) Rear pad-type terminals (short)
(10) Flared busbar extensions

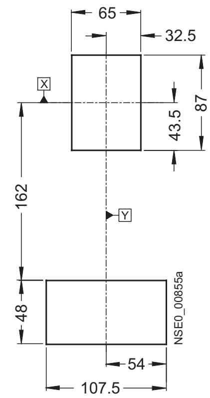
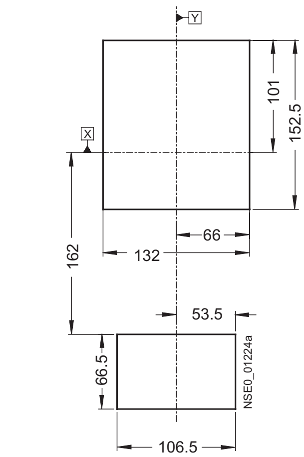
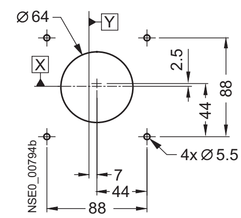
13.9.3 Terminal covers

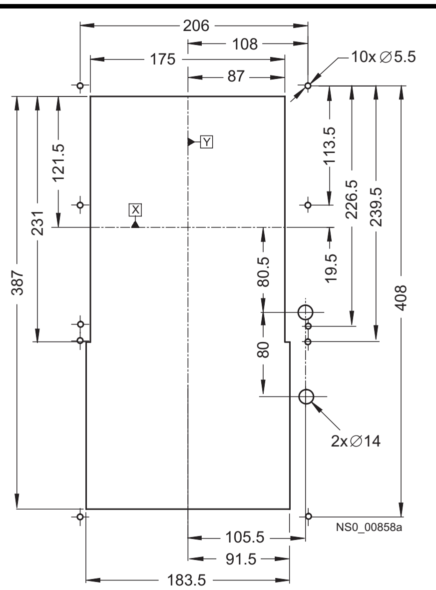
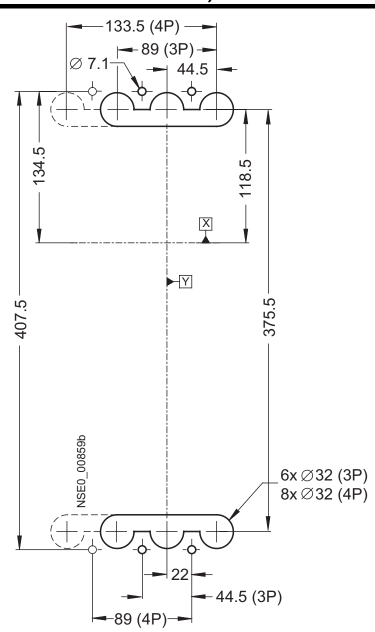
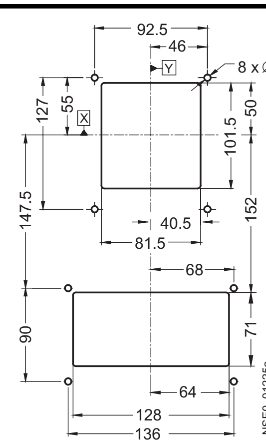




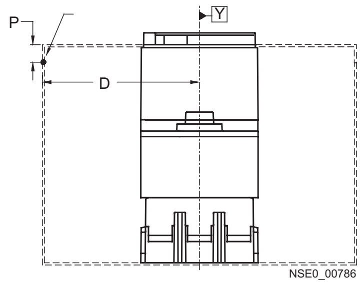
(2)	Front connecting bars
(3)	Terminal covers (standard)
(4)	Terminal covers (extended)
(7)	Mounting level
(8)	Cutout

13.9.4 Door cutouts

Door cutout toggle handle (with cover frame)	Door cutout front rotary operating mechanism (without cover frame)	Door cutout Door coupling rotary operating mechanism
		

Door cutout toggle handle (with cover frame)	Door cutout front rotary operating mechanism (without cover frame)	Door cutout Door coupling rotary operating mechanism
		

Door hinge point (see arrow)



D > A from table + (P × 5)

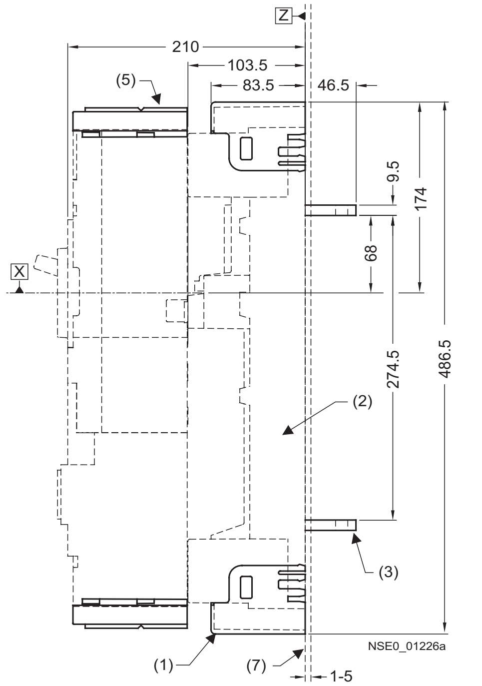
Note

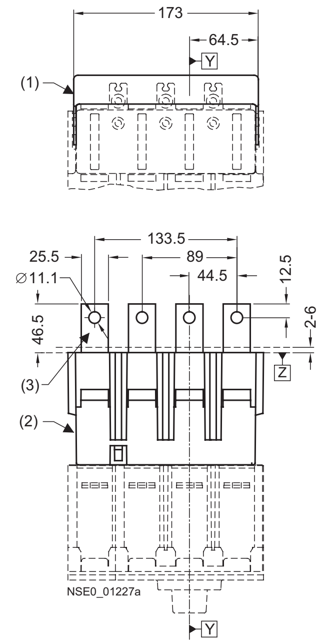
Note:

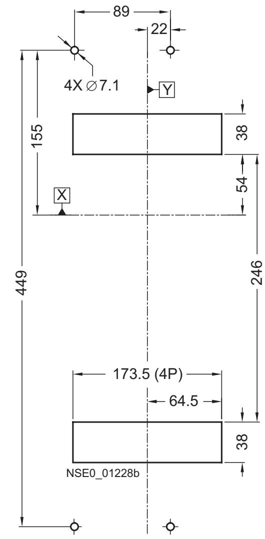
Door cutouts require a minimum clearance between reference point Y and the door hinge.

Combination	A
Circuit breaker only	150
Circuit breaker + plug-in socket + stored-energy motorized operating mechanism	150
Circuit breaker + plug-in socket + front rotary operating mechanism	200
Circuit breaker + withdrawable version	200

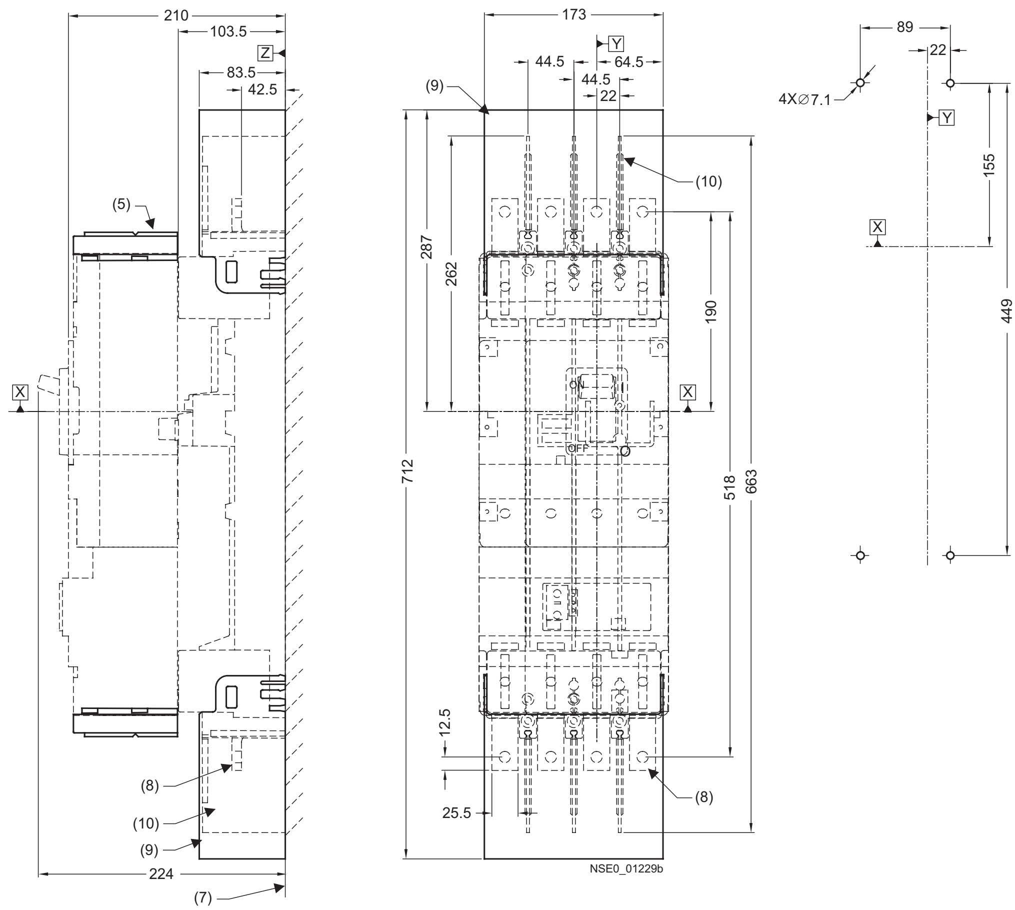
13.9.5 Plug-in socket and accessories



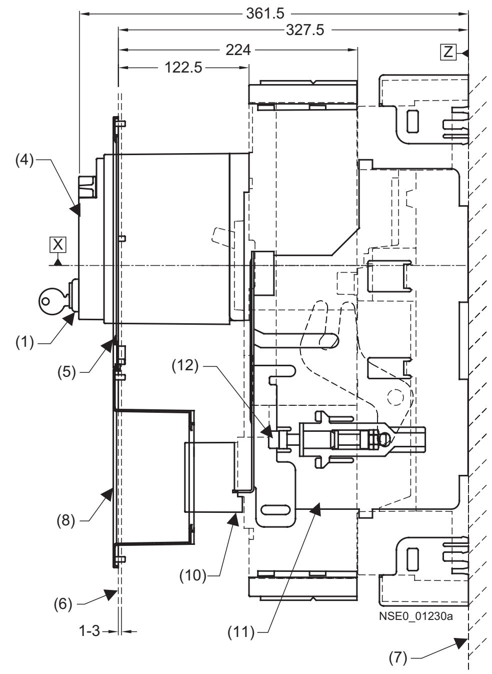
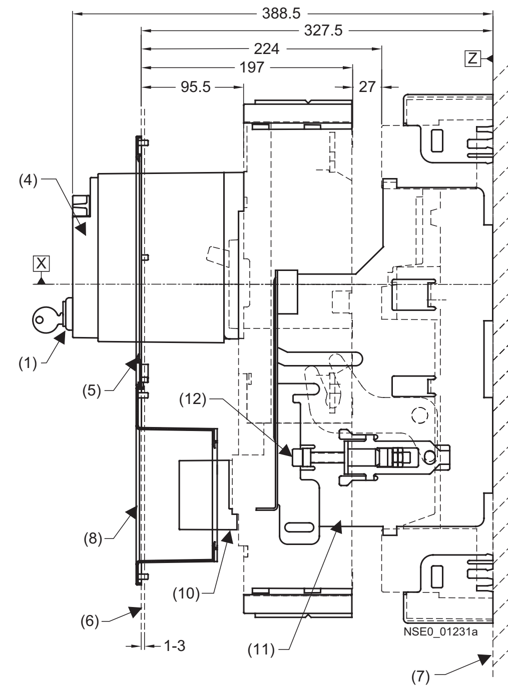


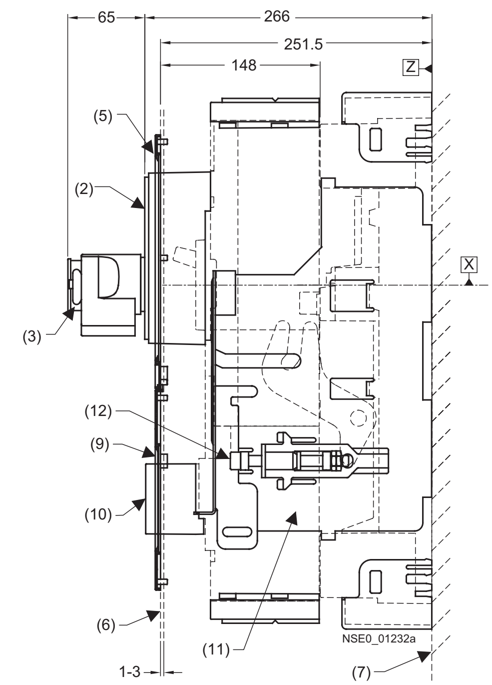
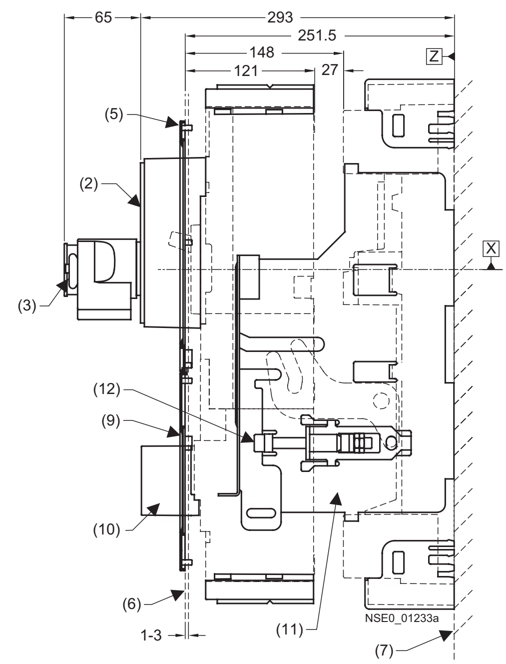


Drilling template and cutout for plug-in socket for rear flat busbar connection

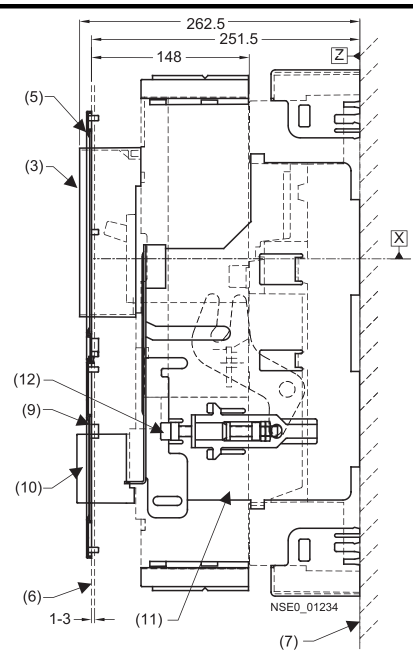


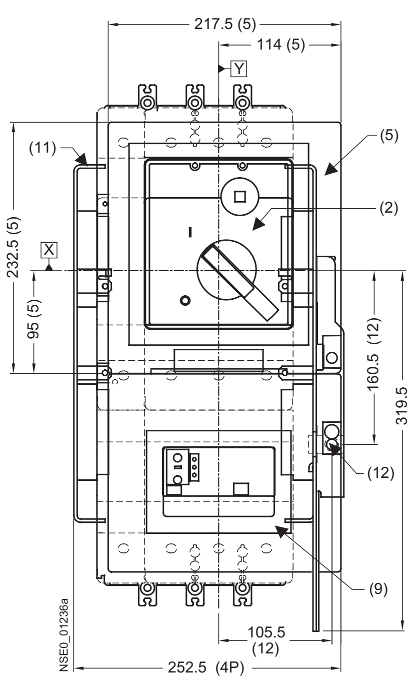
(1)	Plug-in socket with terminal covers
(2)	Socket
(3)	Plug-in socket with rear flat busbar terminals
(5)	Terminal covers (standard)
(7)	Mounting level
(8)	Plug-in socket with front connecting bars
(9)	Plug-in socket with terminal covers on the front
(10)	Interphase barrier

SENTRON VL400 (3VL4) circuit breaker with RCD module, withdrawable version, with stored-energy motorized operating mechanism (connected position)	SENTRON VL400 (3VL4) circuit breaker with RCD module, withdrawable version, with stored-energy motorized operating mechanism (disconnected position)
	

SENTRON VL400 (3VL4) circuit breaker with RCD module, plug-in version, with front rotary operating mechanism (connected position)	SENTRON VL400 (3VL4) circuit breaker with RCD module, plug-in version, with front rotary operating mechanism (disconnected position)
	

(1)	Safety lock
(2)	Front rotary operating mechanism
(3)	Padlock barrier
(4)	Stored-energy motorized operating mechanism
(5)	Cover frame for door cutout (for circuit breakers with operating mechanism)
(6)	External surface of cabinet door
(7)	Mounting level
(8)	Cover frame for door cutout (for circuit breakers with RCD module, motorized operating mechanism)
(9)	Cover frame for door cutout (for circuit breakers with RCD module, toggle handle/rotary operating mechanism)
(10)	RCD extension collar
(11)	Locking device for the racking mechanism
(12)	Racking mechanism

<table>
<tr><th>SENTRON VL400 (3VL4) circuit breaker with RCD module, withdrawable version, with extension collar (connected position)</th><th>SENTRON VL400 (3VL4) circuit breaker with RCD module, withdrawable version, with extension collar (disconnected position)</th></tr>
<tr><td>




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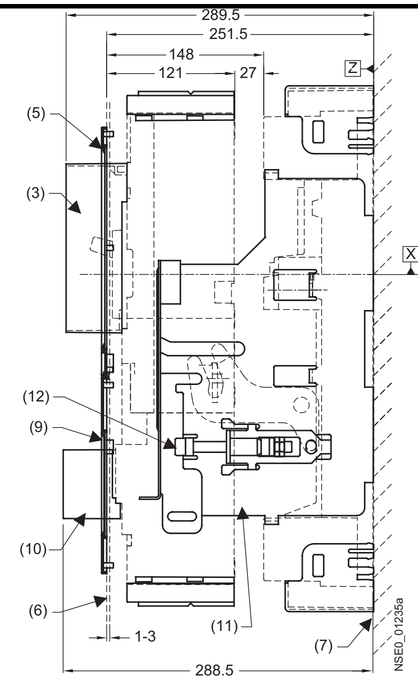
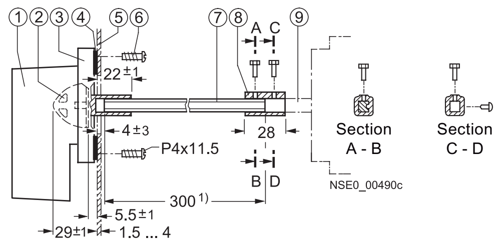
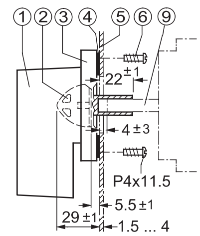
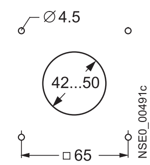


</td></tr>
</table>

(1)	Safety lock
(2)	Front rotary operating mechanism
(3)	Circuit breaker extension collar
(4)	Stored-energy motorized operating mechanism
(5)	Cover frame for door cutout (for circuit breakers with operating mechanism)
(6)	External surface of cabinet door
(7)	Mounting level
(8)	Cover frame for door cutout (for circuit breakers with RCD module, motorized operating mechanism)
(9)	Cover frame for door cutout (for circuit breakers with RCD module, toggle handle/rotary operating mechanism)
(10)	RCD extension collar
(11)	Locking device for the racking mechanism
(12)	Racking mechanism

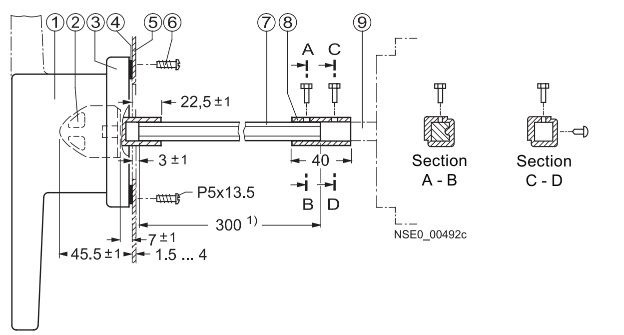
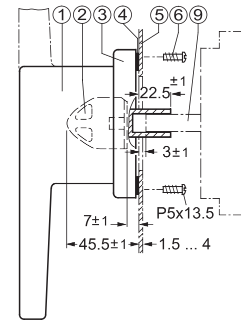
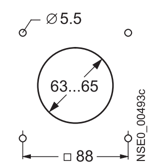
13.10 Door-coupling rotary operating mechanisms 8UC

Door-coupling rotary operating mechanisms 8UC71 and 8UC72, sizes 1 and 2

with extension shaft	without extension shaft	Door cutout with fixing holes
		

(1) Knob
(2) Coupling driver
(3) Cover frame
(4) Seal
(5) Door
(6) Fastening screws, Qty. 4
(7) Extension shaft
(8) Spacer
(9) Actuating shaft of the circuit breaker

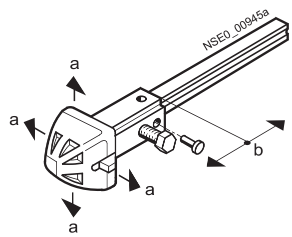
Door-coupling rotary operating mechanisms 8UC73, size 3

with extension shaft	without extension shaft	Door cutout with fixing holes
		

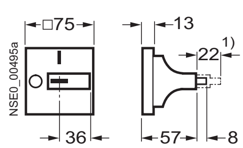
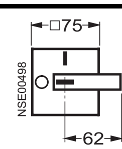
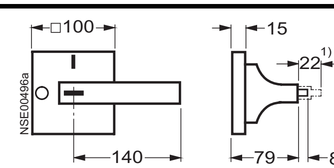
[1] Adjust the length of the extension shaft by reducing the installation depth. Extension shaft also available in 600 mm length.

(1)	Handle or double handle
(2)	Coupling driver
(3)	Cover frame
(4)	Seal
(5)	Door
(6)	Fastening screws, Qty. 4
(7)	Extension shaft
(8)	Spacer
(9)	Actuating shaft of the circuit breaker

Coupling driver 8UC60/8UC70

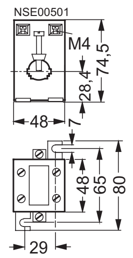
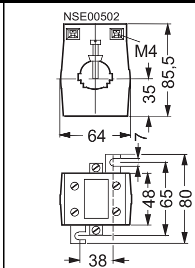
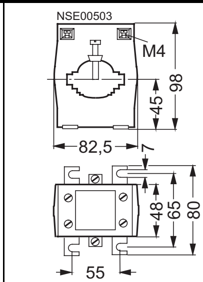
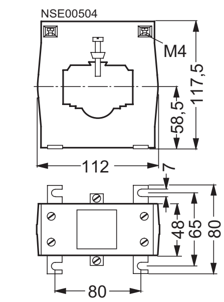
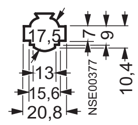
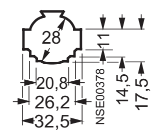
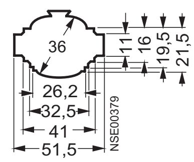
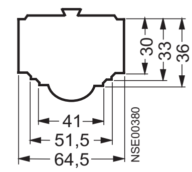


Coupling driver	a	b	Wavelength
with tolerance compensation	+ 5	±5	x
without tolerance compensation	+ 1.5	±2.5	x+23.5

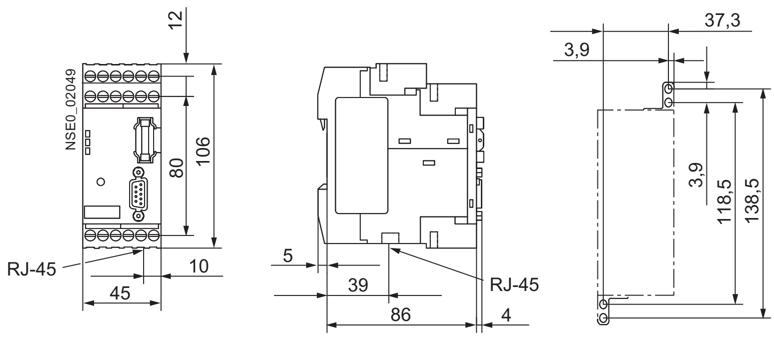
Size 1	Size 2	Size 3
		
Handles with cover frame, sizes 1 to 3		

1) Lock holder of the handle when extended.

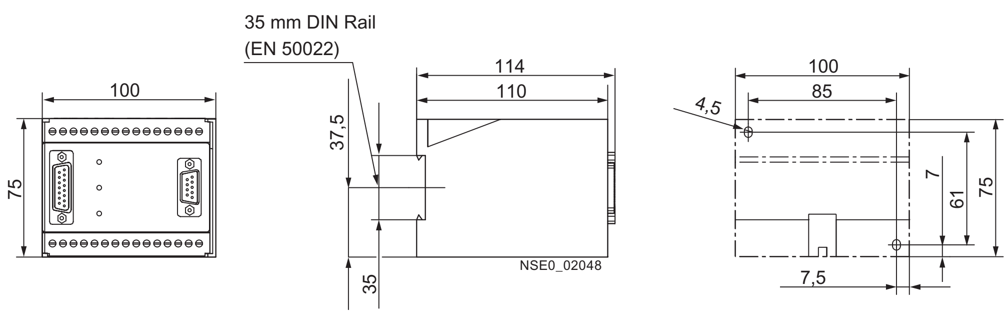
13.11 4NC current transformers for measuring purposes

	4NC51	4NC52		4NC53		4NC54		
4NC current transformers for measuring purposes								
Window openings								
For power rails								
Number	1	1	2	1	2	1	2	3
Width × thickness mm	12 × 5 12 × 10 20 × 5	20 × 5 20 × 10 25 × 5 30 × 5 30 × 10	20 × 5 25 ×	25 × 5 30 × 5 30 × 10 40 × 5 40 × 10 50 × 5 50 × 10	25 × 5 30 × 5 40 × 5	40 × 10 50 × 5 50 × 10 60 × 5 60 × 10	40 × 5 40 × 10 50 × 5 50 × 10 60 × 5 60 × 10	40 × 5 50 × 5 60 × 5
For round conductors								
max. mm	17.5 ∅	28 ∅		36 ∅		45 ∅		

13.12 COM20/COM21 (communications module for SENTRON 3VL)



13.13 COM10/COM 11 (communications module for SENTRON 3VL)



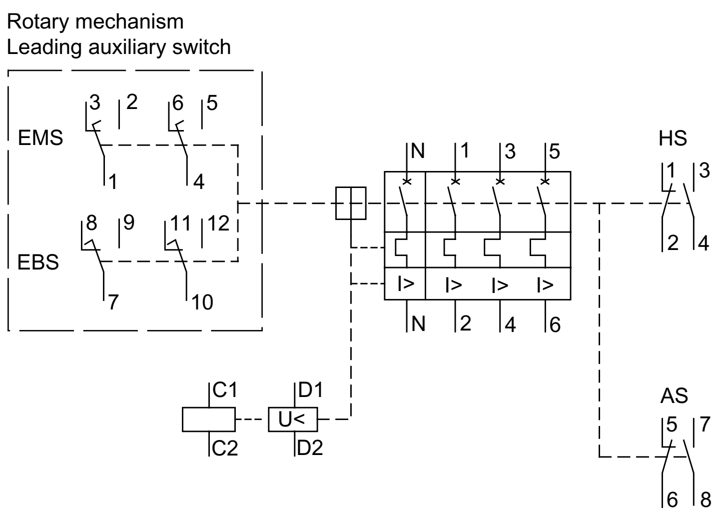
Circuit diagrams

14

The circuit diagram examples below show the most frequent uses of the SENTRON VL circuit breaker:

It is not possible to show combinations here. For versions that differ from those shown, the diagrams must be modified appropriately.

Circuit diagrams are only provided where they are required for improved understanding of the operatin of the device.



HS Auxiliary switches

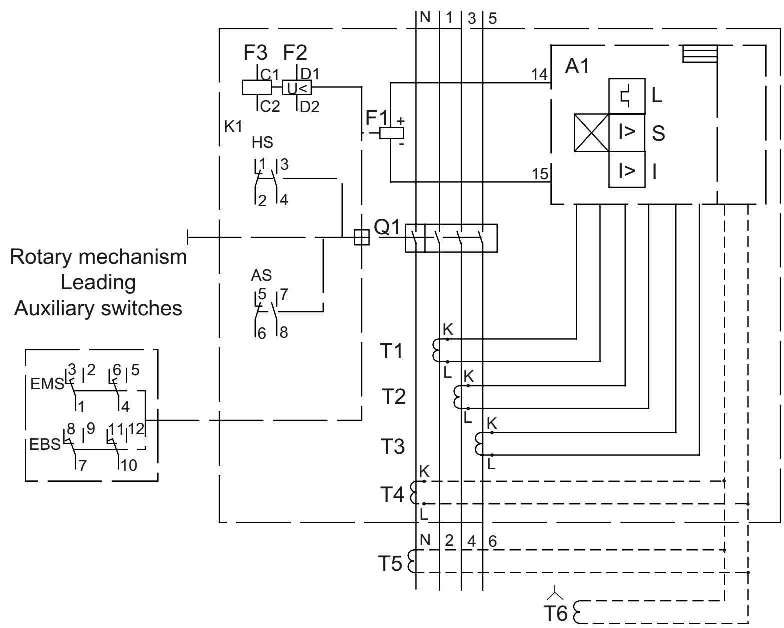
AS Alarm switches

Figure 14-1 Connection diagram for VL160X-VL630

3- and 4-pole circuit breakers for line protection with thermomagnetic overcurrent trip units

Table 14- 1 Terminal assignments for rotary operating mechanism, leading auxiliary switch

Q1	Main contacts	A1	Electronic overcurrent trip unit
F1	Tripping solenoid for A1	F2	Undervoltage release
F3	Shunt release	HS	Auxiliary switches
AS	Alarm switches		
EBS	Leading auxiliary switch ON (integrated into the rotary operating mechanism)		
EMS	Leading auxiliary switch OFF (integrated into the rotary operating mechanism)		
T1 ... T4	Current transformer		

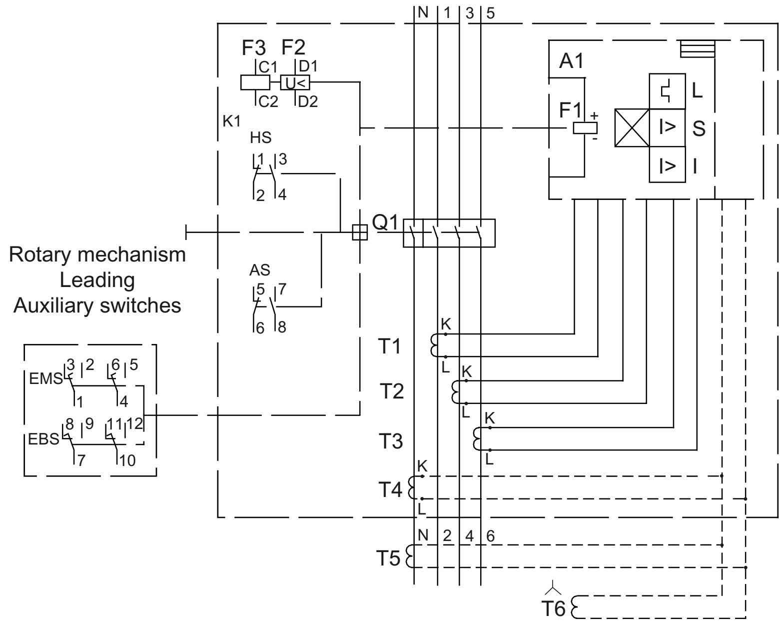


HS Auxiliary switches

AS Alarm switches

Figure 14-2 Internal circuit diagram for VL160-VL250

3- and 4-pole circuit breakers for line and motor protection with electronic overcurrent trip units



HS	Auxiliary switches
AS	Alarm switches

Figure 14-3 Internal circuit diagram for VL400 circuit breaker for motor protection, and VL400-VL1600

3- and 4-pole circuit breakers for line protection with electronic trip units

Circuit diagrams VL1 to 3, with or without undervoltage release

Below are the circuit diagrams for the motorized operating mechanism without stored energy for the circuit breakers VL160X, VL160 and VL250. The functions of the motorized operating mechanisms are described in Chapter ....

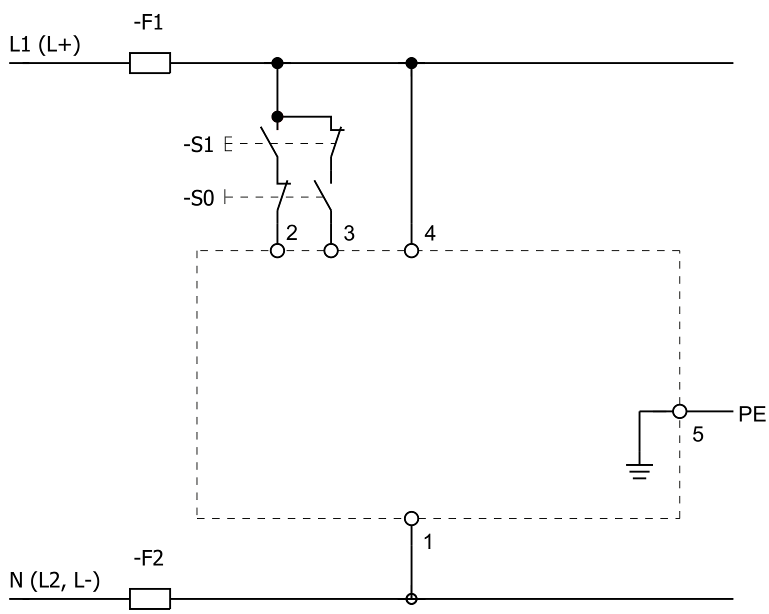


Figure 14-4 Motorized operating mechanism without undervoltage release

Table 14- 2 Motorized operating mechanism without stored energy for VL160X-VL250, without undervoltage release

S0	OFF (to be provided by customer)	S1	ON (to be provided by customer)
-F1, -F2	Control circuit fuse	PE	Protective grounding

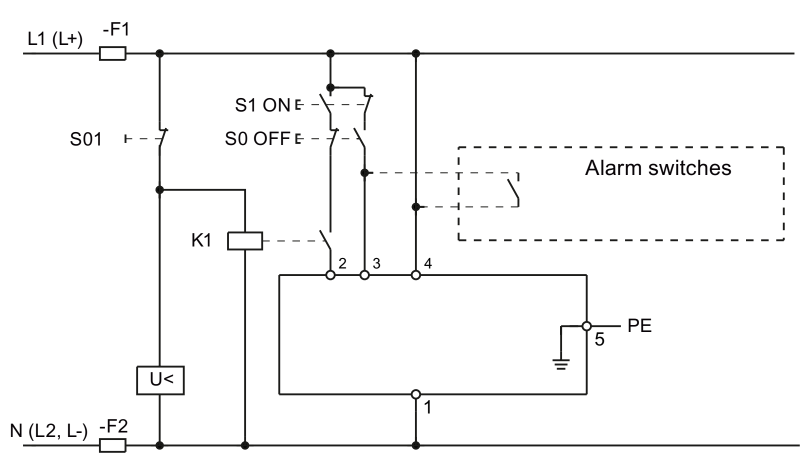


Figure 14-5 Motorized operating mechanism with undervoltage release

Table 14- 3 Motorized operating mechanism without stored energy for VL160X-VL250, without undervoltage release

S0	OFF (to be provided by customer)	S1	ON (to be provided by customer)
S2	Auto/manual selector switch	U<	Undervoltage release
-F1, -F2	Control circuit fuse	S01	Remote command (to be provided by customer)
K1	Auxiliary contactor (to be provided by customer)	PE	Protective grounding

---

Note

Automatic reset/close

A separate alarm switch contact (7-8) can be connected for automatic reset after tripping. To prevent a fault occurring in the protected circuit, automatic closing of a tripped circuit breaker is not recommended.

The contact of auxiliary contactor K1 or K3 prevents no-load operation of the circuit breaker when the undervoltage release "<U" is without power. No load operations represent a high level of stress for the circuit breaker. If the undervoltage release is without power, auxiliary contactor K1 or K3 has not picked up. The contact in the ON circuit (control circuit) of the motorized operating mechanism is thus not closed, that is, the circuit breaker cannot be switched.

This auxiliary contactor is not necessary in principle if the undervoltage release is supplied uninterrupted (e.g. pushbutton S01) from the same source as the motorized operating mechanism itself (e.g. contact 4).

Circuit diagrams VL1 to 6, with or without undervoltage release

Below are the circuit diagrams for the stored-energy motorized operating mechanism for the circuit breakers VL160X, VL160, VL250, VL400, VL630 and VL800. The functions of the motorized operating mechanisms are described in Chapter ....

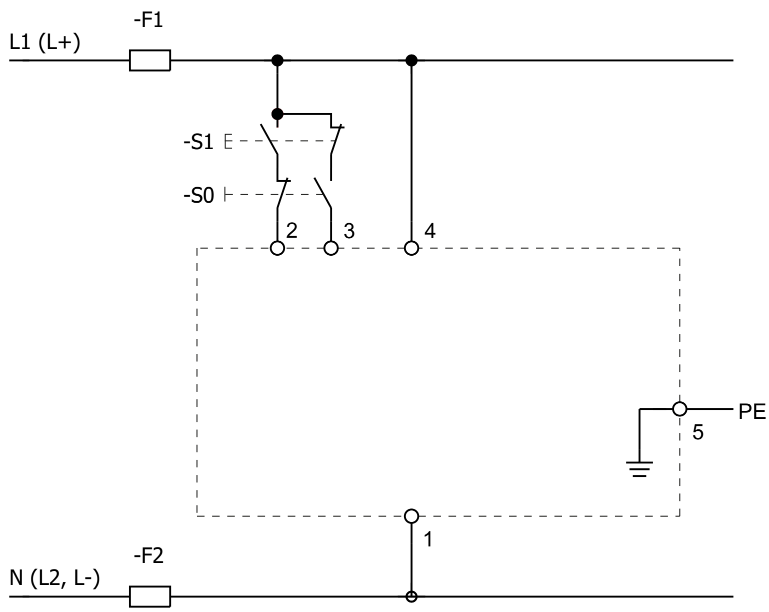


Figure 14-6 Motorized operating mechanism without undervoltage release

Table 14- 4 Stored-energy motorized operating mechanism for VL160X, VL160, VL250, VL400, VL630 and VL800 without undervoltage release

S0	AUS	S1	ON
-F1, -F2	Control circuit fuse	PE	Protective grounding

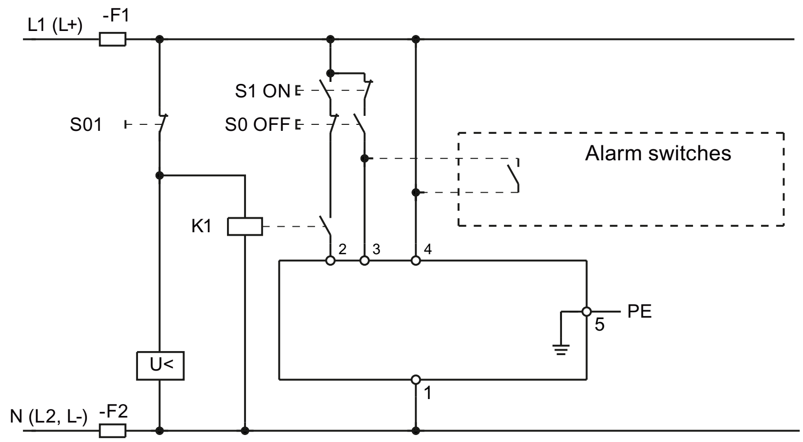


Figure 14-7 Stored-energy motorized operating mechanism

Table 14- 5 Stored-energy operator for VL160X, VL160, VL250, VL400, VL630 and VL800 with undervoltage release

S0	OFF (to be provided by customer)	S1	ON (to be provided by customer)
S01	Remote command (to be provided by customer)	-F1, -F2	Control circuit fuse
K1	Auxiliary contactor (to be provided by customer)	U<	Undervoltage release
PE	Protective grounding		

Note

Automatic charging/close

A separate alarm switch contact (7-8) can be connected for automatic charging after tripping. Automatic switching on of a circuit breaker must be prevented, otherwise a short-circuit could automatically occur after a tripping event.

The contact of auxiliary contactor K1 or K3 prevents no-load operation of the circuit breaker when the undervoltage release "<U" is without power. No load operations represent a high level of stress for the circuit breaker. If the undervoltage release is without power, auxiliary contactor K1 or K3 has not picked up. The contact in the ON circuit (control circuit) of the motorized operating mechanism is thus not closed, that is, the circuit breaker cannot be switched.

This auxiliary contactor is not necessary in principle when the undervoltage release is supplied uninterrupted (e.g. pushbutton S01) from the same source as the motorized operating mechanism itself (e.g. contact 4).

Circuit diagrams VL7 and VL8, with or without undervoltage release

Below are the circuit diagrams for the motorized operating mechanism for the circuit breakers VL1250 und VL1600. The functions of the motorized operating mechanisms are described in the Chapter "Product description, motorized operating mechanisms".

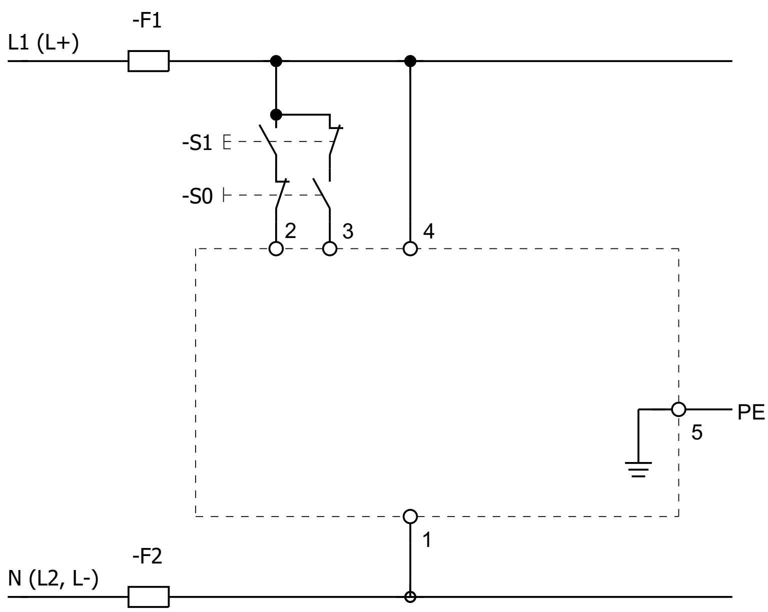


Figure 14-8 Motorized operating mechanism without undervoltage release

Table 14- 6 Motorized operating mechanism for VL1250 and VL1600 without undervoltage release

S0	OFF (to be provided by customer)	S1	ON (to be provided by customer)
-F1, -F2	Control circuit fuse	PE	Protective grounding

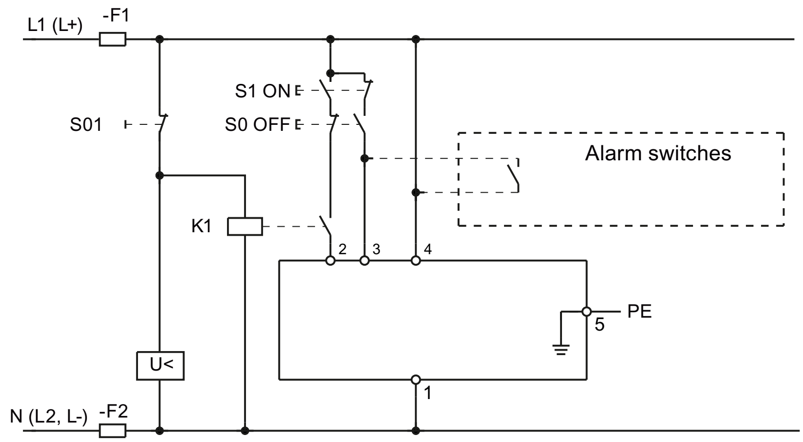


Figure 14-9 Motorized operating mechanism with undervoltage release

Table 14- 7 Motorized operating mechanism for VL1250 and VL1600 wit undervoltage release

S0	OFF (to be provided by customer)	S1	ON (to be provided by customer)
S01	Remote command	K1	Auxiliary contactor
-F1, -F2	Control circuit fuse	U<	Undervoltage release
PE	Protective grounding		

Note

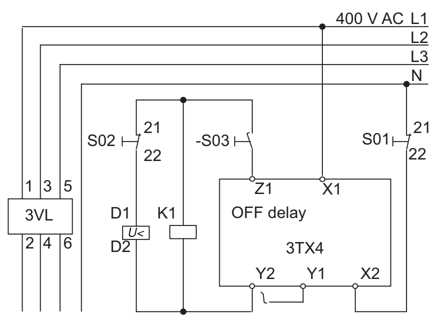
Automatic reset/close

A separate alarm switch contact (7-8) can be connected for automatic reset after tripping. To prevent a fault occurring in the protected circuit, automatic closing of a tripped circuit breaker is not recommended.

The contact of auxiliary contactor K1 or K3 prevents no-load operation of the circuit breaker when the undervoltage release "<U" is without power. No load operations represent a high level of stress for the circuit breaker. If the undervoltage release is without power, auxiliary contactor K1 or K3 has not picked up. The contact in the ON circuit (control circuit) of the motorized operating mechanism is thus not closed, that is, the circuit breaker cannot be switched.

This auxiliary contactor is not necessary in principle when the undervoltage release is supplied uninterrupted (e.g. pushbutton S01) from the same source as the motorized operating mechanism itself (e.g. contact 4).

Undervoltage release and shunt release for VL160X to VL1600



S01 Delayed release

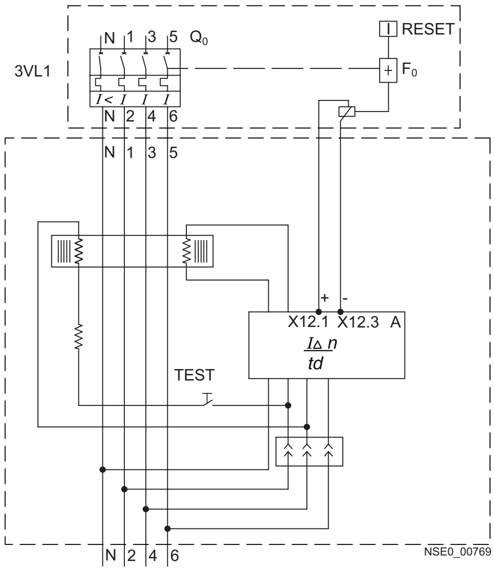
S02 Instantaneous release for EMERGENCY-OFF loop (if required)

S03 Leading auxiliary contact, e.g. 3VL9300-3AS10 "OFF to ON" in the front rotary operating mechanism of the circuit breaker (if required)

K1 Auxiliary contactor 3RH11 (if required)

Figure 14-10 Delay unit (3TX4701-0A) for undervoltage release for VL160X to VL1600

Protective circuit with UVR (220 V to 250 V DC)	Tripping time UVR
Y2 only	3 seconds
Y2 and Y1 bridged	6 seconds



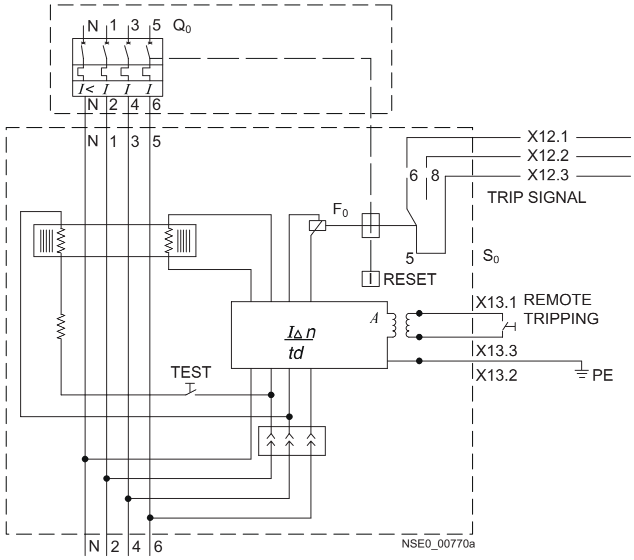
Q_0	Circuit breakers
A	Evaluation electronics
F_0	Closing solenoid with local trip indication and reset
TEST	Test button

Figure 14-11 4-pole 3VL1 with RCD module (shown: 3-pole version is similar but without N pole)

Table 14- 8 4-pole 3VL1 with RCD module

Q_0	Circuit breakers	A	Evaluation electronics
F_0	Tripping solenoid with local trip indication and reset	Test	Test button

shown: 3-pole version is similar but without N pole)



Q_0	Circuit breakers
A	Evaluation electronics
F_0	Closing solenoid with local trip indication and reset
TEST	Test button
S_0	Remote trip (to be provided by customer)

Figure 14-12 4-pole circuit breaker for VL160, VL1250, VL400 circuit breakers with remote trip unit and RCD alarm switch (3-pole version is similar but without N pole)

Table 14- 9 4-pole circuit breaker for VL160, VL1250, VL400

Q_0	Circuit breakers	A	Evaluation electronics
F_0	Tripping solenoid with local trip indication and reset	Test	Test button
S_0	Remote trip (to be provided by customer)		

Circuit breaker with remote trip and RCD alarm switch

3-pole version is similar but without N pole)

Spare parts/accessories 15

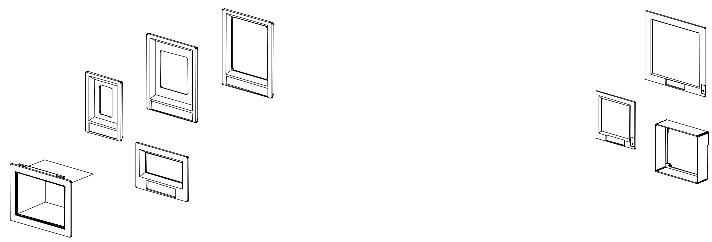
15.1 Installation

The following safety accessory parts are available for installing in the SENTRON VL circuit breaker:

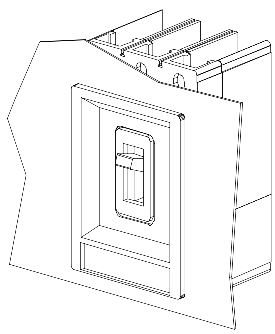
- Cover frames for door cutouts
- Terminal covers/phase barriers
- Phase barriers
- Toggle handle extension

Cover frames for door cutouts:

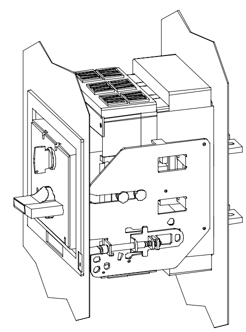
Cover frames for door cutouts are used to increase the IP degree of protection of the circuit breakers and to better adapt them to the control cabinets. Cover frames for door cutouts are available for fixed-mounted, plug-in and withdrawable circuit breakers with rotary operating mechanisms, motorized operating mechanisms and RCD modules. The cover frames for door cutouts are attached to the door with 4 fixing elements.



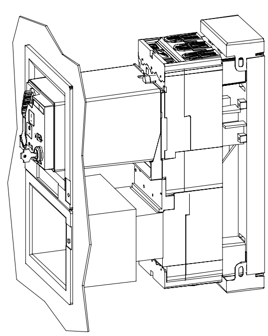
Cover frames for door cutouts



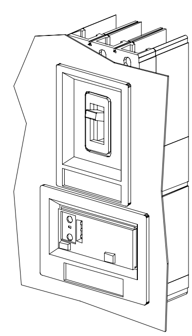
3VL9300-8BC00 (front)



3VL9300-8BG00



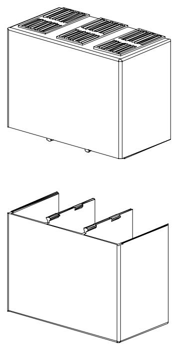
3VL9300-8BC00



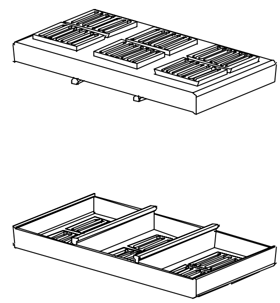
3VL9300-8BJ00 / 3VL9300-8BD00

Terminal covers/phase barriers:

Sealable terminal covers can be installed on the input and output side of the SENTRON VL circuit breakers. They offer degree of protection IP30 for fixed-mounted or withdawable circuit breakers in the connected position. In addition, extended terminal covers provide separation between the phases if uninsulated busbars or cables are used



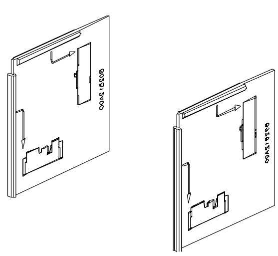
Extended terminal cover



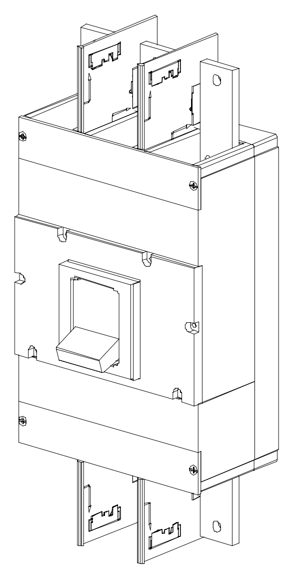
Standard terminal cover

Phase barriers

Phase barriers provide insulation on the input and output side of the circuit breaker. They can be mounted in the specially formed slots on the input and output sides of the circuit breaker. They can be used in conjunction with other connection accessories (except terminal covers). The phase barriers can be used with fixed-mounted, plug-in and withdrawable circuit breakers. Terminal covers must be used if the circuit breakers are mounted immediately next to each other (see the Section Mounting and safety clearances).



Phase barriers

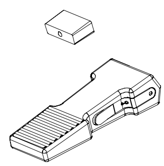


Use of phase barriers

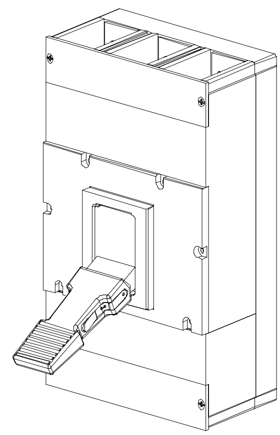
Toggle handle extension

Toggle handle extensions enable user-friendly operation of the circuit breaker toggle handle.

- VL 160X to VL 400: Toggle handle extension not necessary
- VL 630 to VL 800: possible as option
- VL 1250 to VL 1600: possible as option



Toggle handle extension



Use of toggle handle extension

15.2 Electromechanical components

The following electromechanical accessory parts are available for the SENTRON VL circuit breaker:

- Position signaling switch
- Auxiliary conductor plug-in system
- Leading auxiliary switch

Position signaling switch

When a circuit breaker is mounted in a withdrawable or plug-in assembly, the position signaling switch, which is equipped wtih a changeover contact, is used to indicate whether the circuit breaker is in the connected or withdrawn position. Two position signaling switches can be mounted in each withdrawable or plug-in base.

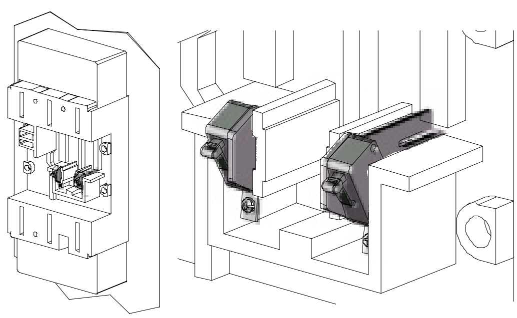


Figure 15-1 Position signaling switch

Auxiliary conductor plug-in system

If a SENTRON VL circuit breaker is installed in a withdrawable or plug-in assembly, the auxiliary conductor connection system connects the internal and external accessories (e.g. auxiliary switch and alarm switch, shunt release, undervoltage release, motorized operating mechanisms) to the terminals on the plug-in socket.

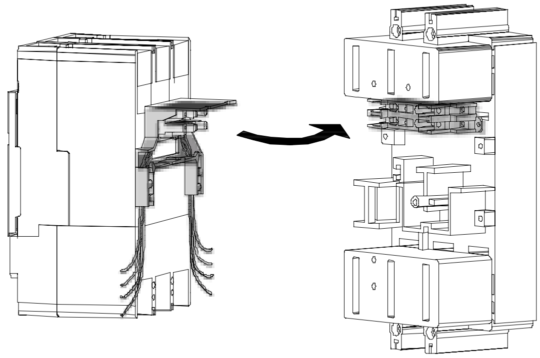


Figure 15-2 Auxiliary conductor plug-in system

This plug-in connection enables two indentically equipped and wired circuit breakers to be interchanged easily. Each socket contains 8 terminals.

The VL160X, VL160, VL250 circuit breakers can be equipped with two sockets or a total of 16 terminals.

The VL400, VL630, VL800, VL1250 and VL1600 circuit breakers with 3 sockets or 24 terminals.

Leading auxiliary switches for switching on and off

The leading auxiliary switches (changeover switches) are available as accessories for front rotary operating mechanisms and door-coupling rotary operating mechanisms.

The following applications are possible:

- Leading auxililary switch for switching from "ON" to "OFF"
- Leading auxililary switch for switching from "OFF" to "ON"

Each version, leading auxiliary switch for switching on and off, can be equipped with one or two changeover switches. The connecting cables of the auxiliary switches are 1.5 m long.

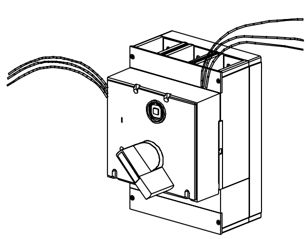


Figure 15-3 Rotary operating mechanism with leading auxiliary switches

15.3 Mechanical components

The following mechanical accessory parts are available for the SENTRON VL circuit breaker:

- Locking options for the guide frame
- Guide frame crank handle
- Trip-to-test button

Locking options for the guide frame

Locking option for the guide frame device support:

The guide frame device support for the SENTRON VL circuit breakers can be locked with up to 3 padlocks (shackles from 4 to 8 mm Ø, padlocks not supplied). The circuit breaker is prevented from moving from the connected to the disconnected position if the device support is secured with a padlock.

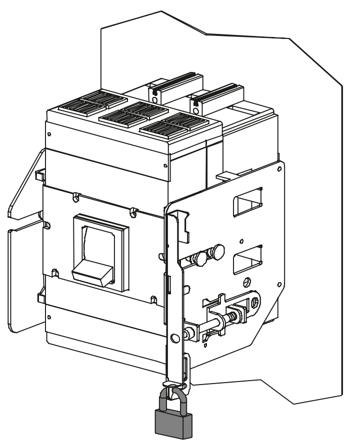


Figure 15-4 Locking the guide frame

Guide frame crank handle

Crank handle for the guide frame:

This crank handle is used to move the circuit breaker into the operating or the disconnected position.

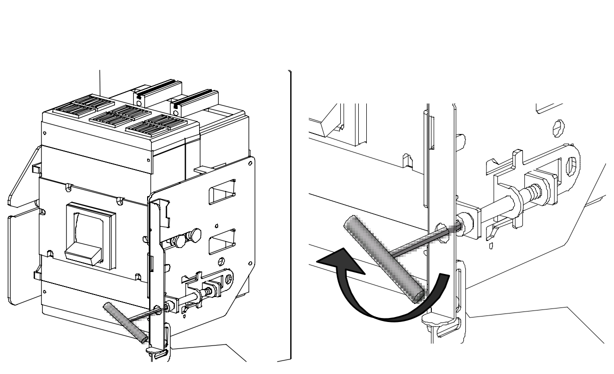


Figure 15-5 Guide frame crank handle

Trip-to-test button

The SENTRON VL circuit breakers are equipped with trip-to-test buttons. When the circuit breaker is in the "ON" position, the user may test the tripping function mechanically by pressing the trip-to-test button. The circuit breaker can be reset afterwards.

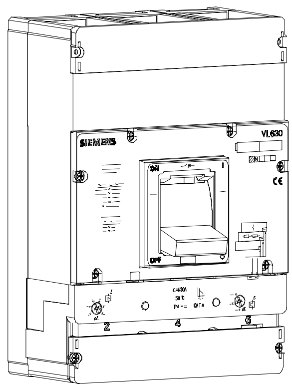


Figure 15-6 Trip-to-test button

15.4 Electrical/electronic engineering

The following electrical/electronic accessories are available for the SENTRON VL circuit breaker:

- Portable tester

Portable tester

The portable tester is used as a local test device for SENTRON VL circuit breakers with electronic trip units. It can also be used as an external voltage supply for the electronic trip unit (ETU and LCD-ETU). The portable tester is powered by three 9-volt batteries (included with device). An optional external voltage supply can also be supplied.

Test functions:

- Current transfomer test, only for the LCD-ETU (ETU40M, ETU40 and ETU42)
- Test release

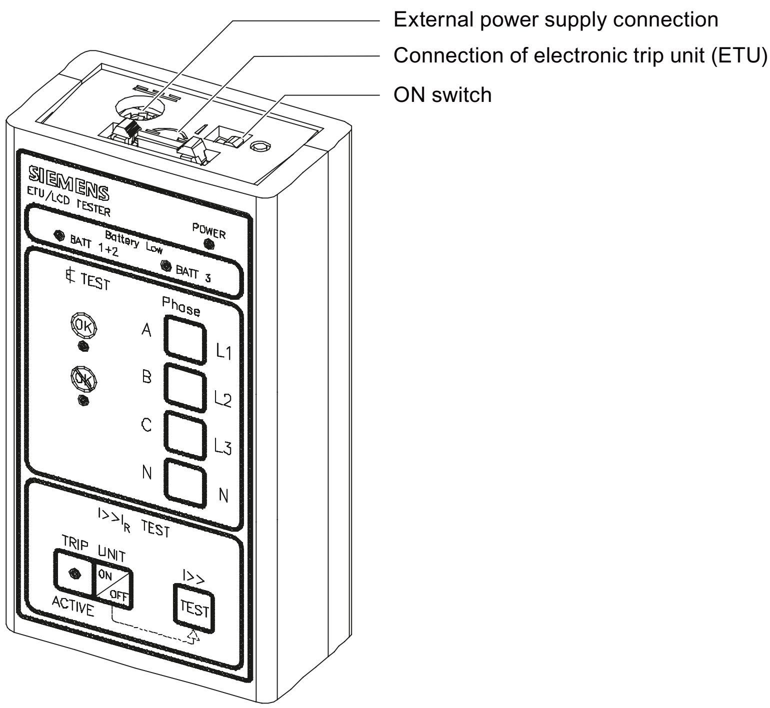


Figure 15-7 Portable tester

Battery supply device

The handheld tester for electronic trip units is used as a local test device for the SENTRON VL circuit breakers with electronic trip unit, and it can be used as an external voltage supply for the electronic trip units (ETU and LCD-ETU). The portable battery power supply is fed by two commercially available 9-V block batteries.

Test function:

- Test release

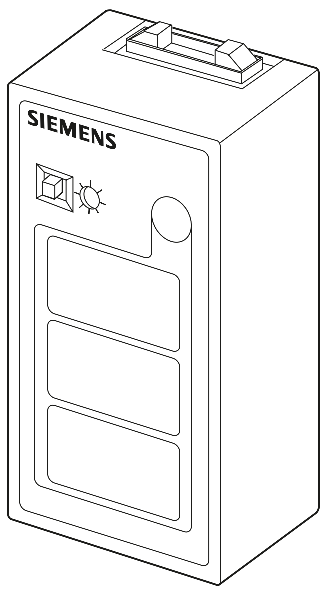


Figure 15-8 Battery supply device

ESD guidelines

A

A.1 ESD Directive

What does ESD mean?

Almost all electronic modules are equipped with highly integrated components and elements in MOS technology. For technological reasons, these electronic components are very sensitive to overvoltages and, consequently, to electrostatic discharge. These components are therefore marked as follows:

- **ESD**: **E**lectrostatically **S**ensitive **D**evices
- **ESD**: Internationally recognized marking for components and modules susceptible to electrostatic discharge

The following symbols on switch cabinets, module carriers or packaging indicate their susceptibility to electrostatic discharge:

ESD components are destroyed by voltage and energy far below the limits of human perception. Voltages of this kind occur as soon as a device or an assembly is touched by a person who is not electrostatically discharged ESD components which were subject to such voltage are usually not recognized immediately as being defective, because the malfunction does not occur until after a longer period of operation.

Note

More information is located on the rating plate. The rating plate is described in the chapter "Planning use."

Precautions against electrostatic discharge

Most plastics can be charged easily. Therefore, keep plastics away from ESD components! When working with electrostatically sensitive components, make sure that the person, the workstation and the packaging are properly grounded. Conduct the electrostatic charge away from your body by touching the mounting plate for the interfaces, for example.

Handling ESD modules

The following applies: Only touch ESD components if unavoidable due to necessary tasks. Only touch the components when the following holds true:

- You are permanently grounded by means of an ESD armband.
- You are wearing ESD shoes or ESD shoes grounding protective strips in connection with ESD floors.

Before you touch an electronic assembly, your body must be discharged. To do this, touch a conductive, grounded object, e.g., a bare metal part of a switch cabinet or the water pipe, immediately before touching the electronic assembly.

Do not allow chargeable, highly insulated materials, e.g. plastic films, insulating tabletops, synthetic clothing fibers, to come into contact with ESD components.

Place ESD components only on conductive surfaces (work surfaces with ESD surface, conductive ESD foam, ESD packing bag, ESD transport container).

Do not expose ESD components to visual display units, monitors or televisions. Maintain a distance of at least 10 cm to screens.

Handle flat components only by their edges. Do not touch component connectors or conductors. This prevents charges from reaching and damaging sensitive components.

Measuring and modifying ESD components

Measure the ESD component under the following conditions only:

- The measuring device is grounded with a protective conductor, for example.
- The probe on the potential-free measuring device has been discharged, e.g. by touching the bare metal of a part of the switch cabinet.
- Your body is discharged. Do so by touching grounded metallic parts.

Solder only with grounded soldering irons.

Shipping ESD modules

Always store or ship ESD components in conductive packaging, e.g. metallized plastic boxes or metal cans. Leave the components and parts in their packaging until installation.

If the packaging is not conductive, wrap the ESD component in a conductive material, e.g. rubber foam, ESD bag, household aluminum foil, or paper, before packing. Do not wrap the ESD component in plastic bags or plastic film.

In ESD components containing installed batteries, make sure that the conductive packaging does not touch the battery connectors or short circuit. Insulate the connectors with suitable material.

Appendix B

B.1 Selectivity

Information about the calculated selectivity limits

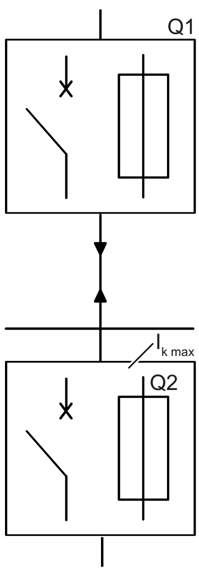


Figure B-1 Circuit breakers connected in series

- The selectivity limits given here refer to
  - the dynamic selectivity.
    That is, the dynamic behavior of the upstream and downstream protective devices in the time period up to 80 ms is shown. This range is the tripping range of the instantaneous short-circuit current release (I release) on the circuit breakers.
  - The rated operating voltage V_e to 415 V and 50 Hz
  - Short-circuit values of a dead three-phase short-circuit.
  - However, in practice approximately 70-80% of short-circuits are single-phase short circuits and significantly lower.
- In addition to taking the dynamic selectivity limits into consideration (= values in this table), selectivity can also be determined by comparing the characteristic curves of the device in the overload range (L range) and the short-time delayed short-circuit range (S range).
  For some combinations, the use of release options such as "selectable characteristic curves" or ZSI can solve the problem of overlap.

- For circuit breakers with LI and LSI characteristics, it is important to ensure that the appropriate I_i and/or I_{sd} protection settings fulfill the protection function in TN/TT/IT networks. This means the I_i or I_{sd} protection setting must switch off a short-circuit within 5 s (for fixed loads) or 0.4 s (for mobile loads).

NOTICE
I_i and I_{sd}protection settings The I_i and I_{sd} protection settings also influence the selectivity behavior of upstream and downstream devices.

- Current selectivity:
  In general, only partial selectivity can be achieved using current grading (current selectivity through the use of LI releases)! That is, complete selectivity can only be achieved up to the setpoint value of the instantaneous short-circuit release of the upstream protection device I_iQ1 minus 20%, rather than up to the maximum short-circuit current I_{kmax}! (See EN 60947 T2)
- Circuit breakers with an adjustable time delay are often required to achieve total selectivity between two protective devices. To calculate the selectivity limit values, the time delay t_{sd} of the LSI releases is always set to the first time level and I_d to the maximum value.
- Time selectivity:
  If the maximum short-circuit currents at the mounting points are approximately the same (e.g. on the main distribution board), the upstream circuit breaker (Q1) requires a delayed short-circuit release (S release). It must not have an instantaneous release (I release) unless the I function must be switched "Off". The microprocessor-controlled "Zone-Selective Interlocking (ZSI)" feature has been developed by SIEMENS to prevent long, undesired tripping times in the case of series-connected circuit breakers. ZSI enables the tripping delay to be reduced to a maximum of 50 ms for the circuit breaker upstream from the location of the short-circuit.
- More information on this topic:
  Additional selectivity limit tables can be obtained on request from our Technical Assistance. Characteristic curves programs Simaris deSign - configuration software

- No characteristics/trips are listed for fuses. The fuse types have the following operating classes:

Type	Operating class
3NA	gL / gG
5SA1	"quick-response"
5SA2	"time-lag"
5SB1/3	"quick-response"
5SB2/4	"time-lag"
5SC1	"quick-response"
5SC2	"time-lag"
5SD4/5	gR
5SD6	"quick-response"
5SE2	gL / gG

- Explanation of the abbreviations:

Abbreviation	Explanation
line	for line protection
motor	for motor protection
starter	for starter combinations
insulation circuit breaker	Switch disconnectors
I_R	Current value of the overload release
I_{sd}	Current value of the short-time delayed short-circuit release
t_{sd}	Delay time of the short-time delayed short-circuit release
I_i	Current value of the instantaneous short-circuit release
I_{cn}	Rated short-circuit breaking capacity
TM	Thermomagnetic trip unit
ETU	Electronic trip unit
Settings of the LI and LSI releases of the upstream and downstream protective devices for calculating the selectivity limits:	
I_R	1 x I_r
I_{sd}	max.
t_{sd}	^{3} 100 ms
I_i	max.

B.2 Conversion tables

The american units can be converted to the corresponding European/metric units using the conversion tables listed.

Metric/US American cross-sections

Metric cross-sections in accordance with VDE (Verband Deutscher Elektroingenieure (Association of German Electrical Engineers)) (mm²) ↔ conductor cross-sections in accordance with AWG (American Wire Gauge) or MCM (Thousand Circular Mils)

Table B- 1 Conversion table AWG / MCM ↔ mm²

	AWG/MCM	mm²
AWG	20	0.52
	18	0.82
	16	1.3
	14	2.1
	12	3.3
	10	5.3
	8	8.4
	6	13.3
	4	21.2
	2	33.6
	1	42.4
	1 / 0	53.5
	2 / 0	67.4
	3 / 0	85.0
	4 / 0	107.2
MCM	250	126
	300	152
	350	177
	400	203
	500	253
	600	304
	800	405
	1000	507
	1500	760
	2000	1010

Other conversions

Table B- 2 Conversion factors of different sizes

Conversion factors	
Power	
1 kilowatt (kW)	1.341 horsepower (hp)
1 horsepower (hp)	0.7457 kilowatt (kW)
Lengths	
1 inch (in.)	25.4 millimeters (mm)
1 centimeter (cm)	0.3937 inch (in.)
Weight	
1 ounce (Oz.)	28.35 grams (g)
1 pound (lb.)	0.454 kilograms (kg)
1 kilogram	2.205 pounds (lb.)
Temperature	
100 degrees Centigrade (°C)	212 degrees Fahrenheit (°C)
80	176
60	140
40	104
20	68
0	32
-5	23
-10	14
-15	5
-20	-4
-25	-13
-30	-22
Tightening torque	
1 Newton-meter (Nm)	8.85 pound-inches (lb.in.)

We cannot guarantee the exhaustiveness of the listed units of measurement.

B.3 Standards and specifications

The SENTRON VL circuit breakers fulfill:

- IEC 60947-2 / DIN EN 60947-2 (VDE 0660-101)
- IEC 60947-1 / DIN EN 60947-1 (VDE 0660-100)

Disconnector properties in accordance with:

- IEC 60947-3 / DIN EN 60947-3 (VDE 0660-107)

Please contact SIEMENS for additional standards.

The overcurrent trip units of the circuit breakers for motor protection additionally fulfill:

- IEC 60947-4-1 / DIN EN 60947-4-1 (VDE 0660-102)

Network disconnecting device (used to be called "main switch" in accordance with:

- IEC 60204-1 / DIN EN 60204-1 (VDE 0113-1) (refer to Application area)

Network disconnecting device for stopping and shutting down in an emergency (used to be called "EMERGENCY-OFF switch") in accordance with:

- IEC 60204-1 / DIN EN 60204-1 (VDE 0113-1) (refer to Application area)

The following certificates are available on request:

CE certificate of conformity

- Type examination certificate IEC 60947
- Type examination certificate CCC (China)
- Shipbuilding approvals (GL, LRS, DNV)
- Certificate of origin
- Halogen-free
- PVC-free

The VL160X–VL400 circuit breakers that are equipped with a SENTRON VL RCD module, correspond to IEC 60947-2 Annex B.

The RCD module SENTRON VL corresponds to IEC 61000-4-2 to 61000-4-6, IEC 61000-4-11 and EN 55011, Class B (corresponds to CISPR 11) with regard to electromagnetic compatibility.

The reference temperature for the RCD modules and the SENTRON VL circuit breakers is 40 °C. The suitability of the RCD module for mounting on the SENTRON VL circuit breakers has no effect on the characteristic key data of the circuit breaker, such as:

- Rated voltage (50 / 60 Hz), switching capacity
- Electrical and mechanical service life
- Connections
- Operating mechanisms (VL160, VL250, VL400)
- Auxiliary switches and trip units

Rated current, see "Use in harsh environments".

In accordance with DIN 40713, the graphical symbols that the internal circuit diagrams contain only provide information on the type, connection and mode of operation of devices, but not on their type of construction.

Shock resistance

All SENTRON VL circuit breakers have shock resistance in accordance with the test procedures defined in IEC 68 Part 2.

B.4 Ordering data

Order number scheme

The table below describes the order number scheme (MLFB) according to which all circuit breakers can be located and combined to suit the individual application:

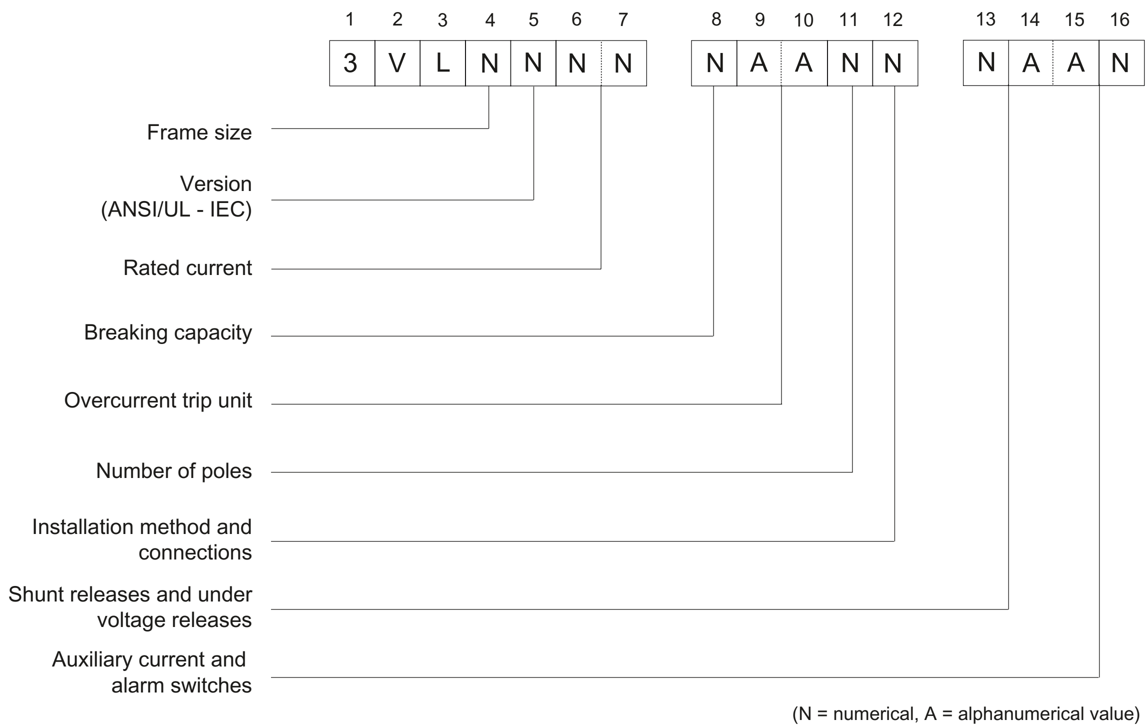


Figure B-2 Overview of the MLFB system

From (please complete):
Name

Company/Department

Address

To
SIEMENS AG
I IA CD MM3
P.O.Box 1954

92220 Amberg / Germany

Fax: 09621 / 80-3337

SENTRON 3VL System Manual

Have you noticed any errors while reading this manual?

If so, please use this form to tell us about them.

We welcome comments and suggestions for improvement.

Glossary

AC (alternating current)

AC current

ATEX Directive

European Union Directive on explosion protection.

Bistable connection

Connection with two stable states.

Contactor

Electromagnetically actuated switch. Control current flows through a magnetic coil and activates the switch. While current flows, the ON position is maintained. A contactor has a higher switching capacity than a relay.

Controller monitoring

Automatically switches a control element in a process sequence when an event has taken place that interrupts the process sequence.

Current protection

Current limiting switching function that prevents excessive current flow or controls current flow.

Dahlander connection

Connection in which the motor stator winding is divided into symmetrical winding sections. A changeover results in changes to the number of poles and thus speed changes in the ratio 1:2.

DC (direct current)

DC current

Delta connection

Connection in which the three phase lines of a three-phase system are connected in series. The end of a phase line is connected with the start of the next phase line.

Differential current

Conductor-related flow measurement over time

Direct starter

Function for switching motors on and off.

ESD

Components sensitive to electrostatic charge

ETU (electronic trip unit)

Electronic trip unit

Frequency converters

Frequency-dependent (voltage/current) infeed for operating AC motors.

Frequency-independent load protection

Compensation of different signals at defined control potentials.

G

Ground fault

GF (ground fault)

Ground fault

GND (ground)

Chassis ground

Ground fault

Fault whereby an external conductor comes into contact with ground or the grounded neutral point.

GSD

Device master data

Harmonic protection

Protection against harmonics at DC voltage levels.

HS

Auxiliary switches

Hysteresis

Continuation of an effect after removal of the cause.

I

Instantaneous (instantaneous short-circuit protection)

INST (instantaneous)

Instantaneous short circuit

IP (International Protection)

International degree of protection

L

Long-time delay (overload protection)

LCD (liquid crystal display)

Liquid crystal display

LCD ETU

Electronic trip unit with LCD display

LED (light emitting diode)

Light emitting diode

Limit monitoring

Checking of measured values for violation of defined upper or lower limits.

Load management

Control of energy consumption through selective connection, disconnection and regulation of loads.

LTD (long-time delay)

Long-time delay

M

Magnetic

MCCB (molded-case circuit breaker)

Molded-case circuit breaker

Measured value display

Graphic or alphanumeric display of a measured quantity.

MLFB

Machine-readable product designation

N

Neutral protection

NC (no connect)

No connect

Neutral conductor

Conductor in AC systems that, in contrast to the outer conductor, is not electrically live. It is used for returning current when the circuit is closed.

NH (normal high)

Normal high

NHL (normal high large)

Normal high large

Outer conductor

AC voltage distribution board (230 V) in three-conductor networks. Also called phase conductor or phase.

Overload protection

Protection against excessive load on the electrical components.

Phase failure

Electrodynamic power loss on multi-phase conductors controlling AC motors.

Position signaling switch

Specifies the status of the circuit breaker tripping.

Power loss

Power dissipated as heat when operating an electrical component.

PROFIBUS

PROFIBUS (Process Field Bus) is a standard for fieldbus communication in automation technology. PROFIBUS exists in three versions:
PROFIBUS-FMS (Fieldbus Message Specification) for networking controllers.
PROFIBUS-DP (distributed I/O) for controlling sensors and actuators via a central controller in production engineering.
PROFIBUS-PA (Process Automation) for controlling field devices via a process control system in process engineering.

PROFIBUS (Process Field Bus)

Standard for fieldbus communication in automation systems

RCD (residual current device)

Resistance Capacitor Wiring

Rectifier

For converting alternating current to direct current.

Relay

Electromagnetically actuated switch. The relay is activated via a galvanically isolated control circuit and can close, open or switch one or more load current circuits. A relay has a lower switching capacity than a contactor.

Remote control

Initiation of a switching operation through a PLC.

Reversing starter

Starting control function for the direction of rotation (CW/CCW) of motors.

S

Short-time delay

Selectivity

Detection of the tripping state of an event.

Soft starters

Function for starting/stopping motors smoothly.

Star connection

Connection in which the three phase elements of a three-phase system are each connected to one end of a motor winding. The connection created in this way forms the center point - also known as the star point.

State detection

Detection and logging of states and state changes in the energy distribution system.

State display

Graphic representation of the states of objects.

STD (short-time delay)

Short-time delay

Stored-energy spring mechanism

Mechanical memory that maintains a defined state over a specific time in response to an event.

Switching capacity

Switching characteristics of a switch that specifies the frequency of switching operations over a specific period of time.

Thermal memory

Checks the motor-size-dependent cooling performance that prevents overheating of the motor when it switches on again following an overload trip.

TM

Thermomagnetic trip unit

Tripping solenoid

Electrically operated magnet that initiates a switching operation (switching relay).

Voltage protection

Voltage switching function that implements a defined state when an event occurs.

Service & Support

Newsletter – always up to date:
www.siemens.com/lowvoltage/newsletter

E-business in the Industry Mall:
www.siemens.com/lowvoltage/mall

OnlineSupport:
www.siemens.com/lowvoltage/support

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e-mail: technical-assistance@siemens.com
www.siemens.com/lowvoltage/technical-assistance

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